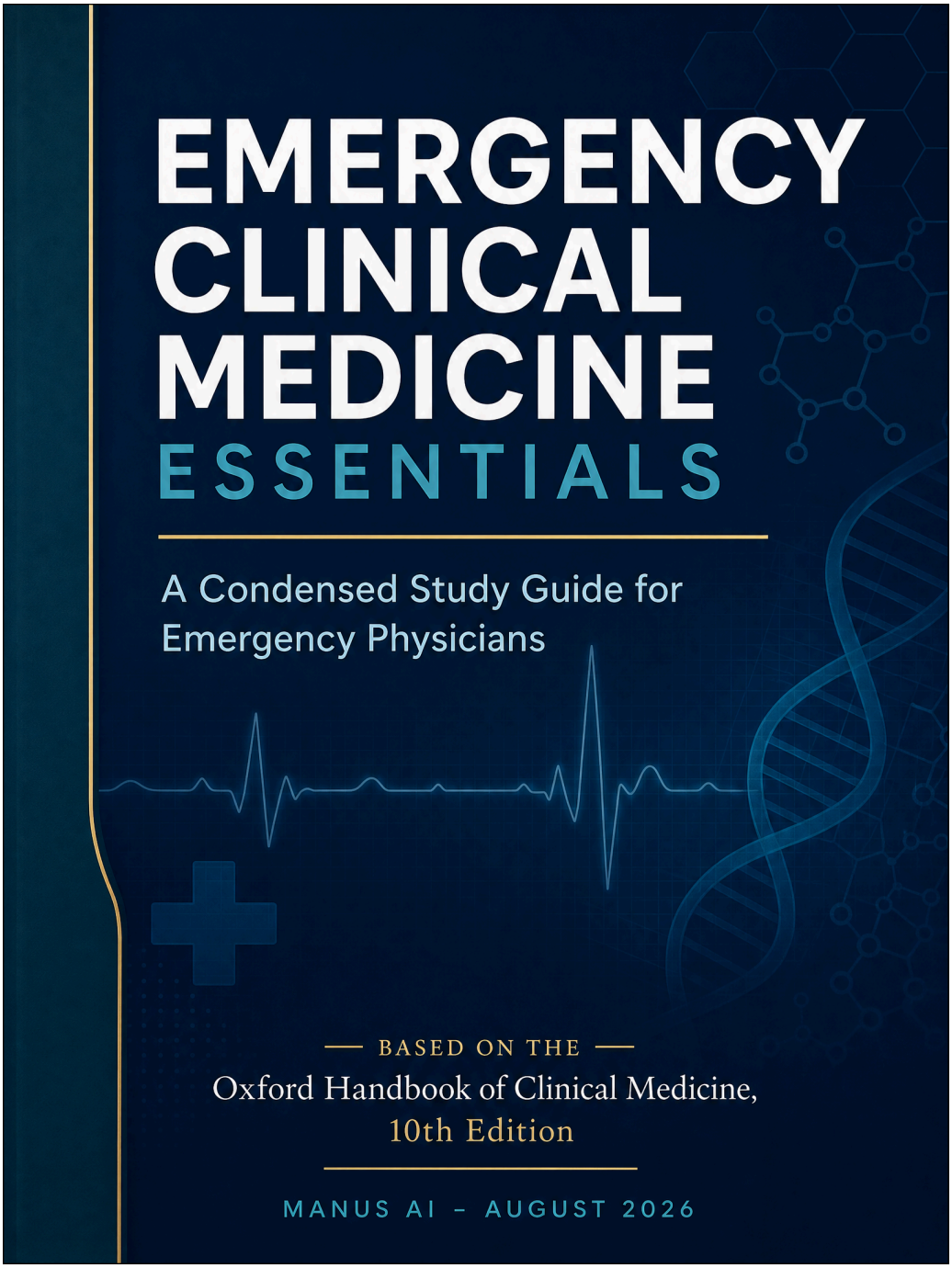

EMERGENCY
CLINICAL
MEDICINE
ESSENTIALS
A Condensed Study Guide for Emergency Physicians
BASED ON THE
Oxford Handbook of Clinical Medicine,
10th Edition
MANUS AI – AUGUST 2026

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Cardiovascular Medicine

Chest pain ►Cardiac-sounding chest pain may have no serious cause, but always think 'Could this be a myocardial infarction (MI), dissecting aortic aneurysm, pericar-ditis, or pulmonary embolism?'.

Character: Constricting suggests angina, oesophageal spasm, or anxiety; a sharp pain may be from the pleura, pericardium, or chest wall. A prolonged (>½h), dull, central crushing pain or pressure suggests MI.

Radiation: To shoulder, either or both arms, or neck/jaw suggests cardiac ischae-mia. The pain of aortic dissection (p654) is classically instantaneous, tearing, and interscapular, but may be retrosternal. Epigastric pain may be cardiac.

Precipitants: Pain associated with cold, exercise, palpitations, or emotion suggests cardiac pain or anxiety; if brought on by food, lying flat, hot drinks, or alcohol, con-sider oesophageal spasm/disease (but meals can also cause angina).

Relieving factors: If pain is relieved within minutes by rest or glyceryl trinitrate (GTN), suspect angina (GTN relieves oesophageal spasm more slowly). If antacids help, suspect GI causes. Pericarditic pain improves on leaning forward.

Pleuritic pain: Pain exacerbated by inspiration. Implies inflammation of the pleura from pulmonary infection, inflammation, or infarction. It causes us to 'catch our breath'. ΔΔ: musculoskeletal pain;1 fractured rib (pain on respiration, exacerbated by gentle pressure on the sternum); subdiaphragmatic pathology (eg gallstones).

2Chest pain & acutely unwell (see p784) • Admit • Check pulse, BP in both arms (unequal in aortic dissection p654), JVP, heart sounds; examine legs for DVT • Give O2

IV line • Relieve pain (eg 5–10mg IV morphine) • Cardiac monitor • 12-lead ECG • CXR

Arterial blood gas (ABG) Famous traps: Aortic dissection; zoster (p404); ruptured oesophagus; cardiac tamponade (p154); opiate addiction.

Palpitation(s) May be due to ectopics, sinus tachycardia, AF, SVT, VT, thyrotoxicosis, anxiety, and rarely phaeochromocytoma. See p26. History: Characterize: do they mean their heart was beating fast, hard, or irregularly? Ask about previous episodes, precipitating/relieving factors, duration of symptoms, associated chest pain, dysp-noea, dizziness, or collapse. Did the patient check their pulse?

25% of non-cardiac chest pain is musculoskeletal: look for pain on specific postures or activity. Aim to reproduce the pain by movement and, sometimes, palpation over the structure causing it. Focal injection of local anaesthetic helps diagnostically and is therapeutic. Tietze's syndrome: self-limiting costochondritis

! costosternal joint swelling. Causes: idiopathic; microtrauma; infection; psoriatic/rheumatoid arthritis. R: NSAIDS or steroid injections. Tenderness is also caused by: fibrositis, lymphoma, chondrosarcoma, myeloma, metastases, rib TB. Imaging: bone scintigraphy; CT.

Chest pain from the patient's perspective 95

Dialogue-transformed symptoms explain one of the junior doctor's main vexations: when patients retell symptoms to a consultant in the light of day, they bear no resemblance to what you originally heard. But do not be vexed: your dialogue may have helped the patient far more than any ward round.

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Reading an ECG

First confirm the patient's name and age, and the ECG date. Then (see fig 2.2):

Rate: At usual speed (25mm/s) each 'big square' is 0.2s; each 'small square' is 0.04s. To calculate the rate, divide 200 by the number of big squares between two con-secutive R waves (table 2.1). The normal rate is 60–100bpm.

Rhythm: If cycles are not clearly regular, use the 'card method': lay a card along the ECG, marking positions of three successive R waves. Slide the card to and fro to check that all intervals are equal. If they are not, note if:

there is slight but regular lengthening and then shortening (with respiration)—sinus arrhythmia, common in the young

there are different rates which are multiples of each other—varying block

it is 100% irregular—atrial fibrillation (AF) or ventricular fibrillation (VF).

Sinus rhythm is characterized by a P wave followed by a QRS complex. AF has no dis-cernible P waves and QRS complexes are irregularly irregular. Atrial flutter (p120, fig 2.25 p121) has a 'sawtooth' baseline of atrial depolarization (r200/min) and regular QRS complexes. Ventricular rhythm has QRS complexes >0.12s with P waves follow-ing them or absent (fig 2.12, p106).

P wave: Normally precedes each QRS complex, and upright in II, III, & aVF but invert-ed in aVR. Absent P wave: AF, P hidden due to junctional or ventricular rhythm. P mi-trale: bifid P wave, indicates left atrial hypertrophy. P pulmonale: peaked P wave, indicates right atrial hypertrophy. Pseudo-P-pulmonale seen if •K+.

PR interval: Measure from start of P wave to start of QRS. Normal range: 0.12–0.2s

(2–5 small squares). A prolonged PR interval implies delayed AV conduction (1st degree heart block). A short PR interval implies unusually fast AV conduction down an accessory pathway, eg WPW (see fig 2.27, p122). See heart block, p98.

R R R R

S Q S Q Q S

Fig 2.2 'QRS' complexes. If the first deflection from the isoelectric line is negative, it is a Q wave. Any positive deflection is an R wave. Any negative deflection after an R is an S.

QT interval: Measure from start of QRS to end of T wave. It varies with rate. The corrected QT interval (QTc) is the QT interval divided by the square root of the R–R interval, ie QTc=QT/√RR. Normal QTc: 0.28–0.42s. For causes of prolonged QT interval see p711. Long QT can lead to VT and sudden death.

ST segment: Usually isoelectric. Planar elevation (>1mm) or depression (>0.5mm) usu-ally implies infarction (p119, figs 2.9, 2.10, pp102–4) or ischaemia, respectively.

T wave: Normally inverted in aVR, V1, and occasionally V2. Normal if inverted in isola-tion in lead III. Abnormal if inverted in I, II, and V4–V6. Peaked in hyperkalaemia (fig 14.4, p675) and flattened in hypokalaemia.

J wave: See p849. The J point is where the S wave finishes and ST segment starts. A

J wave is a notch at this point. Seen in hypothermia, SAH, and •Ca2+.

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Sinus tachycardia All impulses are initiated in the sinoatrial node ('sinus rhythm') hence all QRSS are preceeded by a normal P wave with a normal PR interval. Tachycar-dia means rate >100bpm. See p127.

Sinus bradycardia Sinus rhythm at a rate <60bpm. Causes: Physical fitness, vas-ovagal attacks, sick sinus syndrome, drugs (β-blockers, digoxin, amiodarone), hypo-thyroidism, hypothermia, •intracranial pressure, cholestasis. See p122.

AF (ECG p125) Common causes: IHD, thyrotoxicosis, hypertension, obesity, heart fail-ure, alcohol. See p120.

Heart block (HB) (See fig 2.5.) Disrupted passage of electrical impulse through the

AV node.

1st-degree HB: The PR interval is prolonged and unchanging; no missed beats.

2nd-degree HB: Mobitz I: The PR interval becomes longer and longer until a QRS is missed, the pattern then resets. This is Wenckebach phenomenon.

2nd-degree HB: Mobitz II: QRSs are regularly missed. eg P - QRS - P - - P - QRS - P - - this would be Mobitz II with 2:1 block (2P:1QRS). This is a dangerous rhythm as it may progress to complete heart block.

1st- and 2nd-degree HB may be caused by: normal variant, athletes, sick sinus syn-drome, IHD (esp inferior MI), acute myocarditis, drugs (digoxin, β-blockers).

ST elevation Normal variant (high take-off), acute MI (STEMI), Prinzmetal's angina (p708), acute pericarditis (saddle-shaped), left ventricular aneurysm.

ST depression Normal variant (upward sloping), digoxin toxicity (downward sloping), ischaemic (horizontal): angina, NSTEMI, acute posterior MI (ST depression in V1–V2).

T inversion In V1–V2: normal (black patients and children), right bundle branch block (RBBB), RV strain (eg secondary to PE). In V2–V5: anterior ischaemia, HCM, subarachnoid haemorrhage, lithium. In V4–V6 and aVL: lateral ischaemia, LVH, left bundle branch block (LBBB). In II, III and aVF: inferior ischaemia.

NB: ST- and T-wave changes are often non-specific, and must be interpreted in the light of the clinical context.

Myocardial infarction (See p118 and fig 2.21; example ECGs figs 2.9, 2.10)

Within hours, the T wave may become peaked and ST segments may begin to rise.

Within 24h, the T wave inverts. ST elevation rarely persists, unless a left ventricular aneurysm develops. T-wave inversion may or may not persist.

Within a few days, pathological Q waves begin to form. Q waves usually persist, but may resolve in 10% of patients.

The location of these changes indicates the ischaemic area location, see table 2.2.

Pulmonary embolism (fig 2.11) ECG findings may include: sinus tachycardia (com-monest), RBBB (p100), right ventricular strain pattern (R-axis deviation, dominant R wave and T-wave inversion/ST depression in V1 and V2,).

Rarely, the 'SIQIIITIII' pattern occurs: deep S waves in I, pathological Q waves in III, inverted T waves in III.

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Complete AV block with narrow ventricular complex.

There is no relation between atrial and the slower ventricular activity.

100

Where to place the chest leads (See fig 2.6.) V1 : Right sternal edge, 4th intercostal space. V2 : Left sternal edge, 4th intercostal space. V3 : Half-way between V2 and V4.

V4 : 5th intercostal space, mid-clavicular line; all

subsequent leads are in the same horizontal plane as V4.

V5 : Anterior axillary line.

V6 : Mid-axillary line (V7: posterior axillary line).

Good skin preparation (clean with non-alcoholic wipe, shave if hairy, etc.) will improve ECG qual-ity. Finish 12-lead ECGs with a long rhythm strip in lead II.

QRS complexes: the long and the short

QRS complexes represent ventricular depolariza-tion, and width represents time, so a broader QRS complex means depolarization of the ventricles is

Ventricular depolarization takes longer when depolarization is not initiated in this pattern. For example, if it originates in the ventricles (eg ventricular ectopics, VT) or if one or more branches of the bundles of His are blocked —bundle branch blocks

—meaning depolarization is initiated in one ventricle but not the other, so it has to travel the long (in time and space) path from one ventricle to the other.

Ventricular depolarization also takes longer if all conduction is slowed. This may happen in some electrolyte imbalances, eg hyperkalaemia.

Right bundle branch block: (p102, fig 2.8) QRS >0.12s, 'RSR' pattern in V1; dominant R in V1; inverted T waves in V1–V2 or V4; wide, slurred S wave in V6. Causes: normal variant (isolated RBBB), pulmonary embolism, cor pulmonale.

Left bundle branch block: (p101, fig 2.7) QRS >0.12s, 'M' pattern in V5, dominant S in V1, inverted T waves in I, aVL, V5–V6. Causes: IHD, hypertension, cardiomyopathy, idi-opathic fibrosis. ►NB: if there is LBBB, no comment can be made on the ST segment or T wave. ►►New LBBB may represent a STEMI, see p798.

Bifascicular block: The combination of RBBB and left bundle hemiblock, manifest as an axis deviation, eg left axis deviation in the case of left anterior hemiblock.

Trifascicular block: Bifascicular block plus 1st-degree HB. ►May need pacing (p122).

Suspect left ventricular hypertrophy (LVH) if the R wave in V6 is >25mm or the sum of the S wave in V1 and the R wave in V6 is >25mm (see fig 2.41).

Suspect right ventricular hypertrophy (RVH) if dominant R wave in V1, T wave inver-sion in V1–V2 or V4, deep S wave in V6, right axis deviation.

Other causes of dominant R wave in V1: RBBB, posterior MI, type A WPW syndrome (p122).

See lifeinthefastlane.com for excellent ECG tutorials, cases, and examples.

Fig 2.7 Left bundle branch block: wide QRS with a W pattern in V1 (slight notching in upstroke of S wave—clearer in V3) and the M pattern in V6. WiLLiaM = LBBB.

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Fig 2.8 Right bundle branch block—broad QRS, M pattern in V1 and sloped S wave (with the eye of faith, a 'W' shape) in V5. MaRRoW = RBBB.

Fig 2.9 Acute infero-lateral myocardial infarction: marked ST elevation in the inferior leads (II, III, aVF), but also in V5 and V6, indicating lateral involvement. There is a 'reciprocal change' of ST-segment depression in leads I and aVL; this is often seen with a large inferior myocardial infarction.

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Fig 2.10 Acute anterior myocardial infarction—ST segment elevation and evolving Q waves (the first QRS deflection is negative) in leads V1–4.

Right axis deviation (QRS more negative than positive in lead I);

Positive QRS complexes ('dominant R waves') in V1 and V2 suggesting right ventricular hypertrophy;

ST depression and T-wave inversion in the right precordial leads (V1–2) suggesting right ventricular strain;

Peaked P waves (P pulmonale) suggesting right atrial hypertrophy.

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Fig 2.12 Ventricular tachycardia—regular broad complex tachycardiac indicating a likely ventricular origin for the rhythm.

Fig 2.12 Dual chamber pacemaker. Pacing spikes occur before each P wave and each QRS complex. Paced QRS complexes are broad as the impulse starts in the ventricles.

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Fig 2.14 Two chest X-rays of the same patient, the one on the right was taken 6 months after the one on the left. On the later image, a pericardial effusion has expanded the cardiac shadow and given it a ‘globular’ shape.

University Press.

Fig 2.16 Cardiac MR image demonstrat-ing the asymmetrical left ventricular wall thickening typical of hypertrophic cardio-myopathy.

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This non-invasive technique uses the differing ability of various structures within the heart to reflect ultrasound waves. It not only demonstrates anatomy but also provides a continuous display of the functioning heart throughout its cycle.

Types of scan

M-mode (motion mode): A single-dimension image.

Two-dimensional (real time): A 2D, fan-shaped image of a segment of the heart is produced on the screen (fig 2.17); the moving image may be ‘frozen’. Several views are possible, including long axis, short axis, 4-chamber, and subcostal. 2D echocar-diography is good for visualizing conditions such as: congenital heart disease, LV aneurysm, mural thrombus, LA myxoma, septal defects.

3D echocardiography: Now possible with matrix array probes, and is termed 4D (2D + time) if the images are moving.

Doppler and colour-flow echocardiography: Different coloured jets illustrate flow and gradients across valves and septal defects (p156) (Doppler effect, p726).

Tissue Doppler imaging: This employs Doppler ultrasound to measure the velocity of myocardial segments over the cardiac cycle. It is particularly useful for assessing longitudinal motion—and hence long-axis ventricular function, which is a sensitive marker of systolic and diastolic heart failure.

Transoesophageal echocardiography (TOE): More sensitive than transthoracic echo-cardiography (TTE) as the transducer is nearer to the heart. Indications: diagnosing aortic dissections; assessing prosthetic valves; finding cardiac source of emboli, and IE/SBE. Contraindicated in oesophageal disease and cervical spine instability.

Stress echocardiography: Used to evaluate ventricular function, ejection fraction, myocardial thickening, regional wall motion pre- and post-exercise, and to charac-terize valvular lesions. Dobutamine or dipyridamole may be used if the patient can-not exercise. Inexpensive and as sensitive/specific as a thallium scan (p741).

Uses of echocardiography

Echo is also useful for detecting focal and global hypokinesia, LV aneurysm, mural thrombus, and LVH (echo is 5–10 times more sensitive than ECG in detecting this).

Estimating right heart haemodynamics: Doppler studies of pulmonary artery flow and tricuspid regurgitation allow evaluation of RV function and pressures.

Valve disease: The technique of choice for measuring pressure gradients and valve orifice areas in stenotic lesions. Detecting valvular regurgitation and estimating its significance is less accurate. Evaluating function of prosthetic valves is another role.

Congenital heart disease: Establishing the presence of lesions, and significance. Endocarditis: Vegetations may not be seen if <2mm in size. TTE with colour Doppler is best for aortic regurgitation (AR). TOE is useful for visualizing mitral valve vegeta-

tions, leaflet perforation, or looking for an aortic root abscess.

Pericardial effusion: Best diagnosed by echo. Fluid may first accumulate between the posterior pericardium and the left ventricle, then anterior to both ventricles and anterior and lateral to the right atrium. There may be paradoxical septal motion.

HCM: (p152) Echo features include asymmetrical septal hypertrophy, small LV cavity, dilated left atrium, and systolic anterior motion of the mitral valve.

(a)

(b)

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Septum

Inferolateral

RV

Ascending

aorta AV

wall LV

Right coronary cusp

LA Non coronary cusp

MV

Anterior leaflet Posterior leaflet

(c)

Descending aorta

(d)

Ventricular septum

Lateral wall (papillary muscle usually not seen)

RV LV

TV RA

Lateral leaflet

Septeral leaflet

MV

Posterior leaflet

LA Anterior leaflet

Interatrial septum RUPV

Fig 2.17 Echo images. (a) A normal heart seen with the parasternal long-axis view. (b) Diagram of what can be seen in (a). (c) A normal heart seen in apical four-chamber view. (d) Diagram of what can be seen in (c).

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This involves the insertion of a catheter into the heart via the femoral or radial artery or venous system, and manipulating it within the heart and great vessels to:

Inject radiopaque contrast medium to image cardiac anatomy and blood flow, see

fig 2.18a.

Perform angioplasty (ballooning and stenting), valvuloplasty (eg transcatheter aortic valve implantation (TAVI, fig 2.45)), cardiac biopsies, transcatheter septal defect closure.

Perform electrophysiology studies and radiofrequency ablations.

Sample blood to assess oxygen saturation and measure pressures.

Perform intravascular ultrasound or echocardiography.

During the procedure, ECG and arterial pressures are monitored continuously. In the UK, the majority are performed as day-case procedures.

Indications

Coronary artery disease: diagnostic (assessment of coronary vessels and graft pa-tency); therapeutic (angioplasty, stent insertion), fig 2.18b.

Valvular disease: diagnostic (pressures indicate severity); therapeutic valvuloplas-ty (if the patient is too ill or declines valve surgery).

Congenital heart disease: diagnostic (assessment of severity of lesions by meas-uring pressures and saturations); therapeutic (balloon dilatation or septostomy).

Other: cardiomyopathy; pericardial disease; endomyocardial biopsy.

Pre-procedure checks

Brief history/examination; NB: peripheral pulses, bruits, aneurysms.

Investigations: FBC, U&E, LFT, clotting screen, CXR, ECG.

Consent for procedure, including possible extra procedures, eg consent for angio-plasty if planning to do angiography as you may find a lesion that needs stenting. Explain reason for procedure and possible complications.

IV access, ideally in the left hand.

Patient should be nil by mouth (NBM) from 6h before the procedure.

Patients should take all their morning drugs (and pre-medication if needed)—but withhold oral hypoglycaemics.

Post-procedure checks

Pulse, BP, arterial puncture site (for bruising or swelling), foot pulses.

Investigations: FBC and clotting (if suspected blood loss), ECG.

Complications

Haemorrhage: apply firm pressure over puncture site. If you suspect a false aneu-rysm, ultrasound the swelling and consider surgical repair. Haematomas are high risk for infections.

Contrast reaction: this is usually mild with modern contrast agents.

Loss of peripheral pulse: may be due to dissection, thrombosis, or arterial spasm. Occurs in <1% of brachial catheterizations. Rare with femoral catheterization.

Angina: may occur during or after cardiac catheterization. Usually responds to sublingual GTN; if not, give analgesia and IV nitrates.

Arrhythmias: usually transient. Manage along standard lines.

Pericardial effusion: suspect if unexplained continued chest pain. May need drain depending on severity and haemodynamic status.

Pericardial tamponade: rare, but should be suspected if the patient becomes hypo-tensive and anuric. ☠☠ Urgent pericardial drain.

Infection: post-catheter pyrexia is usually due to a contrast reaction. If it persists for >24h, take blood cultures before giving antibiotics.

Mortality <1 in 1000 patients, in most centres.

Intracardiac electrophysiology This catheter technique can determine types and origins of arrhythmias, and locate and ablate problem areas, eg aberrant pathways in WPW or arrhythmogenic foci. Arrhythmias may be induced, and the effectiveness of control by drugs assessed.

(a) 112

(b) (c)

Images (b) and (c) reproduced from Ramrakha et al., Oxford Handbook of Cardiology, 2012, with permission from Oxford University Press.

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Diuretics

Loop diuretics (eg furosemide) are used in heart failure, and inhibit the Na/2Cl/K co-transporter. SES: dehydration, ‘Na+,’K+, ‘Ca2+, ototoxic

Thiazides and thiazide-like diuretics are used in hypertension (eg indapamide) and heart failure (eg metolazone). SE: ‘K+, •Ca2+, ‘Mg2+, •urate (!gout), impotence (NB: small doses, eg chlortalidone 25mg/24h rarely cause significant SEs)

Potassium-sparing diuretics: aldosterone antagonists (eg spironolactone, eplerenone) directly block aldosterone receptors; amiloride blocks the epithelial sodium channel in the distal convoluted tubules.

Vasodilators Used in heart failure, IHD, and hypertension. Nitrates (p116) preferen-tially dilate veins and the large arteries, filling pressure (pre-load), while hydrala-zine (often used with nitrates) primarily dilates the resistance vessels, thus BP (after-load). Prazosin (an a-blocker) dilates arteries and veins.

Calcium antagonists These cell entry of Ca2+ via voltage-sensitive channels in smooth muscle, thereby promoting coronary and peripheral vasodilation and reduc-ing myocardial oxygen consumption. All current drugs block L-type Ca2+ channels. However, their effects differ because of differential binding properties.

The dihydropyridines, eg nifedipine, amlodipine, are mainly peripheral vasodilators (also dilate coronary arteries) and cause a reflex tachycardia, so are often used with a β-blocker. They are used mainly in hypertension and angina.

The non-dihydropyridines—verapamil and diltiazem—also slow conduction at the

AV and SA nodes and may be used to treat hypertension, angina, and dysrhythmias.

Don’t give non-dihydropyridines with β-blockers (risk of severe bradycardia ! LVF). SES: Flushes, headache, ankle oedema (diuretic unresponsive), ‘LV function, gingival hypertrophy. CI: Heart block.

Digoxin Blocks the Na+/K+ pump. It is used to slow the pulse in fast AF (p120; aim for

appetite, yellow vision, confusion, gynaecomastia. If toxicity is suspected, do an ECG (fig 2.19), digoxin levels, and check K+, Mg2+, and Ca2+. If toxicity is confirmed, stop digoxin, correct electrolyte imbalances, treat arrhythmias, and consider IV DigiFab® (p842). CIS: HCM; WPW syndrome (p122).

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Fig 2.19 This ECG shows the classic ‘reverse tick’ of digoxin toxicity: downsloping ST wave with rapid upstroke back to isoelec-tric line. The bradycardia is also suggestive of digoxin toxicity.

Sodium channel blockers Class I anti-arrhythmics. Procainamide (1a) and lido-caine (1b) can be used to terminate VT. NB QT interval may be prolonged. Flecainide (1c) is useful for AF cardioversion in patients without contraindications,

and for ar-rhythmia prophylaxis in patients with WPW or troublesome paroxysmal AF. CIS: Heart failure, IHD, valve disease, and heart block.

Amiodarone A class III anti-arrhythmic. Amiodarone prolongs the cardiac action potential, reducing the potential for tachyarrhythmias. Used in both supra-ventric-ular and ventricular tachycardias, including during cardiac arrest. Broad range of side effects incl. thyroid disease, liver disease, pulmonary fibrosis and peripheral neuropathy. Monitor TFTs and LFTs every 6 months.

Ivabradine Blocks the pacemaker 'funny current', slowing pulse rate without sig-nificantly dropping blood pressure. Used in angina, heart failure, and (off-licence) in autonomic tachycardia syndromes. CIS: Acute MI, bradycardia, long QT syndrome, shock. Many drug interactions, including with calcium antagonists.

Anti-anginal drugs p116. Antihypertensives p140.

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If ACS is a possible diagnosis (including unstable angina), see pp798–801. Angina12 is symptomatic reversible myocardial ischaemia. Features:

Constricting/heavy discomfort to the chest, jaw, neck, shoulders, or arms.

Symptoms brought on by exertion.

Symptoms relieved within 5min by rest or GTN.

All 2 features = typical angina; 2 features = atypical angina; 0–1 features = non-anginal chest pain.

Other precipitants: emotion, cold weather, and heavy meals. Associated symptoms: dyspnoea, nausea, sweatiness, faintness. Features that make angina less likely: pain that is continuous, pleuritic or worse with swallowing; pain associated with palpita-tions, dizziness or tingling.

Causes Atheroma. Rarely: anaemia; coronary artery spasm; AS; tachyarrhythmias;

HCM; arteritis/small vessel disease (microvascular angina/cardiac syndrome X).

Tests ECG usually normal, but may show ST depression; flat or inverted T waves; signs of past MI. Blood tests: FBC, U&E, TFTs, lipids, HbA1c. Consider echo and chest X-ray. Further investigations are usually necessary to confirm an IHD diagnosis—see BOX.

Management

Address exacerbating factors: Anaemia, tachycardia (eg fast AF), thyrotoxicosis.

Secondary prevention of cardiovascular disease:

Stop smoking; exercise; dietary advice; optimize hypertension and diabetes control.

75mg aspirin daily if not contraindicated.

Address hyperlipidaemia—see p690.

Consider ACE inhibitors, eg if diabetic.

PRN symptom relief: Glyceryl trinitrate (GTN) spray or sublingual tabs. Advise the pa-tient to repeat the dose if the pain has not gone after 5min and to call an ambulance if the pain is still present 5min after the second dose. SE: headaches, BP'.

Anti-anginal medication: (p114) First line: β-blocker and/or calcium channel blocker (►do not combine β-blockers with non-dihydropyridine calcium antagonists). If these fail to control symptoms or are not tolerated, trial other agents.

p-blockers: eg atenolol 50mg BD or bisoprolol 5–10mg OD.

Calcium antagonists: amlodipine—start at 5mg OD; diltiazem—dose depends on formulation.

Long-acting nitrates: eg isosorbide mononitrate—starting regimen depends on formulation. Alternatives: GTN skin patches. SES: headaches, ‘BP.

Ivabradine: reduces heart rate with minimal impact on BP. Patient must be in sinus rhythm. Start with 5mg BD (2.5mg in elderly).

Ranolazine: inhibits late Na+ current. Start at 275mg BD. Caution if heart failure, elderly, weight <60kg or prolonged QT interval.

Nicorandil: a K+ channel activator. Start with 5–10mg BD. CI: acute pulmonary oede-ma, severe hypotension, hypovolaemia, LV failure.

Revascularization: Considered when optimal medical therapy proves inadequate.

CABG: (p122) compared to PCI, patients undergoing CABG are less likely to need re-peat revascularization and those with multivessel disease can expect better out-comes. However, CABG is open heart surgery and so recovery is slower and the patient is left with two large wounds (sternal and vein harvesting).

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Risk factors Non-modifiable: age, ð gender, family history of IHD (MI in 1st-degree relative <55yrs). Modifiable: smoking, hypertension, DM, hyperlipidaemia, obesity, sedentary lifestyle, cocaine use. Controversial risk factors include: stress, type A personality, LVH, fibrinogen•, hyperinsulinaemia, •homocysteine levels, ACE genotype.

Incidence 5/1000 per annum (UK) for ST-segment elevation (declining in UK & USA).

Diagnosis An increase in cardiac biomarkers (eg troponin) and either: symptoms of ischaemia, ECG changes of new ischaemia, development of pathological Q waves, new loss of myocardium, or regional wall motion abnormalities on imaging.

Signs Distress, anxiety, pallor, sweatiness, pulse • or ‘, BP • or ‘, 4th heart sound. There may be signs of heart failure (•JVP, 2rd heart sound, basal crepitations) or a pansystolic murmur (papillary muscle dysfunction/rupture, VSD). Low-grade fever may be present. Later, a pericardial friction rub or peripheral oedema may develop.

Differential diagnosis (p94.) Stable angina, pericarditis, myocarditis, Takotsubo cardiomyopathy (p145), aortic dissection (p655), PE, oesophageal reflux/spasm, pneu-mothorax, musculoskeletal pain, pancreatitis.

Management See p120, pp798–801.

Mortality 50% of deaths occur within 2h of onset of symptoms. Up to 7% die before discharge. Worse prognosis if: elderly, LV failure, and ST changes.[1a]

Fig 2.22 Enzyme changes following acute MI. Increasingly, high-sensitivity troponins are used alone for routine investigation of ACS.

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ACS management depends on whether the ACS is ‘ST elevated’ or not:

ST elevated myocardial infarction (STEMI): this category includes ACS with ST el-evation on ECG (fig 2.9) but also ACS with new LBBB (fig 2.7); and posterior MIs (fig 2.24) where ST elevation may only be seen with extra leads (V7–V9). Urgent revascularization is essential. 2p796.

ACS without ST elevation: serial troponins are needed to differentiate non-ST el-evated MIs (NSTEMIs) (trop rise) from unstable angina (no trop rise). 2p798.

After the immediate actions described on pp796–9, treatment of ACS17 focuses on managing symptoms, secondary prevention of further cardiovascular disease, revas-cularization (if not already undertaken), and addressing complications.

Symptom control

Manage chest pain with PRN GTN and opiates. If this proves insufficient, consider a GTN infusion (monitor BP, omit if recent sildenafil use). If pain is deteriorating, seek senior help. Manage symptomatic heart failure, p126.

Modify risk factors

Patients should be strongly advised and helped to stop smoking (p92).

Identify and treat diabetes mellitus, hypertension, and hyperlipidaemia.

Advise a diet high in oily fish, fruit, vegetables, & fibre, and low in saturated fats.

Encourage daily exercise. Refer to a cardiac rehab programme.

Mental health: flag to the patient's GP if depression or anxiety are present—these are independently associated with poor cardiovascular outcomes.

Optimize cardioprotective medications

Antiplatelets: aspirin (75mg OD) and a second antiplatelet agent (eg clopidogrel) for at least 12 months to ↓vascular events (eg MI, stroke). Consider adding a PPI (eg lansoprazole) for gastric protection.

Anticoagulate, eg with fondaparinux, until discharge.

β-blockade reduces myocardial oxygen demand. Start low and increase slowly, monitoring pulse and BP. If contraindicated, consider verapamil or diltiazem.

ACE-i in patients with LV dysfunction, hypertension, or diabetes unless not tolerated (consider ARB). Titrate up slowly, monitoring renal function.

High-dose statin, eg atorvastatin 80mg.

Do an echo to assess LV function. Eplerenone improves outcomes in MI patients with heart failure (ejection fraction <40%).

Revascularization

STEMI patients and very high-risk NSTEMI patients (eg haemodynamically unstable) should receive immediate angiography ! PCI. NSTEMI patients who are high risk (eg GRACE score >140) should have angiography within 24h; intermediate risk (eg GRACE 109–140) within 2d; low-risk patients may be considered for non-invasive testing.

Patients with multivessel disease may be considered for CABG instead of PCI (p122).

Manage complications See p122.

Discharge Address any questions the patient has. Discuss 'red flag' symptoms and where to seek medical advice should they arise. Ensure the management plan is communicated to the patient's GP. Book clinic and cardiac rehab appointments.

General advice

Driving:18 drivers with group 1 licences (car and motorcycle) can resume driving 1wk after successful angioplasty, or 4wk after ACS without successful angioplasty, if their ejection fraction is >40%. Group 2 licence holders must inform

the DVLA of their ACS and stop driving; depending on the results of functional tests, they may be able to restart after 6wk.

Work: how soon a patient can return to work will depend on their clinical progress and the nature of their work. They should be encouraged to discuss speed of return

! changes in duties (eg to lighter work if manual labour) with their employer. Some occupations cannot be restarted post-MI: eg airline pilots & air traffic controllers. Drivers of public service or heavy goods vehicles will have to undergo functional testing (eg exercise test), as mentioned previously.

122

Cardiac arrest (See p894, fig A2.) Cardiogenic shock (p802.) Left ventricular failure (p126, p800, p802.)

Right ventricular failure (RVF)/infarction Presents with low cardiac output and

JVP. Fluid is key; avoid vasodilators (eg nitrates) and diuretics.20 Inotropes are re-quired in some cases.

Pericarditis Central chest pain, relieved by sitting forwards. ECG: saddle-shaped ST

elevation, see fig 2.51, p155. Treatment: NSAIDS. Echo to check for effusion.

Systemic embolism May arise from LV mural thrombus. After large anterior MI, consider anticoagulation with warfarin for 2 months.

Cardiac tamponade (p802) Presents with low cardiac output, pulsus paradoxus, Kussmaul's sign,2 muffled heart sounds. Diagnosis: echo. Treatment: pericardial aspi-ration (provides temporary relief, 2see p772 for technique), surgery.

Mitral regurgitation May be mild (minor papillary muscle dysfunction) or severe (chordal or papillary muscle rupture secondary to ischaemia). Presentation: pulmo-nary oedema. Treat LVF (p800) and consider valve replacement.

Ventricular septal defect Presents with pansystolic murmur, •JVP, cardiac failure. Diagnosis: echo. Treatment: surgery. 50% mortality in first week.

Late malignant ventricular arrhythmias Occur 1–2wks post-MI and are the cardi-ologist's nightmare. Avoid hypokalaemia, the most easily avoidable cause. Consider 24h ECG monitoring prior to discharge if large MI.

Dressler's syndrome (p698) Recurrent pericarditis, pleural effusions, fever, anae-mia, and •ESR 1–2wks post-MI. Treatment: consider NSAIDS; steroids if severe.

Left ventricular aneurysm This occurs late (4–6wks post-MI), and presents with LVF, angina, recurrent VT, or systemic embolism. ECG: persistent ST-segment elevation. Treatment: anticoagulate, consider excision.

JVP rises during inspiration. Adolf Kussmaul was a prominent 19th-century physician and the first to attempt gastroscopy. Inspired by a sword swallower he passed a rigid tube into the stomach, however light technology was limited and it was not until years later that gastroscopists could visualize the stomach.

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Disturbances of cardiac rhythm (arrhythmias) are:

common

often benign (but may reflect underlying heart disease)

often intermittent, causing diagnostic difficulty see BOX 'Continuous ECG monitoring'

occasionally severe, causing cardiac compromise which may be fatal.

2Emergency management: pp804–9.

Presentation Palpitations, chest pain, presyncope/syncope, hypotension, or pulmo-nary oedema. Some arrhythmias may be asymptomatic, incidental findings, eg AF. History Take a detailed history of palpitations (p26). Ask about precipitating fac-

tors, onset/offset, nature (fast or slow, regular or irregular), duration, associated

symptoms (chest pain, dyspnoea, collapse). Review drug history. Ask about past medical history and family history of cardiac disease and sudden death. Syncope occuring during exercise is always concerning; the patient may have a condition predisposing them to sudden cardiac death (eg long QT syndrome).

2Narrow complex tachycardias: See pp806–7, 126.

2Atrial fibrillation and flutter: See pp806–7, 120.

2Broad complex tachycardias: See pp804–5, 128.

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126

Definition ECG shows rate of >100bpm and QRS complex duration of <120ms. Narrow QRS complexes occur when the ventricles are depolarized via the normal conduction pathways (fig 2.26).

Differential diagnosis

Regular narrow complex tachycardias: See fig 2.27.

Irregular narrow complex tachycardias:

Normal variant: sinus arrhythmia (rate changes with inspiration/expiration); sinus rhythm with frequent ectopic beats.

Atrial fibrillation (AF): p121, fig 2.25.

Atrial flutter with variable block: eg P–P–P–QRS–P–P–QRS (2:1 block then 2:1 block). The atrial rhythm is regular but the ventricular rhythm (hence pulse) is irregular.

Multifocal atrial tachycardia: like focal atrial tachycardia but there are multiple groups of atrial cells taking it in turns to initiate a cardiac cycle. P-wave morphol-ogy and P-P intervals vary. Usually associated with COPD.

Principles of management See p807.

►►If the patient is compromised, use DC cardioversion (p770).

Vagal manoeuvres: carotid sinus massage, Valsalva manoeuvre (eg blowing into a syringe).

IV adenosine: see p806.

In some cases, narrow complex tachyarrhythmias cause symptomatic episodes of sufficient severity and frequency to warrant more invasive treatment, eg ablation therapy for accessory pathways.

Fig 2.28 This patient was given adenosine for tachycardia thought to be due to AVRT or AVNRT. The adenosine has slowed the ventricular rate, revealing flutter waves (sawtooth appearance), disproving an AVRT/AVNRT diagnosis.

Normal conduction

Normal conduction: initiated by the sinoatrial node (SAN), electrical ac-tivity spreads around the atria. The atrioventricular node (AVN) receives this activity, pauses, then passes it on, down the bundle of His which splits into left and right bundle branches. These cause depolariza-tion of the ventricular myocardium from bottom (apex) to top (outflow tracts).

SAN

AVN TV

Right bundle branch

MV

His bundle

Left bundle branch

Left anterior

and

See fig 2.27.

A B

C

autonomic neuropathy (eg inappro- H

group of atrial cells act as a pace-maker, out-pacing the SAN. P-wave

Atrial flutter: Electrical activity circles the atria 200 times per minute, giving a 'sawtooth' baseline, see fig 2.25. The AVN passes some of these impulses on, result-ing in ventricular rates that are factors of 200 (150, 100, 75).

AVN, causing narrow complex tachycardias. This is very common.

Ventricular tachycardia: (VT) This can result from circuits, similar to atrial flutter, or from focuses of rapidly-firing cells. The QRS is broad. When a circuit is in action and its plane rotates, the ECG shows broad complex tachycardia with regularly increasing and decreasing amplitudes; this is called torsades de pointes.

Principles of management

If in doubt, treat as ventricular tachycardia (VT)—the commonest cause.

Differential diagnosis

Torsade de pointes (polymorphic VT)—VT with varying axis (see fig 2.21), may look like VF. •QT interval is a predisposing factor.

Antidromic AVRT (eg WPW), p127.

QRS >160ms.

Fusion beats or capture beats (figs 2.22, 2.22).

Bigeminy—ectopic every other beat, see fig 2.24. ECG machines may disregard the second QRS and so calculate the rate to be half the true value.

Triplet—three ectopics together.

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Causes Heart failure; hypertension; IHD (seen in 22% MI patients);[2a] PE; mitral valve disease; pneumonia; hyperthyroidism; caffeine; alcohol; post-op; 'K+; 'Mg2+. Rare causes: Cardiomyopathy; constrictive pericarditis; sick sinus syndrome; lung cancer; endocarditis; haemochromatosis; sarcoid. 'Lone' AF means no cause found.

Tests ECG shows absent P waves, irregular QRS complexes, fig 2.25. Blood tests: U&E, cardiac enzymes, thyroid function tests. Echo to look for left atrial enlargement, mitral valve disease, poor LV function, and other structural abnormalities.

Managing acute AF

If the patient is stable & AF started <48h ago: rate or rhythm control may be tried. For rhythm control, DC cardiovert or give flecainide (CI: structural heart disease, IHD) or amiodarone. Start heparin in case cardioversion is delayed (see BOX 'Anti-coagulation and AF').

Managing chronic AF

• ‘U&E). Anticoagulation: See BOX ‘Anticoagulation and AF’.

Atrial flutter See pp120–1, fig 2.25. Treatment: Similar to AF regarding rate and rhythm control and the need for anticoagulation.29 DC cardioversion is preferred to pharmacological cardioversion; start with 70–120J. IV amiodarone may be needed if rate control is proving difficult. Recurrence rates are high so radiofrequency ablation is often recommended for long-term management.

(b)

122

Indications for temporary cardiac pacing include

Symptomatic bradycardia, unresponsive to atropine.

Mobitz type I AV block (Wenckebach)

After inferior MI, pacing may not be needed in complete AV block if reasonably stable, rate is >40–50, and QRS complexes are narrow.

See p776 for further details and insertion technique.

Indications for a permanent pacemaker (PPM) include

Mobitz type II AV block (p99).

Heart failure (cardiac resynchronization therapy).

2nd letter the chamber sensed (A=atria, V=ventricles, D=dual chamber, O=none).

5th letter (P means that in tachycardia the pacemaker will pace the patient. S means that in tachycardia the pacemaker shocks the patient. D=dual ability to pace and shock. O=neither of these).

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output. Ejection fraction (EF) is <40%. Causes: IHD, MI, cardiomyopathy.

Right ventricular failure (RVF): Causes: LVF, pulmonary stenosis, lung disease (cor pulmonale, see p194). Symptoms: peripheral oedema (up to thighs, sacrum, abdomi-nal wall), ascites, nausea, anorexia, facial engorgement, epistaxis.

Low-output heart failure: Cardiac output is ‘ and fails to • normally with exertion.

Pump failure: systolic and/or diastolic HF (see above), ‘heart rate (eg β-blockers, heart block, post MI), negatively inotropic drugs (eg most antiarrhythmic agents).

Investigations According to NICE,33 if ECG and B-type natriuretic peptide (BNP; p127) are normal, heart failure is unlikely, and an alternative diagnosis should be consid-ered; if either is abnormal, then echocardiography (p110) is required.

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Acute heart failure ₂This is a medical emergency (p800).

Treat exacerbating factors (anaemia, thyroid disease, infection, ⛄BP).

Drugs:

ACE-i: Consider in all those with left ventricular systolic dysfunction (LVSD); improves symptoms and prolongs life (see p114–5). If cough is a problem, an angiotensin receptor blocker (ARB) may be substituted. SE: •K+.

Minimal exertion; Na+ & fluid restriction (1.5L/24h PO).

Weigh daily. Do frequent U&E (beware ‘K+).

Palliative care Treat/prevent comorbidities (eg ’flu vaccination). Good nutrition (allow alcohol!). Involve GP: continuity of care and discussion of prognosis is much appreciated.’² Dyspnoea, pain (from liver capsule stretching), nausea, constipation, and ‘mood all need tackling.’3 Opiates improve pain and dyspnoea. O2 may help.

Hypertension’’ is the most important risk factor for premature death and CVD; caus-ing r50% of all vascular deaths (8≈106/yr). Usually asymptomatic, so regular screen-ing (eg 2-yrly) is a vital task—most preventable deaths are in areas without universal screening.’™

Primary or ‘essential’ hypertension: (Cause unknown.) r95% of cases.

Endocrine disease: Cushing’s (p224) and Conn’s syndromes (p228), phaeochromo-cytoma (p228), acromegaly, hyperparathyroidism.

AV nipping (narrowing where arteries cross veins, p560, fig 12.19).

Measuring BP with a sphygmomanometer

Inflate the cuff until 20mmHg above systolic pressure, then place stethoscope over the brachial artery. Deflate the cuff at 2mmHg/s.

For advice on using automated sphygmomanometers and a list of validated de-vices see http://www.bhsoc.org/latest-guidelines/how-to-measure-blood-pressure/

Offer ABPM.

Target pressure is <140/90mmHg (150/90 if aged >80), but in diabetes mellitus aim for <120/80mmHg, and diagnosis-and-management-35109454941637.

Treatment goal <140/90mmHg (<120/80 in diabetes, 150/90 if aged >80). Reduce blood pressure slowly; rapid reduction can be fatal, especially in the context of an acute stroke. These may fall—SPRINT49 showed a target of 120/80 was beneficial.

Monotherapy: If ?55yrs, and in black patients of any age, 1st choice is a Ca2+-channel antagonist or thiazide. If <55, 1st choice is ACE-i (or ARB if ACE-i intolerant, eg cough). β-blockers are not 1st line for hypertension, but consider in younger people, particu-larly if: intolerance or contraindication to ACE-i/ARB, she is a woman of child-bearing potential, or there is •sympathetic drive.

Malignant hypertension (fig 2.41) In general, use oral therapy, unless there is encephalopathy or CCF. The aim is for a controlled reduction in BP over days, not hours. Avoid sudden drops in BP as cerebral autoregulation is poor (•stroke risk). Bed rest; there is no ideal hypotensive, but atenolol or long-acting Ca2+ blockers may be used PO.

Fig 2.41 Left ventricular hypertrophy—this is from a patient with malignant hypertension—note the sum of the S-wave in V2 and R-wave in V6 is greater than 35mm.

Diagnosis Use the revised Jones criteria (may be over-rigorous). There must be evidence of recent strep infection plus 2 major criteria, or 1 major + 2 minor.

Rapid streptococcal antigen test +ve.

Recent scarlet fever.

Arthritis: a migratory, ‘flitting’ polyarthritis; usually affects larger joints (75%).

Sydenham’s chorea (St Vitus’ dance): occurs late in 10%. Unilateral or bilateral involuntary semi-purposeful movements. May be preceded by emotional lability and uncharacteristic behaviour.

Raised ESR or CRP.

Previous rheumatic fever.

Management

Benzylpenicillin 0.6–1.2g IV stat, then phenoxymethylpenicillin 250–500mg 4 times daily

Immobilize joints in severe arthritis.

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Symptoms: Dyspnoea; fatigue; palpitations; symptoms of causative factor (eg fever). Signs: AF; displaced, hyperdynamic apex; pansystolic murmur at apex radiating to axilla; soft S1; split S2; loud P2 (pulmonary hypertension). Severity: the more severe, the larger the left ventricle.

Echocardiogram: To assess LV function and MR severity and aetiology (transoesoph-ageal to assess severity and suitability for repair rather than replacement). Cardiac catheterization to confirm diagnosis, exclude other valve disease, and assess coro-nary artery disease (can combine CABG with valve surgery).

Mitral stenosis Causes: Rheumatic fever, congenital, mucopolysaccharidoses, endocardial fibroelastosis, malignant carcinoid (p271; rare), prosthetic valve.

What Becomes of the Broken Hearted?

and perceiveth not his heart to bee coarcted? Whom inflameth ire, and hath not heart-burning?

Thomas Wright, The Passions of the Minde in Generall, 1604

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Aortic stenosis (AS) Causes: Senile calcification is the commonest.™7 Others: con-genital (bicuspid valve, Williams syndrome, p149), rheumatic heart disease.

Differential diagnosis: Hypertrophic cardiomyopathy (HCM, p152); aortic sclerosis.

Pulmonary regurgitation Causes: Any cause of pulmonary hypertension (p194). Signs: Decrescendo murmur in early diastole at the left sternal edge (the Graham Steell murmur if associated with mitral stenosis and pulmonary hypertension).

CABG See p122.

Surgery for congenital heart defects See p156.

Aortic root surgery Replacement/repair if dissection or aneurysmal.

The heart in various, mostly rare, systemic diseases

Amyloidosis: (p270) Restrictive cardiomyopathy. Bright myocardium on echo.

Cushing's syndrome: (p224) Hypertension.

Friedreich's ataxia: (p698) Hypertrophic cardiomyopathy, dilatation over time.

Human immunodeficiency virus: (p298) Myocarditis; dilated cardiomyopathy; effusion; ventricular arrhythmias; SBE/IE; non-infective thrombotic (marantic) en-docarditis; RVF (pulmonary hypertension); metastatic Kaposi's sarcoma.

Klinefelter's syndrome ð: (OHCS p646) ASD. Psychopathy; learning difficulties;

Myotonic dystrophy (p510) Progressive conduction system disease; arrhythmias;

Rheumatoid arthritis: Conduction defects; pericarditis; LV dysfunction; aortic regurgitation; coronary arteritis. Look for arthritis signs, p546.

Systemic lupus erythematosus: (p554) Pericarditis/effusion; myocarditis; Lib-man–Sacks endocarditis; mitral valve prolapse; coronary arteritis.

Turner syndrome φ: Coarctation of aorta. Look for webbed neck. XO.

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Treatment Liaise early with microbiologists and cardiologists.62 Antibiotics: see BOX ‘Antibiotic therapy for infective endocarditis’. Surgery if: Heart failure, valvular ob-struction; repeated emboli; fungal IE; persistent bacteraemia; myocardial abscess; unstable infected prosthetic valve.[a]’

Recommendations Give clear information about prevention, including:

The risks of invasive procedures, including non-medical procedures such as body piercing or tattooing.[a™]

usually red-brown in colour. eg on the palms or soles.

Hypertrophic cardiomyopathy (HCM) LV outflow tract (LVOT) obstruction from asymmetric septal hypertrophy. HCM is the leading cause of sudden cardiac death in the young. Prevalence: 0.2%. Autosomal dominant inheritance, but 50% are spo-radic. 70% have mutations in genes encoding β-myosin, a-tropomyosin, and troponin

Restrictive cardiomyopathy Causes: Idiopathic; amyloidosis; haemochromato-sis; sarcoidosis; scleroderma; Löffler’s eosinophilic endocarditis; endomyocardial fibrosis. Presentation: Is like constrictive pericarditis (p154). Features of RVF pre-dominate: •JVP, with prominent x and y descents; hepatomegaly; oedema; ascites. Diagnosis: Echo, MRI, cardiac catheterization. 9: Treat the cause.

Fig 2.49 Echocardiogram of a 35-yr-old patient who presented with severe exertional dyspnoea and several episodes of syncope. Look at the large mass (cardiac myxoma) in left atrium. Abbreviations: RV: right ventricle; LV: left ventricle; AV: aortic valve; AO: aorta; MV: mitral valve.

Acute pericarditis This is inflammation of the pericardium.68

Bacteria: eg TB—commonest cause worldwide, Lyme disease, Q fever, pneumonia, rheumatic fever, Staphs, Streps, mycoplasma, legionella, MAI in HIV.

Drugs: eg procainamide, hydralazine, penicillin, isoniazid, chemotherapy.

Clinical features: Central chest pain worse on inspiration or lying flat ! relief by sitting forward. A pericardial friction rub (p46) may be heard. Look for evidence of a pericardial effusion or cardiac tamponade (see later in topic). Fever may occur.

Pericardial effusion Accumulation of fluid in the pericardial sac (normally 10–50mL).68 Causes: Pericarditis, myocardial rupture (haemopericardium—surgical, stab wound, post-MI); aortic dissection; pericardium filling with pus; malignancy.

Management: Treat the cause. Pericardiocentesis may be diagnostic (suspected bacterial pericarditis) or therapeutic (cardiac tamponade). See p772. Send pericar-dial fluid for culture, ZN stain/TB culture, and cytology.

Clinical features: These are mainly of right heart failure with •JVP (with prominent x and y descents, p42); Kussmaul’s sign (JVP rising paradoxically with inspiration); soft, diffuse apex beat; quiet heart sounds; S2; diastolic pericardial knock, hepato-splenomegaly, ascites, and oedema.

Cardiac tamponade A pericardial effusion that raises intrapericardial pressure, reducing ventricular filling and thus dropping cardiac output.68 ☠☠ Can lead rapidly to cardiac arrest.

Management: Seek expert help. The pericardial effusion needs urgent drainage (p772). Send fluid for culture, ZN stain/TB culture, and cytology.

Bicuspid aortic valve These work well at birth and go undetected. Many even-tually develop aortic stenosis (needing valve replacement) ! aortic regurgitation predisposing to IE/SBE ! aortic dilatation/dissection. Intense exercise may accelerate complications, so do yearly echocardiograms on affected athletes.7º

Ostium primum defects: associated with AV valve anomalies, eg in Down’s syn-drome; present in childhood.

Angina Driving must cease when symptoms occur at rest or with emotion. Driving may recommence when satisfactory symptom control is achieved. DVLA need not be notified.

Dysrhythmias Including sinoatrial disease, AF/flutter, atrioventricular conduction defects, and narrow or broad complex tachycardias. Driving must cease if the dys-rhythmia has caused or is likely to cause incapacity. Driving may recommence 4wks after successful control provided there is no other disqualifying condition.

6 months have passed since ICD implanted for secondary prevention.

No therapy (shock) in the last 2 years has been accompanied by incapacity (wheth-er caused by the device or arrhythmia)—unless this was a result of device malfunc-tion which has been corrected for at least 1 month or steps have been taken to avoid recurrence (eg ablation) which have been successful for at least 6 months.

There is no other disqualifying condition.

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Chest Medicine

Larynx

Trachea

Superior lobe of right lung

Right primary bronchus

Right secondary bronchus

Right tertiary bronchus

Inferior lobe of right lung

Smaller

bronchi

Superior lobe of left lung

Left primary bronchus

Left secondary bronchus

Left tertiary bronchus

Middle lobe Inferior lobe

of left lung

Smaller bronchi

Fig 4.2 Segmental anatomy of the lungs and main bronchi. The left lung has two lobes and the right has three.

There is no substitute for careful history taking and examination in making the ‘correct’ diagnosis. Tests should help clarify and assess severity. When examining the chest think about the anatomy, and the location of pathology (fig 4.2).

Sputum examination Collect a good sample; if necessary ask a physiotherapist to help. Note the appearance: clear and colourless (chronic bronchitis), yellow-green or brown (pulmonary infection), red (haemoptysis), black (smoke, coal dust), or frothy white-pink (pulmonary oedema). Send the sample to the laboratory for microscopy, culture/sensitivity. If indicated, ask for ZN stain, and PCR.

Peak expiratory flow (PEF) Measured by a maximal forced expiration through a peak flow meter. It correlates well with the forced expiratory volume in 1 second (FEV1) & is used as an estimate of airway calibre in asthma, but is effort-dependent.

Arterial blood gas (ABG) analysis Heparinized blood is usually taken from the radial or femoral artery (see p771). The brachial artery is used less because of median nerve proximity and it is an end artery. pH, PaO2, PaCO2, HCO2 are measured using an auto-mated analyser.

ABG interpretation See pp188–9.

Table 4.1 Spirometry results (data source NICE COPD 2010 guidelines)

| | FEV1 | FVC | FEV1/FVC ratio | | — | — | — | — | | Normal | >80% predicted | >80% predicted | 75–80% | | Restrictive | <80% predicted | <80% predicted | >70% normal | | Obstructive | <80% predicted | Normal or low | <70% predicted |

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Pulmonary alveolar proteinosis causes cough, dyspnoea, and restrictive spirometry. It is caused by ac-cumulation of surfactant-derived acidophilic phospholipid/protein compounds which fill alveoli and distal bronchioles. Diagnosis may require lung biopsy. Cause: primary genetic or antibody problem, or secondary to inflammation caused by inhaling silica, aluminium, or titanium.3

PEF=peak expiratory flow; FEF50=forced expiratory flow at 50% TLC; FEF25=forced expiratory flow at 25% TLC; PIF=peak inspiratory flow; FIF50=forced inspiratory flow at 50% TLC.

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An acute lower respiratory tract infection associated with fever, symptoms and signs in the chest, and abnormalities on the chest x-ray—fig 16.2, p722. Incidence: 5–11/1000, • if very young or old (20% are under 65yrs). Mortality: r21% in hospital.

Classification and causes

Hospital-acquired: Defined as >48h after hospital admission. Most commonly Gram-negative enterobacteria or Staph. aureus. Also Pseudomonas, Klebsiella, Bacteroides, and Clostridia.

Aspiration: Those with stroke, myasthenia, bulbar palsies, ‘consciousness (eg post-ictal or intoxicated), oesophageal disease (achalasia, reflux), or poor dental hygiene risk aspirating oropharyngeal anaerobes.

Immunocompromised patient: Strep. pneumoniae, H. influenzae, Staph. aureus,

M. catarrhalis, M. pneumoniae, Gram –ve bacilli and Pneumocystis jirovecii (for-merly named P. carinii, pp400–1). Other fungi, viruses (CMV, HSV), and mycobacteria.

Tests Assess oxygenation: oxygen saturation, p162 (ABGs if SaO2 <92% or severe pneumonia) and BP. Blood tests: FBC, U&E, LFT, CRP (GPs should consider a point of care CRP to guide antibiotic prescribing where LRTI is suspected, NICE 2014′).

Severity ‘CURB-65’ is a simple, validated severity scoring system.™, [a] 1 point for each of:

Confusion (abbreviated mental test ≤8) Urea >7mmol/L

Respiratory rate ?20/min

BP <90 systolic and/or 60mmHg diastolic) Age ?65.

0–1, PO antibiotic/home treatment; 2, hospital therapy; ?2, severe pneumonia indicates mortality 15–40%—consider ITU. It may ‘underscore’ the young—use clinical judgement. Other features increasing the risk of death are: comorbidity; bilateral/multilobar; PaO2

<8kPa.

Complications (See p170.) Pleural effusion, empyema, lung abscess, respiratory failure, septicaemia, brain abscess, pericarditis, myocarditis, cholestatic jaundice. Repeat CRP and CXR in patients not improving to look for progression/ complications.

Table 4.2 Empirical treatment of pneumonia (check local policy)

| Pneumococcal vaccine At-risk groups: All adults ? 65yrs old. Chronic heart, liver, renal, or lung conditions. Diabetes mellitus not controlled by diet. Immunosuppression, eg ↶spleen function, AIDS, or on chemotherapy or prednis-olone >20mg/d, cochlear implant, occupation risk (eg welders), CSF fluid leaks. Vaccinate every 5yrs. CI: Pregnancy, lactation, ⍙T°, previous anaphylaxis to vaccine or one of its com-ponents. | | — |

Staphylococcal pneumonia May complicate influenza infection or occur in the young, elderly, intravenous drug users, or patients with underlying disease, eg leu-kaemia, lymphoma, cystic fibrosis (CF). It causes a bilateral cavitating bronchopneu-monia. Treatment: flucloxacillin ! rifampicin, MRSA: contact lab; consider vancomycin.

Klebsiella pneumonia Rare. Occurs in elderly, diabetics, and alcoholics. Causes a cavitating pneumonia, particularly of the upper lobes, often drug resistant. Treat-ment: cefotaxime or imipenem.

Pseudomonas A common pathogen in bronchiectasis and CF. It also causes hospital-acquired infections, particularly on ITU or after surgery. Treatment: anti-pseudomonal penicillin, ceftazidime, meropenem, or ciprofloxacin + aminoglycoside. Consider dual therapy to minimize resistance.

Viral pneumonia Influenza commonest (p296 and BOX), but ‘swine flu’ (H1N1) is now considered seasonal and covered by the annual ‘flu vaccine. Others: measles, CMV, varicella zoster.

Pneumocystis pneumonia Causes pneumonia in the immunosuppressed (eg HIV). The organism responsible was previously called Pneumocystis carinii, and now called Pneumocystis jirovecii.8 It presents with a dry cough, exertional dyspnoea,

Therapeutic or prophylactic antivirals are said to be the most effective single intervention followed by vaccine and basic public health measures.[17] But oseltamivir resistance and unavailability of a suitable vac-cine during the early stages of a pandemic make non-drug interventions all the more important.

Hypotension May be due to a combination of dehydration and vasodilation due to sepsis. If systolic BP is <90mmHg, give an intravenous fluid challenge of 250mL colloid/crystalloid over 15min. If BP does not rise, consider a central line and give IV fluids to maintain the systolic BP >90mmHg. If systolic BP remains <90mmHg despite fluid therapy, request ITU assessment for inotropic support.

Atrial fibrillation (p120.) Common in the elderly. It usually resolves with treatment of the pneumonia. β-blocker or digoxin may be required to slow the ventricular re-sponse rate in the short term.

Pleural effusion Inflammation of the pleura by adjacent pneumonia may cause fluid exudation into the pleural space. If this accumulates faster than it is reabsorbed, a pleural effusion develops. If small, it may be of no consequence. If larger and patient symptomatic, or infected (empyema), drainage is required (p192, p766).

Lung abscess A cavitating area of localized, suppurative infection within the lung (see fig 4.6).

Causes: •Inadequately treated pneumonia. •Aspiration (eg alcoholism, oesophageal obstruction, bulbar palsy). •Bronchial obstruction (tumour, foreign body). •Pulmo-nary infarction. •Septic emboli (septicaemia, right heart endocarditis, IV drug use).

Subphrenic or hepatic abscess.

Clinical features: Swinging fever; cough; purulent, foul-smelling sputum; pleuritic chest pain; haemoptysis; malaise; weight loss. Look for: finger clubbing; anaemia; crepitations. Empyema develops in 20–20%.

Tests: Blood: FBC (anaemia, neutrophilia), ESR, CRP, blood cultures. Sputum micros-copy, culture, and cytology. CXR: walled cavity, often with a fluid level. Consider CT scan to exclude obstruction, and bronchoscopy to obtain diagnostic specimens.

Treatment: Antibiotics as indicated by sensitivities; continue until healed (4–6 wks). Postural drainage. Repeated aspiration, antibiotic instillation, or surgical excision may be required.

Septicaemia May occur as a result of bacterial spread from the lung parenchyma into the bloodstream. This may cause metastatic infection, eg infective endocarditis, meningitis. Treat with IV antibiotic according to sensitivities.

Pericarditis and myocarditis May also complicate pneumonia.

Jaundice This is usually cholestatic, and may be due to sepsis or secondary to anti-biotic therapy (particularly flucloxacillin and co-amoxiclav).

Pathology Chronic inflammation of the bronchi and bronchioles leading to perma-nent dilatation and thinning of these airways.[18] Main organisms: H. influenzae; Strep. pneumoniae; Staph. aureus; Pseudomonas aeruginosa.

Causes Congenital: Cystic fibrosis (CF); Young's syndrome; primary ciliary dyskinesia; Kartagener's syndrome (OHCS p646). Post-infection: Measles; pertussis; bronchiolitis; pneumonia; TB; HIV. Other: Bronchial obstruction (tumour, foreign body); allergic bronchopulmonary aspergillosis (ABPA, p177); hypogammaglobulinaemia; rheumatoid arthritis; ulcerative colitis; idiopathic.

Clinical features Symptoms: Persistent cough; copious purulent sputum; intermit-tent haemoptysis. Signs: Finger clubbing; coarse inspiratory crepitations; wheeze (asthma, COPD, ABPA). Complications: Pneumonia, pleural effusion; pneumothorax; haemoptysis; cerebral abscess; amyloidosis.

Fig 4.7 PA chest radiograph showing marked abnormal dilatation of the airways throughout the right upper lobe, subtle similar changes throughout the rest of the lung (particulalry periphery of the left upper zone). The fine background reticular pattern in the lungs suggests that there may also be some interstitial lung disease present.

Diagnosis Sweat test: Sweat sodium and chloride >60mmol/L; chloride usu-ally > sodium. Genetics: Screening for known common CF mutations should be con-sidered. Faecal elastase is a simple and useful screening test for exocrine pancreatic dysfunction.

Tests Blood: FBC, U&E, LFT; clotting; vitamin A, D, E levels; annual glucose tolerance test (p206). Bacteriology: Cough swab, sputum culture. Radiology: CXR; hyperinflation; bronchiectasis. Abdominal ultrasound: Fatty liver; cirrhosis; chronic pancreatitis; Spirometry: Obstructive defect. Aspergillus serology/skin test (20% develop ABPA, p177). Biochemistry: Faecal fat analysis.

Carcinoma of the bronchus Second most common cancer in the UK, accounting for 12% of all new cancer cases and 27% of cancer deaths (40 000 cases/yr in UK).[22] Incidence is increasing in women. Only 5% 'cured'. Risk factors: Cigarette smoking (causes 90% of lung ca). Others: passive smoking, asbestos, chromium, arsenic, iron oxides, and radiation (radon gas).

Symptoms: Cough (80%); haemoptysis (70%); dyspnoea (60%); chest pain (40%); recurrent or slowly resolving pneumonia; lethargy, anorexia; weight loss.

Signs: Cachexia; anaemia; clubbing; HPOA (hypertrophic pulmonary osteoarthropa-thy, causing wrist pain); supraclavicular or axillary nodes. Chest signs: none, or con-solidation; collapse; pleural effusion. Metastases: bone tenderness; hepatomegaly; confusion; fits; focal CNS signs; cerebellar syndrome; proximal myopathy; peripheral neuropathy.

Other lung tumours Bronchial adenoma: Rare, slow-growing. 90% are carcinoid tumours; 10% cylindromas. 9: surgery.

Hamartoma: Rare, benign; CT: lobulated mass

! flecks of calcification; ?excise to exclude malignancy.

Clinical features: Chest pain, dyspnoea, weight loss, finger clubbing, recurrent pleu-ral effusions. Signs of metastases: lymphadenopathy, hepatomegaly, bone pain/ten-derness, abdominal pain/obstruction (peritoneal malignant mesothelioma).

Tests: CXR/CT: pleural thickening/effusion. Bloody pleural fluid.

Diagnosis: Made on histology, usually following a thoracoscopy. Often the diagnosis is only made post-mortem.

Management: Pemetrexed + cisplatin chemotherapy can improve survival.[2′] Surgery is hard to evaluate (few randomized trials). Radiotherapy is controversial. Pleurodesis and indwelling intra-pleural drain may help.

Prognosis: Poor (especially without pemetrexed, eg <2yrs). >650 deaths/yr in UK.

Table 4.2 Non-metastatic extrapulmonary manifestations of bronchial cancer

| Coronaviruses: SARS and MERS Severe acute respiratory syndrome (SARS¹™) is caused by SARS-COV virus—a coronavirus. Major features are persistent fever (>28℃), chills, rigors, myalgia, dry cough, headache, diarrhoea, and dyspnoea—with an abnormal CXR and ‘WCC. Respiratory failure is a complicating feature: r20% progress to acute respiratory distress syndrome requiring invasive ventilation.[1a] Mortality is 1–50%, depend-ing on age, but no cases since 2004. Close contacts, or travel to an area with known cases should raise suspicion. The mechanism of transmission of SARS-CoV is human–human. Management: seek expert help. Largely supportive with good infection control measures. Middle East respiratory syndrome (MERS) is a viral respiratory disease caused by novel coronavirus (MERS CoV) and was first identified in 2012 in Saudi Arabia. Symptoms include fever, cough, shortness of breath, and gastrointestinal upset. Incubation period 14 days. Human-to-human transmission has been reported in most cases, but camels play a pivotal host role in animal-to-human transmission. Large outbreaks linked to healthcare facilities have been reported in the Middle East and South Korea. The World Health Organization has reported mortality as high as 26% in known cases.[1]3 | | — |

Table 4.4 TNM staging for non-small cell lung cancer

| Mutation-specific therapies for cystic fibrosis Ivacaftor and lumacaftor target the CFTR protein. Ivacaftor, a CFTR potentiator, targets gating defects in disease causing CFTR mutations including G551D. Ivacaftor increases the open probability of CFTR channels and has been shown to improve clinical outcomes (lung function, weight, lung disease stability) in CF patients >6 years old.[1a] Lumacaftor is a CFTR corrector, and has been shown to correct F508 del CFTR misprocessing and increase the amount of cell surface–localized protein. Ivacaftor and lumacaftor combination therapy, for patients with F508 del, have shown improved lung function and reduced pulmonary exacerations.[2o] Gene therapy (transfer of CFTR gene using liposome or adenovirus vectors): phase 2b studies show modest but significant improvement in FEV1 in those receiv-ing gene therapy.[21] Further work into vectors for gene transfer is ongoing. | | — |

New York: Springer; 2010.

Prognosis Non-small cell: 50% 2yr survival without spread; 10% with spread. Small cell: median survival is 2 months if untreated; 1–1½yrs if treated.

Prevention Stop smoking (p92). Prevent occupational exposure to carcinogens.

Aspergillus This group of fungi affects the lung in five ways:

Asthma: Type I hypersensitivity reaction to fungal spores (p178).

Extrinsic allergic alveolitis (EAA): See p198.

Other fungal infections Candida and Cryptococcus may cause pneumonia in the immunosuppressed (see p408).

Symptoms Intermittent dyspnoea, wheeze, cough (often nocturnal), and sputum (see table 4.5).

Precipitants: Cold air, exercise, emotion, allergens (house dust mite, pollen, fur), infection, smoking and passive smoking,[2a] pollution, NSAIDS, β-blockers.

Diurnal variation Symptoms or peak flow may vary over the day. Marked morning dipping of peak flow is common and can tip the balance into a serious attack, despite having normal peak flow (fig 4.12) at other times.

Exercise: Quantify the exercise tolerance.

Disturbed sleep: Quantify as nights per week (a sign of severe asthma).

Acid reflux: 40–60% of those with asthma have reflux; treating it improves spirom-etry, but not necessarily symptoms.3º

Other atopic disease: Eczema, hay fever, allergy, or family history?

The home (especially the bedroom): Pets? Carpet? Feather pillows or duvet? Floor cushions and other 'soft furnishings'?

Job: If symptoms remit at weekends or holidays, work may provide the trigger (15% of cases are work-related—more for paint sprayers, food processors, welders, and animal handlers).3¹ Ask the patient to measure their peak flow at intervals at work and at home (at the same time of day) to confirm this (see fig 4.12).

Days per week off work or school.

Tests Initial diagnosis: See figs 4.10, 4.11. Acute attack: PEF, sputum culture, FBC, U&E, CRP, blood cultures. ABG analysis usually shows a normal or slightly ↩PaO2 but

Differential diagnosis Pulmonary oedema ('cardiac asthma'); COPD (may co-exist); large airway obstruction (eg foreign body, tumour); SVC obstruction (wheeze/dys-pnoea not episodic); pneumothorax; PE; bronchiectasis; obliterative bronchiolitis (suspect in elderly).

Treatment Chronic asthma (p182). Emergency treatment (p810).

Associated diseases Acid reflux; polyarteritis nodosa (PAN, p556); Churg–Strauss syndrome (p696); ABPA (p177).

Natural history Most childhood asthmatics (see OHCS p164) either grow out of asthma in adolescence or suffer much less as adults. A significant number of people develop chronic asthma late in life.

Mortality r900 asthma deaths in the UK in 2012, 50% were >65yrs old.

Table 4.5 Clinical features which increase or decrease probability of asthma in adults.

| System | Manifestations | | — | — | | Endocrine | Ectopic secretion; ACTH (Cushing's), ADH (dilutional hyponatraemia), PTH (hypercalcaemia), HCG (gynaecomastia) | | Neurological | Cerebellar degeneration, myopathy, polyneuropathy, myasthenic syndrome | | Vascular Cutaneous | Thrombophlebitis migrans (p562), anaemia, DIC Dermatomyositis, herpes zoster, acanthosis nigricans | | Skeletal | Clubbing, HPOA |

(Data from https://www.brit-thoracic.org.uk/document-library/clinical-information/asthma/btsign-asthma-guideline-quick-
reference-guide-2014)

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Fig 4.10 BTS/SIGN British guideline on the management of asthma in children.

Data from Fig 1, p21: https://www.brit-thoracic.org.uk/document-library/clinical-information/asthma/btssign-asthma-guideline-2014/

Fig 4.11 BTS/SIGN British guideline on the management of asthma in adults.

Data from Fig 2, p25: https://www.brit-thoracic.org.uk/document-library/clinical-information/asthma/btssign-asthma-guideline-2014/

Data from Nunn, AJ, Gregg, I. New regression equations for predicting peak expiratory flow in adults. BMJ 1989;298:1068–70.

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British Thoracic Society guidelines (BTS33) Start at the step most appropriate to severity; moving up if needed, or down if control is good for >2 months. Rescue courses of prednisolone may be used at any time. For drug examples see table 4.6.

Step 1: Occasional short-acting inhaled β2-agonist as required for symptom relief. If used more than once daily, or night-time symptoms, go to Step 2.

Step 2: Add standard-dose inhaled steroid, eg beclometasone 200–800mcg/day, or start at the dose appropriate for disease severity, and titrate as required.

Step 3: Add long-acting β2-agonist (eg salmeterol 50mcg/12h by inhaler). If bene-fit—but still inadequate control—continue and •dose of beclometasone to 800mcg/ day. If no effect then stop LABA and •dose of beclometasone to 800mcg/day. Leu-kotriene receptor antagonist or oral theophylline may be tried.

Step 4: Consider trials of: beclometasone up to 2000mcg/day; modified-release oral theophylline; modified-release oral β2-agonist tablets; oral leukotriene receptor an-tagonist, in conjunction with previous therapy.

Step 5: Add regular oral prednisolone (1 dose daily, at the lowest possible dose). Continue with high-dose inhaled steroids. Refer for specialist input.

Anticholinergics: (Eg ipratropium, tiotropium.) May ‘muscle spasm synergistically with β2-agonists but are not recommended in current guidelines for chronic asthma. They may be of more benefit in COPD.

Cromoglicate (Mast cell stabilizer.) May be used as prophylaxis in mild and exercise-induced asthma (always inhaled), especially in children. It may precipitate asthma.

Leukotriene receptor antagonists: (Eg oral montelukast, zafirlukast.) Block the ef-fects of cysteinyl leukotrienes in the airways by antagonizing the CystLT1 receptor.

Anti-IgE monoclonal antibody: Omalizumab3[a] may be of use in highly selected patients with persistent allergic asthma. Given as a subcutaneous injection every 2–4 wks depending on dose. Specialists prescribe only.

Integrated breathing and relaxation training (Papworth method) is psychological and physical: patients learn to drop their shoulders, relax their abdomen, and breathe calmly and appropriately.

Table 4.6 Adult doses of common inhaled drugs used in bronchoconstriction

| Primary tumour (T) | | — | | TX Malignant cells in bronchial secretions, no other evidence of tumour | | TIS Carcinoma in situ | | T0 None evident | | T1 ≤ 2cm, in lobar or more distal airway | | T2 >2cm and >2cm distal to carina or any size if pleural involvement or obstruc-tive pneumonitis extending to hilum, but not all the lung | | T2 Involves the chest wall, diaphragm, mediastinal pleura, pericardium, or <2cm from, but not at, carina. T >7cm diameter and nodules in same lobe | | T4 Involves mediastinum, heart, great vessels, trachea, oesophagus, vertebral body, carina, malignant effusion, or nodules in another lobe | | Regional nodes (N) | | N0 None involved (after mediastinoscopy) | | N1 Peribronchial and/or ipsilateral hilum | | N2 Ipsilateral mediastinum or subcarinal | | N2 Contralateral mediastinum or hilum, scalene, or supraclavicular | | Distant metastasis (M) | | M0 None | | M1 a) Nodule in other lung, pleural lesions, or malignant effusion; b) distant metastases present | | Stages | | Occult I II IIIa IIIb IV | | TX N0 M0 TIS/T1/T2 N0 M0 T1/T2 N1 M0 T2 N1 M0 T1–4 N2 M0 T1–4 N0–2 M1 | | or T2 N0 M0 or T1–2 N2 M0 or T4 N0–2 M0 |

Prescribe beclometasone by brand name, and state that a CFC-free inhaler should be dispensed. This is because, dose for dose, Qvar® is twice as potent as the other available CFC-free brand (Clenil Modulite®).

Any dose ?250mcg ≈ significant steroid absorption: carry a steroid card; this recom-mendation is being widened, and lower doses (beclometasone) are now said to merit a steroid card (manufacturer’s information).

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Prevalence 10–20% of the over-40s; 2.5≈106 deaths/yr worldwide.3[a]

Symptoms Cough; sputum; dyspnoea; wheeze. Signs Tachypnoea; use of accessory muscles of respiration; hyperinflation; ‘cricosternal distance (<2cm); ‘expansion; resonant or hyperresonant percussion note; quiet breath sounds (eg over bullae); wheeze; cyanosis; cor pulmonale.

Complications Acute exacerbations ! infection; polycythaemia; respiratory failure; cor pulmonale (oedema; ♁JVP); pneumothorax (ruptured bullae); lung carcinoma.

Long-term O2 therapy (LTOT): An MRC trial showed that if PaO2 was maintained

?8.0kPa for 15h a day, 2yr survival improved by 50%. UK NICE guidelines suggest LTOT should be given for: 1 Clinically stable non-smokers with PaO2 <7.2kPa—despite maximal 9. These values should be stable on two occasions >2wks apart. 2 If PaO2

7.2–8.0 and pulmonary hypertension (eg RVH; loud S2), or polycythaemia, or peripheral oedema, or nocturnal hypoxia. 3 O2 can also be prescribed for terminally ill patients.

ARDS, or acute lung injury, may be caused by direct lung injury or occur secondary to severe systemic illness. Lung damage and release of inflammatory mediators cause increased capillary permeability and non-cardiogenic pulmonary oedema, often ac-companied by multiorgan failure.

Causes Pulmonary: Pneumonia; gastric aspiration; inhalation; injury; vasculitis (p556); contusion. Other: Shock; septicaemia; haemorrhage; multiple transfusions; DIC (p252); pancreatitis; acute liver failure; trauma; head injury; malaria; fat embo-lism; burns; obstetric events (eclampsia; amniotic fluid embolus); drugs/toxins (as-pirin, heroin, paraquat).

Clinical features Cyanosis; tachypnoea; tachycardia; peripheral vasodilation; bilat-eral fine inspiratory crackles.

Investigations FBC, U&E, LFT, amylase, clotting, CRP, blood cultures, ABG. CXR shows bilateral pulmonary infiltrates. Pulmonary artery catheter to measure pulmonary capillary wedge pressure (PCWP).

Diagnostic criteria One consensus requires these four to exist:’™ 1 Acute onset. 2 CXR: bilateral infiltrates (fig 4.15). 3 PCWP <19mmHg or a lack of clinical congestive heart failure. 4 Refractory hypoxaemia with PaO2: FiO2 <200 for ARDS. Others include total thoracic compliance <20mL/cmH2O.

Management Admit to ITU; give supportive therapy; treat the underlying cause.

Sepsis: Identify organism(s) and treat. If septic, but no organisms cultured, use empirical broad-spectrum antibiotics (p167). Avoid nephrotoxic antibiotics.

Other: Nutritional support: enteral is best: p584 & p586, with high-fat, antioxidant formulations. Steroids protect those at risk of fat embolization and with pneumo-cystosis and may improve outcome in subacute ARDS. Their role in established ARDS is controversial.™°, ™¹

Prognosis Overall mortality is 50–75%. Prognosis varies with age of patient, cause (pneumonia 86%, trauma 28%), and number of organs involved (three organs in-volved for >1wk is ‘invariably’ fatal).

Respiratory failure occurs when gas exchange is inadequate, resulting in hypoxia. It is defined as a PaO2 <8kPa and subdivided into two types according to PaCO2 level.

Type I respiratory failure Defined as hypoxia (PaO2 <8kPa) with a normal or low PaCO2. It is caused primarily by ventilation/perfusion (V/Q) mismatch, hypoventila-tion, abnormal diffusion, right to left cardiac shunts. Examples of V/Q mismatch:

Pneumonia.

Pulmonary oedema.

PE.

Asthma.

Emphysema.

Pulmonary fibrosis.

ARDS (p186).

Type II respiratory failure Defined as hypoxia (PaO2 <8kPa) with hypercapnia (PaCO2 >6.0kPa). This is caused by alveolar hypoventilation, with or without V/Q mis-match. Causes include:

Pulmonary disease: Asthma, COPD, pneumonia, end-stage pulmonary fibrosis, ob-structive sleep apnoea (OSA, p194).

Reduced respiratory drive: Sedative drugs, CNS tumour or trauma.

Neuromuscular disease: Cervical cord lesion, diaphragmatic paralysis, poliomyeli-tis, myasthenia gravis, Guillain–Barré syndrome.

Thoracic wall disease: Flail chest, kyphoscoliosis.

Clinical features are those of the underlying cause together with symptoms and signs of hypoxia, with or without hypercapnia.

Hypoxia: Dyspnoea; restlessness; agitation; confusion; central cyanosis. If long-standing hypoxia: polycythaemia; pulmonary hypertension; cor pulmonale.

Hypercapnia: Headache; peripheral vasodilation; tachycardia; bounding pulse; tremor/flap; papilloedema; confusion; drowsiness; coma.

Investigations are aimed at determining the underlying cause:

Blood tests: FBC, U&E, CRP, ABG. See table 4.7.

Radiology: CXR.

Microbiology: sputum and blood cultures (if febrile).

Spirometry (COPD, neuromuscular disease, Guillain–Barré syndrome).

Management Depends on the cause:

Type I respiratory failure:

Treat underlying cause.

Give oxygen (24–60%) by facemask.

Assisted ventilation if PaO2 <8kPa despite 60% O2.

Type II respiratory failure: The respiratory centre may be relatively insensitive to CO2 and respiration could be driven by hypoxia.

Treat underlying cause.

Controlled oxygen therapy: start at 24% O2. ►Oxygen therapy should be given with care. Nevertheless, don't leave the hypoxia untreated.

Recheck ABG after 20min. If PaCO2 is steady or lower, increase O2 concentration to 28%. If PaCO2 has risen >1.5kPa and the patient is still hypoxic, consider assisted ventilation (eg NIPPV, p812, ie non-invasive positive pressure ventilation).

If this fails, consider intubation and ventilation, if appropriate.

Causes PES usually arise from a venous thrombosis in the pelvis or legs. Clots break off and pass through the veins and the right side of the heart before lodging in the pulmonary circulation. Rare causes: RV thrombus (post-MI); septic emboli (right-sided endocarditis); fat, air, or amniotic fluid embolism; neoplastic cells; parasites.

Risk factors

Recent surgery, especially abdominal/pelvic or hip/knee replacement.

Thrombophilia, eg antiphospholipid syndrome (p274).

Leg fracture.

Prolonged bed rest/reduced mobility.

Malignancy.

Pregnancy/postpartum; combined contraceptive pill; HRT (lower risk).

Previous PE.

Tests

FBC, U&E, baseline clotting, D-dimers (BOX).

ABG may show ‘PaO2 and ‘PaCO2.

Imaging: CXR may be normal, or show oligaemia of affected segment, dilated pul-monary artery, linear atelectasis, small pleural effusion, wedge-shaped opacities or cavitation (rare). CTPA—see fig 4.16.

ECG may be normal, or show tachycardia, right bundle branch block, right vent-ricular strain (inverted T in V1 to V4). The classical SI QIII TIII pattern (p98) is rare.

Further investigations are shown on p818.

Unprovoked PE In patients with no known provoking risk factors, consider inves-tigation for possible underlying malignany. Undertake full history, examination (including breast), CXR, FBC, calcium, LFTS, urinalysis. Patients >40yrs consider abdo-pelvic CT and mammography in women. Consider antiphospholipid and thrombo-philia testing if family history positive (p274).

Prevention Give heparin to all immobile patients. Stop HRT and the combined con-traceptive pill pre-op (if reliable with another form of contraception).

Investigating suspected PE

Diagnosis of PE is improved by adopting a stepwise approach, combining an objec-tive probability score, with subsequent investigations, as follows.

Assess the clinical probability of a PE: Many systems exist and one of the most frequently used is the modified Wells Criteria (table 4.8).

Table 4.8 Modified two-level PE Wells score

| Increase probability of asthma | Lower probability of asthma | | — | — | | Wheeze, SOB, chest tightness | Prominent dizziness, lightheadedness, tingling | | Diurnal variation | Chronic productive cough with no wheeze | | Response to exercise, allergen, cold air | Normal examination when symptomatic | | Symptoms after aspirin or β-blocker | Change in voice | | History of atopy | Symptoms with colds only | | Family history atopy/asthma | Significant smoking history (>20 pack year) | | Widespread wheeze heard on auscultation | Cardiac disease | | Unexplained low FEV1 or PEF | Normal PEF when symptomatic | | Unexplained peripheral blood eosinophilia | Unexplained peripheral blood eosinophilia |

Wells PS, Anderson DR, Rodger M, et al. ‘Derivation of a Simple Clinical Model to Categorize Patients Probability of Pulmonary Embolism: Increasing the Models Utility with the SimpliRED D-dimer’.

Thromb Haemost 2000; 83: 416–20.

PE suspected

Calculate modified Wells score

Length of treatment

Provoked: 2 months and then reassess risk to benefit profile (depends on whether risk factor persists)

Unprovoked: treatment is usually continued for >2 months (people with no identifiable risk factor)

Malignancy: continue treatment with LMWH for 6 months or until cure of cancer

Pregnancy: LMWH is continued until delivery/end of pregnancy

Fig 4.16 Investigation and management of PE.

Definitions A pleural effusion is fluid in the pleural space. Effusions can be divided by their protein concentration into transudates (<25g/L) and exudates (>25g/L), see BOX. Blood in the pleural space is a haemothorax, pus in the pleural space is an em-pyema, and chyle (lymph with fat) is a chylothorax. Both blood and air in the pleural space is called a haemopneumothorax.

Symptoms Asymptomatic—or dyspnoea, pleuritic chest pain.

Tests CXR: Small effusions blunt the costophrenic angles, larger ones are seen as water-dense shadows with concave upper borders. A completely flat horizontal up-per border implies that there is also a pneumothorax.

Ultrasound is useful in identifying the presence of pleural fluid and in guiding diag-nostic or therapeutic aspiration.

Pleural biopsy: If pleural fluid analysis is inconclusive, consider parietal pleural biopsy. Thoracoscopic or CT-guided pleural biopsy increases diagnostic yield (by enabling direct visualization of the pleural cavity and biopsy of suspicious areas).

Management is of the underlying cause.

Drainage: If the effusion is symptomatic, drain it, repeatedly if necessary. Fluid is best removed slowly (0.5–1.5L/24h). It may be aspirated in the same way as a diagnostic tap, or using an intercostal drain (see p766).

Pleurodesis with talc may be helpful for recurrent effusions. Thorascopic mechani-cal pleurodesis is most effective for malignant effusions. Empyemas (p170) are best drained using a chest drain, inserted under ultrasound or CT guidance.

Intra-pleural alteplase and dornase alfa may help with empyema.

Surgery: Persistent collections and increasing pleural thickness (on ultrasound) requires surgery.™o

Table 4.9 Pleural fluid analysis

| | Inhaled aerosol | Inhaled powder | Nebulized (supervised) | | — | — | — | — | | Salbutamol | Salbutamol | Salbutamol | Salbutamol | | Dose example: | 100–200mcg/6h | 200–400mcg/6h | 2.5–5mg/6h | | Airomir® is a CFC-free example of a breath-actuated inhaler | Airomir® is a CFC-free example of a breath-actuated inhaler | Airomir® is a CFC-free example of a breath-actuated inhaler | Airomir® is a CFC-free example of a breath-actuated inhaler | | Terbutaline | Terbutaline | Terbutaline | Terbutaline | | Single dose | | 500mcg | 2.5mg/mL | | Recommended regimen | | 500mcg/6h | 5–10mg/6–12h | | Salmeterol | Salmeterol | Salmeterol | Salmeterol | | Dose/puff | 25mcg | 50mcg | — | | Recommended regimen | 50–100mcg/12h | 50–100mcg/12h | — | | Tiotropium bromide (COPD) | Tiotropium bromide (COPD) | Tiotropium bromide (COPD) | Tiotropium bromide (COPD) | | Dose/puff | 2.5mcg | 9mcg | — | | Recommended regimen | 25mcg daily | 18mcg daily | — | | Steroids | Steroids | Steroids | Steroids | | (Clenil Modulite®=beclometasone; Pulmicort®=budesonide;* Flixotide®=fluticasone) | (Clenil Modulite®=beclometasone; Pulmicort®=budesonide;* Flixotide®=fluticasone) | (Clenil Modulite®=beclometasone; Pulmicort®=budesonide;* Flixotide®=fluticasone) | (Clenil Modulite®=beclometasone; Pulmicort®=budesonide;* Flixotide®=fluticasone) | | Fluticasone (Flixotide®) | Fluticasone (Flixotide®) | Fluticasone (Flixotide®) | Fluticasone (Flixotide®) | | Doses available/puff | 50, 100, 250, & 500mcg | As for aerosol | 250mcg/mL | | Recommended regimen | 100–250mcg/12h | 100–250mcg/12h max 1mg/12h | 0.5–2mg/12h | | Clenil Modulite® | Clenil Modulite® | Clenil Modulite® | Clenil Modulite® | | Doses available/puff | 50

& 100mcg | — | — | | 250mcg | 250mcg | 250mcg | 250mcg | | Recommended regimen | 200mcg/12h then | | | | 400mcg/12h then | 400mcg/12h then | 400mcg/12h then | 400mcg/12h then | | 1000mcg/12h | 1000mcg/12h | 1000mcg/12h | 1000mcg/12h | | *Available as a Turbohaler®; Autohalers® are an alternative (breath-actuated) and don't need breathing coordination, eg Airomir® (salbutamol) and Qvar® (beclometasone). Accuhalers® deliver dry powders (eg Flixotide®, Serevent®). Systemic absorption (via the throat) is less if inhalation is through a large-volume device, eg Volumatic® or AeroChamber Plus® devices. The latter is more compact. Static charge on some devices reduces dose deliv-ery, so wash in water before dose; leave to dry (don't rub). It's pointless to squirt many puffs into a device: it is best to repeat single doses, and be sure to inhale as soon as the drug is in the spacer. SE: local (oral) candidiasis (p277); •rate of cataract if lifetime dose ?2g beclometasone.37 | *Available as a Turbohaler®; Autohalers® are an alternative (breath-actuated) and don't need breathing coordination, eg Airomir® (salbutamol) and Qvar® (beclometasone). Accuhalers® deliver dry powders (eg Flixotide®, Serevent®). Systemic absorption (via the throat) is less if inhalation is through a large-volume device, eg Volumatic® or AeroChamber Plus® devices. The latter is more compact. Static charge on some devices reduces dose deliv-ery, so wash in water before dose; leave to dry (don't rub). It's pointless to squirt many puffs into a device: it is best to repeat single doses, and be sure to inhale as soon as the drug is in the spacer. SE: local (oral) candidiasis (p277); •rate of cataract if lifetime dose ?2g beclometasone.37 | *Available as a Turbohaler®; Autohalers® are an alternative (breath-actuated) and don't need breathing coordination, eg Airomir® (salbutamol) and Qvar® (beclometasone). Accuhalers® deliver dry powders (eg Flixotide®, Serevent®). Systemic absorption (via the throat) is less if inhalation is through a large-volume device, eg Volumatic® or AeroChamber Plus® devices. The latter is more compact. Static charge on some devices reduces dose deliv-ery, so wash in water before dose; leave to dry (don't rub). It's pointless to squirt many puffs into a device: it is best to repeat single doses, and be sure to inhale as soon as the drug is in the spacer. SE: local (oral) candidiasis (p277); •rate of cataract if lifetime dose ?2g beclometasone.37 | *Available as a Turbohaler®; Autohalers® are an alternative (breath-actuated) and don't need breathing coordination, eg Airomir® (salbutamol) and Qvar® (beclometasone). Accuhalers® deliver dry powders (eg Flixotide®, Serevent®). Systemic absorption (via the throat) is less if inhalation is through a large-volume device, eg Volumatic® or AeroChamber Plus® devices. The latter is more compact. Static charge on some devices reduces dose deliv-ery, so wash in water before dose; leave to dry (don't rub). It's pointless to squirt many puffs into a device: it is best to repeat single doses, and be sure to inhale as soon as the drug is in the spacer. SE: local (oral) candidiasis (p277); •rate of cataract if lifetime dose ?2g beclometasone.37 |

* Light's criteria for defining an exudate: effusion protein/serum protein >0.5; effusion LDH/serum LDH >0.6; effusion LDH > ⅔ upper reference range. 98% sensitive and 82% specific.

Inflammation of the pleura caused by pneumonia may lead to infected pleural fluid (empyema); if it is not infected, the term parapneumonic effusion is used.

This disorder is characterized by intermittent closure/collapse of the pharyngeal airway causing apnoeic episodes during sleep. These are terminated by partial arousal.

Clinical features The typical patient is an obese, middle-aged man who presents because of snoring or daytime somnolence. His partner often describes apnoeic epi-sodes during sleep.

Loud snoring.

Daytime somnolence.

Poor sleep quality.

Morning headache.

Decreased libido.

Nocturia.

‘Cognitive performance.

Complications Pulmonary hypertension; type II respiratory failure (p188). Sleep apnoea is also reported as an independent risk factor for hypertension.™¹

Investigations Simple studies (eg pulse oximetry, video recordings) may be all that are required for diagnosis. Polysomnography (which monitors oxygen saturation, airflow at the nose and mouth, ECG, EMG chest, and abdominal wall movement during sleep) is diagnostic. The occurrence of 15 or more episodes of apnoea or hypopnoea during 1h of sleep, on average, indicates significant sleep apnoea.

Management

Weight reduction.

Avoidance of tobacco and alcohol.

Mandibular advancement device.

CPAP via a nasal mask during sleep is effective and recommended by NICE for those with moderate to severe disease.™²

Surgery to relieve pharyngeal or nasal obstruction, eg tonsillectomy or polypectomy, is occasionally needed.

Cor pulmonale is right heart failure caused by chronic pulmonary arterial hyperten-sion. Causes include chronic lung disease, pulmonary vascular disorders, and neuro-muscular and skeletal diseases (see BOX).

Clinical features Symptoms include dyspnoea, fatigue, and syncope. Signs: cya-nosis; tachycardia; raised JVP with prominent a and v waves; RV heave; loud P2, pan-systolic murmur (tricuspid regurgitation); early diastolic Graham Steell murmur; hepatomegaly and oedema.

Investigations FBC: Hb and haematocrit • (secondary polycythaemia). ABG: hypoxia, with or without hypercapnia. CXR: enlarged right atrium and ventricle, prominent pulmonary arteries (see fig 4.17). ECG: P pulmonale; right axis deviation; right ven-tricular hypertrophy/strain.

Management

Treat underlying cause—eg COPD and pulmonary infections.

Treat respiratory failure—in the acute situation give 24% oxygen if PaO2 <8kPa. Monitor ABG and gradually increase oxygen concentration if PaCO2 is stable (p188). In COPD patients, long-term oxygen therapy (LTOT) for 16h/d increases survival (p184). Patients with chronic hypoxia when clinically stable should be assessed for LTOT.

Treat cardiac failure with diuretics such as furosemide, eg 40–160mg/24h PO. Moni-tor U&E and give amiloride or potassium supplements if necessary. Alternative: spironolactone.

Consider venesection if haematocrit >55%.

Consider heart–lung transplantation in young patients.

Prognosis Poor. 50% die within 5yrs.

Fig 4.17 PA chest radiograph showing enlarged pulmonary arteries from pulmonary artery hyper-tension. When caused by interstitial lung disease and leading to right heart failure, this would be termed cor pulmonale. No signs of interstitial lung disease are identifiable in this image.

A multisystem granulomatous disorder of unknown cause. Prevalence highest in Northern Europe, eg UK: 10–20/105 population. Usually affects adults aged 20–40yrs, more common in women. African–Caribbeans are affected more

frequently and more severely than Caucasians, particularly by extra-thoracic disease. Associated with HLA-DRB1 and DQB1 alleles. For other causes of granuloma see table 4.10.

Clinical features In 20–40%, the disease is discovered incidentally, after a routine CXR, and is thus asymptomatic. Acute sarcoidosis often presents with fever, erythe-ma nodosum (fig 12.21, p562),6 polyarthralgia, and bilateral hilar lymphodenopathy (BHL), also called Löfgren syndrome, which usually resolves spontaneously.

Pulmonary disease: 90% have abnormal CXRs with BHL (fig 4.18) ! pulmonary infil-trates or fibrosis; see later in topic for staging. Symptoms: Dry cough, progressive dyspnoea, ‘exercise tolerance, and chest pain. In 10–20%, symptoms progress, with concurrent deterioration in lung function.

Tests Blood: ↑ESR, lymphopenia, ↑LFT, ↑serum ACE in r60% (non-specific), ↑Ca2+,

Bronchoalveolar lavage (BAL): Shows •lymphocytes in active disease; •neutrophils with pulmonary fibrosis.

Transbrochial biopsy: May be diagnostic.

Ultrasound: May show nephrocalcinosis or hepatosplenomegaly.

Bone X-rays: Show ‘punched out’ lesions in terminal phalanges.

CT/MRI: May be useful in assessing severity of pulmonary disease or diagnosing neurosarcoidosis. Ophthalmology assessment (slit lamp examination, fluorescein angiography) is indicated in ocular disease.

Management ►Patients with BHL alone don’t need treatment as most recover spon-taneously.™3, ™’ Acute sarcoidosis: bed rest, NSAIDS.

Indications for corticosteroids:

Parenchymal lung disease (symptomatic, static, or progressive).

Uveitis.

Hypercalcaemia.

Neurological or cardiac involvement.

Prednisolone (40mg/24h) PO for 4–6 wks, then ‘dose over 1yr according to clinical status. A few patients relapse and may need a further course or long-term therapy.

Other therapy: In severe illness, IV methylprednisolone or immunosuppressants (methotrexate, hydroxychloroquine, ciclosporin, cyclophosphamide) may be needed. Anti-TNFa therapy may be tried in refractory cases, or lung transplantation.

Prognosis 60% of patients with thoracic sarcoidosis resolve over 2yrs. 20% respond to steroid therapy; in the rest, improvement is unlikely despite therapy.7

A detailed history and exam (including for synovitis) + CXR, 2 ASO (antistreptolysin-O) titres and a tuber-culin skin test are usually enough to diagnose erythema nodosum.

ACE is also • in: hyperthyroidism, Gaucher’s, silicosis, TB, hypersensitivity pneumonitis, asbestosis, pneu-mocystosis.™™ •ACE levels in CSF help diagnose CNS sarcoidosis (when serum ACE may be normal).™ª ACE is lower in: Caucasians; and anorexia.™7

Table 4.10 Differential diagnosis of granulomatous diseases

| Severity assessment in COPD Severity assessment has implications for therapy and prognosis. The BODE index (Body mass index, airflow Obstruction, Dyspnoea and Exercise capacity) helps predict outcome and number and severity of exacerbations. The Global Initia-tive for COPD (GOLD) categorizes severity of COPD into four stages (mild, moderate, severe, and very severe) based on post-bronchodilator FEV1% predicted, but it is not useful for predicting total mortality for 2 years of follow-up and onwards.’² | | — |

Fig 4.18 PA chest radiograph showing bilateral hilar lymphadenopathy. The important differ-entials for this appearance are: sarcoidosis, TB, lymphoma, pneumoconioses, and metastatic disease. This patient has sarcoidosis but there are no other stigmata (such as the presence of infiltrates, fibrosis, and honeycombing) on this image.

This is the generic term used to describe a number of conditions that primarily af-fect the lung parenchyma in a diffuse manner.™8 They are characterized by chronic inflammation and/or progressive interstitial fibrosis (table 4.11), and share a number of clinical and pathological features. See table 4.11 and fig 4.19.

Clinical features Dyspnoea on exertion; non-productive paroxysmal cough; ab-normal breath sounds; abnormal CXR or high-resolution CT; restrictive pulmonary spirometry with ↓DLCO (p164).

Pathological features Fibrosis and remodelling of the interstitium; chronic inflam-mation; hyperplasia of type II epithelial cells or type II pneumocytes.

Classification The ILDS can be broadly grouped into three categories:

Those with known cause, eg:

Occupational/environmental, eg asbestosis, berylliosis, silicosis, cotton worker's lung (byssinosis).

Drugs, eg nitrofurantoin, bleomycin, amiodarone, sulfasalazine, busulfan.

Hypersensitivity reactions, eg hypersensitivity pneumonitis.

Infections, eg TB, fungi, viral.

Gastro-oesophageal reflux.

Those associated with systemic disorders, eg:

Sarcoidosis.

Rheumatoid arthritis.

SLE, systemic sclerosis, mixed connective tissue disease, Sjögren's syndrome.

Ulcerative colitis, renal tubular acidosis, autoimmune thyroid disease.

Idiopathic, eg:

Idiopathic pulmonary fibrosis (IPF, p200).

Cryptogenic organizing pneumonia.

Non-specific interstitial pneumonitis.

Causes

Bird-fancier's and pigeon-fancier's lung (proteins in bird droppings).

Farmer's and mushroom worker's lung (Micropolyspora faeni, Thermoactino-myces vulgaris).

Malt worker's lung (Aspergillus clavatus).

Bagassosis or sugar worker's lung (Thermoactinomyces sacchari).

Clinical features 4–6h post-exposure: Fever, rigors, myalgia, dry cough, dyspnoea, fine bibasal crackles. Chronic: Finger clubbing (50%), increasing dyspnoea, ‘weight, exertional dyspnoea, type I respiratory failure, cor pulmonale.

Management Acute: Remove allergen and give O2 (25–60%), PO prednisolone (40mg/24h PO), reducing course. Chronic: Allergen avoidance, or wear a facemask or +ve pressure helmet. Long-term steroids often achieve CXR and physiological im-provement. Compensation (UK Industrial Injuries Act) may be payable.

Fig 4.19 AP chest radiograph showing air-space shadowing in the left upper zone. Although this appearance often represents infection, it is non-specific. Differential diagnosis for this distribution of shadowing include lymphoma, alveolar cell carcinoma (both to be considered if not resolving in appearance on follow-up imaging), and haemorrhage.

Table 4.11 Causes of fibrotic shadowing on a CXR

| Risk factors for ARDS Sepsis • Massive transfusion Hypovolaemic shock • Burns (p846) Trauma • Smoke inhalation (p847) Pneumonia • Near drowning Diabetic ketoacidosis • Acute pancreatitis Gastric aspiration • DIC (p252) Pregnancy • Head injury Eclampsia • ↑ICP Amniotic fluid embolus • Fat embolus Drugs/toxins • Heart/lung bypass Paraquat, heroin, aspirin • Tumour lysis syndrome (p529) Pulmonary contusion • Malaria. | | — |

This is a type of idiopathic interstitial pneumonia. Inflammatory cell infiltrate and pulmonary fibrosis of unknown cause. It is the commonest cause of interstitial lung disease.

Symptoms Dry cough; exertional dyspnoea; malaise; ‘weight; arthralgia.

Signs Cyanosis; finger clubbing; fine end-inspiratory crepitations.

Complications Respiratory failure; •risk of lung cancer.

lymphocytes (good response/prognosis) or •neutrophils and •eosinophils (poor response/prognosis). 99TCm-DTPA scan: (diethylene-triamine-penta-acetic acid) May reflect disease activity.™ª Lung biopsy: May be needed for diagnosis.The histological changes observed on biopsy are referred to as usual interstitial pneumonia (UIP).

Management Supportive care: oxygen, pulmonary rehabilitation, opiates, palliative care input. All patients should be considered for current clinical trials or lung trans-plantation.ªº It is strongly recommended that high-dose steroids are not used except where the diagnosis of IPF is in doubt.

Prognosis 50% 5yr survival rate (range 1–20yrs).

Fig 4.20 Interstitial lung disease due to idiopathic pulmonary fibrosis (a similar appearance to the interstitial oedema of moderate left heart failure, but without a big heart).

Coal worker’s pneumoconiosis (CWP) A common dust disease in countries that have or have had underground coal-mines. It results from inhalation of coal dust par-ticles (1–2μm in diameter) over 15–20yrs. These are ingested by macrophages which die, releasing their enzymes and causing fibrosis.

Clinical features: Asymptomatic, but coexisting chronic bronchitis is common. CXR: many round opacities (1–10mm), especially in upper zone.

Management: Avoid exposure to coal dust; treat co-existing chronic bronchitis; claim compensation (in the UK, via the Industrial Injuries Act).

Progressive massive fibrosis (PMF) Due to progression of CWP, which causes pro-gressive dyspnoea, fibrosis, and, eventually, cor pulmonale. CXR: usually bilateral, upper-mid zone fibrotic masses (1–10cm), develop from periphery towards hilum.

Management: Avoid exposure to coal dust; claim compensation (as for CWP).

Caplan’s syndrome The association between rheumatoid arthritis, pneumoconiosis, and pulmonary rheumatoid nodules.

Silicosis (See fig 4.21.) Caused by inhalation of silica particles, which are very fibro-genic. A number of jobs may be associated with exposure, eg metal mining, stone quarrying, sandblasting, and pottery/ceramic manufacture.

Clinical features: Progressive dyspnoea, •incidence of TB, CXR shows diffuse miliary or nodular pattern in upper and mid-zones and egg-shell calcification of hilar nodes. Spirometry: restrictive ventilatory defect.

Management: Avoid exposure to silica; claim compensation (as for CWP).

Asbestosis Caused by inhalation of asbestos fibres. Asbestos was commonly used in the building trade for fire proofing, pipe lagging, electrical wire insulation, and roofing felt. Degree of asbestos exposure is related to degree of pulmonary fibrosis.

Clinical features: Similar to other fibrotic lung diseases with progressive dyspnoea, clubbing, and fine end-inspiratory crackles. Also causes pleural plaques, •risk of bronchial adenocarcinoma and mesothelioma.

Management: Symptomatic. Patients are often eligible for compensation through the UK Industrial Injuries Act.

Mesothelioma See p174.

Endocrinology

We thank Dr Stephen Gilbey, our Specialist Reader for this chapter.

The essence of endocrinology

For scientists

Define a syndrome, and match it to a gland malfunction.

Measure the gland's output in the peripheral blood. Define clinical syndromes associated with too much or too little secretion (hyper- and hypo-syndromes, respectively; eu- means normal, neither • nor ', as in euthyroid). Note factors that may make measurement variable, eg diurnal release of cortisol.

If suspecting hormone deficiency, test by stimulating the gland that produces it (eg short ACTH stimulation test or Synacthen® test in Addison's). If the gland is not functioning normally, there will be a blunted response to stimulation.

If suspecting hormone excess, test by inhibiting the gland that produces it (eg dexamethasone suppression test in Cushing's). If there is a hormone-secreting tumour then this will fail to suppress via normal feedback mechanisms.

Find a way to image the gland. NB: non-functioning tumours or 'incidentalomas' may be found in health, see p224.

Imaging alone does not make the diagnosis.

For those doing exams

Thyrotoxicosis: Hair loss; pretibial myxoedema (confusing term, p218); onycho-lysis (nail separation from the nailbed); bulging eyes (exophthalmos/proptosis).

Hypothyroidism: Hair loss; eyebrow loss; cold, pale skin; characteristic face. You might, perhaps should, fail your exam if you blurt out 'Toad-like face'.

Cushing's syndrome: Central obesity and wasted limbs (='lemon on sticks' see fig 5.2); moon face; buffalo hump; supraclavicular fat pads; striae.

Addison's disease: Hyperpigmentation (face, neck, palmar creases).

Acromegaly: Acral (distal) + soft tissue overgrowth; big jaws (macro-gnathia), hands and feet; the skin is thick; facial features are coarse.

Hyperandrogenism (φ): Hirsutism; temporal balding; acne. Hypopituitarism: Pale or yellow tinged thinned skin, resulting in fine wrinkling around the eyes and mouth, making the patient look

older.

Hypoparathyroidism: Dry, scaly, puffy skin; brittle nails; coarse hair.

Pseudohypoparathyroidism: Short stature, short neck, and short

4th and 5th metacarpals.

Fig 5.4 Neuroregulation and integration of endocrine axes makes us who we are—and who we are and what we do feeds back into our hormonal milieu. Multifactorial disruptions within the growth hormone (GH), luteinizing hormone (LH)–testosterone, adrenocorticotropin (ACTH)–cortisol and insu-lin axes play a major role in healthy maturation and ageing.

Categories of diabetes and dyslipidaemia

Impaired glucose tolerance (IGT): Fasting plasma glucose <7mmol/L and OGTT (oral glucose tolerance) 2h glucose ? 7.8mmol/L but <11.1mmol/L.

Impaired fasting glucose (IFG): Fasting plasma glucose ? 6.1mmol/L but <7mmol/L (WHO criteria). Do an OGTT to exclude DM. The cut-off point is somewhat arbitrary.

IGT and IFG denote different abnormalities of glucose regulation (post-prandial and fasting). There may be lower risk of progression to DM in IFG than IGT. Manage both with lifestyle advice (p92) + annual review. Incidence of DM if IFG and HbA1C at high end of normal (27–46mmol/mol) is r25%.™

Other causes of DM ►Steroids; anti-HIV drugs; newer antipsychotics.

Pancreatic: pancreatitis; surgery (where >90% pancreas is removed); trauma; pancreatic destruction (haemochromatosis, cystic fibrosis); pancreatic cancer.

Cushing's disease; acromegaly; phaeochromocytoma; hyperthyroidism; pregnancy.

Others: congenital lipodystrophy; glycogen storage diseases.

Chicken or egg? Most type 2 diabetes-associated genes have a function in the vasculature, and stress in β-cells can result from vascular defects in the pancreas, so maybe vascular events trigger DM.[a]

Type 1 DM Insulin (see BOX 'Using insulin').

Type 2 DM See fig 5.5.

Fig 5.5 Management of type 2 diabetes. Aim for HbA1C 48mmol/mol or 53 if two or more agents.

Data from Algorithm for blood glucose lowering therapy in adults with type 2 diabetes, http://www.nice.org.uk/guidance/ng28/chapter/1-Recommendations#drug-treatment-2

Prospective studies show that good control of hyperglycaemia is key to preventing microvascular complications in type 1 and 2 DM.[1o] ►Find out what problems are be-ing experienced (eg glycaemic control, morale, erectile dysfunction—p220).

Assess vascular risk BP control (see BOX 'Controlling blood pressure in diabetes') is crucial for preventing macrovascular disease and 'mortality. Refer to smoking ces-sation services. Check plasma lipids.

Look for complications •Check injection sites for infection or lipohypertrophy (fatty change): advise on rotating sites of injection if present.

Diabetic retinopathy: Blindness is preventable. ►Annual retinal screening man-datory for all patients. Refer to an ophthalmologist if pre-proliferative changes or if any uncertainty at or near the macula (the only place capable of 6/6 vision).

Pre-symptomatic screening enables laser photocoagulation to be used, aimed to stop production of angiogenic factors from the ischaemic retina. Indications: macu-lopathy or proliferative retinopathy. See figs 5.6–5.9.

Background retinopathy: Microaneurysms (dots), haemorrhages (blots), and hard exudates (lipid deposits). Refer if near the macula, eg for intravitreal tri-amcinolone.

Pre-proliferative retinopathy: Cotton-wool spots (eg infarcts), haemorrhag-es, venous beading. These are signs of retinal ischaemia. Refer to a specialist.

Proliferative retinopathy: New vessels form. Needs urgent referral.

Cataracts: May be juvenile 'snowflake' form, or 'senile'—which occur earlier in diabetic subjects. Osmotic changes in the lens induced in acute hyperglycaemia reverse with normoglycaemia (so wait before buying glasses).

Rubeosis iridis: New vessels on iris: occurs late and may lead to glaucoma.

Metabolic complications: p822.

Diabetic feet: p212.

Neuropathy: p212.

As DM has so many vascular events, particularly encourage statin use (p690), esp. if LDL >2mmol/L or systolic BP >140. Even consider a statin whatever the pre-treatment cholesterol; discuss with your patient.

University Press.

Refer early to foot services (podiatry, imaging, vascular surgery).

Amputations are common (125/week)—and preventable: good care saves legs. Examine feet regularly. Distinguish between ischaemia (critical toes ! absent foot pulses and worse outcome) and peripheral neuropathy (injury or infection over pres-sure points, eg the metatarsal heads). In practice, many have both.

Foot ulceration Typically painless, punched-out ulcer (fig 5.10) in an area of thick callus ! superadded infection. Causes cellulitis, abscess ! osteomyelitis.

Assess degree of:

Neuropathy (clinically).

Ischaemia (clinically + Doppler ! angiography).

Bony deformity, eg Charcot joint (clinically + X-ray). See fig 5.11.

Infection (swabs, blood culture, X-ray for osteomyelitis, probe ulcer to reveal depth).

Absolute indications for surgery: Abscess or deep infection; spreading anaerobic infection; gangrene/rest pain; suppurative arthritis.

Pituitary insufficiency.

Liver failure, plus some rare inherited enzyme defects.

Addison's disease.

Islet cell tumours (insulinoma) and immune hypoglycaemia (eg anti-insulin receptor antibodies in Hodgkin's disease).

Non-pancreatic neoplasms, eg fibrosarcomas and haemangiopericytomas.

When to investigate:

Whipple answered this (Whipple's triad): symptoms or signs of hypoglycemia +

'plasma glucose + resolution of symptoms or signs post glucose rise.

Document BM during attack and lab glucose if in hospital (monitors often not reli-able at low readings).

Take a drug history and exclude liver failure.

72h fasting may be needed (monitor closely). Bloods: glucose, insulin, C-peptide, and plasma ketones if symptomatic.

If endogenous hyperinsulinism suspected, do insulin, C-peptide, proinsulin, β-hydroxybutyrate.

Interpreting results:

Hypoglycaemic hyperinsulinaemia (HH): Causes: insulinoma, sulfonylureas, insulin injection (no detectable C-peptide—only released with endogenous insulin); non-insulinoma pancreatogenous hypoglycaemia syndrome, mutation in the insulin-re-ceptor gene. Congenital HH follows mutations in genes involved in insulin secretion (ABCC8, KCNJ11, GLUD1, CGK, HADH, SLC16A1, HNF4A, ABCC8, & KCNJ11).¹™

Insulin low or undetectable, no excess ketones. Causes: non-pancreatic neoplasm; anti-insulin receptor antibodies.

'Insulin, •ketones. Causes: alcohol, pituitary insufficiency, Addison's disease.

Post-prandial hypoglycaemia May occur after gastric/bariatric surgery ('dump-ing', p622), and in type 2 DM.

Investigation: Prolonged OGTT (5h, p206).

Treatment 2See p824. If episodes are often, advise many small high-starch meals. If post-prandial 'glucose, give slowly absorbed carbohydrate (high fibre). In diabet-ics, rationalize insulin therapy (p209).

This often benign (90–95%) pancreatic islet cell tumour is sporadic or seen with MEN-1 (p222). It presents as fasting hypoglycaemia, with Whipple's triad:

Symptoms associated with fasting or exercise.

Recorded hypoglycaemia with symptoms.

Symptoms relieved with glucose.

Screening test Hypoglycaemia + •plasma insulin during a long fast.

Suppressive tests Give IV insulin and measure C-peptide. Normally exogenous insu-lin suppresses C-peptide production, but this does not occur in insulinoma.

Imaging CT/MRI ! endoscopic pancreatic US ! IACS (see BOX; all fallible, so don't waste too much time before proceeding to intra-operative visualization ! intra-operative ul-trasound). 18F-L-2,4-dihydroxyphenylalanine PET-CT can help guide laparoscopic surgery.

Treatment Excision.

Nesidioblastosis See BOX. If this doesn't work, options are: •diet, diazoxide, dextrose IVI, enteral feeding, everolimus.

2 Diabetic ketoacidosis Results from insulin deficiency (eg unknown diagnosis, intercurrent illness, interuption of insulin therapy). See p822.

Physiology

TRH from the hypothalamus acts on the pituitary gland

Hyperthyroidism suspected: Ask for T2, T4, and TSH. All will have 'TSH (except the rare TSH-secreting pituitary adenoma). Most have ΔT4, but r1% have only raised T2.

Hypothyroidism suspected or monitoring replacement 9: Ask for only T4 and TSH. T2 does not add any extra information. TSH varies through the day: trough at 2PM; 20% higher during darkness, so during monitoring, try to do at the same time.

Sick euthyroidism: In any systemic illness, TFTS may become deranged. The typical pattern is for 'everything to be low'. The test should be repeated after recovery.

Assay interference is caused by antibodies in the serum, interfering with the test.

Other tests

Thyroid autoantibodies: Antithyroid peroxidase (TPO; formerly called microsomal) antibodies or antithyroglobulin antibodies may be increased in autoimmune thy-roid disease: Hashimoto's or Graves' disease. If +ve in Graves', there is an increased risk of developing hypothyroidism at a later stage.

TSH receptor antibody: May be • in Graves' disease (useful in pregnancy).

Serum thyroglobulin: Useful in monitoring the treatment of carcinoma (p600), and in detection of factitious (self-medicated) hyperthyroidism, where it is low.

Ultrasound: This distinguishes cystic (usually, but not always, benign) from solid (possibly malignant) nodules. If a solitary (or dominant) large nodule, in a multinod-ular goitre, do a fine-needle aspiration to look for thyroid cancer; see fig 12.22, p601.

Fig 5.12 The images are from an isotope scan, with and without markers placed over the ster-nal notch. We can see on the left that the nodule is metabolically inactive ('cold'). The hot nod-ule (right pair) is a very avid nodule causing background thyroid suppression.

The clinical effect of excess thyroid hormone, usually from gland hyperfunction.

Symptoms Diarrhoea; ↓weight; ↑appetite (if ↑↑, paradoxical weight gain in 10%); over-active; sweats; heat intolerance; palpitations; tremor; irritability; labile emotions; oligomenorrhoea ! infertility. Rarely psychosis; chorea; panic; itch; alopecia; urticaria.

Tests ↓TSH (suppressed), T4, and ↑T2. There may be mild normocytic anaemia, mild neu-tropenia (in Graves'), ↑ESR, ↑Ca2+, ↑LFT. Also: Check thyroid autoantibodies. Isotope scan if the cause is unclear, to detect nodular disease or subacute thyroiditis. If ophthal-mopathy, test visual fields, acuity, and eye movements (see BOX 'Thyroid eye disease').

Causes

Toxic multinodular goitre: Seen in the elderly and in iodine-deficient areas.There are nodules that secrete thyroid hormones. Surgery is indicated for compressive symptoms from the enlarged thyroid (dysphagia or dyspnoea).

Toxic adenoma: There is a solitary nodule producing T2 and T4. On isotope scan, the nodule is 'hot' (p216), and the rest of the gland is suppressed.

Ectopic thyroid tissue: Metastatic follicular thyroid cancer, or struma ovarii: ovar-ian teratoma with thyroid tissue.

Exogenous: Iodine excess, eg food contamination, contrast media (2thyroid storm, p824, if already hyperthyroid).

Levothyroxine excess causes •T4, 'T2, 'thyroglobulin.

Others: 1 Subacute de Quervain's thyroiditis: self-limiting post-viral with painful goitre, •T° ! •ESR. Low isotope uptake on scan. 9: NSAIDS. 2 Drugs: amiodarone (p220), lithium (hypothyroidism more common). 3 Postpartum. 4 TB (rare).

Treatment

Radioiodine (131I): Most become hypothyroid post-treatment. There is no evi-dence for •cancer, birth defects, or infertility in women. CI: pregnancy, lactation. Caution in active hyperthyroidism as risk of thyroid storm (p825).

Thyroidectomy (usually total): Carries a risk of damage to recurrent laryngeal nerve (hoarse voice) and hypoparathyroidism. Patients will become hypothyroid, so thyroid replacement needed.

In pregnancy and infancy: Get expert help. See OHCS p24 & OHCS p182.

Complications Heart failure (thyrotoxic cardiomyopathy, • in elderly), angina, AF (seen in 10–25%: control hyperthyroidism and warfarinize if no contraindication), osteoporosis, ophthalmopathy, gynaecomastia. 2Thyroid storm (p825).

Table 5.2 Manifestations of Graves' disease—and pathophysiology

| Differentiating type 1 and 2 diabetes Occasionally it may be difficult to differentiate whether a patient has type 1 or 2 DM, although they can present differently (see table 5.1). Features of type 1 include weight loss; persistent hyperglycaemia despite diet and medications; presence of autoantibodies: islet cell antibodies (ICA) and anti-glutamic acid decarboxylase (GAD) antibodies; ketonuria. Table 5.1 Differences between type 1 and type 2 diabetes | Differentiating type 1 and 2 diabetes Occasionally it may be difficult to differentiate whether a patient has type 1 or 2 DM, although they can present differently (see table 5.1). Features of type 1 include weight loss; persistent hyperglycaemia despite diet and medications; presence of autoantibodies: islet cell antibodies (ICA) and anti-glutamic acid decarboxylase (GAD) antibodies; ketonuria. Table 5.1 Differences between type 1 and type 2 diabetes | | — | — | | | Type 1 DM Type 2 DM | | | Epidemiology Often starts before puberty Older patients (usually) | | | Genetics HLA D2 and D4 linked No HLA association | | | Cause Autoimmune β-cell destruction Insulin resistance/β-cell dysfunction | | |

Presentation Polydipsia, polyuria, ‘weight, ketosis Asymptomatic/complications, eg MI | | Not all new-onset DM in older people is type 2: if ketotic ! a poor response to oral hy-poglycaemics (and patient is slim or has a family or personal history of autoimmunity), think of latent autoimmune diabetes in adults (LADA) and measure islet cell antibodies. | Not all new-onset DM in older people is type 2: if ketotic ! a poor response to oral hy-poglycaemics (and patient is slim or has a family or personal history of autoimmunity), think of latent autoimmune diabetes in adults (LADA) and measure islet cell antibodies. |

*Emotional or tactile stimuli cause spasms; seen in any autoimmune state (eg type 1 DM); R: baclofen! IV Ig.

The clinical effect of lack of thyroid hormone. It is common (4/1000/yr). If treated, prognosis is excellent; untreated it is disastrous (eg heart disease, dementia). ►As it is insidious, both you and your patient may not realize anything is wrong, so be alert to subtle, non-specific symptoms, esp. in women —40yrs old (φ: ð≈ 6: 1).

Symptoms Tiredness; sleepy, lethargic; ‘mood; cold-disliking; •weight; constipation; menorrhagia; hoarse voice; ‘memory/cognition; dementia; myalgia; cramps; weakness.

Signs BRADYCARDIC; reflexes relax slowly; ataxia (cerebellar); dry thin hair/skin; yawning/drowsy/coma (p824); cold hands ! ‘T°; ascites ! non-pitting oedema (lids; hands; feet) ! pericardial or pleural effusion; round puffy face/ double chin/obese; de-feated demeanour; immobile ! ileus; CCF. Also: neuropathy; myopathy; goitre (fig 5.16).

Diagnosis (p216) ►Have a low threshold for doing TFTs! •TSH (eg ?4mU/L);2 ‘T4 (in rare secondary hypothyroidism: ‘T4 and ‘TSH or — due to lack from the pituitary, p222). Cholesterol and triglyceride•; macrocytosis (less often normochromic anaemia too).

Causes of primary autoimmune hypothyroidism

Primary atrophic hypothyroidism: φ:ð≈6:1. Common. Diffuse lymphocytic infilt-ration of the thyroid, leading to atrophy, hence no goitre.

Hashimoto’s thyroiditis: Goitre due to lymphocytic and plasma cell infiltration. Commoner in women aged 60–70yrs. May be hypothyroid or euthyroid; rarely initial period of hyperthyroid (‘Hashitoxicosis’). Autoantibody titres are very high.

Other causes of primary hypothyrodism

World-wide the chief cause is iodine deficiency.

Post-thyroidectomy or radioiodine treatment.

Drug-induced: Antithyroid drugs, amiodarone, lithium, iodine.

Subacute thyroiditis: Temporary hypothyroidism after hyperthyroid phase.

Secondary hypothyroidism Not enough TSH (due to hypopituitarism); very rare.

Pregnancy problems Eclampsia, anaemia, prematurity, ‘birthweight, stillbirth, PPH.

Treatment

Healthy and young: Levothyroxine (T4), 0–100mcg/24h PO; review at 12wks. Ad-just 6-weekly by clinical state and to normalize but not suppress TSH (keep TSH

>0.5mU/L). Thyroxine’s t½ is r7d, so wait r4wks before checking TSH to see if a dose change is right. NB: small changes in serum free T4 have a logarithmic effect on TSH. Once normal, check TSH yearly. Enzyme inducers (p689) •metabolism of levothyroxine.

Elderly or ischaemic heart disease: Start with 25mcg/24h; •dose by 25mcg/4wks according to TSH (►cautiously, as levothyroxine may precipitate angina or MI).

If diagnosis is in question and T4 already given: Stop T4; recheck TSH in 6 weeks.

►►Myxoedema coma The ultimate hypothyroid state before death. See p824.

►Treat the patient not the blood level! No exact cut-off in TSH can be given partly because risk of death from heart disease mirrors TSH even when in the normal range in women. Risk ≈ 1.4 if TSH 1.5–2.4 vs 1.7 if TSH 2.5–2.5. If TSH >2.65 and possibly symptomatic, a low dose of levothyroxine may be tried. Monitor symptoms, TSH and T4 carefully. Over-exposure to thyroxine may cause osteoporosis ! AF.

Fig 5.16 Facial appearance in hypothyroidism. Look for: pallor; coarse, brittle, diminished hair (scalp, axillary, and pubic); dull or blank expression lacking sparkle; coarse features; puffy lids. These signs are subtle: ►have a low threshold for measuring TSH.

Parathyroid hormone (PTH) is normally secreted in response to low ionized Ca2+ levels, by four parathyroid glands situated posterior to the thyroid (p601). The glands are controlled by –ve feedback via Ca2+ levels. PTH acts by: ••osteoclast activity releasing Ca2+ and PO 2– from bones ••Ca2+ and ‘PO 2– reabsorption in the kidney •active 1,25 dihydroxy-vitamin D2 production is •. Overall effect is •Ca2+ and ‘PO 2–.

Primary hyperparathyroidism Causes: r80% solitary adenoma, r20% hyperplasia of all glands, <0.5% parathyroid cancer. Presentation: Often ‘asymptomatic’ (►not in retro-spect!), with •Ca2+ on routine tests. Signs relate to: 1 •Ca2+ (p676): weak, tired, depressed, thirsty, dehydrated-but-polyuric; also renal stones, abdominal pain, function, age ≡50yrs. Complications: Hypoparathyroidism, recur-rent laryngeal nerve damage (m hoarse), symptomatic Ca2+’ (hungry bones syndrome; check Ca2+ daily for ?14d post-op). Pre-op US and MIBI scan may localize an adenoma; intra-operative PTH sampling is used to confirm removal. Recurrence: r8% over 10yrs.[1a]

Cinacalcet

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(a ‘calcimimetic’) •sensitivity of parathyroid cells to Ca2+ (m ‘PTH secretion); monitor Ca2+ within 1 week of dose changes; SE: myalgia; ‘testosterone.

Secondary hyperparathyroidism ‘Ca2+, •PTH (appropriately). Causes: ‘vit D intake, chronic renal failure. 9: Correct causes. Phosphate binders; vit D; cinacalcet if PTH ?85pmol/L and parathyroidectomy tricky.

Tertiary hyperparathyroidism •Ca2+, ••PTH (inappropriately). Occurs after pro-longed secondary hyperparathyroidism, causing glands to act autonomously having undergone hyperplastic or adenomatous change. This causes •Ca2+ from ••secretion of PTH unlimited by feedback control. Seen in chronic renal failure.

Malignant hyperparathyroidism Parathyroid-related protein (PTHrP) is produced by some squamous cell lung cancers, breast and renal cell carcinomas. This mimics PTH resulting in •Ca2+ (PTH is ‘, as PTHrP is not detected in the assay).

Primary hypoparathyroidism PTH secretion is ‘ due to gland failure. Tests: ‘Ca2+, PO 2–or —, —ALP. Signs: Those of hypocalcaemia, p678 ! autoimmune comorbidities (BOX ‘Autoimmune polyendocrine syndromes’). Causes: Autoimmune; congenital (Di George syn., OHCS p642). 9: Ca2+ supplements + calcitriol (or synthetic PTH /12h SC: it prevents hypercalciuria).

Secondary hypoparathyroidism Radiation, surgery (thyroidectomy, parathyroid-ectomy), hypomagnesaemia (magnesium is required for PTH secretion).

Pseudohypoparathyroidism Failure of target cell response to PTH. Signs: Short metacarpals (esp. 4th and 5th, fig 5.18), round face, short stature, calcified basal ganglia (fig 5.19), ‘IQ. Tests: ‘Ca2+, •PTH, — or •ALP. 9: As for 1° hypoparathyroidism.

Pseudopseudohypoparathyroidism The morphological features of pseudohypo-parathyroidism, but with normal biochemistry. The cause for both is genetic.

Fig 5.19 Cerebral calcification in pseudohypoparathyroid-ism: periventricular (left) and basal ganglia (right).

Cushing’s disease: Bilateral adrenal hyperplasia from an ACTH-secreting pituitary adenoma (usually a microadenoma, p224). φ: ð>1: 1. Peak age: 20–50yrs. A low-dose dexamethasone test (BOX) leads to no change in plasma cortisol, but 8mg may be enough to more than halve morning cortisol (as occurs in normals).

Ectopic ACTH production: Especially small cell lung cancer and carcinoid tum-ours, p271. Specific features: pigmentation (due to ••ACTH), hypokalaemic metabolic alkalosis (••cortisol leads to mineralocorticoid activity), weight loss, hyperglycaemia. Classical features of Cushing’s are often absent. Dexamethasone even in high doses (8mg) fails to suppress cortisol production.

Rarely, ectopic CRF production: Some thyroid (medullary) and prostate cancers.

ACTH-independent causes (↶ACTH due to Ωve feedback)

Iatrogenic: Pharmacological doses of steroids (common).

Adrenal adenoma/cancer: (May cause abdo pain ! virilization in φ, p220.) Because the tumour is autonomous, dexamethasone in any dose won’t suppress cortisol.

Adrenal nodular hyperplasia: (As for adrenal adenoma, no dexamethasone sup-pression.)

Rarely: Carney complex, p222. McCune–Albright syndrome, see OHCS p650.

Symptoms ⚠Weight; mood change (depression, lethargy, irritability, psychosis); proximal weakness; gonadal dysfunction (irregular menses; hirsutism; erectile dys-function); acne; recurrent Achilles tendon rupture; occasionally virilization if φ.

Signs Central obesity; plethoric, moon face; buffalo hump; supraclavicular fat distri-bution; skin & muscle atrophy; bruises; purple abdominal striae (fig 5.20); osteoporo-sis; •BP; •glucose; infection-prone; poor healing. Signs of the cause (eg abdo mass).

Tests Random plasma cortisols may mislead, as illness, time of day, and stress (eg venepuncture) influence results. Also, don’t rely on imaging to localize the cause: non-functioning ‘incidentalomas’ occur in r5% on adrenal CT and r10% on pituitary MRI. MRI detects only r70% of pituitary tumours causing Cushing’s (many are too small).

Treatment Depends on the cause.

Iatrogenic: Stop medications if possible.

Cushing’s disease: Selective removal of pituitary adenoma (trans-sphenoidally). Bilateral adrenalectomy if source unlocatable, or recurrence post-op (complica-tion: Nelson’s syndrome: •skin pigmentation due to ••ACTH from an enlarging pituitary tumour, as adrenalectomy removes –ve feedback; responds to pituitary radiation).

Adrenal adenoma or carcinoma: Adrenalectomy: ‘cures’ adenomas but rarely cures cancer. Radiotherapy & adrenolytic drugs (mitotane) follow if carcinoma.

Ectopic ACTH: Surgery if tumour is located and hasn’t spread. Metyrapone, ketocon-azole, and fluconazole ‘cortisol secretion pre-op or if awaiting effects of radia-tion. Intubation + mifepristone (competes with cortisol at receptors) + etomidate (blocks cortisol synthesis) may be needed, eg in severe ACTH-associated psychosis.

Prognosis Untreated Cushing's has •vascular mortality.[22] Treated, prognosis is good (but myopathy, obesity, menstrual irregularity, •BP, osteoporosis, subtle mood chang-es and DM often remain—so follow up carefully, and manage individually).

Physiology:

Causes: 80% are due to autoimmunity in the UK. Other causes: TB (commonest cause worldwide), adrenal metastases (eg from lung, breast, renal cancer), lymphoma, op-portunistic infections in HIV (eg CMV, Mycobacterium avium, p400); adrenal haem-orrhage (2Waterhouse–Friderichsen syndrome p714; antiphospholipid syndrome; SLE), congenital (late-onset congenital adrenal hyperplasia).

Secondary adrenal insufficiency The commonest cause is iatrogenic, due to long-term steroid therapy leading to suppression of the pituitary–adrenal axis. This only becomes apparent on withdrawal of the steroids. Other causes are rare and include hypothalamic–pituitary disease leading to 'ACTH production. Mineralocorticoid pro-duction remains intact, and there is no hyperpigmentation as 'ACTH.

21-Hydroxylase adrenal autoantibodies: +ve in autoimmune disease in >80%

Plasma renin & aldosterone: to assess mineralocortocoid status. AXR/CXR: Any past TB, eg upper zone fibrosis or adrenal calcification? If no autoantibodies, con-sider further tests (eg adrenal CT) for TB, histoplasma, or metastatic disease.

Treatment 2See p826 for Addisonian crisis (shocked). Replace steroids: r15–25mg hydrocortisone daily, in 2–2 doses, eg 10mg on waking, 5mg lunchtime. Avoid giving late (may cause insomnia). Mineralocorticoids to correct postural hypotension, 'Na+,

K+: fludrocortisone PO from 50–200mcg daily. Adjust both on clinical grounds. If

there is a poor response, suspect an associated autoimmune disease (check thyroid, do coeliac serology: p266).

Steroid use Advise wearing a bracelet declaring steroid use. Add 5–10mg hydrocor-tisone to daily intake before strenuous activity/exercise. Double steroids in febrile illness, injury, or stress. Give out syringes and in-date IM hydrocortisone, and show how to inject 100mg IM if vomiting prevents oral intake (seek medical help; admit for IV fluids if dehydrated).

Follow-up Yearly (BP, U&E); watch for autoimmune diseases (pernicious anaemia).4

Prognosis (treated) Adrenal crises and infections do cause excess deaths: mean age at death for men is r65yrs (11yrs <estimated life expectancy; women lose r2yrs).

Autoimmune polyglandular syndromes types 1–4: 1 Monogenic syndrome (AIRE gene on chromosome 21); signs: candidiasis, hypoparathyroidism + Addison's. 2 (Schmidt syndrome.) Adrenal insufficiency + auto immune thyroid disease ! DM ! pleuritis /pericarditis. 2 Autoimmune thyroid disease + other autoimmune conditions but not Addison's. 4 Autoimmune combinations not included in 1–2.

Primary hyperaldosteronism Excess production of aldosterone, independent of the renin–angiotensin system, causing •sodium and water retention, and 'renin re-lease. Consider if: hypertension, hypokalaemia, or alkalosis in someone not on diuret-ics. Sodium tends to be mildly raised or normal.

Symptoms: Often asymptomatic or signs of hypokalaemia (p674): weakness (even quadriparesis), cramps, paraesthesiae, polyuria, polydipsia. •BP but not always.

Secondary hyperaldosteronism Due to a high renin from 'renal perfusion, eg in renal artery stenosis, accelerated hypertension, diuretics, CCF, or hepatic failure.

2Emergency 9 : p827. If malignant, chemotherapy or therapeutic radiolabelled MIBG

may be used. Follow-up: Lifelong: malignant recurrence may present late, genetic screening.

Tumours from the zona glomerulosa, zona fasciculata, or zona reticularis associate with syndromes of • mineralocorticoids, glucocorticoids, or androgens respectively, usually; remember 'GFR≈miner GA'.

Takotsubo cardiomyopathy (=stress- or catecholamine-induced cardiomyopathy/broken heart syn-drome) may cause sudden chest pain mimicking MI, with •ST segments, and its signature apical ballooning on echo (also 'ejection fraction) occurring during catecholamine surges. It is a cause of MI in the presence of normal arteries. The stress may be medical (SAH, p478) or psychological.

Management: Healthy eating, optimize weight, shaving; laser photoepilation; wax; creams, eg eflornithine, or electrolysis (expensive/time-consuming, but effective); bleach (1:10 hydrogen peroxide).

Metformin (helps with insulin resistance) and spironolactone are sometimes tried.

Clomifene is used for infertility (a fertility expert should prescribe).

Virilism Onset of amenorrhoea, clitoromegaly, deep voice, temporal hair recession + hirsutism. Look for an androgen-secreting adrenal or ovarian tumour.

Workup: After a full sexual and psychological history do: U&E, LFT, glucose, TFT, LH, FSH, lipids, testosterone, prolactin ! Doppler. Is penile arterial pressure enough for inflow? Is penile sensation OK (if not, ?CNS problem)? Is the veno-occlusive mechanism OK?

Features are due to:

- Hormone lack: •GH: central obesity, atherosclerosis, dry wrinkly skin, 'strength, 'balance, 'well-being, 'exercise ability, 'cardiac output, osteoporosis, 'glucose.

Gonadotropin (FSH; LH): φ oligomenorrhoea or amenorrhoea, 'fertility, 'libido, os-teoporosis, breast atrophy, dyspareunia. ð Erectile dysfunction, 'libido, 'muscle bulk, hypogonadism ('hair, all over; small testes; 'ejaculate volume; 'spermatogenesis).

Thyroid: as for hypothyroidism (p220). •Corticotropin: as for adrenal insufficiency (p226). NB: no •skin pigmentation as 'ACTH. •Prolactin: rare: absent lactation.

- Causes: Eg pituitary tumour (p224), causing mass effect, or hormone secretion with 'secretion of other hormones—eg prolactinoma, acromegaly, rarely Cushing's.

Tests (The triple stimulation test is now rarely done.)

Basal tests: LH and FSH (' or —), testosterone or oestradiol ('); TSH (' or —), T4 ('); prolactin (may be •, from loss of hypothalamic dopamine that normally inhibits its release), insulin-like growth factor-1 (IGF-1; '—used as a measure of GH axis, p228), cortisol ('). Also do U&E ('Na+from dilution), 'Hb (normochromic, normocytic).

Dynamic tests: 1 Short Synacthen® test: (p226) to assess the adrenal axis.

Arginine + growth hormone-releasing hormone test.

Glucagon stimulation test is alternative when ITT is contraindicated.

Investigate cause: MRI scan to look for a hypothalamic or pituitary lesion.

Treatment Refer to an endocrinologist for assessment of pituitary fuction and to oversee hormone replacement and treatment of underlying cause.

Hydrocortisone for 2° adrenal failure (p226) ►before other hormones are given.

Thyroxine if hypothyroid (p220, but TSH is useless for monitoring).

Gonadotropin therapy is needed to induce fertility in both men and women.

Growth hormone (GH). Somatotrophin mimics human GH. It addresses problems of •fat mass, 'bone mass, 'lean body mass (muscle bulk), 'exercise capacity, and problems with heat intolerance.

Autoimmune hypophysitis (=inflamed pituitary) mimics pituitary adenoma. It may be triggered by preg-nancy or immunotherapy blocking CTLA-4. No pituitary auto-antigen is yet used diagnostically.

Snake bite is a common cause in India (eg when associated with acute kidney injury).

Sheehan's syndrome is pituitary necrosis after postpartum haemorrhage.

Fig 5.22 Neuroendocrinology: emotions ↬ thoughts ↬ actions. As Michelangelo foretold (in his Creation of Adam) 'all gods and demons that have ever existed are within us as possibilities, de-sires, and ways of escape'.Within the dark red vault of our skull we see human and god-like forms reaching out, as thoughts escape into actions—with legs extending into our brainstem (B) and a

fist is pushing from our hypothalamus into the pituitary stalk (P). Above the pituitary we have thoughts, ideas, impulses, and neurotransmitters. Below we have hormones. Between is the realm of neuroendocrinology—the neurosecretory cells which turn emotions into the releasing factors for the pituitary hormones (fig 5.4).

the Brain, Too?

224

Pituitary tumours (almost always benign adenomas) account for 10% of intracranial tumours (see figs 5.22, 5.24). They may be divided by size: a microadenoma is a tumour

<1cm across, and a macroadenoma is >1cm. There are three histological types (table 5.4): 1 Chromophobe 70%. Many are non-secretory,10 some cause hypopituitarism. Half produce prolactin (PRL); a few produce ACTH or GH. Local pressure effect in 20%.

Acidophil 15%. Secrete GH or PRL. Local pressure effect in 10%.

Basophil 15%. Secrete ACTH. Local pressure effect rare.

Symptoms are caused by pressure, hormones (eg galactorrhoea), or hypopituitarism (p222). FSH-secreting tumours can cause macro-orchidism in men, but are rare.

Tests MRI defines intra- and supra-sellar extension; accurate assessment of visual fields; screening tests: PRL, IGF-1 (p228), ACTH, cortisol, TFTS, LH/FSH, testosterone in ♂, short Synacthen® test. Glucose tolerance test if acromegaly suspected (p228). If Cushing's suspected, see p225. Water deprivation test if DI is suspected (p240).

Treatment Start hormone replacement as needed (p222). Ensure steroids are given before levothyroxine, as thyroxine may precipitate an adrenal crisis. For Cushing's disease see p225, prolactinoma p226, acromegaly p228.

Radiotherapy: (Eg stereotactic.) Good for residual or recurrent adenomas (good rates of tumour control and normalization of excess hormone secretion).[2]7

Post-op Recurrence may occur late after surgery, so life-long follow-up is required. Fertility should be discussed: this may be reduced post-op due to 'gonadotropins.

! cabergoline (dopamine agonist, if prolactinoma) ! surgery; find the cause, eg a predisposition to thrombosis, from antiphospholipid syndrome.

Table 5.4 Frequency of hormones secreted by pituitary adenomas based on immunohistochemistry

| What is the best diet for obese patients with type 2 diabetes? Dietary carbohydrate is a big determinant of postprandial glucose levels, and low-carbohydrate diets improve glycaemic control. How do low-carbohydrate, ke-togenic diets (<20g of carbohydrate daily; LCKD) compare with low-glycaemic in-dex, reduced-calorie diet (eg 500kcal/day deficit from weight maintenance diet)? In one randomized study over 24 weeks, LCKD had greater improvements in HbA1c (−15 vs −5mmol/L), weight (−11kg vs −7kg), and HDL. Diabetes drugs were reduced or eliminated in 95% of LCKD vs 62% of LGID participants.[a] NB: effects on renal func-tion and mortality are unknown so these diets remain controversial. Monitoring glucose control Fingerprick glucose if on insulin (type 1/2). NB: before a

meal informs about long-acting insulin doses; after meals inform about the dose of short-acting insulin. Glycated haemoglobin (HbA1c) relates to mean glucose level over previous 8wks (RBC t½). Targets are negotiable, eg 48–57mmol/mol (depends on pa-tient's wish and arterial risk, eg past MI or stroke). If at risk from the effects of hypoglycaemia, eg elderly patients prone to falls, consider less tight control. Tight control may not alter all-cause mortality.7 Complications rise with rising HbA1c, so any improvement helps. Be sure to ask about hypoglycaemic attacks (and whether symptomatic). Hypoglycaemic awareness may diminish if control is too tight, or with time in type 1 DM, due to ↓glucagon secretion. It may return if control is loosened. | | — |

*Sensitive methods of TSH measurement have improved recognition of TSH-secreting tumours. These are now more frequently found at microadenoma stage, medially located, and without associated hormone hypersecretion. In these tumours, somatostatin analogues (p228) are very helpful. See also Socin et al. Eur J Endocrinol. 2002;148:422–42.

If <1cm, usually 'incidentaloma'; most non-functioning macroadenomas are revealed by mass effect and/ or hypopituitarism. Here, recurrence after surgery is common, so follow carefully with MRIS.

Fig 5.22 Sagittal T1-weighted MRI of the brain (no gadolinium contrast) showing a lesion in the pituitary fossa, most likely a haemorrhagic pituitary adenoma. Differential diagnosis in-cludes a Rathke's cleft cyst.

Fig 5.24 Coronal T1-weighted MRI of the brain (no gadolinium contrast) showing a lesion in the pituitary fossa (see fig 5.22).

This is the commonest hormonal disturbance of the pituitary. It presents earlier in women (menstrual disturbance) but later in men (eg with erectile dysfunction and/or mass effects). Prolactin stimulates lactation.11,12 Raised levels lead to hypo-gonadism, infertility, and osteoporosis, by inhibiting secretion of gonadotropin-re-leasing hormone (hence 'LH/FSH and 'testosterone or oestrogen).

Physiological: Pregnancy; breastfeeding; stress. Acute rises occur post-orgasm.11

Drugs (most common cause): Metoclopramide; haloperidol; methyldopa; oes-trogens; ecstasy/MDMA;12 antipsychotics (a reason for 'non-compliance': sustained hyperprolactinaemia may cause 'libido, anorgasmia, and erectile dysfunction).

Diseases: Prolactinoma: micro- or macroadenoma; Stalk damage: pituitary ad-enomas, surgery, trauma; Hypothalamic disease: craniopharyngioma, other tu-mours; Other: hypothyroidism (due to •TRH), chronic renal failure ('excretion).

Symptoms φ: Amenorrhoea or oligomenorrhoea; infertility; galactorrhoea (fig 5.27). Also: 'libido, •weight, dry vagina.

ð: Erectile dysfunction, 'facial hair, galactorrhoea. May present late with osteoporosis or local pressure effects from the tumour (p224).

Tests Basal PRL: non-stressful venepuncture between 09.00 and 16.00h. Do a preg-nancy test, TFT, U&E. MRI pituitary if other causes are ruled out.

Management Refer to a specialist endocrinology clinic. Dopamine agonists (bro-mocriptine or cabergoline) are 1st line.

Follow-up Monitor PRL. If headache or visual loss, check fields (? do MRI). Medication can be decreased after 2yrs, but recurrence of hyperprolactinaemia and expansion of the tumour may occur, and so these patients should be monitored carefully.

The prolactin increase (ð and φ) after coitus is r400% greater than after masturbation; post-orgasmic prolactin is part of a feedback loop decreasing arousal by inhibiting central dopaminergic processes. The size of post-orgasmic prolactin increase is a neurohormonal index of sexual satisfaction.

MDMA also •oxytocin; prolactin + oxytocin are thought to mediate post-orgasmic well-being.

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This is due to •secretion of GH (growth hormone) from a pituitary tumour (99%) or hyperplasia, eg via ectopic GH-releasing hormone from a carcinoid tumour. φ:ð≈1:1. Incidence: UK 2/million/yr. r5% are associated with MEN-1 (p222). GH stimulates bone and soft tissue growth through •secretion of insulin-like growth factor-1 (IGF-1).

Symptoms Acroparaesthesia (akron=extremities); amenor-rhoea; ‘libido; headache; •sweating; snoring; arthralgia; backache; fig 5.28: ‘My rings don’t fit, nor my old shoes, and now I’ve got a wonky bite (malocclusion) and curly hair. I put on lots of weight, all muscle and looked good for a while; now I look so haggard’.

Signs (BOX ‘Signs of acromegaly’.) Often predate diagnosis by

>4yrs. If acromegaly occurs before bony epiphyses fuse (rare), gigantism occurs.

Complications (May present with CCF or ketoacidosis.)

Impaired glucose tolerance (r40%), DM (r15%).

Vascular: •BP, left ventricular hypertrophy (!dilatation/CCF), cardiomyopathy, arrhythmias. There is •risk of ischaemic heart disease and stroke (?due to •BP ! insulin resistance and GH-in-duced increase in fibrinogen and decrease in protein S).

Neoplasia: •colon cancer risk; colonoscopy may be needed.[21]

Acromegaly in pregnancy (Subfertility is common.) Pregnancy may be normal; signs and chemistry may remit. Monitor glucose.

Tests •Glucose, •Ca2+, and •PO 2–. GH: Don’t rely on random GH as secretion is pulsatile and during peaks acromegalic and normal levels overlap. GH also • in: stress, sleep, puberty, and pregnancy. Normally GH secretion is inhibited by high glucose, and GH hardly detectable. In acromegaly GH release fails to suppress.

MRI scan of pituitary fossa. • Look for hypopituitarism (p222).

Visual fields and acuity. • ECG, echo. Old photos if possible.

Prognosis May return to normal (any excess mortality is mostly vascular). 16% get diabetes with SSAs vs r12% after surgery.

My life with acromegaly. (http://odelrio.blogspot.com)

14 GH variants made by the placenta rise exponentially until 27wks’ gestation; pituitary GH gradually drops to near-undetectable levels. ‘Gestational acromegaly’ probably develops to foster fetoplacental growth; its side-effects include facial oedema, carpal tunnel symptoms, and nose enlargement.

15 Puberty sees GH- and gonad-mediated rises in bone and muscle mass + other ‘acromegalic’ effects.

Physiology This is the passage of large volumes (>2L/day) of dilute urine due to impaired water resorption by the kidney, because of reduced ADH secretion from the posterior pituitary (cranial DI) or impaired response of the kidney to ADH (nephro-genic DI).

Symptoms Polyuria; polydipsia; dehydration; symptoms of hypernatraemia (p672). Polydipsia can be uncontrollable and all-consuming, with patients drinking anything and everything to hand: in such cases, if beer is on tap, disaster will ensue!

Causes of cranial DI •Idiopathic (≡50%). •Congenital: defects in ADH gene, DIDMOAD.16

Tumour (may present with DI + hypopituitarism): craniopharyngioma, metastases, pituitary tumour. •Trauma: temporary if distal to pituitary stalk as proximal nerve endings grow out to find capillaries in scar tissue and begin direct secretion again.

Hypophysectomy. •Autoimmune hypophysitis (p222). •Infiltration: histiocytosis, sarcoidosis.17 •Vascular: haemorrhage.18 •Infection: meningoencephalitis.

Causes of nephrogenic DI •Inherited. •Metabolic: low potassium, high calcium.

Drugs: lithium, demeclocycline. •Chronic renal disease. •Post-obstructive uropathy.

Treatment Cranial DI: MRI (head); test anterior pituitary function (p222). Give desm-opressin, a synthetic analogue of ADH (eg Desmomelt® tablets).

Nephrogenic: Treat the cause. If it persists, try bendroflumethiazide 5mg PO/24h. NSAIDS lower urine volume and plasma Na+ by inhibiting prostaglandin synthase: prostaglandins locally inhibit the action of ADH.

2Emergency management

Do urgent plasma U&E, and serum and urine osmolalities. Monitor urine output carefully and check U&E twice a day initially.

IVI to keep up with urine output. If severe hypernatraemia, do not lower Na+ rapidly as this may cause cerebral oedema and brain injury. If Na+ is ?170, use 0.9% saline initially—this contains 150mmol/L of sodium. Aim to reduce Na+ at a rate of less than 12mmol/L per day. Use of 0.45% saline can be dangerous.

Desmopressin 2mcg IM (lasts 12–24h) may be used as a therapeutic trial.

16 DIDMOAD is a rare autosomal recessive disorder: Diabetes Insipidus, Diabetes Mellitus, Optic Atrophy and Deafness (also known as Wolfram's syndrome).

17 Suspect neurosarcoidosis if •CSF protein (seen in 24%), facial nerve palsy (25%), CSF pleocytosis (22%), diabetes insipidus (21%), hemiparesis (17%), psychosis (17%), papilloedema (15%), ataxia (12%), seizures (12%), optic atrophy (12%), hearing loss (12%), or nystagmus (9%).

18 Sheehan's syndrome is pituitary infarction from shock, eg postpartum haemorrhage. It is rare.

19 Most of us could drink 20L/d and not be hyponatraemic; some get hyponatraemic drinking 5L/d; they may have Psychosis, Intermittent hyponatraemia, and Polydipsia (PIP syndrome), ?from •intravascular volume leading to •atrial natriuretic peptide, p127, hence natriuresis and hyponatraemia.

Gastroenterology

We thank Dr Simon Campbell, our Specialist Reader for this chapter.

- Lumen is Latin for light (hence its medical meaning of a tubular cavity open to the world at both ends), as well as being the SI unit of light flux falling on an object—ie the power to illuminate. All doctors have this power, whether by insightfully interpreting patients' lives and illnesses to them, or by acts of kindness—even something so simple as bringing a cup of tea.

'There's a lot of people in this world who spend so much time watching their health that they haven't the time to enjoy it.' Josh Billings (1818–85).

Current recommendations must take into account three facts

Obesity costs health services as much as smoking—1 in 4 UK adults is obese.

Diabetes mellitus is burgeoning: in some places prevalence is >7% (p206).

Past advice has not changed eating habits in large sections of the population.

Advice is likely to focus on the following

Body mass index: (BMI; table 6.1.) Aim for 18.5–25. Controlling quantity may be more important than quality. In hypertension, eating the 'right' things lowers BP only mar-ginally, but controlling weight causes a more significant reduction.

Base meals on starch: (Bread, rice, potatoes, pasta.) These provide a slower re-lease form of carbohydrate compared to diets containing refined sugar, and be-ware of the high sugar contents of, eg soft drinks.

Eat enough fruit and vegetables: Aiming for 5 portions a day.

Eat foods high in fat, salt, or sugar infrequently.

Eat some meat, fish, eggs, and beans: Aim for 2 portions of fish a week, includ-ing oily fish (those rich in omega-3 fatty acid, such as mackerel, herring, pilchards, salmon). Non-dairy sources of protein include beans and nuts. Aim to reduce intake of red or processed meat to <70g/day.

Eat some milk and dairy products: Select those options with lower fat, sugar and salt where possible.

Moderate alcohol use (adults <65yrs): ≡14U/wk for both men and women, spread over 2 or more days. There is no 'safe' level.

This diet is not appropriate for all: • <5yrs old. • Need for low residue (eg Crohn's, UC, p264) or special diet (coeliac disease, p266). Emphasis may be differ-ent in: Dyslipidaemia (p690); DM (p206); obesity; constipation (p260); liver failure (p274); chronic pancreatitis (p270); renal failure (less protein)(p204); •BP (see p140).

Difficulties It is an imposition to ask us to change our diet (children often refuse point-blank); a more subtle approach is to take a food we enjoy (crisps) and make it healthier (eg low-salt crisps made from jacket potatoes and fried in sunflower oil).

The diagnosis will often come out of your patient's mouth, so open it! So many GI investigations are indirect…now is your chance for direct observation.

Leucoplakia (fig 6.2) Is an oral mucosal white patch that will not rub off and is not attributable to any other known disease. It is a premalignant lesion, with a transformation rate, which ranges from 0.6% to 18%. Oral hairy leucoplakia is a shaggy white patch on the side of the tongue seen in HIV, caused by EBV. ►When in doubt, refer all intra-oral white lesions (see BOX).

Candidiasis (thrush) (fig 6.4) Causes white patches or erythema of the buccal mu-cosa. Patches may be hard to remove and bleed if scraped. Risk factors: Extremes of age; DM; antibiotics; immunosuppression (long-term corticosteroids, including in-halers; cytotoxics; malignancy; HIV). 9: Nystatin suspension 400 000U (4mL swill and swallow/6h). Fluconazole for oropharyngeal thrush.

Cheilitis (angular stomatitis) Fissuring of the mouth's corners is caused by denture problems, candidiasis, or deficiency of iron or riboflavin (vitamin B2). (fig 8.5, p227.)

Gingivitis Gum inflammation ! hypertrophy occurs with poor oral hygiene, drugs (phenytoin, ciclosporin, nifedipine), pregnancy, vitamin C deficiency (scurvy, p268), acute myeloid leukaemia (p256), or Vincent's angina (p712).

Microstomia (fig 6.5) The mouth is too small, eg from thickening and tightening of the perioral skin after burns or in epidermolysis bullosa (destructive skin and mucous membrane blisters ! ankyloglossia) or systemic sclerosis (p552).

Teeth (fig 6.6) A blue line at the gum–tooth margin suggests lead poisoning. Pre-natal or childhood tetracycline exposure causes a yellow–brown discolouration.

Tongue This may be furred or dry (xerostomia) in dehydration, drug therapy,2 after radiotherapy, in Crohn's disease, Sjögren's (p710), and Mikulicz's syndrome (p706).

Glossitis: Means a smooth, red, sore tongue, eg caused by iron, folate, or B12 defi-ciency (fig 8.27, p225). If local loss of papillae leads to ulcer-like lesions that change in colour and size, use the term geographic tongue (harmless migratory glossitis).

Macroglossia: The tongue is too big. Causes: myxoedema; acromegaly; amyloid (p270). A ranula is a bluish salivary retention cyst to one side of the frenulum, named after the bulging vocal pouch of frogs' throats (genus Rana).

Tongue cancer: Appears as a raised ulcer with firm edges. Risk factors: smoking, alcohol.4 Spread: anterior ⅓ of tongue drains to submental nodes; middle ⅓ to sub-mandibular nodes; posterior ⅓ to deep cervical nodes (see BOX, p599). Treatment: Radiotherapy or surgery. 5yr survival (early disease): 80%. ►When in doubt, refer.

Drugs causing xerostomia: ACE-i; antidepressants; antihistamines; antipsychotics; antimuscarinics/ anticholinergics; bromocriptine; diuretics; loperamide; nifedipine; opiates; prazosin; prochlorperazine, etc.

Betel nut (Areca catechu) chewing, common in South Asia, may be an independent risk factor.

Fig 6.2 Leucoplakia on the underside of the tongue. It is important to refer leucoplakia be-cause it is premalignant.

Fig 6.4 White fur on an erythematous tongue caused by oral candidiasis. Oropharyngeal can-didiasis in an apparently fit patient may sug-gest underlying HIV infection.

Fig 6.2 An aphthous ulcer inside the cheek. The name is tautological: aphtha in Greek means ulceration.

Fig 6.5 Microstomia (small, narrow mouth), eg from hardening of the skin in scleroderma which narrows the mouth. It is cosmetically and functionally disabling.[1]

Consent is needed for all these procedures; see p568.

ture device worn by patient. Normal activity can take place for the day. Complications: Capsule retention in 1% (endo-scopic or surgical removal is needed)—avoid MRI for 2 wks after unless AXR confirms capsule has cleared; obstruction,

incomplete exam (eg slow transit, achalasia). Problems: No therapeutic options; poor localization of lesions; may miss more subtle lesions.

Table 6.2 Indications for upper GI endoscopy

| Calculating body mass index Table 6.1 BMI=(weight in kg)/(height in m)2 Caveats: BMI does not take into account the distribution of body fat, and is harder to interpret for children and adolescents. ►Waist circumference >94cm in men and >80cm in women reflects omental fat and correlates better with risk than does BMI. BMI is still a valid way of comparing populations: average BMI in the USA is 28.8; in Japan, 22. A nation can be lean without being poor. As nations continue to adopt the lifestyle trends of the USA, this impacts sustainability. | | — |

Table 6.2 Indications for colonoscopy

| Diagnostic indications | Therapeutic indications | | — | — | | Haematemesis/melaena | Treatment of bleeding lesions | | Dysphagia | Variceal banding and sclerotherapy | | Dyspepsia (—55yrs old + alarm symptoms or treatment refractory, p252) | Argon plasma coagulation for suspected vascular abnormality | | Duodenal biopsy (?coeliac) | Stent insertion, laser therapy | | Persistent vomiting | Stricture dilatation, polyp resection | | Iron deficiency (cancer) | Stricture dilatation, polyp resection |

Fig 6.7 A big polyp seen on colonoscopy. An advantage of colonoscopy over barium enema is the ability to biopsy or intervene at the same time—in this case, polypectomy.

Fig 6.9 Angiodysplasia lesion seen at colonosco-py. Bleeding may be brisk. R: endoscopic oblitera-tion. It is associated with aortic stenosis (Heyde's syndrome).[2]

Fig 6.8 Colonoscopic image of an adenocar-cinoma—p616. Compared with a colonic polyp (fig 6.7), the carcinoma is irregular in shape and colour, larger and more aggressive.

Fig 6.10 Colonic mucosa in active UC (p262); it is red, inflamed, and friable (bleeds on touch-ing). Signs of severity: mucopurulent exudate, mucosal ulceration ! spontaneous bleeding. If quiescent, there may only be a distorted or absent mucosal vascular pattern.

Dysphagia is difficulty in swallowing and should prompt urgent investigation to exclude malignancy (unless of short duration, and associated with a sore throat). Causes Oral, pharyngeal, or oesophageal? Mechanical or motility related? (See BOX

'Causes of dysphagia'.)

Five key questions to ask

Was there difficulty swallowing solids and liquids from the start? Yes: motility disorder (eg achalasia, CNS, or pharyngeal causes). No: Solids then liquids: suspect a stricture (benign or malignant).

Is it difficult to initiate a swallowing movement?

Yes: Suspect bulbar palsy, especially if patient coughs on swallowing.

Is swallowing painful (odynophagia)?

Yes: Suspect ulceration (malignancy, oesophagitis, viral infection or Candida in immunocompromised, or poor steroid inhaler technique) or spasm.

Is the dysphagia intermittent or is it constant and getting worse?

Intermittent: suspect oesophageal spasm.

Constant and worsening: suspect malignant stricture.

Does the neck bulge or gurgle on drinking?

Yes: Suspect a pharyngeal pouch (see OHCS p570).

Signs Is the patient cachectic or anaemic? Examine the mouth; feel for supraclavic-ular nodes (left supraclavicular node = Virchow's node—suggests intra-abdominal ma-lignancy); look for signs of systemic disease, eg systemic sclerosis (p552), CNS disease. Tests FBC (anaemia); U&E (dehydration). Upper GI endoscopy ! biopsy (fig 6.12). If suspi-

cion of pharyngeal pouch consider contrast swallow (! ENT opinion). Video fluoroscopy may help identify neurogenic causes. Oesophageal manometry for dysmotility.

Consider pregnancy where appropriate! Other causes, p56.

What's in the vomit? Reports of 'coffee grounds' may indicate upper GI bleeding but represent one of the most over-called signs in clinical medicine — always verify yourself and look for other evidence of GI bleeding; recognizable food ≈ gastric stasis; feculent ≈ small bowel obstruction.

Timing Morning ≈ pregnancy or •ICP; 1h post food ≈ gastric stasis/gastroparesis (DM); vomiting that relieves pain ≈ peptic ulcer; preceded by loud gurgling ≈ GI obstruction.

Tests Bloods: FBC, U&E, LFT, Ca2+, glucose, amylase. ABG: A metabolic (hypochlor-aemic) alkalosis from loss of gastric contents (pH >7.45, •HCO2 –) indicates severe vomiting. Plain AXR: If suspected bowel obstruction (p728). Upper GI endoscopy: (See p248.) If suspicion of bleed or persistent vomiting. Consider head CT in case •ICP.

Treatment Identify and treat underlying causes. Symptomatic relief: table 6.4. Be pre-emptive, eg pre-op for post-op symptoms. Try oral route first. 20% need a 2nd-line anti-emetic, so be prepared to prescribe more than one. Give IV fluids with K+ replace-ment if severely dehydrated or nil-by-mouth, and monitor electrolytes and fluid balance.

Fig 6.12 A malignant lower oesophageal stricture seen at endoscopy. Note the asymmetry and heaped edges. Benign strictures have a smoother appearance and tend to be cir-cumferential.

University Press.

Non-propulsive contractions manifest as tertiary contractions or 'corkscrew oesophagus' (fig 16.24, p742) and suggest a motility disorder and may lead to 'acid clearance. Symptoms and radiology may not match. Nutcracker oesophagus denotes distal peristaltic contractions >180mmHg. It may cause pain, relieved by nitrates or sublingual nifedipine.

Symptoms Epigastric pain often related to hunger, specific foods, or time of day, fullness after meals, heartburn (retrosternal pain); tender epigastrium. Beware Alarm symptoms: Anaemia (iron deficiency); loss of weight; anorexia; recent on-set/progressive symptoms; melaena/haematemesis; swallowing difficulty.

H. pylori (See table 6.5.) If ≡55yrs old: 'Test and treat' for H. pylori;6 if +ve give appropriate PPI and 2 antibiotic combination, eg lansoprazole 20mg/12h PO, clarithromycin 250mg/12h PO, and amoxicillin 1g/12h PO for 1wk. If –ve give acid suppression alone. ►Refer for urgent endoscopy (p248) all with dysphagia, as well as those —55 with alarm symptoms or with treatment-refractory dyspepsia.

Duodenal ulcer (DU, fig 6.15.) 4-fold commoner than GU. Major risk factors: H. py-lori (90%); drugs (NSAIDS; steroids; SSRI). Minor: •Gastric acid secretion; •gastric emptying ('duodenal pH); blood group O; smoking.

Symptoms: Asymptomatic or epigastric pain (relieved by antacids) ! 'weight. Signs: Epigastric tenderness. Diagnosis: Upper GI endoscopy. Test for H. pylori. Measure gastrin concentrations when off PPIS if Zollinger–Ellison syndrome (p716) is sus-pected. ΔΔ: Non-ulcer dyspepsia; duodenal Crohn's; TB; lymphoma; pancreatic cancer (p270). Follow-up: None; if good response to R (eg PPI).

Gastritis Risk factors: Alcohol, NSAIDS, H. pylori, reflux/hiatus hernia, atrophic gastritis, granulomas (Crohn's; sarcoidosis), CMV, Zollinger–Ellison syndrome & Mé-nétrier's disease (p716 & 706). Symptoms: Epigastric pain, vomiting; Tests: Upper GI endoscopy only if suspicious features (fig 6.14).

Treatment Lifestyle: ↶Alcohol and tobacco.

H. pylori eradication: Triple therapy is 80–85% effective at eradication.7

Drugs to reduce acid: PPIS are effective, eg lansoprazole 20mg/24h PO for 4 (DU) or 8

(GU) wks. H2 blockers have a place (ranitidine 200mg each night PO for 8wks).

Drug-induced ulcers: Stop drug if possible. PPIS may be best for treating and pre-venting GI ulcers and bleeding in patients on NSAID or antiplatelet drugs. Misopros-tol is an alternative with different SE. If symptoms persist, re-endoscope, retest for

H. pylori, and reconsider differential diagnoses (eg gallstones). Surgery: See p622.

Complications 2Bleeding (p256), 2perforation (p606), malignancy, ↶gastric outflow. Functional (non-ulcer) dyspepsia Common. H. pylori eradication (only after a +ve result) may help. Some evidence favours PPIS and psychotherapy. Low-dose ami-

triptyline (10–20mg each night PO) may help. Antacids, antispasmodics, H2 blockers, misoprostol, prokinetic agents, bismuth, or sucralfate all have less evidence.

H. pylori is the commonest bacterial pathogen found worldwide (>50% of the world population over 40yrs has it). It's a class I carcinogen causing gastritis (p252), duodenal/gastric ulcers & gastric cancer/ lymphoma (MALT, p262), also associated with coronary artery disease, B12 and iron deficiency.

- week of therapy sufficient; 2 weeks increases erradication rates by r5% but also increases SE. For resist-ant infections switch to a 2nd-line antibiotic combination (see BNF).

Table 6.5 Tests (other than serology) should be performed after >2 wks off ppi

| Diagnostic indications | Therapeutic indications | | — | — | | Rectal bleeding—when settled, if acute | Haemostasis (eg by clipping vessel) | | Iron-deficiency anaemia (bleeding cancer) | Bleeding angiodysplasia lesion (argon beamer

photocoagulation) | | Persistent diarrhoea | Bleeding angiodysplasia lesion (argon beamer photocoagulation) | | Positive faecal occult blood test | Colonic stent deployment (cancer) | | Assessment or suspicion of IBD | Volvulus decompression (flexi sig) | | Colon cancer surveillance | Pseudo-obstruction Polypectomy |

* The 12C breath test is the most accurate non-invasive Helicobacter test.

*Nothing magical happens on the 55th birthday—this is simply an inflection point in population risk data. We should not be overly rigid in applying these rules to the patient in front of us—though those who hold the purse strings may at time seek to reduce costs by strict enforcement of such guidelines.

†Don't treat +ve cases of H. pylori more than twice. If still +ve refer for specialist opinion.

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Complications Oesophagitis, ulcers, benign stricture, iron-deficiency. Metaplasia

dysplasia neoplasia: GORD may lead to Barrett's oesophagus (p695; distal oe-sophageal epithelium undergoes metaplasia from squamous to columnar, fig 6.17). 0.1–0.4%/yr of those with Barrett's progress to oesophageal cancer (higher if dys-plasia is present).

Oesophagitis from corrosives, NSAIDS, herpes, Candida; duodenal or gastric ul-cers or cancers; non-ulcer dyspepsia; oesophageal spasm; cardiac disease.

Tests Endoscopy if dysphagia, or if ≥55yrs old with alarm symptoms (p252) or with treatment-refractory dyspepsia. 24h oesophageal pH monitoring ! manom-etry help diagnose GORD when endoscopy is normal.

Treatment Lifestyle: Weight loss; smoking cessation; small, regular meals; reduce hot drinks, alcohol, citrus fruits, tomatoes, onions, fizzy drinks, spicy foods, caffeine, chocolate; avoid eating < 2h before bed. Raise the bed head.

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The reflux of duodenal fluid, pancreatic secretions and bile may be as important as acid; it may re-spond to similar lifestyle measures, sucralfate (2g/12h PO), domperidone, or metoclopramide.

H. pylori association with GORD controversial, but eradication may help symptoms.

Fig 6.19 CT chest (IV contrast) showing the roll-ing components of a hiatus hernia anterior to the oesophagus. Between the oesophagus and the vertebral column on the left-hand side is the aorta.

©Dr S Golding.

Haematemesis is vomiting of blood. It may be bright red or look like coffee grounds. Melaena (Greek melas = black) means black motions, often like tar, and has a char-acteristic smell of altered blood. Both indicate upper GI bleeding. Take a brief history and examine to assess severity. Ask about past GI bleeds; dyspepsia/known ulcers; known liver disease or oes-ophageal varices (p257); dysphagia; vomiting; weight loss. Check drugs (see BOX on common and rare causes) and alcohol use. Is there serious co-morbidity (bad prognosis), eg cardiovascular disease, respiratory disease, hepatic or renal impairment, or malignancy? Look for signs of chronic liver disease (p276) and do a PR to check for melaena. Is the patient shocked? Also:

Peripherally cool/clammy; capillary refill time >2s; urine output <0.5mL/kg/h.

'GCS (tricky to assess in decompensated liver disease) or encephalopathy (p275).

Tachycardic (pulse >100bpm).

Systolic BP <100mmHg; postural drop >20mmHg.

Calculate the Rockall score (tables 6.6, 6.7).

Acute management (p820.) Skill in resuscitation determines survival, so get good at this! Summary:™ start by protecting the airway and giving high-flow O2, then:

2 Insert 2 large-bore (14–16G) IV cannulae and take blood for FBC (early Hb may be normal because haemodilution has not yet taken place), U&E (•urea out of propor-tion to creatinine indicative of massive blood meal), LFT, clotting, and crossmatch.

2 Give IV fluids (p821) to restore intravascular volume while waiting for cross-matched blood. If haemodynamically deteriorating despite fluid resuscitation, give group O Rh–ve blood. Avoid saline if cirrhotic/varices.

2 Insert a urinary catheter and monitor hourly urine output.

2 Organize a CXR, ECG, and check ABG.

2 Consider a CVP line to monitor and guide fluid replacement.

2 Transfuse (with crossmatched blood if needed) if significant Hb drop (<70g/L).

2 Correct clotting abnormalities (vitamin K (p274), FFP, platelets).

2 If suspicion of varices then give terlipressin IV eg 1–2mg/6h for ≤2d; relative risk of death ‘ by 24%.Initiate broad-spectrum IV antibiotic cover.

2 Monitor pulse, BP, and CVP (keep >5cmH2O) at least hourly until stable.

2 Arrange an urgent endoscopy (p248).

2 If endoscopic control fails, surgery or emergency mesenteric angiography/ embolization may be needed. For uncontrolled oesophageal variceal bleeding, a Sengstaken–Blakemore tube may compress the varices, but should only be placed by someone with experience.

Further management ►Anatomy is important in assessing risk of rebleeding. Posterior DUs are highest risk as they are nearest to the gastroduodenal artery.

Re-examine after 4h and consider the need for FFP if >4 units transfused.

Hourly pulse, BP, CVP, urine output (4hrly if haemodynamically stable may be OK).

Transfuse to keep Hb >70g/L; ensure a current valid group & save sample.

Check FBC, U&E, LFT, and clotting daily.

Keep nil by mouth if at high rebleed risk (see BOX ‘Management of peptic ulcer bleeds’ and p256)—ask the endoscopist.

A patient with an aortic graft repair and upper GI bleeding is considered to have an aorto-enteric fistula until proven otherwise: CT abdomen is usually required as well as endoscopy.

A Dieulafoy lesion is the rupture of an unusually big arteriole, eg in the fundus of the stomach.

Diarrhoea is characterized by increased stool frequency and volume and decreased consistency—though patients’ perspectives of these may vary wildly.

Mucus: Occurs in IBS (p266), colorectal cancer, and polyps. Frank pus: Suggests IBD, diverticulitis, or a fistula/abscess.

Explosive: Eg cholera; Giardia; Yersinia (p425); Rotavirus.

Steatorrhoea: Characterized by gas, offensive smell, and floating, hard-to-flush stools—consider pancreatic insufficiency (p267) or biliary obstruction.

Look for Dehydration—dry mucous membranes, ‘skin turgor; capillary refill >2s; shock. Any fever, weight, clubbing, anaemia, oral ulcers (p246), rashes or abdominal mass or scars? Any goitre/hyperthyroid signs? Do rectal exam for masses (eg rectal cancer) or impacted faeces:

Blood: FBC: ↓MCV/Fe deficiency, eg coeliac or colon ca; ↑MCV if alcohol abuse or ↓B12 ab-sorption, eg in coeliac or Crohn's; eosinophilia if parasites. ↑ESR/CRP: infection, Crohn's/ UC, cancer. U&E: ↓K+ ≈ severe D&V. ↓TSH: thyrotoxicosis. Coeliac serology: p266.

Stool: MC&S: bacterial pathogens, ova cysts, parasites, C. diff toxin (CDT, see BOX 'Causes of diarrhoea'), viral PCR. Faecal elastase: if suspect chronic pancreatitis (ma-labsorption, steatorrhoea).

Lower GI endoscopy: (Malignancy? colitis?) ►If acutely unwell, limited flexible sim-goidoscopy with biopsies. Full colonoscopy (including terminal ileum) can assess for more proximal disease If normal, consider small bowel radiology or video capsule.

Erythromycin is prokinetic, others cause overgrowth of bowel organisms, or alter bile acids.

Think of this in any chronic watery diarrhoea; diagnosis by biopsy. Associated with a range of drugs including NSAIDS and PPIS. Stop the offending drug where possible. Treat with budesonide.

Vasoactive intestinal polypeptide-secreting tumour; suspect if ' K+and acidosis; • Ca2+; ' Mg2+.

Managing infective diarrhoea*

Stool culture not needed

Routine culture and microscopy. Ask microbiologist about R

*Be aware of local pathogens, and be prepared to close wards and hospitals if contagion is afoot.

†Prompt specific R: eg ciprofloxacin 500mg/12h PO for 6d may be needed before sensitivities are known. Metronidazole is also tried, as Giardia is a common cause of watery diarrhoea in travellers.

Tests None in young, mildly affected patients. Threshold for investigation di-minishes with age; triggers include:[a] ↓weight, abdominal mass, +PR blood, iron deficiency anaemia. Blood: FBC, ESR, U&E, Ca2+, TFT. Colonoscopy: If suspected colo-rectal malignancy. Transit studies; anorectal physiology; biopsy for Hirschprung's are occasionally needed.

If these don't help Prucalopride is an elective 5HT4 agonist with prokinetic prop-erties; Lubiprostone is a chloride-channel activator that increases intestinal fluid secretion; Linaclotide is a guanylate cyclase-C agonist that also increases fluid se-cretion and decreases visceral pain. A multidisciplinary approach with behaviour therapy, habit training ! sphincter biofeedback may help.

Rectocele: front wall of the rectum bulges into the back wall of the vagina.

Dantron causes colon & liver tumours in animals, so reserve use for the very elderly or terminally ill.

Symptoms Episodic or chronic diarrhoea (! blood & mucus); crampy abdominal discomfort; bowel frequency relates to severity (see table 6.8); urgency/tenes-mus ≈ proctitis. Systemic symptoms in attacks: fever, malaise, anorexia, 'weight.

Signs May be none. In acute, severe UC there may be fever, tachycardia, and a ten-der, distended abdomen. Extraintestinal signs: Clubbing; aphthous oral ulcers; ery-thema nodosum (p265); pyoderma gangrenosum; conjunctivitis; episcleritis; iritis; large joint arthritis; sacroiliitis; ankylosing spondylitis; PSC (p282); nutritional deficits.

Table 6.8 Assessing severity in UC (Truelove & Witts criteria modified to include CRP)

| Remembering your anti-emetics One way of recalling anti-emetics involves using (simplified) pharmacology. Table 6.4 Pharmacology of common anti-emetics All antidopaminergics can cause dystonias and oculogyric crisis, especially in younger patients. | | — |

Data from Truelove et al., 'Cortisone in ulcerative colitis', BMJ; 2(4947): 1041–8.

Complications Acute: Toxic dilatation of colon (mucosal islands, colonic diameter

Treatment Goals are to induce, then maintain disease remission.7

Moderate UC: If 4–6 motions/day, but otherwise well, induce remission with oral prednisolone 40mg/d for 1wk, then taper by 5mg/week over following 7wks. Then maintain on 5-ASA (SES: rash, haemolysis, hepatitis, pancreatitis, paradoxical wors-ening of colitis ►monitor FBC and U&E at start, then at 2 months, then annually).

5-aminosalicylic acid (5-ASA or mesalazine) must be stabilized in oral preparations to survive gastric pH. Alternatively, olsalazine is a dimer of 5-ASA or balsalazide is a prodrug, both of which are cleaved in the colon. Rare hypersensitivity reactions: worsening colitis, pancreatitis, pericarditis, nephritis.

Severe UC: If unwell and —6 motions/d, admit for: IV hydration/electrolyte replace-ment; IV steroids, eg hydrocortisone 100mg/6h or methylprednisolone 40mg/12h; rectal steroids, eg hydrocortisone 100mg in 100mL 0.9% saline/12h PR; thromboem-bolism prophylaxis (p250); ensure multiple stool MC&S/CDT to exclude infection.

Monitor T°, pulse, and BP—and record stool frequency/character on a stool chart.

Twice-daily exam: document distension, bowel sounds, and tenderness.

Daily FBC, ESR, CRP, U&E ! AXR. Consider blood transfusion (eg if Hb <80g/L).

If on day 2–5 CRP >45 or >6 stools/d, 2action is needed.18 Rescue therapy with ciclo-sporin or infliximab, can avoid colectomy, but involve surgeons early in shared care.

If improving, transfer to prednisolone PO (40mg/24h). Schedule maintenance inflixi-mab if used for rescue, or azathioprine if ciclosporin rescue.

If fails to improve then urgent colectomy by d7–10—the challenge is not to delay surgery so long as to accumulate significant steroid exposure and debilitation that will delay post-surgical recovery.

It's time for immunomodulation if… Patients flare on steroid tapering or require ≥2 courses of steroids/year eg azathioprine (2–2.5mg/kg/d PO). 20% of patients will develop SES requiring treatment cessation including abdominal pain, nausea, pan-creatitis, leucopenia, abnormal LFTS. Monitor FBC, U+E, LFT weekly for 4 wks, then every 4 wks for 2 months, then at least 2-monthly.

Biologic therapy For patients intolerant of immunomodulation, or developing symp-toms despite an immunomodulator, monoclonal antibodies to TNFa (infliximab, adali-mumab, golimumab) or to adhesion molecules involved in gut lymphocyte trafficking (vedolizumab) play an important role (see BOX 'Therapies in Crohn's disease' p265).

Day 2 stool frequency >8≈/day or frequency 2–8≈/day & CRP >45 =85% chance of colectomy this admission.

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risk ≈2–4; NSAIDS may exacerbate disease.

Symptoms Diarrhoea, abdominal pain, weight loss/failure to thrive. Systemic symptoms: fatigue, fever, malaise, anorexia.

Signs Bowel ulceration (fig 6.22); abdominal tenderness/mass; perianal abscess/ fistulae/skin tags; anal strictures. Beyond the gut: (fig 6.21) Clubbing, skin, joint, & eye problems.

Complications Small bowel obstruction; toxic dilatation (colonic diameter >6cm, toxic dilatation is rarer than in UC); abscess formation (abdominal, pelvic, or peri-anal); fistulae (present in r10%), eg entero-enteric, colovesical (bladder), colovagi-nal, perianal, enterocutaneous; perforation; colon cancer; PSC (p282), malnutrition.

Treatment (See BOX.8) Find out how your patient deals with what may be a brutal disease (no intimacy…no sex…no hope…'I live with this alone and will die alone'). With a collaborative approach, courage, attention to detail, and a dose of humour, this can change. Help quit smoking. ►Optimize nutrition. Assess severity: •T°, •pulse, •ESR, WCC, •CRP, + 'albumin may merit admission for IV steroids.

Mild–moderate: Symptomatic but systemically well. Prednisolone 40mg/d PO for 1wk, then taper by 5mg every wk for next 7wks. An alternative dietary approach based upon 'elemental' or 'polymeric' diets is effective in children but less used for adults. Plan maintenance therapy (see BOX).

Severe: Admit for IV hydration/electrolyte replacement; IV steroids, eg hydrocorti-sone 100mg/6h or methylprednisolone 40mg/12h; thromboembolism prophylaxis (p250); ensure multiple stool MC&S/CDT to exclude infection.

Monitor T°, pulse, BP, and record stool frequency/character on a stool chart.

Physical examination daily. Daily FBC, ESR, CRP, U&E, and plain AXR.

Consider need for blood transfusion (if Hb <80g/L) and nutritional support.

If improving switch to oral prednisolone (40mg/d). If not, biologics have a role.

Consider abdominal sepsis complicating Crohn's disease especially if abdominal pain (ultrasound, CT, & MRI are often required to assess this). Seek surgical advice. Perianal disease: Occurs in about 50%. MRI and examination under anaesthetic

(EUA) are an important part of assessment. Treatment includes oral antibiotics, im-

munosuppressant therapy ! anti-TNFa, and local surgery ! seton insertion.

Much of the genetic risk is shared with UC—small differences in genetics combined with environmental modifiers may explain the very different phenotypes.

Fig 6.21 Beyond the gut… 'I hate how this stupid illness is crippling me…' As well as erythema nodosum on the shins (above; also caused by sarcoid, drugs, streptococci, and TB), Crohn's can associate with sero –ve arthritis of large or small joints, spondyloarthropathy, an-kylosing spondylitis, sacroiliitis, pyoderma gan-grenosum, conjunctivitis, episcleritis, and iritis.

Fig 6.22 Deep fissured ulcers seen at colon-oscopy. The end result? 'My family does not or will not even talk to me about the disease … I don't know when urgency to race for the bath-room will happen so I don't go out and have been living a hermit life…'

©Dr A Mee.

Causes See BOX 'Causes of gastrointestinal malabsorption'.

Symptoms Diarrhoea; 'weight; lethargy; steatorrhoea; bloating.

Deficiency signs Anaemia ('Fe, B12, folate); bleeding disorders ('vit K); oedema ('protein); metabolic bone disease ('vit D); neurological features, eg neuropathy.

Tests FBC (' or •MCV); 'Ca2+; 'Fe; 'B12 + folate; •INR; lipid profile; coeliac tests (see 'Coeliac disease'). Stool: Sudan stain for fat globules; stool microscopy (infestation); elastase. Breath hydrogen analysis: For bacterial overgrowth.20 Take samples of end-expired air; give glucose; take more samples at ½h intervals; early •exhaled hydrogen

overgrowth. Endoscopy + small bowel biopsy.

Infectious malabsorption Giardia, Cryptosporidium, Isospora belli, Cyclospora cayetanensis, microsporidia. Tropical sprue: Villous atrophy + malabsorption occur-ring in the Far and Middle East and Caribbean—the cause is unknown. Tetracycline 250mg/6h PO + folic acid 5mg/d PO for 2–6mnths may help.

Presentation Stinking stools/steatorrhoea; diarrhoea; abdominal pain; bloating; nau-sea + vomiting; aphthous ulcers; angular stomatitis (p227, fig 8.5); 'weight; fatigue; weakness; osteomalacia; failure to thrive (children). r20% less severe: may mimic IBS.

Diagnosis ↓Hb; ↑RCDW (p225); ↓B12, ↓ferritin. Antibodies: anti-transglutaminase is single preferred test (but is an IgA antibody—check IgA levels to exclude subclass defi-ciency). Where serology positive or high index of suspicion proceed to duodenal biopsy while on a gluten-containing diet: expect subtotal villous atrophy, •intra-epithelial WBCS

1. crypt hyperplasia. Where doubt persists, HLA DQ2 and DQ8 genotyping may help.

Treatment Lifelong gluten-free diet—patients become experts. Rice, maize, soya, potatoes, and sugar are OK. Limited consumption of oats (≤50g/d) may be toler-ated in patients with mild disease. Gluten-free biscuits, flour, bread, and pasta are prescribable. Monitor response by symptoms and repeat serology.[a]

Complications Anaemia; dermatitis herpetiformis (OHCS p588); osteopenia/os-teoporosis; hyposplenism (offer ‘flu and pneumococcal vaccinations); GI T-cell lymphoma (rare; suspect if refractory symptoms or ‘weight); • risk of malignancy (lymphoma, gastric, oesophageal, colorectal); neuropathies.

IBS denotes a mixed group of abdominal symptoms for which no organic cause can be found. Most are probably due to disorders of intestinal motility, enhanced visceral perception (the ‘brain–gut’ axis: see BOX ‘Managing IBS’), or microbial dysbiosis. Sev-eral diagnostic criteria exist (see BOX ‘Defining gastrointestinal dysfunction’ p261).

Prevalence 10–20%; age at onset: ≡-40yrs; φ:ð ?2:1.

Bacterial overgrowth proximal to the colon causes diarrhoea, abdominal pain, and vitamin malabsorp-tion. Causes: old age, autonomic neuropathy (eg diabetic), ileocaecal valve resection; PPI usage; amyloidosis.

Always consider that more than one nutritional disorder is likely to be present (table 6.9).

Beriberi There is heart failure with general oedema (wet beriberi) or neuropathy (dry beriberi) due to lack of vitamin B1 (thiamine). For treatment and diagnostic tests, see Wernicke’s encephalopathy (p714).

Pellagra = lack of nicotinic acid. Classical triad: diarrhoea, dementia, dermatitis (Casal’s necklace) ! neuropathy, depression, insomnia, tremor, rigidity, ataxia, fits. It may occur in carcinoid syndrome and anti-TB drugs (isoniazid). It is endemic in China and Africa. 9: Education, electrolyte replacement, nicotinamide 100mg/4h.

Xerophthalmia This vitamin A deficiency syndrome is a big cause of blindness in the Tropics. Conjunctivae become dry and develop oval/triangular spots (Bitôt’s spots). Corneas become cloudy and soft. Give vitamin A (OHCS p460).

►Get special help if pregnant: vitamin A embryopathy must be avoided. Re-educate and monitor diet.

Table 6.9 Deficiency syndromes and the sites of nutrient absorption

| | | Sensitivity | Specificity | | — | — | — | — | | Invasive tests | CLO test | 95% | 95% | | | Histology | 95% | 95% | | | Culture | 90% | 100% | | Non-invasive | 12C breath test* | 95% | 95% | | | Stool antigen | 95% | 94% | | | Serology | 92% | 82% |

F = fat-soluble vitamin, thus deficiency is likely if there is fat malabsorption.

That oranges and lemons prevent ‘the scurvy’ was noted by the naval surgeon James Lind in 1752. In what may rank as the first ever clinical trial, he randomly divided 12 sailors with scurvy into 6 groups, given the same basic diet but each group received a unique dietary intervention. The 2 sailors receving oranges and lemons both made a good recovery, unlike the other 10.

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Epigastric pain ‘bores’ through to the back, eg relieved by sitting forward or hot water bottles on epigastrium/back (look for erythema ab igne’s mottled dusky greyness); bloating; steatorrhoea; ‘weight; brittle diabetes. Symptoms relapse and worsen.

Causes Alcohol; smoking; autoimmune; rarely: familial; cystic fibrosis; haemochro-matosis; pancreatic duct obstruction (stones/tumour); congenital (pancreas divisum). Tests Ultrasound ! CT: pancreatic calcifications confirm the diagnosis, MRCP; AXR.

Speckled calcification; faecal elastase.

Complications Pseudocyst; diabetes; biliary obstruction; local arterial aneurysm; splenic vein thrombosis; gastric varices; pancreatic carcinoma.

Prognosis Often dismal. Mean survival <6 months. 5yr survival: 2%. Overall 5yr sur-vival after Whipple's procedure 5–14%. Prognosis is better if: tumour <2cm; no nodes involved; –ve resection margins at surgery; ampullary or islet cell tumours.

Carcinoid syndrome Occurs in r5% and implies hepatic involvement.

Symptoms and signs: Bronchoconstriction; paroxysmal flushing especially in upper body (! migrating weals); diarrhoea; CCF (tricuspid incompetence and pulmonary stenosis from 5HT–induced fibrosis). ►Carcinoid crisis: See BOX 'Carcinoid crisis'.

Median survival 5–8yrs (r2yrs if metastases are present, but may be up to 20yrs; so beware of giving up too easily, even in metastatic disease).

Some are never clinically detected: 1 in 200 autopsies have a small bowel carcinoid tumour.

Jaundice refers to yellowing of skin, sclerae, and mucosae from •plasma bilirubin (visible at —60μmol/L fig 6.24).

Jaundice is classified by the site of the problem (pre-hepatic, hepatocellular, or cholestatic/obstructive) or by the type of circulating bili-rubin (conjugated or unconjugated, fig 6.25).

Unconjugated hyperbilirubinaemia As unconjugated bilirubin is water-insoluble, it does not enter urine, resulting in unconjugated hyperbilirubinaemia.

Overproduction: Haemolysis (p228, eg malaria/DIC, etc); ineffective erythropoiesis. Impaired hepatic uptake: Drugs (paracetamol, rifampicin), ischaemic hepatitis. Impaired conjugation: Eponymous syndromes: Gilbert's, p700; Crigler–Najjar, p696. Physiological neonatal jaundice: Caused by a combination of the above, OHCS p115.

Conjugated hyperbilirubinaemia As conjugated bilirubin is water-soluble, it is excreted in urine, making it dark. Less conjugated bilirubin enters the gut and the faeces become pale. When severe, it can be associated with an intractable pruri-tus which is best treated by relief of the obstruction.

What to do? ₂Treat the cause promptly. Ensure adequate hydration; broad-spec-trum antibiotics if obstruction. Monitor for ascites, encephalopathy; call a hepatologist.

Multiple segmental cystic or saccular dilatations of intrahepatic bile ducts with congenital hepatic fibrosis. It may present in 20yr-olds, with portal hypertension ! recurrent cholangitis/cholelithiasis.

Pancreatic or gallbladder cancer is more likely, as stones lead to a fibrotic, unexpandable gallbladder.

Albumin & INR are the best indicators of hepatic synthetic function. •Transaminases (ALT, AST) indi-cate hepatocyte damage. •ALP suggests obstructive jaundice, but also occurs in hepatocellular jaundice, malignant infiltration, pregnancy (placental isoenzyme), Paget's disease, and childhood (bone isoenzyme).

Table 6.10 Examples of drug-induced jaundice

| Differential diagnosis of dyspepsia Non-ulcer dyspepsia • Duodenal/gastric ulcer • Duodenitis Oesophagitis/GORD • Gastric malignancy • Gastritis (p257) | | — |

Fig 6.26 Amanita phalloides (Latin for 'phallic toadstool'; also known as the 'death cap') is a lethal cause of jaundice. It is the most toxic mushroom known. After ingestion (its benign appearance is confusing), amatoxins induce hepatic necrosis leaving few options other than transplantation.

©Ian Herriott. NB: don't use this image for identification!

Definitions Liver failure may be recognized by the development of coagulopathy (INR>1.5) and encephalopathy. This may occur suddenly in the previously healthy liver

acute liver failure (hyperacute = onset ≤ 7d; acute = 8–21d; subacute = 4–26wks.) More often it occurs on a background of cirrhosis = chronic liver failure. Fulminant hepatic failure is a clinical syndrome resulting from massive necrosis of liver cells leading to severe impairment of liver function.

Signs Jaundice, hepatic encephalopathy (see BOX 'Hepatic encephalopathy'), fetor hepaticus (smells like pear drops), asterixis/flap (p50), constructional apraxia (can-not copy a 5-pointed star?). Signs of chronic liver disease (p276) suggest acute-on-chronic hepatic failure.

Management 2Beware sepsis, hypoglycaemia, GI bleeds/varices, & encephalopathy:

Nurse with a 20° head-up tilt in ITU. Protect the airway with intubation and in-sert an NG tube to avoid aspiration and remove any blood from stomach.

Insert urinary and central venous catheters to help assess fluid status.

Monitor T°, respirations, pulse, BP, pupils, urine output hourly. Daily weights.

Check FBC, U&E, LFT, and INR daily.

10% glucose IV, 1L/12h to avoid hypoglycaemia. Do blood glucose every 1–4h.

Treat the cause, if known (eg GI bleeds, sepsis, paracetamol poisoning, p844).

If malnourished, get dietary help: good nutrition can decrease mortality. Give thiamine and folate supplements (p714).

Treat seizures with phenytoin (p826).[11]

Haemofiltration or haemodialysis, if renal failure develops (BOX 'What is hepato-renal syndrome?').

Try to avoid sedatives and other drugs with hepatic metabolism (BOX 'Prescrib-ing in liver failure' and BNF).

Consider PPI as prophylaxis against stress ulceration, eg omeprazole 40mg/d IV/PO.

Liaise early with nearest transplant centre regarding appropriateness.

Treat complications Cerebral oedema: On ITU: 20% mannitol IV; hyperventilate.

Ascites: Restrict fluid, low-salt diet, weigh daily, diuretics (p276).

Bleeding: Vitamin K 10mg/d IV for 2d, platelets, FFP + blood as needed ! endoscopy.

Blind 9 of infection: Ceftriaxone 1–2g/24h IV, not gentamicin (•risk of renal failure).

Blood glucose: If ≡2mmol/L or symptomatic, R 50mL of 50% glucose IV; check often. Encephalopathy: Avoid sedatives; 20° head-up tilt in ITU; correct electrolytes; lactulose 20–50mL/8h (aim for 2–4 soft stools/d) is catabolized by bacterial flora

to short-chain fatty acids which 'colonic pH and trap NH2 in the colon as NH4+ ; Rifaximin 550mg/12h is a non-absorbable oral antibiotic that ' numbers of nitrogen-forming gut bacteria.

Worse prognosis if Grade III–IV encephalopathy, age >40yrs, albumin <20g/L, •INR, drug-induced liver failure, late-onset hepatic failure worse than fulminant failure.

Neutrophilic leucocytosis need not mean a secondary infection: alcoholic hepatitis may be the cause.

Urea is synthesized in the liver, so is a poor test of renal function in liver failure; use creatinine instead.

Cirrhosis (Greek kirrhos = yellow) implies irreversible liver damage. Histologically, there is loss of normal hepatic architecture with bridging fibrosis and nodular regeneration.

Causes Most often chronic alcohol abuse, HBV, or HCV infection. Others: see BOX

'Causes of cirrhosis'.

Prognosis Overall 5yr survival is r50%. Poor prognostic indicators: encephalopa-thy; serum Na+ <110mmol/L; serum albumin <25g/L; •INR.

Liver transplantation is the only definitive treatment for cirrhosis (p277). Acute indications: Acute liver failure meeting King's College criteria (see BOX 'King's Col-lege Hospital criteria in acute liver failure' p275) Chronic indications: Advanced cir-rhosis of any cause; hepatocellular cancer (1 nodule <5cm or ≤5 nodules <2cm).

Fig 6.27 Spider naevi: a central arteriole, from which numerous vessels radiate (like the legs of a spider). These fill from the centre unlike tel-angiectasias that fill from the edge. They occur most commonly in skin drained by the superior vena cava. ≤5 are normal (especially in φ). Caus-es include liver disease, OCP, and pregnancy (ie changes in oestrogen metabolism).

Fig 6.28 Gross ascites. Note the umbilical her-nia (p613), gynaecomastia, and veins visible on the anterior abdominal wall.

Clues as to which patients with chronic HCV will get cirrhosis: platelet count ≤140 x 109/L, globulin/albu-min ratio ? 1, and AST/ALT ratio ?1—100% +ve predictive value but lower sensitivity (r20%).

Online calculators available, eg at www.odt.nhs.uk

9: Supportive. Avoid alcohol. Rarely, interferon alfa for fulminant hepatitis.

Active immunization: With inactivated viral protein. 1 IM dose gives immunity for

1yr (20yrs if further booster is given at 6–12 months).

Prognosis: Usually self-limiting. Fulminant hepatitis is rare. Chronicity doesn't occur.

Endemic in: Far East, Africa, Mediterranean. Incubation: 1–6 months.

Signs: Resemble hepatitis A but arthralgia and urticaria are commoner.

Complications: Fulminant hepatic failure, cirrhosis, HCC, cholangiocarcinoma, cryo-globulinaemia, membranous nephropathy, polyarteritis nodosa (p556).

9: Avoid alcohol. Immunize sexual contacts. Refer all with chronic liver inflamma-tion (eg ALT —20IU/L), cirrhosis, or HBV DNA >2000IU/mL for antivirals (choice is 48 wks pegylated (PEG) interferon alfa-2a vs long-term but better tolerated nucleos(t) ide analogues, eg tenofovir, entecavir). The aim is to clear HBSAg and ►prevent cir-rhosis and HCC (risk is •• if HBSAg and HBeAg +ve).

Hepatitis C virus (HCV) RNA flavivirus. Spread: Blood: transfusion, IV drug abuse, sexual contact. UK prevalence: >200 000. Early infection is often mild/asympto-matic. r85% develop silent chronic infection; r25% get cirrhosis in 20yrs—of these,

9: BOX; quit alcohol. Other complications: Glomerulonephritis; cryoglobulinaemia; thyroiditis; autoimmune hepatitis; PAN; polymyositis; porphyria cutanea tarda.

Hepatitis D virus (HDV) Incomplete RNA virus (needs HBV for its assembly). HBV vac-cination prevents HDV infection. 5% of HBV carriers have HDV co-infection. It may cause acute liver failure/cirrhosis. Tests: Anti-HDV antibody (only ask for it if HBsAg

+ve). 9: As interferon alfa has limited success, liver transplantation may be needed. Hepatitis E virus (HEV) RNA virus. Similar to HAV; common in Indochina (commoner in older men and also commoner than hepatitis A in UK); mortality is high in pregnancy.

It is associated with pigs. Epidemics occur (eg Africa). Vaccine is available in China

(not Europe). ⚹: Serology. 9: Nil specific.

Other infective causes of hepatitis EBV; CMV; leptospirosis; malaria; Q fever; syph-ilis; yellow fever.

Table 6.11 Serological markers of HBV infection

| | Stop drugs causing dyspepsia, eg NSAIDS Lifestyle changes (p254) Over-the-counter antacids, eg magnes-ium trisilicate 10mL/8h PO Review after 4wks | Stop drugs causing dyspepsia, eg NSAIDS Lifestyle changes (p254) Over-the-counter antacids, eg magnes-ium trisilicate 10mL/8h PO Review after 4wks | | — | — | — | | Improvement | Stop drugs causing dyspepsia, eg NSAIDS Lifestyle changes (p254) Over-the-counter antacids, eg magnes-ium trisilicate 10mL/8h PO Review after 4wks | Stop drugs causing dyspepsia, eg NSAIDS Lifestyle changes (p254) Over-the-counter antacids, eg magnes-ium trisilicate 10mL/8h PO Review after 4wks | | | | No improvement |

HBS=hep. B surface; HBC=hep. B core; HBe=hep. B e antigen; DNAP=DNA polymerase.

An alcoholic is one whose problematic pattern of alcohol use leads to clinically sig-nificant impairment or distress, manifested by multiple psychosocial, behavioural, or physiological features. Other addictions may coexist. Lifetime prevalence: ðr10% (φ ≈ 4%). ►Denial is a leading feature, so be sure to question relatives.

Organs affected ►Don't forget the risk of trauma while intoxicated.

CNS: Self neglect; ↷memory/cognition: high-potency vitamins IM may reverse it (p714; don't delay!); cortical atrophy; retrobulbar neuropathy; fits; falls; wide-based gait; neuropathy; confabulation/Korsakoff 's (p704) ! Wernicke's encephalopathy (p714).

Gut: Obesity; D&V; gastric erosions; peptic ulcers; varices (p257); pancreatitis (acute or chronic); cancer (many types); oesophageal rupture (m vomiting against a closed glottis; suspect if shock and surgical emphysema in the neck: Boerhaave's syndrome).

Blood: •MCV; anaemia from: marrow depression, GI bleeding, alcoholism-associated folate deficiency, haemolysis; sideroblastic anaemia. See p226.

Heart: Arrhythmias; •BP; cardiomyopathy; sudden death in binge drinkers.

Reproduction: Testicular atrophy; 'testosterone/progesterone; •oestrogen; fetal al-cohol syndrome—'IQ, short palpebral fissure, absent philtrum, and small eyes.

Withdrawal starts 10–72h after last drink. Signs: •Pulse; 'BP; tremor; confusion; fits; hallucinations (delirium tremens) —may be visual or tactile, eg animals crawling all over skin. Consider it in any new (≡2d) ward patient with acute confusion.

ffkGT is ⍋ in 52% of alcoholics; it is also ⍋ in 50% of those with non-alcoholic fatty livers. Its best use is not in diagnosing alcoholism but in seeing if a raised ALP is likely to be from liver, not bone.

Tests Blood: •ALP, •ffkGT, and mildly •AST & ALT; late disease: •bilirubin, 'albumin,

Prognosis Highly variable. The Mayo survival model is a validated predictor of survival that combines age, bilirubin, albumin, PT time, oedema, and need for diuretics.

Progressive cholestasis with bile duct inflammation and strictures (figs 6.20, 6.21).

Symptoms/signs Pruritus ! fatigue; if advanced: ascending cholangitis, cirrhosis, and hepatic failure. Associations: •ð sex. • HLA-A1; B8; DR2. • AIH (p284); >80% of Northern European patients also have IBD, usually UC; this combination is associated with ••risk of colorectal malignancy.

Cancers Bile duct, gallbladder, liver, and colon cancers are more common, so do yearly colonoscopy + ultrasound; consider cholecystectomy for gallbladder polyps.22

Tests •ALP, then •bilirubin; hypergammaglobulinaemia and/or •IgM; AMA –ve, but ANA, SMA, and ANCA may be +ve; see BOX and p552. ERCP (fig 6.20) or MRCP (fig 6.21) reveal duct anatomy and damage. Liver biopsy shows a fibrous, obliterative cholangitis.

Other causes of liver granulomas: TB, sarcoid, infections with HIV (eg toxoplasmosis, CMV, mycobacte-ria), PAN, SLE, granulomatosis with polyangiitis, lymphoma, syphilis, isoniazid, quinidine, carbamazepine, al-lopurinol. Signs: PUO; ΔLFT.

Usually gallbladder polyps are an incidental finding on ultrasound, and they can often be left if <1cm diameter, but in PSC they are much more likely to become malignant.

Fig 6.20 ERCP showing many stric-tures in the biliary tree with a charac-teristic ‘beaded’ appearance.
© Dr Anthony Mee.

Fig 6.21 MRCP showing features of PSC. The intrahe-patic ducts show multifocal strictures. Strictures can be hard to differentiate from cholangiocarcinoma (coexistence of UC may promote this development). Stenting may be needed.
©Norwich Radiology Department.

Tests Serum bilirubin, AST, ALT, and ALP all usually •, hypergammaglobulinaemia (esp. IgG), +ve autoantibodies (see table 6.12). Anaemia, ‘WCC, and ‘platelets indicate hy-persplenism. Liver biopsy: (See p248.) Mononuclear infiltrate of portal and peripor-tal areas and piecemeal necrosis ! fibrosis; cirrhosis ≈ worse prognosis. MRCP: (See p742.) Helps exclude PSC if ALP • disproportionately.

Diagnosis Depends on excluding other diseases (no lab test is pathognomonic). Di-agnostic criteria based on IgG levels, autoantibodies, and histology in the absence of viral disease are helpful. Sometimes diagnosis is a challenge—there is overlap with other chronic liver disease: eg PBC (p282), PSC (p282) and chronic viral hepatitis.

Table 6.12 Classifying autoimmune hepatitis: types I–II

| Improvement | 9 to eradicate H. pylori;† review after 4 wks | 9 to eradicate H. pylori;† review after 4 wks | | — | — | — | | | 9 to eradicate H. pylori;† review after 4 wks | 9 to eradicate H. pylori;† review after 4 wks | | No improvement | No improvement | |

Liver transplantation: (See p277.) Indicated for decompensated cirrhosis or if there is failure to respond to medical therapy, but recurrence may occur. It is ef-fective (actuarial 10yr survival is 75%).

Prognosis Appears not to matter whether symptomatic or asymptomatic at presentation (10yr survival r80% for both). The presence of cirrhosis at presenta-tion reduces 10yr survival from 94% to 62%. Overlap syndromes: AIH-PBC (primary biliary cholangitis) overlap is worse than AIH-AIC (autoimmune cholangitis).

Hepatotropic viruses (eg measles, herpes viruses) and some drugs appear to trigger AIH in geneti-cally predisposed individuals exposed to a hepatotoxic milieu intérieur. Viral interferon can inactivate cytochrome P450 enzymes (m ‘ metabolism of ex- or endogenous hepatotoxins). Resulting modifications to proteins may generate autoantigens driving CD4 T-helper cell activation.

Wilson’s disease is a rare (2/100 000) inherited disorder of copper excretion with excess deposition in liver and CNS (eg basal ganglia). It is treatable, so screen all with cirrhosis.

Signs Children present with liver disease (hepatitis, cirrhosis, fulminant liver fail-ure); young adults often start with CNS signs: tremor; dysarthria, dysphagia; dyski-nesias; dystonias; dementia; Parkinsonism; ataxia/clumsiness.

Mood: Depression/mania; labile emotions; •'libido; personality change. ►Ignoring these may cause years of needless misery: often the doctor who is good at combin-ing the analytical and integrative aspects will be the first to make the diagnosis.

Cognition: 'Memory; slow to solve problems; 'IQ; delusions; mutism.

Kayser–Fleischer (KF) rings: Copper in iris (see 6 in following list); they are not invariable.

Also: Haemolysis; blue lunulae (nails); arthritis; hypermobile joints; grey skin.

Tests ►Equivocal copper studies need expert interpretation.

Urine: 24h copper excretion is high, eg >100mcg/24h (normal <40mcg).

⍙LFT: non-specific (but ALT >1500 is not part of the picture).

Serum copper: typically <11μmol/L.

Serum caeruloplasmin: <200mg/L (<140mg/L is pathognomonic)—beware incidental low values in protein-deficiency states (eg nephrotic syndrome, mal-absorption).

Molecular genetic testing can confirm the diagnosis.

MRI: degeneration in basal ganglia, fronto-temporal, cerebellar, and brainstem.

The commonest (90%) liver tumours are metastases (see fig 6.22), eg from breast, bronchus, or the gastrointestinal tract (see table 6.14). Primary hepatic tumours are much less common and may be benign or malignant (see table 6.12).

pain, upper GI haemorrhage, and jaundice. It may be life-threatening.

Milan criteria for liver transplantation in HCC: 1 nodule <5cm or 2–2 nodules <2cm.

Table 6.14 Origins of secondary liver tumours

| Pre-endoscopy | 0 pts | 1 pt | 2 pts | 3 pts | | — | — | — | — | — | | Age | <60yrs | 60–79yrs | ?80yrs | | | Shock: systolic BP & pulse rate | BP >100mmHg Pulse <100/min | BP >100mmHg Pulse >100/min | BP <100mmHg | | | Comorbidity | Nil major | Heart failure; ischaemic heart disease | Renal failure Liver failure | Metastases | | Post-endoscopy Diagnosis | Mallory–Weiss tear; no lesion; no sign of recent bleeding | All other diagnoses | Upper GI malig-nancy | | | Signs of recent haemorrhage on endoscopy | None, or dark red spot | | Blood in upper GI tract; adherent clot; visible vessel | |

An inherited disorder of iron metabolism in which •intestinal iron absorption leads to iron deposition in joints, liver, heart, pancreas, pituitary, adrenals, and skin. Middle-aged men are more frequently and severely affected than women, in whom the dis-ease tends to present r10yrs later (menstrual blood loss is protective).

Tests Blood: ⍙LFT, ⍙ferritin (ð>200/φ>150ng/mL; but inflammation will also ⍙ferritin);

Liver biopsy: Perl's stain quantifies iron loading40 and assesses disease severity.

Over-the-counter drugs: Ensure vitamin preparations contain no iron.

Prognosis Venesection returns life expectancy to normal if non-diabetic and non-cir-rhotic (and liver histology can improve). Arthropathy may improve or worsen. Gonadal failure may reverse in younger men. ►If cirrhosis, 22–20% get hepatocellular cancer, especially if: age >50yrs (risk ⍙≈ 12), HBsAg +ve (risk ⍙ ≈ 5), or alcohol abuse (risk ⍙ ≈ 2).

Fig 6.22 Haemochromatosis causes stressed joints to deteriorate faster than resting joints: the 2nd and 3rd MCP joints have osteophytes and narrowed joint spaces compared to the normal hand (right image) in this man who only used his dominant hand for his production line job.

The patient Symptomatic patients usually have the Pi ZZ genotype: dyspnoea from emphysema; cirrhosis; cholestatic jaundice. Cholestasis often remits in adolescence.

Prognosis Some patients have life-threatening symptoms in childhood, whereas others remain asymptomatic and healthy into old age. Worse prognosis if male, a smoker, or obese. Emphysema is the cause of death in most, liver disease in r5%. In adults, cirrhosis ! HCC affect 25% of A1AT-deficient adults >50yrs.

Renal Medicine

Renal disease presents as:

Asymptomatic disease

Non-visible haematuria: (NVH, microscopic haematuria.) Detected on urine dipstick on repeated testing. Most is not due to renal disease and urological investigation is first-line for all those aged >40 years. See p294.

Asymptomatic proteinuria: Normal renal protein excretion is less than 150mg/24 hours (non-pregnant). Quantification by 24h urine collection is unreli-able and rarely used in clinical practice. A spot urinary protein to creatinine ratio (P:CR) >15mg/mmol or urinary albumin to creatinine ratio (A:CR) >2.5(♂) or 2.5(♀)mg/mmol may signify either glomerular (common) or tubular (rare) pathology.

High blood pressure: A renal aetiology should be excluded if hypertension oc-curs with any indicators of renal disease: haematuria, proteinuria, ↓eGFR.

Electrolyte abnormalities: Disorders of sodium, potassium, and acid–base bal-ance (pp201 and 670–5) may be due to underlying renal disease.

With renal tract symptoms

Loin pain: Ureteric colic is severe and radiates anteriorly and to the groin. It is caused by a renal stone, clot, or a sloughed papilla. For pain confined to the loin consider pyelonephritis, renal cyst pathology, and renal infarct.

Visible haematuria: (VH, macroscopic.) Urological investigation is required to exclude renal tract malignancy. Nephrological causes include polycystic kidney disease and glomerular disease (IgA p211, anti-glomerular basement membrane (anti-GBM) disease p211, Alport syndrome p220).

Nephrotic syndrome: Proteinuria >2g/24 hours (=P:CR >200mg/mmol) with hypoalbuminaemia (<20g/L) and peripheral oedema. Renal biopsy is usually indicated in adults (p210).

Symptomatic chronic kidney disease: Dyspnoea, anorexia, weight loss, pruri-tus, bone pain, sexual dysfunction, cognitive decline (pp202–5).

A systemic disorder with renal involvement

DM: (p214.)

Metabolic: Sickle cell disease (p215), tuberous sclerosis (p220), Fabry disease (p220), cystinosis (p221).

Auto-immune: ANCA-associated vasculitis (p214, 556), SLE (p214), Henoch–Schonlein purpura (p211), systemic sclerosis (p215), sarcoid (p218), Sjögren's syndrome (p218, 710).

Infection: Sepsis is a common cause of AKI. Specific renal involvement may oc-cur with TB (p292), malaria, chronic hepatitis (p278), HIV (pp298–402).

Malignancy: Obstruction, hypercalcaemia, direct toxicity, eg myeloma (p214).

Pregnancy: Pre-eclampsia, obstruction.

Drugs used in systemic disorders: NSAIDS, ACE-i, ARB, aminoglycosides, chemo-therapy.

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Perform dipstick urinalysis whenever you suspect renal disease. This is a crude way of checking whether the urine contains anything that it should not, eg protein, blood, glucose. Abnormalities can indicate intrinsic renal disease or renal tract abnormali-ties and usually require further investigation. However, a dipstick-positive catheter sample is difficult to interpret.

Look for a urine dip result (before the catheter was inserted). A positive result indicates the need for specialist advice from nephrology/urology. A negative result may help to reassure you about the absence of intrinsic renal disease.

Proteinuria

Requires quantification. 24h collections are rarely used due to inaccuracy. Albumin:creatinine ratio (A:CR) or protein:creatinine ratio (P:CR) is performed on a random spot urine sample. Normal A:CR is <2.5(♂) or <2.5(♀). P:CR is <15. Approximate equivalent levels of proteinuria are given in table 7.1.

Table 7.1 Conversion factors

| Protein excretion g/24h | A:CR mg/mmol | P:CR mg/mmol | | — | — | — | | 0.15 (physiological) | 2.5 ♂ or 2.5 ♀ | 15 | | 0.5 | 20 | 50 | | 1 | 70 | 100 | | 2 (nephrotic range) | 250 | 200 |

Causes of raised A:CR/P:CR: The higher the proteinuria, the more chance it is caused by glomerular disease, eg glomerulonephritis, DM, amyloidosis, myeloma (though dipsticks do not detect light chains). Proteinuria is associated with an ♁risk of cardio-vascular disease and death. False positive: Postural (repeat using an early morning sample), post-exercise, fever, heart failure.

Microalbuminuria: Ultra-sensitive dipsticks are available to measure albuminuria (albumin excretion 20–200mg/24h). Suggests early glomerular disease, eg DM, •BP.

Haematuria

Blood in the urine may arise from anywhere in the renal tract. Transient causes should be excluded, eg UTI, menstruation. Classified as:

visible (VH): previously known as macroscopic, frank

non-visible (NVH): found on dipstick/microscopy, previously known as microscopic. NVH is subdivided according to the presence of urinary tract symptoms: sympto-matic (SNVH) or asymptomatic (aNVH).

Causes: Malignancy (kidney, ureter, bladder), calculi, IgA nephropathy, Alport syn-drome (p220), other glomerulonephritis (p210), polycystic kidney disease (p220), schistosomiasis. Do not attribute haematuria to anticoagulation without investiga-tion. False positive: Myoglobin triggers same dipstick reaction—check microscopy.

Management: VH, SNVH, and aNVH >40yr should undergo urological assessment, im-aging, and cystoscopy to exclude renal tract malignancy and calculi.

A renal aetiology should be considered, and renal referral made, for NVH with:

eGFR <60

coexistent proteinuria (A:CR >20 or P:CR >50)

hypertension >140/90mmHg

family history of renal disease.

A cause is not established in 19–68% of patients with NVH. These patients should be monitored via BP, eGFR, and repeat A:CR/P:CR every 6 months–1 year. Increasing pro-teinuria and deteriorating eGFR warrant repeat referral and investigation.

Others

Urine microscopy

Cells:

Red blood cells:

>2 red cells/mm2 is abnormal.

White blood cells: (fig 7.2a)

>10 white cells/mm2 in an unspun specimen is abnormal.

Causes include UTI, glomerulonephritis, tubulointerstitial nephritis, renal trans-plant rejection, and malignancy.

Squamous epithelial cells:

Often seen, not pathological.

Casts:

Casts are cylindrical bodies formed in the lumen of distal tubules. They are formed of Tamm-Horsfall protein combined with cells.

Hyaline cast (fig 7.2b)—seen in normal urine.

Red cell cast (fig 7.2c)—signify an inflammatory process in the glomerulus, eg glomerulonephritis (p211).

White cell cast (fig 7.2d)—pyelonephritis, interstitial nephritis (p218), glomeru-lonephritis (p211).

Granular cast (fig 7.2e)—formed from degenerated tubular cells, seen in any chronic kidney disease.

Crystals:

Crystals are common in old or cold urine and may not signify pathology. They are important in stone formers.

Uric acid (fig 7.2f)(p680)—uric acid stones, tumour lysis syndrome.

Calcium oxalate (fig 7.2g)—stones (p628), high oxalate diet, ethylene glycol poi-soning.

Cystine (fig 7.2h)—seen in cystinuria (p221).

(a)

(b)

(c)

(d)

(e)

(f)

(g) (h)

Fig 7.2 (a) White cells; (b) hyaline cast; (c) red cell cast; (d) white cell cast; (e) granular cast; (f) uric acid crystals; (g) calcium oxalate crystals; (h) cystine crystal.

Images (a) to (h) reproduced from Turner et al., Oxford Textbook of Clinical Nephrology, 2005, with permission from Oxford University Press.

Incidence Annual incidence of UTI in women is 10–20%. 10% of men and 20% of women >65 years have asymptomatic bacteriuria (>65 years MSU is no longer diagnos-tic and clinical assessment is mandatory). Pyelonephritis = 2 per 1000 patient years.

Classification

Uncomplicated: normal renal tract structure and function.

Complicated: structural/functional abnormality of genitourinary tract, eg obstruc-tion, catheter, stones, neurogenic bladder, renal transplant.

Risk factors

- Bacterial inoculation: Sexual activity, urinary incontinence, faecal incontinence, constipation.

Binding of uropathogenic bacteria: Spermicide use, ‘oestrogen, menopause.

Urine flow: Dehydration, obstructed urinary tract (p640).

Bacterial growth: DM, immunosuppression, obstruction, stones, catheter, renal tract malformation, pregnancy.

Symptoms

Cystitis: Frequency, dysuria, urgency, suprapubic pain, polyuria, haematuria.

Acute pyelonephritis: Fever, rigor, vomiting, loin pain/tenderness, costovertebral pain, associated cystitis symptoms, septic shock.

Prostatitis: Pain: perineum, rectum, scrotum, penis, bladder, lower back. Fever, malaise, nausea, urinary symptoms, swollen or tender prostate on PR. See p645.

Signs Fever, abdominal or loin tenderness. Check for a distended bladder, enlarged prostate. If vaginal discharge, consider PID, see p412.

Do not rely on classical symptoms and signs in a catheterized patient.

Tests In non-pregnant women, if ? 2 (or one severe) symptoms of cystitis, and no vaginal discharge, treat empirically without further tests.

Dipstick: Use in non-pregnant women <65 years with less than three symptoms. A negative dipstick reduces probability of UTI to <20%. Do not use in pregnant women. Limited data for men. No diagnostic value in catheterized sample.

MSU culture: Conventional cut off >105 colony-forming units (cfu)/mL (but best di-agnostic criterion may be >102–102cfu/mL). Use in pregnant women, men, children, and if fail to respond to empirical antibiotics. Catheterized sample only if septic.

Blood tests: If systemically unwell: FBC, U&E, CRP, and blood culture (positive in only 10–25% of pyelonephritis). Consider fasting glucose.

Imaging: Consider USS and referral to urology for assessment (cystoscopy, urody-namics, CT) in men with upper UTI; failure to respond to treatment; recurrent UTI (>2/year); pyelonephritis; unusual organism; persistent haematuria.

Organisms Usually anaerobes and Gram-negative bacteria from bowel and vaginal flora. E. coli is the main organism (75–95% in community but ‘ in hospital). Staphylo-coccus saprophyticus (a skin commensal) in 5–10%. Other enterobacteriaceae such as Proteus mirabilis and Klebsiella pneumonia. For sterile pyuria see table 7.2.

Table 7.2 Causes of sterile pyuria (numbers of white cells but sterile on standard culture)

| Infection related | Non-infection related | Non-infection related | | — | — | — | | TB | Calculi | Polycystic kidney | | Recently treated UTI | Renal tract tumour | Recent catheter | | Inadequately treated UTI | Papillary necrosis | Pregnancy | | Fastidious culture requirement Appendicitis, prostatitis, chlamydia | Tubulointerstitial nephritis Chemical cystitis | SLE Drugs, eg steroids |

- Uroscopy in Early Modern Europe by Michael Stolberg, Routledge, 2016, p52–6.

Definition

Acute kidney injury (AKI) is a syndrome of decreased renal function, measured by serum creatinine or urine output, occurring over hours–days. It includes different aetiologies and may be multifactorial.

Different definitions of AKI exist. In 2012, there was an attempt to amalgamate different diagnostic criteria into a single definition and staging system. The Kidney Diseases: Improving Global Outcomes (KDIGO) guidelines[2] define AKI as:

rise in creatinine >26μmol/L within 48h.

rise in creatinine >1.5 ≈ baseline (ie before the AKI) within 7 days.

urine output <0.5mL/kg/h for >6 consecutive hours.

The severity of AKI is then staged according to the highest creatinine rise or longest period/severity of oliguria (table 7.2).

Table 7.2 KDIGO staging system for AKI

| Urinary tract tuberculosis A cause of sterile pyuria: dysuria, frequency, suprapubic pain but negative stand-ard culture. Ask about malaise, fever, night sweats, weight loss, back/flank pain, visible haematuria (p292). Can also cause an interstitial nephritis (p218) and renal amyloidosis (p215). Glo-merulonephritis is rare. Diagnose by microscopy with acid-fast techniques and mycobacterial culture of an early morning MSU and/or urinary tract tissue. Treat with rifampicin and isoniazid for 6 months in conjunction with pyrazina-mide and ethambutol for 2 months (see p294). | | — |

Although there are limitations to the use of serum creatinine, including the effects of muscle mass and dilution, no other biomarker has been able to supersede it (yet). The clinical approach to AKI is shown in fig 7.2.3

Epidemiology

AKI is common, occurring in up to 18% of hospital patients and r50% of ICU patients. Risk factors for AKI include pre-existing CKD, age, male sex, and comorbidity (DM, cardiovascular disease, malignancy, chronic liver disease, complex surgery).

Causes

Commonest causes:

Sepsis.

Major surgery.

Cardiogenic shock.

Other hypovolaemia.

Drugs.

Hepatorenal syndrome.

Obstruction.

Aetiology can be divided according to site (table 7.4) as:

pre-renal: ↓perfusion to the kidney.

renal: intrinsic renal disease.

post-renal: obstruction to urine.

Table 7.4 Aetiology of AKI

| The ‘Piss Prophets’ Beware the fallacies, deceit and juggling of the piss-pot science used by all those who pretend knowledge of diseases by the urine. Thomas Brian, 1655. Medieval texts1 give the following maxims regarding urinary change and disease: White or straw coloured urine = weak and cold liver and stomach Foamy urine = eructation (belching) Light coloured, turbid urine = mucus Lead circle on thin urine = pathological melancholy

Bubbles on the surface = disease of the head Watery urine = love sickness Swampy, black, stinking urine = fatal Lead coloured urine = a disintegrating uterus Reddish, cloudy urine with bubbles = asthma or an irregular heart beat. | | — |

Is there a life-threatening complication?

Examine Heart rate, BP, JVP, capillary refill, palpate for

bladder

Monitor

Fluid balance—consider urinary catheter and hourly urine output

K+—check response to treatment and at least daily until creatinine falls

Observations—minimum every 4 hours

Lactate if signs of sepsis

Daily creatinine until ‘ (lags r24 hours behind clinical response)

Investigate

Urine dipstick (pre-catheter) and quantification of any proteinuria. Haematuria/ proteinuria may suggest intrinsic renal disease

USS within 24 hours (unless cause obvious or AKI improving). Small kidneys (<9cm) suggest CKD. Asymmetry may suggest renal vascular disease

Check liver function (hepatorenal)

Check platelets—if low need blood film to check for haemolysis (HUS/TTP)

Investigate for intrinsic renal disease if indicated: immunoglobulins, paraprotein, complement, autoantibodies (ANA, ANCA, anti-GBM)

Support

Treat sepsis (p792)

Stop nephrotoxic medication—NSAIDS, ACE-i, ARB, aminoglycosides

Stop drugs that may ♙complications: diuretics (especially K+-sparing), met-formin, antihypertensives

Check all drug dosages are appropriate for renal impairment

Consider gastroprotection (H2 antagonist, PPI) and nutritional support

Avoid radiological contrast (or ☙risk: p219).

Management of AKI requires diagnosis and treatment of the underlying aetiology:

Pre-renal: correct volume depletion and/or •renal perfusion via circulatory/cardiac support, treat any underlying sepsis.

Renal: refer for likely biopsy and specialist treatment of intrinsic renal disease.

Post-renal: catheter, nephrostomy, or urological intervention.

Common to all aetiologies of AKI is the need to manage fluid balance, acidosis, hyper-kalaemia, and the timely recognition of those who may require renal replacement.

- Calcium chloride contains 2x calcium than the same volume of gluconate. Concern exists about the ☙bioavailability of calcium gluconate. Both salts carry a risk of tissue necrosis with extravasation.

Definition Abnormal kidney structure or function, present for >2 months, with im-plications for health.[a]

Classification Based on GFR category (table 7.5), the presence of albuminuria as a marker of kidney damage (table 7.6), and the cause of kidney disease (table 7.7). (Problems using formula to grade renal disease by eGFR p669).

Table 7.5 Classification of CKD by GFR (mL/min/1.73m2)

| Stage Serum creatinine Urine output | | — | | 1 >26.5μmol/L (0.2mg/dL) or <0.5mL/kg/h for 6–12h 1.5–1.9 ≈ baseline | | 2 2.0–2.9 ≈ baseline <0.5mL/kg/h for >12h | | 2 >252.6μmol/L (4.0mg/dL) or <0.2mL/kg/h for >24h or >2.0 ≈ baseline or anuria for >12h renal replacement therapy |

Table 7.6 Classification of CKD by albuminuria

| Where? | Pathology | Example | | — | — | — | | Pre-renal | ↓Vascular volume | Haemorrhage, D&V, burns, pancreatitis | | Pre-renal | ↓Cardiac output | Cardiogenic shock, MI | | Pre-renal | Systemic vasodilation | Sepsis, drugs | | Pre-renal | Renal vasoconstriction | NSAIDs, ACE-i, ARB, hepatorenal syndrome | | Renal | Glomerular (pp210–12) | Glomerulonephritis, ATN (prolonged renal hypo-perfusion causing intrinsic renal damage) | | Renal | Interstitial (p218) | Drug reaction, infection, infiltration (eg sarcoid) | | Renal | Vessels (pp214–5) | Vasculitis, HUS, TTP, DIC | | Post-renal | Within renal tract | Stone, renal tract malignancy, stricture, clot | | Post-renal | Extrinsic compression | Pelvic malignancy, prostatic hypertrophy, retro-peritoneal fibrosis |

Table 7.7 Classification of CKD based on underlying disease

| Referring to the renal team Request that advice/review is necessary due to: AKI not responding to treatment AKI with complications: ↑K+, acidosis, fluid overload stage 2 AKI (table 7.2) AKI with difficult fluid balance (eg hypoalbuminaemia, heart failure, pregnancy) AKI due to possible intrinsic renal disease (table 7.4) AKI with hypertension. Know: Creatinine trend and pre-morbid result if available, K+, bicarbonate/lactate, Hb, platelet count (+film), urine dipstick (before catheter), clinical observations (NEWS) since admission, fluid input/output, examination findings (hypo/hypervol-aemia), USS result, comorbidity (eg DM), drugs given (what? when? nephrotoxic?). | | — |

The most common causes of CKD in the UK are diabetes (24%), glomerulonephritis (12%), and ↑BP/renovascular disease (11%).7

Prognosis ↓GFR and albuminuria are independently associated with a higher risk of:

all-cause mortality

cardiovascular mortality

progressive kidney disease and kidney failure

AKI.

Patients with CKD are much more likely to die of CVD than to need renal replacement therapy. The risk of adverse outcome in CKD can be represented as a ‘heat map’ ac-cording to GFR and albuminuria categories (fig 7.5).

Low risk Moderate risk High risk Very high risk

Extrapolated data

Reprinted from Kidney International, 80, AS Levey et al., Chronic kidney disease: definition, classification, and prognosis, 17–28, 2011, with permission from Elsevier.

The patient with CKD: a clinical approach

History

Does the patient really have CKD? Does the eGFR reflect the true GFR (p669)? Is the eGFR corrected for ethnicity/ drugs (eg trimethoprim alters creatinine con-centration but not GFR)? Evidence of chronicity, ie >2 months—is there a previous creatinine on record?

Possible cause: Ask about previous UTI, lower urinary tract symptoms, PMH of

BP, DM, IHD, systemic disorder, renal colic. Check drug history including when medications started. Family history including renal disease and subarachnoid haemorrhage. Systems review: look out for more than is immediately

obvious, consider rare causes, ask about eyes, skin, joints, ask about symptoms suggestive of systemic disorder ('When did you last feel well?') and malignancy.

Current state: Patients may have symptomatic CKD if GFR <20. Includes symp-toms of fluid overload (SOB, peripheral oedema), anorexia, nausea, vomiting, rest-less legs, fatigue, weakness, pruritus, bone pain, amenorrhoea, impotence.

Examination

Periphery: Peripheral oedema. Signs of peripheral vascular disease or neuropa-thy. A vasculitic rash. Gouty tophi. Joint disease. Arteriovenous fistula (thrill, bruit, recently needling?). Signs of immunosuppression: bruising from steroids, skin malignancy. Uraemic flap/encephalopathy if GFR <15.

Face: Anaemia, xanthelasma, yellow tinge (uraemia), jaundice (hepatorenal), gum hypertrophy (ciclosporin), Cushingoid (steroids), periorbital oedema (ne-phrotic syndrome), taut skin/telangiectasia (scleroderma), facial lipodystrophy (glomerulonephritis).

Neck: JVP for fluid state, tunnelled line (if removed, look for small scar over in-ternal jugular, and a larger scar in 'breast pocket' area), scar from parathyroid-ectomy, lymphadenopathy.

Cardiovascular: BP, sternotomy, cardiomegaly, stigmata of endocarditis. If right-sided heart failure/tricuspid regurgitation, JVP does not reflect fluid state.

Respiratory: Pulmonary oedema or effusion.

Abdomen: PD catheter or scars from previous catheter (small scars just below umbilicus and to side of midline), signs of previous transplant (scar, palpable graft), ballotable polycystic kidneys ! palpable liver.

Investigation

Blood: U&E (compare with previous), Hb (normochromic, normocytic anaemia), glucose (DM), ↓Ca2+, ↑PO 2 −, ↑PTH (renal osteodystrophy). Directed investigation of intrinsic renal disease: ANA, ANCA, antiphospholipid antibodies, paraprotein, complement, cryoglobulin, anti-GBM, hepatitis serology, anti-PLA2R (membranous nephropathy). Note: ESR is not helpful as ↑ in CKD and proteinuric states.

Urine: Dipstick, MC&S, A:CR or P:CR (p294), Bence Jones.

Imaging: USS for size, symmetry, anatomy, corticomedullary differentiation, and to exclude obstruction. In CKD kidneys may be small (<9cm) except in infiltrative disorders (amyloid, myeloma), APKD, and DM. If asymmetrical consider renovas-cular disease. Scarring may be seen on USS but isotope scans are more sensitive.

Histology: Consider renal biopsy (p210) in progressive disease, nephrotic syn-drome, systemic disease, AKI without recovery. Biopsy is unlikely to change treat-ment if GFR stable and P:CR <150. DM with neuropathy/retinopathy may not need biopsy unless atypical, ie nephrotic, haematuria, other systemic symptoms.

CKD encompasses a range of disease from mild disease without progression to ad-vanced, symptomatic disease requiring renal replacement.

Management of CKD[a],8 requires:

Appropriate referral to nephrology.

Treatment to slow renal disease progression.

Treatment of renal complications of CKD.

Treatment of other complications of CKD.

Preparation for renal replacement therapy (dialysis/transplantation) (p206).

Referral to nephrology

Consider referral for:

stage G4 and G5 CKD (table 7.5)

moderate proteinuria A:CR >70mg/mmol unless due to DM and already treated

proteinuria A:CR >20mg/mmol with haematuria

declining eGFR:

‘eGFR by ?25% + ‘GFR category (table 7.5)

sustained ‘eGFR ?15% within 12 months

- BP poorly controlled despite ?4 antihypertensive drugs at therapeutic dose

known or suspected rare or genetic cause of CKD.

Treatment to slow renal disease progression

BP: Target systolic BP is <140mmHg (range 120–129mmHg) and diastolic <90mmHg. If DM or A:CR >70 then systolic target is <120mmHg (range 120–129) and diastolic

<80mmHg.

Renin–angiotensin system: Offer treatment with a renin–angiotensin system an-tagonist (ACE-i, ARB) to:

DM and A:CR >2mg/mmol.

Hypertension and A:CR >20mg/mmol.

Any CKD with A:CR >70mg/mmol.

Do not combine renin–angiotensinsin antagonists due to risk of hyperkalaemia and hypotension. Check K+ and renal function prior to, and 1–2 weeks after, starting treat-ment or changing dose. Stop if K+ >6mmol/L, eGFR >25%, or creatinine >20%: ex-clude other possible causes and consider a lower dose.

Glycaemic control: Target HbA1C of r52mmol/mol (7.0%) unless risk of hypo-glycaemia, comorbidity or limited life expectancy.

Lifestyle: Offer advice about exercise, healthy weight, and smoking cessation. Salt intake should be reduced to <2g of sodium/day (=<5g sodium chloride/day).

Treatment of renal complications of CKD

Anaemia: Check Hb when eGFR <60. Investigate (especially if anaemic with eGFR >20) and treat other deficiencies: iron (hypochromic red cells >6%, transferrin saturation

Acidosis: Consider sodium bicarbonate supplements for patients with eGFR <20 and low serum bicarbonate (<20mmol/L). Caution in patients with hypertension and fluid overload due to sodium component.

Oedema: Restrict fluid and sodium intake. High doses of loop diuretics may be need-ed. Combination of a loop and thiazide diuretic can have a powerful effect: distal tubule sodium excretion (and its inhibition with a thiazide) is more significant when already treated with a loop diuretic (fig 7.12). Diuretic treatment should only be given with careful monitoring of fluid state and renal function.

CKD bone-mineral disorders: CKD causes in serum phosphate and reduced hy-droxlyation of vitamin D by the kidney. Measure calcium, phosphate, ALP, PTH, and 25-OH vit D if eGFR <20.

Treat if phosphate >1.5mmol/L (>1.7mmol/L if RRT) with dietary restriction ! phos-phate binders. The use of binders which do not contain calcium may be beneficial in preventing vascular calcification.

Give vitamin D supplements (colecalciferol, ergocalciferol) if deficient. If ⇧PTH per-sists or is increasing, treat with an activated vit D analogue eg 1a-calcidol or calci-triol. Paricalcitol suppresses PTH with less effect on gut absorption of calcium and phosphate and is less likely to cause ⇧calcium/phosphate.

Restless legs/cramps: Exclude iron deficiency as a possible exacerbating factor. Give sleep hygiene advice. Treatment for severe cases with gabapentin/pregabalin/ dopamine agonists is off licence and may be complicated by side-effects (falls, cogni-tive impairment, impulse-control disorder).

Diet: Expert dietary advice should be available regarding protein intake, K+ if hyper-kalaemic, and phosphate restriction (eg dairy products).

Treatment of other complications of CKD

Cardiovascular disease: CKD confers ⇧risk of cardiovascular disease due to ⇧BP, vas-cular stiffness, inflammation, oxidative stress, and abnormal endothelial function (CV risk often higher than the risk of kidney failure).

Antiplatelets (low-dose aspirin) for CKD at risk for atherosclerotic events unless bleeding risk outweighs benefit (mortality benefit unclear in CKD).

Atorvastatin 20mg (and higher if GFR >20) for primary and secondary prevention of cardiovascular disease.

CKD should not affect treatment for heart failure but ⇧monitoring of GFR and K+.

GFR <60 may affect troponin and BNP values. Interpret results cautiously with con-sideration for the GFR.

Preparation for renal replacement therapy (RRT)

Planning for RRT should begin in progressive CKD when the risk of renal failure is 10–20% within a year. Referral to nephrology less than 1 year before RRT is required is considered a late referral.

All suitable patients should be listed for a deceased donor transplantation 6 months before the anticipated start of RRT. All suitable patients should be informed about the advantages of a pre-emptive living kidney transplant and efforts made to find a donor (p208).

Long-term dialysis is started when it is necessary to manage one or more symptoms of renal failure including:

inability to control volume status, including pulmonary oedema

inability to control blood pressure

serositis

acid–base or electrolyte abnormalities

pruritus

nausea/vomiting/deterioration in nutritional status

cognitive impairment.

RRT is a misnomer for dialysis: renal function is not replaced, rather there is provi-sion of just enough clearance to ameliorate the symptoms of kidney failure.

GFR at commencement of dialysis is usually r5–10. When transplantation is awaited or not possible, there are two main options: haemodialysis and peritoneal dialysis.

Haemodialysis (HD) (fig 7.6) Blood is passed over a semi-permeable membrane against dialysis fluid flowing in the oppo-site direction. Diffusion of solutes occurs down the concentration gradient. A hy-drostatic gradient is used to clear excess

Blood in

Waste

Blood

Dialysate

Blood returned

Fluid

fluid as required (ultrafiltration). Access is preferentially via an arteriovenous fistula which provides ↓blood flow and longevity. This should be created prior to need for RRT

removed in

Haemofiltration (fig 7.7) Water cleared by positive pressure, dragging solutes into the waste by convection. The ultrafiltrate (waste) is replaced with an appropriate volume of ('clean') fluid either before (pre-dilution) or after (post-dilution) the mem-brane. 'Haemodynamic instability so used in critical care when HD not possible due

Pre-dilution

Blood Blood

in Blood returned

Dialysate

Waste out

Post dilution

to 'BP. Not used for chronic RRT unless in combination with HD (haemodiafiltration)

for haemodynamic stability and ↓middle molecule clearance, eg β2-microglobulin. Complications of RRT Annual mortality is significant, mostly due to cardiovascu-lar disease: ↓BP, calcium/phosphate dysregulation, vascular stiffness, inflammation,

oxidative stress, abnormal endothelial function. Protein-calorie malnutrition: In-

creases morbidity and mortality. Renal bone disease: High bone turnover, renal osteodystrophy, osteitis fibrosa. Infection: Uraemia causes granulocyte and T-cell dysfunction with ↓sepsis-related mortality. Amyloid: β2-microglobulin accumulates in long–term dialysis causing carpal tunnel syndrome, arthralgia, visceral effects. Conservative management is for those who opt not to receive RRT due to lack of benefit on quality or quantity of life. Focus is on preserving residual renal function, symptom control, and advanced planning with patient and family for end-of-life care.

Transplantation (figs 7.8, 7.9) should be considered for every patient with, or pro-gressing towards, stage G5 kidney disease (p202). It is the treatment of choice for kidney failure provided risks do not exceed benefits. Many will not make the trans-plant list due to comorbidity or frailty.

Contraindications

Absolute: cancer with metastases.

Temporary: active infection, HIV with viral replication, unstable CVD.

Relative: congestive heart failure, CVD.

Types of graft

Living donor: Best graft function and survival, especially if HLA matched.

Deceased donor: (See organ donation p12.)

Donor after brain death (DBD, heart-beating donor).

Expanded criteria donor (ECD) is from an older kidney or from a patient with a history of CVA, BP, or CKD. This impacts on the long-term prognosis of the trans-plant but offers a better outcome than remaining on dialysis.
Donor after cardiac death (DCD, non-heart-beating donor) with risk of delayed graft function.

Immunosuppression

A combination of drugs are used. Aim is to use the minimal effective dose with the lowest drug-related toxicity.
Protocol used depends upon the immunological risk of the recipient and type of donated kidney.
BP, cholesterol, NODAT (new-onset diabetes after transplantation). Antimetabolites: Eg mycophenolic acid (MPA), azathioprine. MPA is now used pref-erentially due to better prevention of acute rejection and graft survival (not in preg-nancy, MPA is teratogenic). Side effects: anaemia, leucopenia, and GI toxicity.
Glucorticosteroids: Transcription of inflammatory cytokines. First-choice treat-ment for acute rejection. Significant side-effects (BP, hyperlipidaemia, DM, impaired wound healing, osteoporosis, cataracts, skin fragility) have led to protocols with ear-ly withdrawal of steroids and the use of steroid-free immunosuppression regimens.

Complications

Surgical: Bleed, thrombosis, infection, urinary leaks, lymphocele, hernia.
Delayed graft function: Affects up to 40% of grafts, more common in DCD. Rejection: Acute or chronic. Acute is divided into antibody mediated (rare un-less known pre-sensitized recipient) or cellular (most common). Causes renal function, diagnosed on graft biopsy. Treatment with high-dose steroids and immunosuppression. Chronic antibody-mediated rejection causes progressive dys-function of the graft. Most graft loss is now thought to be due to an immune re-sponse by donor-specific antibodies causing damage to the kidney microcirculation. Complex pathology and lack of controlled studies mean treatment is not clear. The results of monoclonal antibody trials are awaited.
Infection: Risk of all infections. Typically hospital acquired/donor derived in month 1, opportunistic in months 1–6 (therefore prophylactic treatment for CMV and Pneu-mocystis jirovecii given), usual spectrum of community-acquired infection after 6–12 months. Late viral infection should always be considered: eg CMV, HSV.
Malignancy: Up to 25≈ risk of cancer with immunosuppression, particularly skin, post-transplant lymphoproliferative disorder (PTLD), and gynaecological.
CVD: 2–5≈ risk of premature CVD compared to general population (but r80% less than dialysis). BP, NODAT, rejection, and renal history (uraemic cardiomyopathy) contribute.

Prognosis

Acute rejection <15%, 1-year graft survival >90%. Longer-term graft loss r4%/year. Factors contributing to graft loss:
Donor factors: age, comorbidity, living/deceased, DBD/DCD.
Rejection.
Infection.
BP/CVD.
Recurrent renal disease in graft.
Most common outcome is death with a functioning transplant (ie transplant ‘out-lives’ the patient).
The term glomerulonephritis (GN) encompasses a number of conditions which:
are caused by pathology in the glomerulus

present with proteinuria, haematuria, or both

are diagnosed on a renal biopsy

cause CKD

can progress to kidney failure (except minimal change disease).

The names of the diseases come from either the histological appearance (eg membra-nous glomerulonephritis), or the associated systemic condition (eg lupus nephritis).

Nephrotic or nephritic?

(nephritis).

Figure adapted from Turner et al., Oxford Textbook of Clinical Nephrology, 2015, with permission from

Oxford University Press

Management General management as for CKD (pp204–5) including BP control and in-hibition of renin-angiotensin axis. Specific treatment including immunosuppression de-pends on histological diagnosis, disease severity, disease progression, and comorbidity.

Oxford University Press.

Fig 7.12 Crescentic GN: a proliferation of epi-thelial cells and macrophages with rupture of Bowman's capsule.

University Press.

If there is oedema, dipstick the urine to avoid missing renal disease.

Definition The nephrotic syndrome is a triad of:

proteinuria >2g/24h (P:CR >200mg/mmol, A:CR >250mg/mmol, p294)

hypoalbuminaemia (usually <20g/L, can be <10g/L)

oedema.

Aetiology Primary renal disease or secondary to a systemic disorder.

Primary renal disease: Minimal change disease, membranous nephropathy (may be associated with underlying inflammation/malignancy), focal segmental glo-merulosclerosis (FSGS), membranoproliferative GN.

Secondary causes: DM, lupus nephritis, myeloma, amyloid, pre-eclampsia.

Pathophysiology The filtration barrier of the kidney is formed by podocytes, the glomerular basement membrane (GBM), and endothelial cells. Proteinuria results from podocyte pathology: abnormal function in minimal change disease, immune-mediated damage in membranous nephropathy, and podocyte injury/death in FSGS; or pathology in the GBM/endothelial cell: membranoproliferative GN.

Presentation Generalized, pitting oedema, which can be rapid and severe. Look in dependent areas (ankles if mobile, sacral pad/elbows if bed-bound) and areas of low tissue resistance, eg periorbitally. History: Ask about systemic symptoms, eg joint, skin. Consider malignancy and chronic infection. ☆☆: CCF (•JVP, pulmonary oedema), liver disease (‘albumin).

Management

Reduce oedema

Treat underlying cause

Adults need a renal biopsy (p210). This is technically more difficult when there is gross oedema so diuresis may be required first. Treatment known to induce remission should be given, eg corticosteroids in minimal change disease. Look for and treat any underlying systemic disease, infection, or malignancy.

In children, minimal change disease is the commonest aetiology and steroids induce remission in the majority. Biopsy is therefore avoided in children unless there is no response to steroids, or if clinical features suggest another cause: age <1yr, family history, extrarenal disease (eg arthritis, rash, anaemia), renal failure, haematuria.

Reduce proteinuria

ACE-i/ARB reduce proteinuria (may not be needed in minimal change disease).

Complications

Thromboembolism. Hypercoagulable due to •clotting factors, ‘anti-thrombin III, and platelet abnormalities. ♁Risk of VTE including DVT/PE (r10% adult pa-tients) and renal vein thrombosis (loin pain, haematuria, •LDH, AKI if bilateral). Treat with heparin (may need to dose adjust low-molecular-weight heparin if ‘GFR) and warfarin. If low bleeding risk, consider prophylaxis when albumin

Hyperlipidaemia. •Cholesterol (>10mmol/L), •LDL, •triglycerides, ‘HDL. Thought due to hepatic synthesis in response to ‘oncotic pressure and defective lipid breakdown. Abnormalities are proportional to proteinuria. The benefits of statins in CKD are extrapolated to nephrotic syndrome where there is ‘evidence.

Minimal change disease

Focal segmental glomerulosclerosis (FSGS)

Treatment: ACE-i/ARB and blood pressure control in all. Corticosteroids only in primary (idiopathic) disease: remission in r25%, partial remission in up to 50%. Calcineurin inhibitors may be considered second line. Plasma exchange and rituxi-mab have been used for recurrence in transplants.

Membranous nephropathy

malignancy: lung, breast, GI, prostate, haematological

drugs: gold, penicillamine.

Membranoproliferative glomerulonephritis

immune-complex associated: driven by increased or abnormal immune com-plexes which deposit in the kidney and activate complement. An underlying cause can be found in most adult cases, eg infection, cryoglobulinaemia, mono-clonal gammopathy, autoimmunity

Treatment: ACE-i/ARB and blood pressure control in all. Underlying cause in im-mune-complex disease. Trial of immunosuppression if no underlying cause found and progressive decline in renal function. Treatments to block or modify C2 activa-tion are awaited.

Diabetic nephropathy

Intensive DM control prevents microalbuminuria and reduces risk of progression to macroalbuminuria (‘severely increased albuminuria’) = A:CR >20mg/mmol. HbA1C of 52mmol/mol (7%) reduces the development of all microvascular complications. However, less impact on CVD risk and hard renal outcomes including progression to kidney failure. Consider risk of hypoglycaemia.

Statins to reduce CV risk (p205). Unclear benefit once on dialysis: do not initiate but do not need to discontinue if tolerated.

Lupus nephritis

Small vessel vasculitis

Amyloid

Haemolytic uraemic syndrome (HUS)

Thrombotic thrombocytopenic purpura (TTP)

Atherosclerotic renovascular disease

Part of a systemic atheromatous vascular disease including cardio-, cerebro-, and peripheral vascular disease (ask about claudication, check foot pulses), ↑BP, and

Scleroderma renal crisis

>150/85mmHg) and AKI (↓eGFR by >20%). Biopsy: collapsed glomeruli, onion-skin thickening of arterioles.

Treatment: ACE-i/ARB. IV vasodilators to ↓vascular resistance and for digital ischaemia. Care with β-blockers as ↑HR compensating for ↓stroke volume. May recover renal function after many months.

Sickle cell nephropathy

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Fig 7.12 The nephron divided into segments (proximal tubule, thick ascending loop of Henle, distal tubule, collecting duct) with key solute movement (red).

Acute tubulointerstitial nephritis (ATIN)

Take a full drug history including over-the-counter and herbal preparations.

Treatment: Stop causative agent or treat underlying cause. Steroids are used de-spite a paucity of RCT evidence.

Chronic tubulointerstitial nephritis (CTIN)

drugs: NSAIDs (p219), lithium, calcineurin inhibitors, aminosalicylates (eg mesala-zine, sulfasalazine), chemotherapy (eg cisplatin)

specific nephrotoxins: lead, cadmium, mercury, aristolochic acid (p219)

Treatment: Stop causative agent or treat underlying cause. Reduce risk of progres-sion as per CKD management: ACE-i/ARB, BP control, glucose, lipids (pp204–5). Future: antifibrotic agents?

Analgesic nephropathy

Aminoglycosides (gentamicin >tobramycin >amikacin >streptomycin)

Radiocontrast nephropathy

Rhabdomyolysis

Urate nephropathy

Radiation nephritis

Renal impairment due to ionizing radiation. Presents 6 months–years after total body irradiation, local field radiotherapy, or targeted radionucleotide therapy. Presents with •BP, proteinuria/haematuria, progression to renal failure. Prognosis linked to •BP. Treatment: 'Radiation dose with shielding. As CKD (pp204–5) with strict BP control.

Aristolochic acid nephropathy

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Autosomal dominant polycystic kidney disease (ADPKD)

>20) may suppress cyst growth. •BP should be treated to target <120/80mmHg: 1st-

Autosomal recessive polycystic kidney disease

- in 20 000, chromosome 6. Presents ante/perinatally with renal cysts ('salt and pep-per' appearance on USS), congenital hepatic fibrosis↬portal hypertension. Poor prog-nosis if neonatal respiratory distress. No specific therapy. (See OHCS p122.)

Renal phakomatoses

Alport syndrome

Fabry disease

- in 40 000–120 000. X-linked. Lysosomal storage disorder due to a deficiency of the enzyme a-galactosidase-A. Causes proteinuria and progressive renal failure in most men and some female 'carriers'. Lipid deposits are seen in urine and on renal biopsy ('zebra body'). Treatment with IV enzyme replacement can stabilize kidney function if proteinuria controlled to <1g/24h.

Cystinuria

Cystinosis

Haematology

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- Compare these personalities with those of the 2015 anthropomorphic Pixar film 'Inside Out'.

Symptoms Due to the underlying cause or to the anaemia itself: fatigue, dyspnoea, faintness, palpitations, headache, tinnitus, anorexia—and angina if there is pre-exist-ing coronary artery disease.

Signs May be absent even in severe anaemia. There may be pallor (eg of the con-junctivae, see fig 8.2, although this is not a reliable sign). In severe anaemia (Hb

<80g/L), there may be signs of a hyperdynamic circulation, eg tachycardia, flow murmurs (ejection-systolic loudest over apex), and cardiac enlargement; or retinal haemorrhages (rarely). Later, heart failure may occur: here, rapid blood transfusion can be fatal.

Types of anaemia The first step in diagnosis is to look at the mean cell volume (MCV). Normal MCV is 76–96 femtolitres (1015 fL = 1L).

Low MCV (microcytic anaemia):

- Iron-deficiency anaemia (IDA), the most common cause: see p226.
- Thalassaemia (suspect if the MCV is 'too low' for the Hb level and the red cell count is raised, though definitive diagnosis needs DNA analysis): see p242.

3 Sideroblastic anaemia (very rare): p226.

NB: there is iron accumulation in the last two conditions, and so tests will show in-creased serum iron and ferritin with a low total iron-binding capacity (TIBC).

Normal MCV (normocytic anaemia):

- Acute blood loss. 5 Hypothyroidism (or •MCV).
- Anaemia of chronic disease (or 'MCV). 6 Haemolysis (or •MCV).

3 Bone marrow failure. 7 Pregnancy.

4 Renal failure.

NB: if 'wcc or 'platelet in normocytic anaemia, suspect marrow failure: see p264.

High MCV (macrocytic anaemia):

- B12 or folate deficiency. 5 Myelodysplastic syndromes.
- Alcohol excess—or liver disease. 6 Marrow infiltration.

3 Reticulocytosis (p228, eg with haemolysis). 7 Hypothyroidism.

4 Cytotoxics, eg hydroxycarbamide. 8 Antifolate drugs (eg phenytoin).

Haemolytic anaemias: These do not fit into the above-mentioned classification as the anaemia may be normocytic or, if there are many young (hence larger) RBCS and reticulocytes, macrocytic (p222). Suspect if there is a reticulocytosis (>2% of RBCS; or reticulocyte count >100≈109/L), mild macrocytosis, 'haptoglobin, •bilirubin,

LDH, or •urobilinogen. These patients will often be mildly jaundiced (but note that haemolysis causes pre-hepatic jaundice so there will be no bilirubin in their urine).

In severe anaemia with heart failure, transfusion is vital to restore Hb to a safe level, eg 60–80g/L, but this must be done with great care. Give it slowly with 10–40mg furosemide IV/PO with alternate units (dose depends on previous exposure to diuretics; do not mix with blood). Check for signs of worsening overload: rising JVP and basal crackles: in this eventuality, stop and treat.

Fig 8.2 'Conjunctival pallor', the classic sign of anaemia, is a confusing term as the conjunctiva is translucent, transmitting the colour of structures under it. The 'pallor' refers to the vasculature on the inner surface of the lid which is lacking Hb. It is this colour but it should be:

This is common (seen in up to 14% of menstruating women).

Causes •Blood loss, eg menorrhagia or GI bleeding2 (upper p256; lower p629).

Poor diet or poverty may cause IDA in babies or children (but rarely in adults).

Malabsorption (eg coeliac disease) is a cause of refractory IDA.

In the tropics, hookworm (GI blood loss) is the most common cause.

Signs Chronic IDA (signs now rare): koilonychia (fig 8.4 and p76), atrophic glossitis, angular cheilosis (fig 8.5), and, rarely, post-cricoid webs (Plummer–Vinson syndrome).

ic disease; or misdiagnosis, eg when thalassaemia is to blame.

The commonest anaemia in hospital patients (and the 2nd commonest, after IDA, worldwide). It arises from three problems (in which the polypeptide, hepcidin, plays a key role): 1 Poor use of iron in erythropoiesis. 2 Cytokine-induced shortening of RBC survival. 3 ↜Production of and response to erythropoietin.

Causes Many, eg chronic infection, vasculitis, rheumatoid, malignancy, renal failure. Tests Ferritin normal or • in mild normocytic or microcytic anaemia (eg Hb >80g/L; see table 8.1). Check blood film, B12 , folate, TSH, and tests for haemolysis (p226).

Treatment Treating the underlying disease may help (eg in 60% of patients with

RA), as may erythropoietin (SE: flu-like symptoms, hypertension, mild rise in the platelet count and thromboembolism). Also effective in improving quality of life in malignant disease. IV iron can safely overcome the functional iron deficiency. Hepcidin inhibitors and inflammatory modulators show promise.

Microcytic anaemia does not always mean iron deficiency! 20% of older people with an MCV <75fL are not iron deficient. ►Think of sideroblastic anaemia whenever microcytic anaemia is not responding to iron. This condition is characterized by ineffective erythropoiesis, leading to ⍙iron absorption, iron loading in marrow ! hae-mosiderosis (endocrine, liver, and heart damage due to iron deposition).

- In one study, 11% presenting to their GP with IDA had GI carcinoma. Plan both upper and lower GI investiga-tion: there may be abnormalities on both.

Table 8.1 Interpreting plasma iron studies

| Red cell distribution width (RCDW or RDW) In health or in unifactorial anaemia, all the red cells in a sample are about the same size, and the graph of their volume distribution forms a narrow peak. In mixed anaemias, however, this peak broadens, reflecting an abnormally large RDW—this may be the first clue to dual pathology. In coeliac disease, for example, poor ab-sorption of iron (‘MCV) and folate (•MCV) may occur simultaneously, resulting in a combination of microcytes and macrocytes in the circulation. The visual analogue of this is anisocytosis (p328) on a blood film. The laboratory measure is a ⍙RDW, where RDW = the standard deviation of MCV divided by the mean MCV, multiplied by 100. Reference interval: 11.5–14.6%. If the MCV is high and the RDW is normal, the cause is likely to be alcohol, liver disease, or a marrow problem (chemotherapy or aplastic anaemia). | | — |

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Fig 8.8 Ring sideroblasts in the marrow, with a perinuclear ring of iron granules, found in sidero-blastic anaemia.

Christine Lawrence.

Fig 8.5 Angular cheilosis (also known as sto-matitis): ulceration at the side of the mouth. Also a feature of vitamin B12 and B2 (ribofla-vin) deficiency, and glucagonoma (p223).

Courtesy of Dr Joseph Thompson: AskAnOrtho-

dontist.com.

Christine Lawrence.

Fig 8.9 Two ringed sideroblasts showing how the distribution of perinuclear mito-chondrial ferritin can vary. The problem in congenital sideroblastic anaemia is disordered mitochondrial haem synthesis.

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Many haematological (and other) diagnoses are made by careful examination of the peripheral blood film. It is also necessary for interpretation of the FBC indices.

Features Include:

Blasts: Nucleated precursor cells. They should not normally appear in peripheral blood but do in myelofibrosis, leukaemia, and malignant marrow infiltration.

Howell–Jolly bodies: DNA nuclear remnants in RBCS, which are normally removed by the spleen (fig 8.14). Seen post-splenectomy and in hyposplenism (eg sickle-cell disease, coeliac disease, congenital, UC/Crohn's, myeloproliferative disease, amyloid). Also in dyserythropoietic states: myelodysplasia, megaloblastic anaemia.

Hypochromia: (p226.) Less dense staining of RBCS due to 'Hb synthesis, seen in IDA, thalassaemia, and sideroblastic anaemia (iron stores unusable, p266).

Left shift: Immature neutrophils released from the marrow, eg in infection (fig 8.15). Leukoerythroblastic film: Immature cells (myelocytes, promyelocytes, metamyelo-cytes, normoblasts) ! tear-drop RBCS from haemolysis or marrow infiltration/infec-tion (malignancy; TB; brucella; visceral leishmaniasis; parvovirus B19).

Leukaemoid reaction: A marked leucocytosis (WCC >50≈109/L). Seen in severe illness, eg with infection or burns, and also in leukaemia.

Pappenheimer bodies: (fig 8.16) Granules of siderocytes containing iron. Seen in lead poisoning, carcinomatosis, and post-splenectomy.

Poikilocytosis: Variation in RBC shape, eg in IDA, myelofibrosis, thalassaemia. Polychromasia: RBCS of different ages stain unevenly (young are bluer). This is a re-sponse to bleeding, haematinic replacement (ferrous sulfate, B12, folate), haemolysis, or marrow infiltration. Reticulocyte count is raised.

Reticulocytes: (Normal range: 0.8–2%; or <85≈109/L.) (fig 8.17) Young, larger RBCS (contain RNA) signifying active erythropoiesis. Increased in haemolysis, haemor-rhage, and if B12, iron, or folate is given to marrow that lack these.

Right shift: Hypermature white cells: hypersegmented polymorphs (>5 lobes to nu-cleus) seen in megaloblastic anaemia, uraemia, and liver disease. See p222, fig 8.25. Rouleaux formation: (fig 8.18) Red cells stack on each other (causing a raised ESR; p272). Seen with chronic inflammation, paraproteinaemia, and myeloma.

Schistocytes: Fragmented RBCS sliced by fibrin bands, in intravascular haemolysis (p229, fig 8.21). Look for microangiopathic anaemia, eg DIC (p252), haemolytic urae-mic syndrome, thrombotic thrombocytopenic purpura (TTP: p215), or pre-eclampsia. Spherocytes: Spherical cells found in hereditary spherocytosis and autoimmune haemolytic anaemia. See p228.

Target cells: (Also known as Mexican hat cells, fig 8.14 and fig 8.41, p242.) These are RBCS with central staining, a ring of pallor, and an outer rim of staining seen in liver disease, hyposplenism, thalassaemia—and, in small numbers, in IDA.

Tear-drop RBCS: Seen in extramedullary haematopoiesis; see leukoerythroblastic film.

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Fig 8.14 Film in hyposplenism: target cell (short arrow), acanthocyte (long arrow), and a Howell–Jolly body (arrow head).

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Top image ©Prof. Christine Lawrence, staining (azure B; cresyl blue) is needed.

bottom image ©Crookston Collection.

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Neutrophils (figs 8.19, 8.20) 2–7.5 ≈ 109/L (40–75% of white blood cells: but absolute values are more meaningful than percentages).

Increased in (ie neutrophilia):

Bacterial infections.

Inflammation, eg myocardial infarction, polyarteritis nodosa.

Myeloproliferative disorders.

Drugs (steroids).

Disseminated malignancy.

Stress, eg trauma, surgery, burns, haemorrhage, seizure.

Decreased in (ie neutropenia—see p252):

Viral infections.

Drugs: post-chemotherapy, cytotoxic agents, carbimazole, sulfonamides.

Severe sepsis.

Neutrophil antibodies (SLE, haemolytic anaemia)—•destruction.

Hypersplenism (p272), eg Felty's syndrome (p698).

Bone marrow failure—'production (p264).

Other neutrophil responses to infection: These include: • vacuoles in the cytoplasm (the most specific sign of bacterial infection); • Döhle bodies: inconspicuous grey-blue areas of cytoplasm (residual ribosomes). Up to 17% of neutrophils from females show a drumstick-shaped Barr body (arrow, fig 8.20d). It is the inactivated X chromosome.

Lymphocytes (fig 8.21) 1.5–4.5 ≈ 109/L (20–45%).

Increased in (ie lymphocytosis):

Acute viral infections.

Chronic infections, eg TB, brucellosis, hepatitis, syphilis.

Leukaemias and lymphomas, especially chronic lymphocytic leukaemia (CLL). Large numbers of abnormal ('atypical') lymphocytes are characteristically seen with EBV infection: these are T cells reacting against EBV-infected B cells. They have a large amount of clearish cytoplasm with a blue rim that flows around neighbouring RBCS. Other causes of 'atypical' lymphocytes: see p405.

Decreased in (ie lymphopenia):

Steroid therapy; SLE; uraemia; Legionnaire's disease; HIV infection; marrow infil-tration; post chemotherapy or radiotherapy.

T-lymphocyte subset reference values: CD4 count: 527–1571/mm2 (low in HIV infection).

CD8 count: 225–752/mm2; CD4/CD8 ratio: 1.2–2.8. Eosinophils (fig 8.22) 0.04–0.4 ≈ 109/L (1–6%). Increased in (ie eosinophilia):

Drug reactions, eg with erythema multiforme, p562.

Allergies: asthma, atopy.

Parasitic infections (especially invasive helminths).

Skin disease: especially pemphigus, eczema, psoriasis, dermatitis herpetiformis. Also seen in malignant disease (including lymphomas and eosinophilic leukaemia), PAN, adrenal insufficiency,irradiation, Löffler's syndrome (p704), Churg–Strauss sym-drome (p696) and during the convalescent phase of any infection.

Monocytes (fig 8.22) 0.2–0.8 ≈ 109/L (2–10%). Increased in (ie monocytosis): the aftermath of chemo- or radiotherapy, chronic infections (eg malaria, TB, brucellosis, protozoa), malignant disease (including M4 and M5 acute myeloid leukaemia (p256), and Hodgkin's disease), myelodysplasia.

Basophils (fig 8.24) 0–0.1 ≈ 109/L (0–1%). Increased in (ie basophilia): myeloprolif-erative disease, viral infections, IGE-mediated hypersensitivity reactions (eg urticaria, hypothyroidism), and inflammatory disorders (eg UC, rheumatoid arthritis).

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Fig 8.21 Lymphocyte: divided into T & B types, which have important roles in cell-mediated im-munity and antibody production.

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Fig 8.20 Neutrophils: (a) 'toxic granulation' seen in infection or pregnancy; (b) normal ap-pearances; (c) 'left shift': immature forms are released with few lobes to their nuclei, seen in infection; (d) Barr body (arrow, see text).

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Fig 8.24 Basophil. The cytoplasm is filled with dark-staining granules, containing histamine, myeloperoxidase and other enzymes. On bind-ing IgE, histamine is released from the basophil.

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Macrocytosis (MCV >96fL) is common, and may not always be accompanied by anae-mia (eg in alcohol excess).

Causes of macrocytosis (MCV >96fL)

Megaloblastic: (fig 8.25) a megaloblast is a cell in which nuclear maturation is delayed compared with the cytoplasm. This occurs with B12 (p224) and folate de-ficiency: both are required for DNA synthesis. Another cause is cytotoxic drugs.

Non-megaloblastic: Alcohol excess, reticulocytosis (eg in haemolysis), liver dis-ease, hypothyroidism, pregnancy.

Other haematological disease: Myelodysplasia (fig 8.26), myeloma, myeloprolif-erative disorders, aplastic anaemia.

Tests B12 and folate deficiency result in similar blood film and bone marrow biopsy appearances.

Blood film: Hypersegmented neutrophils (fig 8.25) in B12 and folate deficiency. Target cells if liver disease; see fig 8.14, p229 and fig 8.41, p242.

Other tests: LFT (include ffkGT), TFT, serum B12, and serum folate (or red cell folate—a more reliable indicator of folate status, as serum folate only reflects recent intake). Bone marrow biopsy is indicated if the cause is not revealed by the above tests. It is likely to show one of the following four states:

Megaloblastic marrow.

Normoblastic marrow (eg in liver disease, hypothyroidism).

Abnormal erythropoiesis (eg sideroblastic anaemia, p226, leukaemia, aplasia).

Increased erythropoiesis (eg haemolysis).

Folate Found in green vegetables, nuts, yeast, and liver; it is synthesized by gut bacteria. Body stores can last for 4 months. Maternal folate deficiency causes fe-tal neural tube defects. It is absorbed by duodenum/proximal jejunum.

Causes of deficiency:

Poor diet, eg poverty, alcoholics, elderly.

Increased demand, eg pregnancy or •cell turnover (seen in haemolysis, malig-nancy, inflammatory disease, and renal dialysis).

Malabsorption, eg coeliac disease, tropical sprue.

Alcohol.

Drugs: anti-epileptics (phenytoin, valproate), methotrexate, trimethoprim.

NB: in unwell patients (eg CCF) with megaloblastic anaemia, it may be necessary to treat before serum B12 and folate results are known. Do tests then treat with large doses of hydroxocobalamin, eg 1mg/48h IM—see BNF, with folic acid 5mg/24h PO. Blood transfusions are very rarely needed (see p224).

Fig 8.25 Megaloblastic anaemia: peripheral blood film showing many macrocytes and one hypersegmented neutrophil (normally there should be ≤5 segments).

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Medical Society.

Fig 8.26 Oval macrocytes seen here in myelod-ysplastic syndromes. Note aniso- and poikilocy-tosis with small fragmented cells (schistocytes). NB: B12 and folate deficiencies also cause oval macrocytes, but macrocytes caused by alcohol and liver disease are usually round.

Tangün and Dr Körog˘ lu.

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Vitamin B12 deficiency is common, occurring in up to 15% of older people. B12 helps synthesize thymidine, and hence DNA, so in deficiency RBC production is slow. Untreated, it can lead to megaloblastic anaemia (p222) and irreversible CNS compli-cations. —— Body stores of B12 are sufficient for 4yrs.

Features General: Symptoms of anaemia (p224), 'lemon tinge' to skin due to combination of pallor (anaemia) and mild jaundice (due to haemolysis), glossitis (beefy-red sore tongue; fig 8.27), angular cheilosis (p226).

Neuropsychiatric: Irritability, depression, psychosis, dementia.

Neurological: Paraesthesiae, peripheral neuropathy. Also subacute combined degeneration of the spinal cord, a combination of peripheral sensory neuropa-thy with both upper and lower motor neuron signs due to 'B12. The patient may display the classical triad of: • extensor plantars (UMN) • absent knee jerks (LMN)

—— The neurological signs of B12 deficiency can occur without anaemia.

Pernicious anaemia (PA) This is an autoimmune condition in which atrophic gastritis leads to a lack of IF secretion from the parietal cells of the stomach. Dietary B12 there-fore remains unbound and conseqeuently cannot be absorbed by the terminal ileum. Incidence: 1:1000; φ:ð≈1.6:1; usually >40yrs; higher incidence if blood group A.

Associations: Other autoimmune diseases (p552): thyroid disease (r25%), vitiligo, Addison's disease, hypoparathyroidism. Carcinoma of stomach is r2-fold more common in pernicious anaemia, so have a low threshold for upper GI endoscopy.

Tests: • ‘Hb. • •MCV. • WCC and ‘platelets if severe. • ‘Serum B 4. • Reticulocytes may be ‘ as production impaired. • Hypersegmented neutrophils (p222). • Megaloblasts in the marrow. • Specific tests for PA: 1 Parietal cell antibodies: found in 90% with PA, but also in 2–10% without. 2 IF antibodies: specific for PA, but lower sensitivity.

Practical hints: • Beware of diagnosing PA in those under 40yrs old: look for GI ma-labsorption (small bowel biopsy, p266).

Watch for hypokalaemia due to uptake into new haematopoietic cells.

Transfusion is best avoided, but PA with high-output CCF may require transfusion, after doing tests for FBC, folate, B12, and marrow sampling.

As haematopoiesis accelerates on treatment, additional iron may be needed.

Hb rises r10g/L per week; WCC and platelet count should normalize in 1wk.

Prognosis: Supplementation usually improves peripheral neuropathy within the first 2–6 months, but has little effect on cord signs. Patients do best if treated as soon as possible after the onset of symptoms: don’t delay!

4 Serum B12 levels are normal in many patients with subclinical B12 deficiency. Measuring homocysteine or methylmalonic acid (• if B12 low) may be helpful, but these are non-standard tests.

Fig 8.27 The big, beefy tongue of B12 deficiency glossitis. Other causes of glossitis: iron (or Zn) deficiency, pellagra, contact dermatitis/specific food intolerances, Crohn’s disease, drugs (minocy-cline, clarithromycin, some ACE-i), TB of the tongue. Glossitis may be the presenting feature of coeliac disease or alcoholism.

An approach is first to confirm haemolysis and then find the cause—try to answer these four questions:

Is there increased red cell breakdown?

Anaemia with normal or •MCV.

- Bilirubin: unconjugated, from haem breakdown (pre-hepatic jaundice).
- Urinary urobilinogen (no urinary conjugated bilirubin).
- Serum LDH, as it is released from red cells.

Is there increased red cell production?

- Reticulocytes, causing •MCV (reticulocytes are large immature RBCS) and poly-chromasia.

Is the haemolysis mainly extra- or intravascular?

Extravascular haemolysis may lead to splenic hypertrophy and splenomegaly. Features of intravascular haemolysis are:

- Free plasma haemoglobin: released from RBCS.

Methaemalbuminaemia: some free Hb is broken down in the circulation to pro-duce haem and globin; haem combines with albumin to make methaemalbumin.

‘Plasma haptoglobin: mops up free plasma Hb, then removed by the liver.

Haemoglobinuria: causes red-brown urine, in absence of red blood cells.

Haemosiderinuria: occurs when haptoglobin-binding capacity is exceeded, causing free Hb to be filtered by the renal glomeruli, with absorption of free Hb via the renal tubules and storage in the tubular cells as haemosiderin. This is detected in the urine as sloughed tubular cells by Prussian blue staining r1 week after onset (implying a chronic intravascular haemolysis).

Why is there haemolysis? Causes are on p228.

History Family history, race, jaundice, dark urine, drugs, previous anaemia, travel.

Examination Jaundice, hepatosplenomegaly, gallstones (pigmented, due to •bili-rubin from haemolysis), leg ulcers (due to poor blood flow).

Tests FBC, reticulocytes, bilirubin, LDH, haptoglobin, urinary urobilinogen. Thick and thin films for malaria screen if history of travel. The blood film may show poly-chromasia and macrocytosis due to reticulocytes, or point to the diagnosis:

Hypochromic microcytic anaemia (thalassaemia).

Sickle cells (sickle-cell anaemia).

Schistocytes (fig 8.20, p229; microangiopathic haemolytic anaemia).

Abnormal cells in haematological malignancy.

Spherocytes (hereditary spherocytosis or autoimmune haemolytic anaemia).

Elliptocytes (fig 8.26, p229; hereditary elliptocytosis).

Heinz bodies, 'bite' cells (fig 8.22, p229; glucose-6-phosphate dehydrogenase deficiency).

Further tests (if the cause is still not obvious)

Osmotic fragility testing will confirm the presence of membrane abnormalities which have been identified on the film.

Hb electrophoresis will detect haemoglobinopathies.

The direct antiglobulin (Coombs) test (DAT, fig 8.28) identifies red cells coated with antibody or complement, the presence of which indicates an immune cause.

Enzyme assays are reserved for when other causes have been excluded.

Fig 8.28 The direct Coombs test detects antibodies on RBCS. The indirect Coombs test is used in pre-natal testing and before blood transfusion. It detects antibodies against RBCs that are free in serum: serum is incubated with RBCS of known antigenicity. If agglutination occurs, the indirect Coombs test is positive.

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Acquired

Immune-mediated/direct antiglobulin test +ve: (Coombs test, p227.)

Drug-induced: causing formation of RBC autoantibodies from binding to RBC membranes (eg penicillin) or production of immune complexes (eg quinine).

Paroxysmal cold haemoglobinuria: seen with viruses/syphilis. It is caused by Donath–Landsteiner antibodies sticking to RBCS in the cold, causing self-limit-ing complement-mediated haemolysis on rewarming.

Isoimmune: acute transfusion reaction (p249); haemolysis of the newborn.

Direct antiglobulin/Coombs –ve AIHA: (2% of all AIHA.) Autoimmune hepatitis; hepatitis B & C; post flu and other vaccinations; drugs (piperacillin, rituximab).

Microangiopathic haemolytic anaemia (MAHA): Mechanical damage to RBCS in circula-tion, causing intravascular haemolysis and schistocytes (figs 8.20, 8.21). Causes include haemolytic–uraemic syndrome (HUS), TTP (p215), DIC, pre-eclampsia, and eclampsia. Prosthetic heart valves can also cause mechanical damage.

Infection: Malaria (p416): RBC lysis and 'blackwater fever' (haemoglobinuria). ——All infections can exacerbate haemolysis.

Paroxysmal nocturnal haemoglobinuria: Rare acquired stem cell disorder, with haemolysis (esp. at night→haemoglobinuria, fig 15.8, p705), marrow failure + thrombophilia. Tests: urinary haemosiderin +ve; if suspect in Coombs -ve intravascular haemolysis, seek confirmation by flow cytometry. 9: anticoagulation; monoclonal anticomplement antibodies (eg eculizumab); stem cell transplantation.

Hereditary

Enzyme defects:

Pyruvate kinase deficiency (AUTOSOMAL RECESSIVE): ‘ATP production causes ‘RBC survival. Homozygotes have neonatal jaundice; later, haemolysis with splenomeg-aly ! jaundice. Tests: enzyme assay. 9: often not needed; splenectomy may help.

Membrane defects: All are Coombs –ve; all need folate; splenectomy helps some.

Hereditary spherocytosis (AUTOSOMAL DOMINANT): prevalence: 1:2000. Less de-formable spherical RBCS, so trapped in spleen extravascular haemolysis. Signs: Splenomegaly, jaundice. Tests: Mild if Hb >110g/L and reticulocytes <6%; film: fig

8.25. • Bilirubin (gallstones).

Hereditary elliptocytosis (AUTOSOMAL DOMINANT): film: fig 8.26. Mostly asympto-matic (somewhat protects from malaria). 10% display a more severe phenotype (! death in utero).

Hereditary ovalocytosis and stomatocytosis are rarer. Refer to a haematologist.

Haemoglobinopathy: • Sickle-cell disease (p240). • Thalassaemia (p242).

Fig 8.29 Autoimmune haemolytic anaemia: antibody-coated red cells undergoing phagocy-tosis by monocytes.

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TTP (p315), slicing up RBCS (microangiopathy).

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Fig 8.22 A bite-cell in G6PD, after removal of a Heinz body by the spleen; these are formed from denatured Hb during oxidative crises.

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Fig 8.22 Blister-cells (arrows) in G6PD, following removal of Heinz bodies. Also contracted red cells (arrowheads).

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Fig 8.25 Hereditary spherocytosis. Osmotic fragility tests: RBCS show •fragility in hypotonic solutions.

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Pathogenesis HbS polymerizes when deoxygenated, causing RBCS to deform, pro-ducing sickle cells, which are fragile and haemolyse, and also block small vessels.

Prevalence 1:700 people of African descent.

Signs/symptoms Chronic haemolysis is usually well tolerated (except in crises; see

BOX ‘Managing sickle-cell crises’).

>12h prompt cavernosus–spongiosum shunting can prevent later impotence).

Aplastic crisis: This is due to parvovirus B19, with sudden reduction in marrow production, especially RBCs. Usually self-limiting <2wks; transfusion may be needed. Sequestration crisis: Mainly affects children as in adults the spleen becomes atrophic. There is pooling of blood in the spleen ! liver, with organomegaly, severe anaemia, and shock. Urgent transfusion is needed.

Complications • Splenic infarction occurs before 2yrs old, due to micro-vascular occlusion, leading to •susceptibility to infection (40% of childhood sickle deaths are caused this way). • Poor growth. •Chronic renal failure. • Gall-stones. • Retinal disease. • Iron overload (see BOX 'A 7-year-old …'). •Lung damage: hypoxia↬fibrosis↬pulmonary hypertension.

Management of chronic disease —— Get help from a haematologist.

Hydroxycarbamide if frequent crises (•production of fetal haemoglobin, HbF). Dose example: 20mg/kg/d if eGFR >60mL/min.

Splenic infarction leads to hyposplenism and immunocompromise. Prophylaxis, in terms of antibiotics and immunization, should be given (p272).

Febrile children risk septicaemia: repeated admission may be avoided by early rescue out-patient antibiotics, eg ceftriaxone (eg 2 doses, 50mg/kg IV on day 0 and 1). Consider admission if Hb <50g/L, WCC <5 or >20 ≈ 109/L, T° >40℃, severe pain, dehydration, lung infiltration. Seek expert advice.

Bone marrow transplant can be curative but remains controversial.

Prevention Genetic counselling; prenatal tests (OHCS pp154–5). Parental education can help prevent 90% of deaths from sequestration crises.

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This was necessary until a once-daily oral iron chelator came along: deferasirox.

The thalassaemias are genetic diseases of unbalanced Hb synthesis, with underpro-duction (or no production) of one globin chain (see table 8.2 and BOX). Unmatched globins precipitate, damaging RBC membranes, causing their haemolysis while still in the marrow. They are common in areas from the Mediterranean to the Far East.

The β thalassaemias Usually caused by point mutations in β-globin genes on chromosome 11, leading to 'β chain production (β+) or its absence (β0). Various combinations of mutations are possible (eg β0/β0, β+/β+, or β+/β0).

Tests: FBC, MCV, film, iron, HbA2, HbF, Hb electrophoresis. MRI where myocardial siderosis suspected (from iron overload).

p thalassaemia minor or trait (eg p/p+; heterozygous state): this is a carrier state, and is usually asymptomatic. Mild, well-tolerated anaemia (Hb >90g/L) which may worsen in pregnancy. MCV <75fL, HbA2 >2.5%, slight •HbF. Often con-fused with iron-deficiency anaemia.

Treatment: ►Promote fitness; healthy diet. Folate supplements help.

Splenectomy if hypersplenism persists with increasing transfusion requirements (p272)—this is best avoided until >5yrs old due to risk of infections.

Hormonal replacement or treatment for endocrine complications, eg diabetes mellitus, hypothyroidism. Growth hormone treatment has had variable success.

A histocompatible marrow transplant can offer the chance of a cure.

Prevention: Approaches include genetic counselling or antenatal diagnosis using fetal blood or DNA, then 'therapeutic' abortion.

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- Vascular defects Congenital: Osler–Weber–Rendu syndrome (p708), connective tissue disease (eg Ehlers–Danlos syndrome, OHCS p642, pseudoxanthoma elasticum). Acquired: Senile purpura, infection (eg meningococcal, measles, dengue fever), steroids, scurvy (perifollicular haemorrhages), Henoch–Schönlein purpura (p702).

3 Coagulation disorders Congenital: Haemophilia, von Willebrand's disease (p712). Acquired: Anticoagulants, liver disease, DIC (p252), vitamin K deficiency.

Haemophilia B (Christmas disease) Factor IX deficiency (inherited, X-linked re-cessive); behaves clinically like haemophilia A. Treat with recombinant factor IX.

Acquired haemophilia is a bleeding diathesis causing big mucosal bleeds in males and females caused by suddenly appearing autoantibodies that interfere with factor VIII.Tests: •APPT; •VIII autoantibody; factor VIII activity <50%. 9: Steroids.

Liver disease Produces a complicated bleeding disorder with 'synthesis of clot-ting factors, 'absorption of vitamin K, and abnormalities of platelet function.

Malabsorption Leads to less uptake of vitamin K (needed for synthesis of factors II, VII, IX, and X). Treat with IV vitamin K (10mg). In acute haemorrhage, use hu-man prothrombin complex or FFP.

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There are three sets of questions to be answered:

1 Is there an emergency needing immediate resuscitation or senior help?

Is the patient about to exsanguinate (dizzy on sitting up, shock, coma)?

Is there hypovolaemia (postural hypotension, oliguria)?

Is there CNS bleeding (meningism, CNS and retinal signs)?

Is there an underlying condition which escalates apparently minor bleeding into an evolving catastrophe? For example:

Bleeding in pregnancy or the puerperium.

GI bleeding in a jaundiced patient (ie coagulation factors already depleted).

Bleeding in someone who is already anaemic (esp if other comorbidities).

- Why is the patient bleeding? Is bleeding normal, given the circumstances (eg surgery, trauma, parturition), or does the patient have a bleeding disorder (BOX 'Is this pre-op patient at risk of excessive bleeding?')?

Is there a secondary cause, eg drugs (warfarin), alcohol, liver disease, sepsis?

Is there unexplained bleeding, bruising, or purpura?

Past or family history of excess bleeding, eg during trauma, dentistry, surgery?

Is the pattern of bleeding indicative of vascular, platelet, or coagulation prob-lems (p244)? Are old venepuncture or cannula sites bleeding (DIC, p252)? Look for associated conditions (eg with DIC).

Is a clotting screen abnormal (table 8.2)? Check FBC, platelets, PT, APTT, and thrombin time. Consider D-dimers, bleeding time, and a factor VIII assay. ►If both PT and APTT are very raised, with low platelets and •D-dimers, consider DIC (p252).

3 In cases of bleeding disorders, what is the mechanism? Investigate with

FBC, film, and coagulation screen (citrate tube; false results if under-filled):

Prothrombin time (PT): Thromboplastin is added to test the extrinsic system. PT is expressed as a ratio compared to control (international normalized ratio (INR), normal range = 0.9–1.2). It tests for abnormalities in factors I, II, V, VII, X. Prolonged by: warfarin, vitamin K deficiency, liver disease, DIC.

Activated partial thromboplastin time (APTT): Kaolin is added to test the intrinsic system. Tests for abnormalities in factor I, II, V, VIII, IX, X, XI, XII. Normal range 25–45s. Prolonged by: heparin treatment, haemophilia, DIC, liver disease.

Thrombin time: Thrombin is added to plasma to convert fibrinogen to fibrin. Normal range: 10–15s. Prolonged by: heparin treatment, DIC, dysfibrinogenaemia.

D-dimers are a fibrin degradation product, released from cross-linked fibrin during fibrinolysis (p245). This occurs during DIC, or in the presence of venous thromboembolism—deep vein thrombosis (DVT) or pulmonary embolism (PE). D-dimers may also be raised in inflammation, eg with infection or malignancy.

Table 8.2 Clotting screen abnormalities in coagulopathies

| A 7-year-old tells us what it's like to have sickle-cell disease 'I have been hospitalized over 50 times for complications from this disease. To keep it controlled I started having monthly transfusions. After repeated transfusions my body began to get too much iron so I had to start getting infusions. I was taking the medication desferal 5 which my mummy had to insert a needle in my belly hooked up to a pump which I had to carry on my back in my neat Spiderman backpack. I was hooked up to the machine for 10 hours a day 5 days a week but it was okay I still got to play‼! I suffered from pain crisis which makes my legs and back hurt like someone is hitting me with a hammer. You may notice that I may move slow or look tired when it is time for my blood transfusion. That is because the transfusions are like a heartbeat for my body, without it I can't survive. When I'm in pain the only thing that helps is morphine… I tell my mummy when she's crying I WILL BE OK‼' | | — |

Blood should only be given if strictly necessary and there is no alternative. Out-comes may be worse after an inappropriate transfusion.

Know and use local procedures to ensure that the right blood gets to the right patient at the right time.

Take blood for crossmatching from only one patient at a time. Label immedi-ately. This minimizes risk of wrong labelling of samples.

When giving blood, monitor TPR and BP every ½h.

Use a dedicated line where practicable (or dedicated lumen of multilumen line).

Group-and-save (G&S) requests Know your local guidelines for elective surgery. Having crossmatched blood to hand may not be needed if a blood sample is al-ready in the lab, with group determined, without any atypical antibodies (ie G&S).

Complications of transfusion ►Management of acute reactions:[2] see BOX 'Trans-fusion reactions' and table 8.4.

Early (within 24h): Acute haemolytic reactions (eg ABO or Rh incompatibility); anaphylaxis; bacterial contamination; febrile reactions (eg from HLA antibodies); allergic reactions (itch, urticaria, mild fever); fluid overload; transfusion-related acute lung injury (TRALI, ie ARDS due to antileucocyte antibodies in donor plasma).

Delayed (after 24h): Infections (eg viruses: hepatitis B/C, HIV; bacteria; protozoa; prions); iron overload (treatment, p242); GVHD; post-transfusion purpura—poten-tially lethal fall in platelet count 5–7d post-transfusion requiring specialist treat-ment with IV immunoglobulin and platelet transfusions.

Transfusing patients with heart failure If Hb =50g/L with heart failure, trans-fusion with packed red cells is vital to restore Hb to a safe level, eg 60–80g/L, but must be done with care. Give each unit over 4h with furosemide (eg 40mg slow IV/PO; don't mix with blood) with alternate units. Check for •JVP and basal lung crackles; consider CVP line.

Main indications

Therapeutic: Venous thromboembolic disease: DVT and PE.

Prophylactic: Prevention of DVT/PE in high-risk patients (p275), eg post-op. Prevention of stroke, eg in chronic AF or prosthetic heart valves.

- Unfractionated heparin (UFH): IV or SC. Binds antithrombin (an endogenous in-hibitor of coagulation), increasing its ability to inhibit thrombin, factor Xa, and IXa. Rapid onset and has a short t½. Monitor and adjust dose with APTT (p246).

SE for both: •Bleeding (eg at operative site, gastrointestinal, intracranial), heparin-induced thrombocytopenia (HIT), osteoporosis with long-term use. HIT and osteo-porosis are less common with LMWH than UFH. Beware hyperkalaemia.

Table 8.7 Principal subtypes of leukaemia

| Type | Peptide chains | % in adult blood | % in fetal blood | | — | — | — | — | | HbA | a2 β2 | 97 | 10–50 | | HbA2 | a2 62 | 2.5 | Trace | | HbF | a2 ffk2 | 0.5 | 50–90 |

Neutropenic regimen (For when neutrophil count ≤0.5 ≈ 109/L.)3 ►Close liaison with a microbiologist and haematologist is vital. Abide by infection control proce-dures! Use a risk-assessment tool (eg MASCC, see BOX).

Look for infection (mouth, axillae, perineum, IVI site). Take swabs.

Wash perineum after defecation. Swab moist skin with chlorhexidine. Avoid unnecessary rectal examinations. Oral hygiene (eg hydrogen peroxide mouth washes/2h) and Candida prophylaxis are important (p246).

Hyperviscosity: (p272). If WCC is >100 ≈ 109/L WBC thrombi may form in brain, lung, and heart (leukostasis). Avoid transfusing before lowering WCC, eg with hydroxycarbamide or leukapheresis, as viscosity rises (•risk of leukostasis).

Morphological: The FAB system (French, American, British) divides ALL into three types (L1, L2, L2) by microscopic appearance. Provides limited information (figs 8.45–8.48).

Cytogenetic: Chromosomal analysis. Abnormalities are detected in up to 85%, which are often translocations.6 Useful for predicting prognosis, eg poor with Philadelphia chromosome (p258), and for detecting disease recurrence.

Infiltration: hepato- and splenomegaly, lymphadenopathy—superficial or medi-astinal, orchidomegaly, CNS involvement—eg cranial nerve palsies, meningism.

CXR and CT scan to look for mediastinal and abdominal lymphadenopathy.

Support: Blood/platelet transfusion, IV fluids, allopurinol (prevents tumour lysis syndrome). Insert a subcutaneous port system/Hickman line for IV access.

Remission induction: eg vincristine, prednisolone, L-asparaginase + daunorubicin.

Maintenance: prolonged chemotherapy, eg mercaptopurine (daily), methotrex-ate (weekly), and vincristine + prednisolone (monthly) for 2yrs. Relapse is common in blood, CNS, or testis (examine these sites at follow-up). More details: OHCS p194.

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Neoplastic proliferation of blast cells derived from marrow myeloid elements. It progresses rapidly (death in r2 months if untreated; r20% 2yr survival after R).

AML with recurrent genetic abnormalities.

AML, other (not fitting above-listed; further subclassified as M0–M7 by maturation).

Autologous mobilized peripheral blood stem cell transplantation may offer faster haemopoietic recovery and less morbidity.

Christine Lawrence.

Christine Lawrence.

Symptoms Mostly chronic and insidious: ‘weight, tiredness, fever, sweats. There may be features of gout (due to purine breakdown), bleeding (platelet dysfunction), and abdominal discomfort (splenic enlargement). r20% are detected by chance.

Natural history Variable, median survival 5–6yrs. There are three phases: Chronic, lasting months or years of few, if any, symptoms. • Accelerated phase, with increas-ing symptoms, spleen size, and difficulty in controlling counts. • Blast transforma-tion, with features of acute leukaemia ! death. Treatment See BOX.

Symptoms Often none, presenting as a surprise finding on a routine FBC. Patients may be anaemic or infection-prone, or have ‘weight, sweats, anorexia if severe.

Complications 1 Autoimmune haemolysis. 2 •Infection due to hypogammaglobuli-naemia (=’IgG), bacterial, viral especially herpes zoster. 3 Marrow failure.

Signs Lymphadenopathy. Also, cachexia, anaemia, spleno- or hepatomegaly.

! marrow biopsy if B symptoms, or stage III–IV disease.

Involvement of nodes on both sides of the diaphragm.

Emergency presentations Infection; SVC obstruction—•JVP, sensation of fullness in the head, dyspnoea, blackouts, facial oedema (seek expert help; see p528).

Table 8.9 HL subtypes

| Fibrinolysis The fibrinolytic system works by generating plasmin, which causes fibrin dissolution. The process starts with the release of tissue plasminogen activa-tor (t-PA) from endothelial cells, a process stimulated by fibrin formation. t-PA converts inactive plasminogen to plasmin which can then cleave fibrin, as well as several other factors. t-PA and plasminogen both bind fibrin thus localizing fibrinolysis to the area of the clot. Fibrinolytic agents activate this system and can be utilized in order to break down pathological thrombi, eg in: acute MI, acute ischaemic stroke, DVT, PE, and central retinal venous or arterial thrombosis. In all cases the risk of adverse effects of thrombolysis (eg haemorrhage) must be outweighed by the poten-tial benefits. Streptokinase, a streptococcal exotoxin that binds and activates plasminogen, was the first licensed agent but risks anaphylaxis on repeat dosing. Alteplase is recombinant t-PA. Newer agents include tenecteplase and reteplase. | | — |

Diagnosis/management is multidisciplinary, synthesizing details from clinical evaluation, histology, immunology, molecular genetics, and imaging. Generally:

>50% for low-grade lymphomas, but the picture is very variable.

Marrow support Red cells survive for r120d, platelets for r8d, and neutrophils for

Platelets: Traumatic bleeds, purpura, and easy bruising occur if platelets

Neutrophils: Use neutropenic regimen if the count <0.5≈109/L (p252).

Table 8.10 Classification of myeloproliferative disorders

| Immune thrombocytopenia (ITP) ITP is caused by antiplatelet autoantibodies. It is acute (usually in children, 2wks after infection with sudden self-limiting purpura: OHCS p197) or chronic (seen mainly in women). Chronic ITP runs a fluctuating course of bleeding, purpura (esp. dependent pressure areas), epistaxis, and menorrhagia. There is no spleno-megaly. Tests: •Megakaryocytes in marrow, antiplatelet autoantibodies often present. 9: None if mild. If symptomatic or platelets <20 ≈ 109/L, prednisolone 1mg/kg/d, and reduce after remission; aim to keep platelets >20 ≈ 109/L—takes a few days to work. Platelet transfusions are not used (except during splenecto-my or life-threatening haemorrhage) as these are quickly destroyed by the au-toantibodies. IV immunoglobulin may temporarily raise the platelet count, eg for surgery, pregnancy. If relapse, choices include splenectomy or B-cell depletion with rituximab.

Eltrombopag (an oral thrombopoietin-receptor agonist) and romiplostim (an injectable thrombopoietin analogue) are alternative options for those with refractory disease. | | — |

Treatment: Aim to keep HCT <0.45 to ‘risk of thrombosis. In younger patients at low risk, this is done by venesection. If higher risk (age >60yrs, previous thrombo-sis), hydroxycarbamide (=hydroxyurea) is used. a-interferon is preferred in women of childbearing age. Aspirin 75mg daily is also given.

Fig 8.69 Marrow trephine in myelofibrosis: the streaming effect is caused by intense fi-brosis. Other causes of marrow fibrosis: any myeloproliferative disorder, lymphoma, second-ary carcinoma, TB, leukaemia, and irradiation.

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Incidence 5/100 000. Peak age: 70yrs. ♂:♀ ≈ 1:1. Afro-Caribbeans:Caucasians ≈ 2:1.

Anaemia, neutropenia, or thrombocytopenia may result from marrow infiltra-tion by plasma cells, leading to symptoms of anaemia, infection, and bleeding.

Prognosis Worse if: >2 osteolytic lesions, β2-microglobulin >5.5mg/L, Hb <11g/L; albumin <20g/L. Risk stratification increasingly based upon detection of specific cytogenetic abnormalities associated with high risk of progression.

Causes of death: infection, renal failure.

Fig 8.72 An IGG kappa paraprotein monoclonal band (immunofixation electrophoresis; a con-trol sample has run on the left).

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Paraproteinaemia denotes the presence in the circulation of immunoglobulins pro-duced by a single clone of plasma cells. The paraprotein is recognized as a monoclo-nal band (M band) on serum electrophoresis.9 There are six major categories:

Primary amyloidosis See following topic.

Paraproteinaemia in lymphoma or leukaemia Eg seen in 5% of CLL.

AL amyloid (primary amyloidosis) Proliferation of plasma cell clone ➪ Amyloi-dogenic monoclonal immunoglobulins ➪ Fibrillar light chain protein deposition ➪ Organ failure ➪ Death. Associations: myeloma (15%); Waldenström’s, lymphoma. Organs involved:

Nerves: peripheral and autonomic neuropathy; carpal tunnel syndrome.

9: optimize nutrition; PO melphalan + prednisolone extends survival. High-dose IV

Familial amyloidosis (Autosomal dominant, eg from mutations in transthyretin, a transport protein produced by the liver.) Usually causes a sensory or autonomic neuropathy ! renal or cardiac involvement. Liver transplant can cure.

Electrophoresis and plasmapheresis look as though they should share endings, but they do not: Greek

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Causes of a raised ESR Any inflammation (eg infection, rheumatoid arthritis, ma-lignancy, myocardial infarction), anaemia, and macrocytosis.

Some conditions lower the ESR, eg poly-cythaemia (due to •red cell concentration), microcytosis, and sickle-cell anaemia. Even a slightly raised ESR in these patients should

Symptoms Lethargy; confusion; ‘cognition; CNS disturbance; chest pain; abdominal pain (and sometimes spontaneous GI or GU bleeding); faints; visual disturbance (eg ‘vision, amaurosis fugax, retinopathy—eg engorged retinal veins, haemorrhages, exudates; and a blurred disc as seen in fig 8.76). The visual symptoms are like ‘looking through a watery car windscreen’.

The spleen and splenectomy

When is a mass in the left upper quadrant a spleen: (Main differential: en-larged left kidney.) The spleen: • Is dull to percussion. • Enlarges towards the RIF. • Moves down on inspiration. • You may feel a medial notch. • 'You can't get above it' (ie the top margin disappears under the ribs).

Splenectomy: Main indications: splenic trauma, hypersplenism, autoimmune haemolysis: in ITP (p245), warm autoimmune haemolytic anaemia (p228), or congenital haemolytic anaemias. Mobilize early post-splenectomy as transient

Immunizations:

Meningococcal vaccination course, including Men B, Men C, and Men ACWY.

Pendants, bracelets, or patient-held cards to alert medical staff.

The advice given here also applies to hyposplenic patients, eg in sickle-cell anae-mia or coeliac disease.

Antithrombin deficiency: Antithrombin is a co-factor of heparin, and inhibits thrombin. Less common, affects 1:500. Heterozygotes' thrombotic risk is greater than protein C or S deficiency by r4-fold. Homozygosity is incompatible with life.

! APC resistance test, lupus anticoagulant and anticardiolipin antibodies, and assays for antithrombin and proteins C & S deficiency (! DNA analysis by PCR for the factor V Leiden mutation if APC resistance test is +ve, and for prothrombin gene mutation).

unprovoked VTE (ie at <40yrs with no risk factors) • VTE with oral contraceptives/ pregnancy • unexplained recurrent VTE • unusual site, eg mesenteric or portal vein thrombosis • recurrent fetal loss (?2) • neonatal thrombosis. other risk factors, consider lifelong anticoagulation. Recurrence whilst on treat-ment should be treated by increasing treatment intensity (eg ♁target INR to 2–4). In antithrombin deficiency, high doses of heparin may be needed; liaise with a hae-

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Long-term steroids (>2 weeks, or repeated courses) must not be stopped sud-dently. 2Risk of Addisonian crisis due to adrenal insufficiency, see p826. Plan a gradual taper over weeks (with the advice of an endocrinologist if needed).

Interactions: efficacy is reduced by anti-epileptics (see later in topic) and ri-fampicin.

Do not stop steroids suddenly (p826).

Avoid over-the-counter drugs, eg NSAIDS: aspirin and ibuprofen (•risk of DU).

Ciclosporin, tacrolimus Calcineurin inhibitors with important roles in reducing rejection in organ and marrow transplant. The main SE is dose-related nephrotox-icity: check blood levels.

Interactions are legion: potentiated by: ketoconazole, diltiazem, verapamil, the Pill, erythromycin, grapefruit juice. Efficacy is reduced by: barbiturates, carbamazepine, phenytoin, rifampicin. Avoid concurrent nephrotoxics: eg gen-tamicin. Concurrent NSAIDs augment hepatotoxicity—monitor LFT.

Table 8.11 Side-effects of steroid use

| Towards a better assay for clotting function Bleeding time, a barbaric and unreliable test (the clue is in the name), is no longer used. Amongst the range of techniques to replicate the clotting process in vitro is thromboelastography (TEG). TEG permits rapid and more precise assays of clotting function under massive transfusion situations (eg major surgery, especially trauma). In particular, advances in this field have been driven by recent military usage. | | — |

Infectious Diseases

We thank Dr Chris Conlon, our Specialist Reader for this chapter.

What is life?

Fig 9.2 Tree of life based on subunit RNA sequences sampled from r3000 species out of the 1.7 million species that are formally named. The image on the right is a close up of the 'animal' seg-ment of the diagram (upper left quadrant) showing 'You are here'.

Hillis, Derrick Zwickl and Robin Gutell, University of Texas.

http://www.zo.utexas.edu/faculty/antisense/downloadfilesToL.html

Because micro-organisms kill our friends we think of them as bad:

'I have no philosophy, nor piety, no art of reflection, no theory of compensa-tion to meet things so hideous, so cruel, and so mad, they are just unspeak-ably horrible and irremediable to me and I stare at them with angry and almost blighted eyes.'

Henry James, 1915, describing the death of Rupert Brooke from septicaemia.

Table 9.1 Infectious disease by pathogen (illustrative, not exhaustive)

| Bacteria | Viruses | | — | — | | Gram positive | RNA viruses | | Staphylococci: Staph. aureus (coagulase +ve) Staph. epidermidis (coagulase –ve) | Picornavirus ('tiny RNA'): Rhinovirus Poliovirus | | Streptococci: a-haemolytic, eg Strep. pneumoniae β-haemolytic, eg Strep. pyogenes | Calicivirus ('cup'), eg norwalk | | Streptococci: a-haemolytic, eg Strep. pneumoniae β-haemolytic, eg Strep. pyogenes | Flavivirus ('yellow'): Dengue • Zika Yellow fever Coronavirus ('crown'): URTI | | Enterococci Clostridium species: C. botulinum (botulism) C. perfringens(gas gangrene) C. tetani (tetanus) C. difficile (diarrhoea) | Flavivirus ('yellow'): Dengue • Zika Yellow fever Coronavirus ('crown'): URTI | | Enterococci Clostridium species: C. botulinum (botulism) C. perfringens(gas gangrene) C. tetani (tetanus) C. difficile (diarrhoea) | Rhabdovirus ('rod'), eg rabies | | Enterococci Clostridium species: C. botulinum (botulism) C. perfringens(gas gangrene) C. tetani (tetanus) C. difficile (diarrhoea) | Filovirus ('thread'), eg Ebola/Marburg | | Enterococci Clostridium species: C. botulinum (botulism) C. perfringens(gas gangrene) C. tetani (tetanus) C. difficile (diarrhoea) | Paramyxovirus ('near mucus'), eg mumps | | Gram negative | DNA viruses | | Neisseria: N. meningitidis (meningitis) N. gonorrheae (gonorrhea) | Hepadnavirus ('liver DNA'): hepatitis | | Neisseria: N. meningitidis (meningitis) N. gonorrheae (gonorrhea) | Parvovirus ('small'): gastroenteritis | | Neisseria: N. meningitidis (meningitis) N. gonorrheae (gonorrhea) | Herpesvirus ('creeping'): HSV VZV CMV EBV | | Helicobacter pylori | Herpesvirus ('creeping'): HSV VZV CMV EBV | | Escherichia coli | Herpesvirus ('creeping'): HSV VZV CMV EBV | | Shigella species | Herpesvirus ('creeping'): HSV VZV CMV EBV | | Salmonella species | Herpesvirus ('creeping'): HSV VZV CMV EBV | | Campylobacter jejuni | Fungi | | Klebsiella pneumoniae | Candida | | Pseudomonas aeruginosa Haemophilus influenzae | Pneumocystis jirovecii Cryptococcus | | Bordetella pertussis (whooping cough) | Parasites | | Vibrio cholerae (cholera) | Protozoa | | Yersinia pestis (plague) Mycobacteria | Entamoeba histolytica Giardia lamblia | | M. tuberculosis M. leprae | Cryptosporidium species Toxoplasma gondii | | Intracellular bacteria Chlamydia | Plasmodium species (malaria) Leishmania species (leishmaniasis) | | Rickettsia (rickettsial disease) Coxiella burnetii | Trypanosoma species (trypanosomiasis) Nematodes | | Spirochaetes | Soil-transmitted helminths | | Borrelia burgdorferi (Lyme disease) | Filarial disease | | Treponema (syphilis, yaws) | Trematodes | | Leptospira (Weil's disease) | Schistosoma (schistosomiasis), flukes | | | Cestodes | | | Hydatid disease, tapeworm |

Table 9.2 Infectious disease by organ system

| System | Infection | Page | | — | — | — | | Respiratory | Pneumonia | pp166–70 | | Respiratory | Empyema—infected pleural effusion | p170 | | Respiratory | Fungal infections of the lung | p177 | | GI GU and gynaecology | Peptic ulcer disease | p252 | | GI GU and gynaecology | Gastroenteritis | pp428–22 | | GI GU and gynaecology | Colitis, proctitis, diverticulitis, appendicitis | Colitis, proctitis, diverticulitis, appendicitis | | GI GU and gynaecology | Viral hepatitis | p278 | | GI GU and gynaecology | Tropical liver disease | pp424–5 | | GI GU and gynaecology | Cholecystitis,

cholangitis, gallbladder empyema | p634 | | GI GU and gynaecology | Peritonitis Lower urinary tract infection, cystitis, pyelonephritis | p606 pp296–7 | | GI GU and gynaecology | Cervicitis, vulvovaginitis | pp412–2 | | GI GU and gynaecology | Genital ulceration | pp412–2 | | GI GU and gynaecology | Genital warts | p406 | | GI GU and gynaecology | Pelvic inflammatory disease, endometritis | OHCS p286 | | Cardiovascular | Infective endocarditis | pp150–1 | | Cardiovascular | Myocarditis | pp152–4 | | Cardiovascular | Pericarditis | p154 | | Nervous system | Meningitis, encephalitis, subdural empyema | pp822–4 | | Nervous system | Infective neuropathy | pp504–5 | | Skin and soft tissue | Skin ulcers, gangrene | pp660–1 | | Skin and soft tissue | Tropical skin disease | pp422–2 | | Skin and soft tissue | Surgical wound infection | p571, p576 | | Bone and joint | Osteomyelitis | OHCS p696 | | Bone and joint | Septic arthritis | p544 | | ENT | Pharyngitis, laryngitis, otitis media | OHCS p564 | | Eye | Tropical eye disease | pp428–9 |

Acute encephalitis

Acute infectious hepatitis

Acute meningitis

Acute poliomyelitis

Anthrax

Botulism

Brucellosis

Cholera

Diphtheria

Enteric fever

Food poisoning

HUS

Infectious dysentery

Invasive group A strep

Legionnaire's disease

Leprosy

Malaria

Measles

Meningococcal sepsis

Mumps

Plague

Rabies

Rubella

SARS

Scarlet fever

Small pox

Tetanus

Tuberculosis

Typhus

Viral haemorrhagic fever

Whooping cough

Yellow fever.

Infectious disease resources

The Hillis plot (fig 9.2, p279) tells us that ID chapters will always fail to be exhaustive. We therefore direct you to the following excellent resources:

Public Health England, https://www.gov.uk/topic/health-protection/infectious-diseases

World Health Organization (WHO), http://www.who.int/topics/en/

Centers for Disease Control and Prevention, http://www.cdc.gov

Humans and bacteria are symbiotes, with each of us host to ten times as many bacterial cells as our own human cells. Our gut, skin, and mucosal linings are covered with bacteria. We rely on this for nutrition, functioning vitamin K, anti-inflammatory effects, and immune system regulation.

See ‘Sepsis’, p792.

Antibiotics: action and resistance

utilizing a target unique to bacteria, eg cell wall

selectively targeting bacterial-specific components, eg enzymes, ribosomes

preventing transport of the drug into human cells, eg metronidazole can only be transported into anaerobic bacteria.

The mechanism of action of different classes of antibiotic is shown in fig 9.4.

Folate synthesis

Sulfamethoxazole

Nucleic acid synthesis

Cell wall synthesis

–lactams

Penicillins Cephalosporins

Trimethoprim DNA synthesis Fluoroquinolones Metronidazole

Carbapenems

PABA

THF A

DHF A

RNA polymerase

Rifamycins

Glycopeptides Vancomycin Teicoplanin

Polymyxins

50S

30S

50S subunit Macrolides Clindamycin Linezolid Chloramphenicol

Colistin 30S subunit

Aminoglycosides

Tetracyclines

Fusidic acid

Protein synthesis

Antibiotic resistance can be:

Intrinsic: due to inherent structural or functional characteristics, eg vancomycin cannot cross the outer membrane of Gram-negative organisms.

Acquired: bacteria have been evolving to resist antibacterial agents for billions of years through mutation and/or the transfer of resistance properties. This evo-lutionary phenomenon is accelerated by selection pressure from antibiotic use (including agriculture, aquaculture, and horticulture) which provides a competi-tive advantage for mutated, resistant strains.

Resistance has emerged for all known antibiotics causing morbidity, mortality, and a huge cost burden worldwide.[1] Misadventure is evident. Quinolones are syn-thetic, resistance cannot be acquired in nature, and yet it is epidemic.

A guide to antibiotic prescribing

2 Give antibiotics immediately in patients with a systemic inflammatory response to infection. See 'Sepsis', p792.

Start smart:

Do not prescribe[2] antibiotics in the absence of clinical evidence of bacterial infec-tion, or for a self-limiting condition.

Take time to discuss:

why an antibiotic is not the best option

alternative options, eg symptomatic treatment, delayed prescribing

the views and expectations of the patient

safety-netting advice: what the patient should do if their condition deteriorates.

Take microbiological samples before prescribing,1 especially for:

hospital in-patients: review your prescription as soon as MC&S result is available

recurrent or persistent infection

non-severe infection: consider if your prescription can wait for MC&S results.

Follow local guidelines first: best practice is informed by local epidemiology and sensitivities.

Consider benefit and harm for each individual patient:

Allergies: clarify the patient's reaction—the true incidence of penicillin allergy in patients who report that they are allergic is <10%. In those with a confirmed penicillin allergy, cross-reactivity with 2rd-generation cephalosporins and car-bapenems is possible but rare (<1%).

Dose adjust for renal function and weight: use ideal body weight in extremes of

BMI (or ideal weight plus a % of excess weight—see local guidelines).

Check for medication interactions.

In pregnancy and lactation, see p17.

Prescribe the shortest effective course. Most antibiotics have good oral avail-ability. Use IV antibiotics only if in line with local or national (sepsis) guidelines.

Then focus:

Review the clinical diagnosis and continuing need for antibiotics at 48h for all in-patients and all patients prescribed IV antibiotics:

Stop antibiotics if there is no evidence of infection.

Switch from IV to oral whenever possible.

Change to a narrower spectrum antibiotic whenever possible.

Continue regular clinical review whilst antibiotics are prescribed.

- Clinical diagnosis of low-severity community-acquired pneumonia is an exception, see also UTI p296.

Inhibitors of cell wall synthesis

See fig 9.4. The bacterial cell wall is unique in nature and therefore acts as a selective target for antibiotics. Antibiotics which act on the cell wall include:

β-lactam antibiotics

others: glycopeptides, polymyxins.

β-lactams: penicillins, cephalosporins, carbapenems, monobactam

Contain a β-lactam ring which inhibits the formation of peptidoglycan cross-links in the bacterial cell wall. Resistance occurs when the bacteria (eg staphylococci) produce a β-lactamase enzyme.

Penicillins:

See table 9.2. Include natural penicillins (penicillin G and V) and synthetic penicillins which are chemically modified to extend their spectrum of activity, eg amoxicillin, piperacillin.

In an attempt to overcome β-lactamase resistance, penicillins have been combined with β-lactamase inhibitors to create β-lactam-β-lactamase inhibitor combinations eg co-amoxiclav (amoxicillin+clavulanic acid), Tazocin® (piperacillin+tazobactam).

Staphylococcal resistance is conventionally defined by stability to meticillin, an acid-labile and IV-only equivalent of flucloxacillin (see MRSA p288).

Cephalosporins:

See table 9.4. Contain a β-lactam ring attached to a six-membered nuclear structure (five in penicillin), which allows synthetic modification at two sites (one in penicil-lin). This means that cephalosporins are the largest groups of available antibiotics. Classification into ‘generations’ is not standardized: as a rough rule, the higher the generation, the wider the spectrum.

Carbapenems:

See table 9.5. Broadest spectrum of all β-lactam antibiotics. Seek expert microbiol-ogy advice before use.

Monobactam:

Aztreonam is only active against Gram-negative species including Neisseria men-ingitidis, Haemophilus influenzae, Pseudomonas. Given IV/IM. Inhaled preparation for chronic pulmonary Pseudomonas (cystic fibrosis). Dose adjust for renal function. SEs: N&V, GI bleed, rash, ↑LFTS, ‘plts, paraesthesia, seizures, bronchospasm.

Non-β-lactam cell wall inhibitors

See fig 9.4 and table 9.6. Includes glycopeptides, eg vancomycin, teicoplanin, and polymyxins, eg colistin.

Inhibitors of protein synthesis See fig 9.4 and table 9.7. Includes:

aminoglycosides

macrolides

tetracyclines and derivatives of tetracycline

others: clindamycin, linezolid, chloramphenicol, fusidic acid.

Inhibitors of nucleic acid synthesis See fig 9.4 and table 9.8. Includes:

folate synthesis inhibitors: trimethoprim, co-trimoxazole

fluoroquinolones

others: metronidazole, rifampicin.

Nitrofurantoin is unique. Metabolites interfere with cell growth via ribosomes, DNA, RNA, and cell wall. Multiple sites of attack means ↓resistance. Concentrates in the urine (but not if ↓GFR), used in uncomplicated UTI. SES: haemolysis, pulmonary fibrosis, hepatotoxicity.

Antibiotics for TB, see pp294–5.

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Table 9.2 Penicillins

| Bacterial glossary Bacteria: Prokaryotic micro-organism without a membrane-bound nucleus. Classification of bacteria: By microscopy and culture of infected samples. Informs antibiotic choice. Includes: Gram stain: a staining technique. Bacteria with thick, exposed peptidoglycan layers will stain 'Gram positive' (purple/blue). Bacteria with a protected peptidoglycan layer will counterstain pink/red and are 'Gram negative' (fig 9.2). Shape: cocci = round; bacilli = rod-shaped; spirochaete = spiral. Aerobes/anaerobes: some bacteria cannot sur-vive without oxygen (obligate aerobes), whilst others cannot grow in its presence (obligate an-aerobes). Many more can survive in either en-vironment (facultative anaerobes). Some types of infection are more likely to involve aerobic or anaerobic bacteria, eg GI infections are typi- Fig 9.2 (a) Gram-positive versus cally anaerobic. (b) Gram-negative cell membranes. Bacteraemia: Bacteria circulating in the Reprinted by permission from Macmil-bloodstream. lan Publishers Ltd: Nature Reviews Bacteriocidal: Kills bacteria both in and out Microbiology, Cabeen et al., 3(8), 601–610, of the replication cycle. copyright 2005. Bacteriostatic: Stops replication without killing existing bacteria. Capsulate bacteria: Bacteria with a thick outer capsule, eg Haemophilus influen-zae, Neisseria meningitidis, and Streptococcus pneumoniae. These are destroyed in the spleen. Following splenectomy (or splenic infarction, eg sickle cell anaemia) there is an increased risk of infection by capsulate bacteria and prophylactic vac-cination should be offered (p407). Commensal: An organism that lives in/on a host without causing harm. Endotoxin: A lipopolysaccharide complex found on the outer membrane of Gram-negative bacteria. Can elicit an inflammatory response. Activates complement via the alternative pathway. Enterotoxin: Exotoxin that targets the gut, eg Clostridium difficile toxin (p411). Exotoxin: Toxins secreted by bacteria acting at a site distant from bacterial growth. Production of an exotoxin can determine virulence, eg botulinum, tetanus, diphtheria, shiga toxins. Flagella: A tail-like appendage that moves to propel the bacterium, eg Helico-bacter pylori. Nosocomial: Acquired in a hospital/healthcare setting (pp410–1). Obligate intracellular: Bacteria that can only survive in host cells m induce a cell-mediated immune response and will not grow on standard culture media. Ziehl–Neelsen stain: Mycolic acid in the cell wall of mycobacteria resists Gram staining but will appear red with acid-fast techniques (= acid-fast stain). | | — |

Table 9.4 Cephalosporins

| Antimicrobial stewardship In England between 2010 and 2014, antibiotic prescribing rose by 4% in general practice, 12% in hospitals, and 22% in other healthcare settings; E. coli resistance to ciprofloxacin increased by 18%, to cephalosporins by 28%, and to gentamicin by 27%. 25 000 people in Europe die every year from antibiotic-resistant bacteria. Each year r500 000 develop drug-resistant TB. Cost is the only barrier to buying carbapenems over-the-counter in Egypt, India, and Pakistan. 'This will be a post-antibiotic era. In terms of new replacement antibiotics, the pipeline is virtually dry, especially for Gram negative bacteria. The cupboard is nearly bare. Prospects for turning this situation around look dim.' Dr M Chan, Director-General of WHO, March 2012. Antimicrobial stewardship2 is necessary in all healthcare settings: monitoring, evaluation, and feedback on antimicrobial prescribing, bench-marked against up-to-date local and national guidelines evaluation of high/low levels of prescribing and prescribing outside of guidelines review of patient safety events: avoidable infection, drug reactions, complica-tions of antibiotic therapy, eg MRSA (p288), C. difficile (p259, p411) education and decision support systems for all antibiotic prescribers antibiotic pack sizes that correspond to appropriate course lengths regular review of all antimicrobial policy, treatment, and prophylaxis guidelines. | | — |

Table 9.5 Carbapenems

| Antibiotic | Indications | Considerations | | — | — | — | | Penicillin G | Gram +ve: streptococci (chest, | Give IV, poor oral absorp- | | (benzylpenicillin, | throat, endocarditis, cellulitis), | tion. Dose adjust for GFR. | | ‘penicillin’) | meningococcus, diphtheria, an-thrax, leptospirosis, Lyme disease. | SE: allergy, rash, N&V, C. difficile, cholestasis. | | Penicillin V (phenoxymethylpenicillin) | Prophylaxis: splenectomy/hypo-splenism, rheumatic heart disease. | Oral bioavailability may vary. | | Ampicillin/amoxicillin | Amino acid side chain extends penicillin spectrum to include enterobacteria (but ↓ activity against Gram +ve): URTI, sinusitis, chest, otitis media, UTI, H. pylori. | Ampicillin IV, amoxicillin PO. Dose adjust for GFR. SE: as per penicillin G, rash with EBV. | | Amoxicillin+clavulanic acid (co-amoxiclav) | Used if resistance to narrower-spectrum antibiotics: chest, pyelonephritis, cellulitis, bone. | Dose adjust for GFR. SE: as per amoxicillin. | | Piperacillin+tazobactam | Broad spectrum including Gram +ve, Gram –ve, Pseudomonas: neutropenic sepsis, hospital-acquired/complicated infection. | Tazobactam has ↓penetration of blood–brain barrier. Dose adjust for GFR. SE: as per penicillin G. Myelosuppression with prolonged use (rare). | | Flucloxacillin | β-lactamase resistant, Staphy-lococcus: skin, bone, post-viral pneumonia. | Dose adjust for GFR. SE: allergy, rash, N&V, cholestasis. |

Table 9.6 Lipopeptides and polymyxins

| Antibiotic | Indications | Considerations | | — | — | — | | Imipenem | Broad spectrum (Gram +ve, Gram –ve, aerobes, anaerobes): hospital-acquired/ventilator-associated/ complicated infection, neutropenic sepsis. | Dose adjust for GFR. Imipenem given with cilastatin to ‘renal metabolism. SE: N&V, C. difficile, rash, eosino-philia, ‘plts, ↑LFTs, seizures. | | Meropenem | Broad spectrum (Gram +ve, Gram –ve, aerobes, anaerobes): hospital-acquired/ventilator-associated/ complicated infection, neutropenic sepsis. | Dose adjust for GFR. Imipenem given with cilastatin to ‘renal metabolism. SE: N&V, C. difficile, rash, eosino-philia, ‘plts, ↑LFTs, seizures. | | Ertapenem | Broad spectrum (Gram +ve, Gram –ve, aerobes, anaerobes): hospital-acquired/ventilator-associated/ complicated infection, neutropenic sepsis. | Dose adjust for GFR. Imipenem given with cilastatin to ‘renal metabolism. SE: N&V, C. difficile, rash, eosino-philia, ‘plts, ↑LFTs, seizures. |

Table 9.7 Inhibitors of protein synthesis

| Antibiotic | Indication | Considerations | | — | — | — | | Lipopeptides | Lipopeptides | Lipopeptides | | Vancomycin | Complicated Gram +ve Dose IV to trough serum concentration. SEs: including MRSA. Oral for C. nephrotoxic (monitor creatinine, care with difficile (not absorbed). other nephrotoxics) ototoxic, ‘plts. | Complicated Gram +ve Dose IV to trough serum concentration. SEs: including MRSA. Oral for C. nephrotoxic (monitor creatinine, care with difficile (not absorbed). other nephrotoxics) ototoxic, ‘plts. | | Teicoplanin | Complicated Gram +ve Dose IV to trough serum concentration. SEs: including MRSA. Oral for C. nephrotoxic (monitor creatinine, care with difficile (not absorbed). other nephrotoxics) ototoxic, ‘plts. | Complicated Gram +ve Dose IV to trough serum concentration. SEs: including MRSA. Oral for C. nephrotoxic (monitor creatinine, care with difficile (not absorbed). other nephrotoxics) ototoxic, ‘plts. | | Polymyxins | Polymyxins | Polymyxins | | Colistin, polymyxin B | Multi-resistant Gram –ve. | Nephrotoxicity in r50%. Inhaled colistin for ventilator-associated pneumonia. |

Table 9.8 Inhibitors of nucleic acid synthesis

| Antibiotic | Indications | Considerations | | — | — | — | | Aminoglycosides | Aminoglycosides | Aminoglycosides | | Gentamicin | Gram –ve infection (↓activity against most Gram +ve and anaerobes). Tobramycin has ↑activity against Pseudomonas. Amikacin has least resistance. | SEs: nephrotoxic (monitor drug levels and serum creatinine), vestibular toxicity, ototoxicity. | | Tobramycin | Gram –ve infection (↓activity against most Gram +ve and anaerobes). Tobramycin has ↑activity against Pseudomonas. Amikacin has least resistance. | SEs: nephrotoxic (monitor drug levels and serum creatinine), vestibular toxicity, ototoxicity. | | Amikacin | Gram –ve infection (↓activity against

most Gram +ve and anaerobes). Tobramycin has ↓activity against Pseudomonas. Amikacin has least resistance. | SEs: nephrotoxic (monitor drug levels and serum creatinine), vestibular toxicity, ototoxicity. | | Macrolides | Macrolides | Macrolides | | Azithromycin | Gram +ve cocci (not enterococci and staphylococci), syphilis, chla-mydia. | SEs (↓ with erythromycin): GI, cholestasis, ↓QT. Cytochrome P450 inhibition (↑ with azithromycin): ↓warfarin, rhabdomyolysis with statins, ↓calcineurin inhibitor levels. | | Clarithromycin | Gram +ve cocci (not enterococci and staphylococci), syphilis, chla-mydia. | SEs (↓ with erythromycin): GI, cholestasis, ↓QT. Cytochrome P450 inhibition (↑ with azithromycin): ↓warfarin, rhabdomyolysis with statins, ↓calcineurin inhibitor levels. | | Erythromycin | Gram +ve cocci (not enterococci and staphylococci), syphilis, chla-mydia. | SEs (↓ with erythromycin): GI, cholestasis, ↓QT. Cytochrome P450 inhibition (↑ with azithromycin): ↓warfarin, rhabdomyolysis with statins, ↓calcineurin inhibitor levels. | | Tetracyclines and derivatives Tetracycline Exacerbation COPD, chlamydia, Lyme disease, mycoplasma, rick- | Tetracyclines and derivatives Tetracycline Exacerbation COPD, chlamydia, Lyme disease, mycoplasma, rick- | CI: pregnancy, <8y (teeth/bones). SEs: N&V, C. difficile, fatty liver, | | Doxycycline | ettsiae, brucella, anthrax, syphi- lis, MRSA, malaria prophylaxis. | idiopathic intracranial hyperten-sion. | | Tigecycline Other | Gram +ve and Gram –ve includ-ing β-lactam-resistant strains. | Dose adjust in liver dysfunction. SEs: N&V, photosensitivity, ↓LFTs. | | Clindamycin Linezolid | Gram +ve cocci (not enterococci), MRSA, anaerobes. Gram +ve cocci, MRSA, VRE, anaerobes, mycobacteria. | risk C. difficile. MAOI: check interactions, myelo-suppression, optic neuropathy. | | Chloramphenicol Fusidic acid | Gram +ve, Gram –ve, anaerobes, mycoplasma, chlamydia, conjunc-tivitis (topical). Staphylococci. | Systemic use limited by myelo-suppression. SES: GI, ↓LFTs. |

Gram-positive cocci

Staphylococci:

Staphylococci are skin/nasal commensals in r80% of adults. They can also cause infectious disease. This produces a diagnostic challenge: are the detected organisms causing infection or a contaminating commensal? The answer may lie in the pres-ence or absence of coagulase, an enzyme which coagulates plasma.

Coagulase-negative staphylococci: eg Staphylococcus epidermidis are less viru-lent. Pathogenicity is likely only if there is underlying immune system dysfunction or foreign material (prosthetic valve/joint, IV line, PD catheter, pacemaker).

Staphylococcus aureus is coagulase positive. Presentation:

Toxin release causes disease distant from infection. Includes:

scalded skin syndrome—bullae and desquamation due to epidermolytic toxins (no mucosal disease, ↑skin loss compared to toxic epidermal necrolysis)

preformed toxin in food—sudden D&V (p428)

toxic shock—fever, confusion, rash, diarrhoea, ↑BP, AKI, multiorgan dysfunction. Tampon associated or occurs with (minor) local infection.

Local tissue destruction: impetigo, cellulitis, mastitis, septic arthritis, osteomy-elitis, abscess, pneumonia, UTI.

Streptococci:

Classification based on Lancefield group persists in terminology (fig 9.5). Includes:

Streptococcus agalactiae (β-haemolytic group B): neonatal and peurperal infec-tion, skin, soft tissue. Invasive disease (bacteraemia, endocarditis, osteomyelitis, septic arthritis, meningitis) usually has risk factors: DM, malignancy, chronic dis-ease. Treatment: penicillin, macrolide, cephalosporin, chloramphenicol.

Streptococcus milleri: if found in blood culture look for an abscess—mouth, liver, lung, brain. Treatment: penicillin.

Streptococcus pneumoniae: pneumonia (pp166–8), otitis media, meningitis, septi-caemia. Treatment: penicillin.

Vaccination: childhood, hyposplenism, >65y (p407).

Viridans streptococci: commonest cause of oral/dental origin endocarditis (p150).

Streptococcus bovis: bacteraemia↬endocarditis. Look for colon/liver disease.

Enterococci:

Gut commensal. Resistance to cephalosporins and quinolones leads to nosocomial colonization and infection. Most common is Enterococcus faecalis: if found in blood culture, assume endocarditis until proven otherwise. Treatment: intrinsic and acquired resistance including vancomycin-resistant enterococci (VRE). Seek expert help.

Gram-positive bacilli

Listeria:

Clostridia:

Clostridium difficile, see p259, p411.

Clostridium perfringens:

Gastroenteritis, see p420.

Clostridium botulinum, see p426.

Clostridium tetani, see p426.

Diphtheria:

Caused by Corynebacterium diphtheriae toxin. Preventable with vaccine. Pres-entation: tonsillar pseudomembrane with fever, painful dysphagia, cervical lym-phadenopathy (see OHCS p158). Diagnosis: culture, toxin detection, PCR. Treatment: antitoxin within 48h. Benzylpenicillin/erythromycin. Airway support.

Actinomycosis:

Due to Actinomyces israelii, a mucous membrane commensal. Presentation: suba-cute granulomatous/suppurative infection adjacent to mucous membrane. Diag-nosis: culture. ‘Sulphur’ granules in pus/tissue are pathognomonic. Treatment: antibiotics covering actinomycetes and concomitant microbes.

Nocardia:

Rare cause of disease. Presentation: tropical skin abscess, lung/brain abscess, dis-seminated infection if immunosuppressed. Treatment: usually co-trimoxazole.

Anthrax: See p424.

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Gram-negative cocci

Neisseria:

Meningitis (r50% cases). Main proliferation of bacteria is in CSF. Insidious onset

with malaise, nausea, headache, vomiting. May be misdiagnosed as gastroenteri-tis, URTI, or childhood viral illness. Later meningism: headache, vomiting, nuchal/ back rigidity, photophobia, altered consciousness. Complications in up to 20%: sensorineural hearing loss, impaired vestibular function, epilepsy, diffuse brain injury.

University Press.

University Press.

Neisseria gonorrhoea: see pp412–2.

Moraxella catarrhalis:

Gram-negative bacilli

Enterobacteriaceae:

Enterobacteriaceae family is large: >50 genera, >170 named species. In the clinical setting, 2 species make up 80–95% of isolates:

Escherichia coli: part of normal colonic flora. Pathogenic forms can cause: Intestinal disease:

Enterotoxigenic: a major cause of traveller's diarrhoea (pp428–9). Enterohaemorrhagic: diarrhoea, haemorrhagic colitis eg O157:H7 (p421). Enteropathogenic: infants in areas of poor sanitation.

Enteroinvasive: dysentery-like syndrome.

Enteroadherent: traveller's diarrhoea, chronic diarrhoea in children/HIV.

Extra-intestinal disease: usually patient's own flora that is not pathogenic in the intestine but causes disease elsewhere: UTI (pp296–7); neonatal meningitis; noso-comial infection: pneumonia, meningitis, sepsis. Treat according to sensitivity: trimethoprim, ampicillin, cephalosporin, ciprofloxacin, aminoglycoside.

Klebsiella pneumoniae: colonizes skin, nasopharynx, GI tract, hospitalized patients. Associated with antibiotic exposure, in-dwelling catheters, immuno-suppression. Causes pneumonia (necrotizing disease and sepsis if immunosup-pressed). Also UTI, nasopharyngeal inflammation. Treat according to sensitivity: aminoglycoside, cephalosporin, carbapenem, quinolone.

Proteus mirabilis: gut commensal. Causes UTI (pp296–7). Stone formation due to urease production: breaks down urea to produce ammonia, struvite stones ('infection stones') then form in the presence of magnesium, calcium, and phos-phate (pp628–9).

Other Enterobacteriaceae include Salmonella, Shigella, Yersinia: see enteric fever (p415), gastroenteritis (pp428–21), plague (p425).

Resistance: widespread antibiotic use has led to the development of highly virulent, multiple resistant E. coli and Klebsiella species including:

extended-spectrum β-lactamase (ESBL) producing Enterobacteriaceae. Resistant to penicillins, cephalosporins, fluoroquinolones, trimethoprim, tetracycline, with pos-sible extension to other antibiotic groups

carbapenem-resistant Enterobacteriaceae (CRE).

Resistance requires antimicrobial stewardship (p284), surveillance, robust infection control, research into resistance risk and transmission (p282).

Pseudomonas aeruginosa:

Haemophilus influenzae:

Whooping cough:

Bordetella pertussis. Presentation: catarrhal phase 1–2wk, then paroxysmal cough-ing. 'Whoop' is a breath through partially closed vocal cords, seen mainly in children. Cough is prolonged ('100 day cough'). Infants have ⛄complications/mortality. Diag-nosis: PCR nasal/throat swab. Culture sensitivity 10–60%. Treatment: macrolides ↩infectivity, but may not alter disease course. Routine childhood vaccination. Vac-cination in pregnancy ⛄placental antibody transfer to protect neonate (p407).

Other:

Brucellosis (p424), cholera (p420), melioidosis (p414).

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Epidemiology

9.6 million new cases/yr of which 27% are unreported/undiagnosed (fig 9.8).

2.2% of new cases, and 20% of previously treated cases are drug resistant (p295).

Co-infection with HIV in 12% of new cases.

Leading cause of death worldwide, 1.5 million deaths/yr.

Effective diagnosis and treatment saved 42 million lives between 2000 and 2014.

UK: r8000/yr, r12 per 100 000. 72% born outside UK, 70% in deprived areas, 20% with pulmonary disease wait >4 months from symptoms to treatment.

Health Organization 2016. http://www.who.int/tb/publications/global_report/en/

Pathophysiology

Caused by infection with Mycobacterium tuberculosis.

Active infection occurs when containment by the immune system (T-cells/mac-rophages) is inadequate. It can arise from primary infection, or re-activation of previously latent disease. Transmission of TB is via inhalation of aerosol droplets containing bacterium. This means only pulmonary disease is communicable.

Presentation

TB, or not TB—that is the question. Maintain a high index of suspicion. TB can affect any organ in the body (table 9.9).

Table 9.9 UK TB case reports by site of disease

| Antibiotic | Indications | Considerations | | — | — | — | | Folate synthesis inhibitors Trimethoprim Gram –ve: UTI, prostatitis. Co-trimoxazole Pneumocystis jirovecii, GI (sulfamethoxa- infection (eg Shigella, E. coli), zole+trimetho- protozoans (eg Cyclospora), prim) listeria, nocardia, MRSA. | Folate synthesis inhibitors Trimethoprim Gram –ve: UTI, prostatitis. Co-trimoxazole Pneumocystis jirovecii, GI (sulfamethoxa- infection (eg Shigella, E. coli), zole+trimetho- protozoans (eg Cyclospora), prim) listeria, nocardia, MRSA. | Inhibits creatinine secretion: ↑serum creatinine without ↓GFR. Synergistic combination. Good oral absorption and tissue/ CSF penetration. SEs: folate deficiency, ↑K+, rash, myelosup-pression, haemolysis with G6PD deficiency. | | Fluoroquinolones Ciprofloxacin | Broad including Pseudomonas: UTI, prostatitis, atypical and hospital-acquired chest infection, infectious diarrhoea. | SEs: GI irritation, CNS effects (↓ seizure threshold, headache, drowsiness, mood change), pe-ripheral neuropathy, tendinopa-thy (Achilles), ↑QT, C. difficile. | | Levofloxacin | Broad including Pseudomonas: UTI, prostatitis, atypical and hospital-acquired chest infection, infectious diarrhoea. | SEs: GI irritation, CNS effects (↓ seizure threshold, headache, drowsiness, mood change), pe-ripheral neuropathy, tendinopa-thy (Achilles), ↑QT, C. difficile. | | Moxifloxacin | Broad including Pseudomonas: UTI, prostatitis, atypical and hospital-acquired chest infection, infectious diarrhoea. | SEs: GI irritation, CNS effects (↓ seizure threshold, headache, drowsiness, mood change), pe-ripheral neuropathy, tendinopa-thy (Achilles), ↑QT, C. difficile. | | Others Metronidazole | Anaerobic infection: intra-ab-dominal, pelvic, oral, soft-tissue. Bacterial vaginosis. C. difficile. | Good oral absorption. Dose adjust for liver function. SEs: Disulfiram reaction with alcohol, inhibits warfarin metabolism. | | Rifamycins: rifampicin, rifabutin, rifapentine | Mycobacteria (TB, atypical mycobacteria, leprosy), some staphylococci, Legionella, menin-gococcal prophylaxis. | SEs: hepatitis (monitor LFTs), GI, CNS effects, myelosuppression, red secretions (urine, saliva, sweat, sputum, tears). |

From: Tuberculosis in the UK

England. www.gov.uk/phe

Clinical features of TB

Systemic features: Low-grade fever, anorexia, weight loss, malaise, night sweats, clubbing (bronchiectasis), erythema nodosum (p562).

Pulmonary TB: Cough (in r50%, >2–2 weeks, dry then productive), pleurisy, haemoptysis (uncommon, seen with bronchiectasis m not always active disease), pleural effusion. An aspergilloma/mycetoma (p177) may form in the cavities. Presentation varies and may be silent or atypical, especially with immunosup-pression, eg HIV, post-transplantation.

Gastrointestinal TB: Most disease is ileocaecal. Causes colicky abdominal pain and vomiting. Bowel obstruction can occur due to bowel wall thickening, stric-ture formation, or inflammatory adhesions. Biopsy is required for diagnosis. Ca-seation necrosis and an absence of transmural cracks/fissures distinguish from Crohn's disease.

Spinal TB: Local pain and bony tenderness for weeks–months. Slow, insidious progression. May not present until deformity or neurological symptoms. Look for bony destruction, vertebral collapse, and soft tissue abscess (see Pott's ver-tebra p708).

CNS TB: Haematogenous spread lead-

ing to foci of infection in brain and spinal cord. Foci can enlarge to form tuberculomas. Foci rupture leads to

©Dr Vijay Sadasivam, Radiologist, SKS Hosp, Salem,

Tamil Nadu, India.

meningitis. Risk with immune suppression, HIV, aged <2y. Headache, meningism, confusion, seizures, focal neurological deficit, and systemic symptoms. Needs LP and examination of CSF (leucocytosis, raised protein, CSF: plasma glucose <50%, AFB stain, PCR and culture). Look for TB elsewhere (CXR, etc), test for HIV. CT/ MRI may show hydrocephalus, basal exudates. Tuberculomas are ring-enhancing.

2All rapid diagnostic tests (p294) have sensitivity, so treat on suspicion.

Genitourinary TB: Symptoms may be chronic, intermittent, or silent. Include dysuria, frequency, loin pain, haematuria, sterile pyuria (see p296). Granuloma may cause fibrosis, strictures, infertility, and genital ulceration.

Skin: Lupus vulgaris = persistent, progressive, cutaneous TB: red-brown, 'apple-jelly' nodules. Scrofuloderma: skin lesion extended from underlying infection eg lymph node, bone; causes ulceration and scarring.

Treatment

Antibiotics used in the treatment3 of TB are detailed in table 9.10.

Table 9.10 Antibiotics used in the treatment of TB

| Resistant Staphylococcus aureus: MRSA, VISA, VRSA Staph. aureus which produces β-lactamase, or an altered enzyme responsible for cell wall formation, will be resistant to β-lactam antibiotics (penicillins, cepha-losporins, carbapenems, see p285). Resistance is usually defined by stability to meticillin, ie meticillin-resistant Staph. aureus (MRSA). Vancomycin resistance also exists and is classified according to the amount of vancomycin needed to inhibit bacterial growth: vancomycin-intermediate Staph. aureus (VISA) and vancomy-cin-resistant Staph. aureus (VRSA). Resistant staphylococci cause morbidity and mortality compared to sensitive strains. Risk factors for colonization include: antibiotic exposure, hospital stay, surgery, nursing home residence. Treatment of infection (not colonization): vancomycin (for MRSA), teicoplanin. Oral agents with activity against MRSA include clindamycin, co-trimoxazole, doxycycline, linezolid. Prevention: surveillance, barrier precautions, hand-hygiene, decolonization (mupi-rocin 2%, chlorhexidine, tea tree oil), antimicrobial stewardship. See p284, pp410–11. | | — |

Latent TB

Treat with 2 months of isoniazid (with pyridoxine) and rifampicin OR 6 months of isoniazid (with pyridoxine).

If concerns about hepatotoxicity then 2 months of isoniazid and rifampicin may be preferred. In severe liver disease, seek specialist advice. If interactions with rifamy-cins are a concern (eg HIV, transplant) then 6 months of isoniazid may be preferred.

Active TB

All forms of active TB are statutorily notifiable in UK. This includes both clinical and culture diagnoses. Notification is via your local public health protection team (www. gov.uk/health-protection-team).Treatment is given under the care of a specialist TB clini-cian/service according to table 9.10. Exceptions include:

active CNS disease (including spinal cord involvement): continuation phase of treat-ment is extended from 4 to 10 months

CNS and pericardial disease: use adjunctive high-dose steroids (with weaning and withdrawal during the intensive treatment period)

drug-resistant TB.

Adherence is important for treatment to be effective and to prevent drug resistance. Directly observed therapy (DOT) should be considered if: previous treatment for TB, homelessness, drug/alcohol misuse, prison, psychiatric or cognitive disorder, multi-drug resistant disease, patient request.

Universal access to diagnosis and treatment of TB is part of social justice. WHO has developed an ‘End TB’ strategy aiming to reduce TB deaths by 90% by 2020, and TB incidence by 90% by 2025 (www.who.int/tb/strategy/en).

Drug-resistant TB

NAAT (p294) for drug resistance should be requested for all patients with risk fac-tors for drug-resistance: previous TB treatment, contact with drug-resistant disease, birth or residence in a country where ≥5% new cases are drug resistant (fig 9.10). Drug resistance may be:

to any single agent in table 9.10.

multidrug-resistant TB (MDR-TB): resistant to rifampicin and isoniazid.

extensively drug-resistant TB (XDR-TB): resistant to rifampicin, isoniazid, one inject-able agent (capreomycin, kanamycin or amikacin) and one fluoroquinolone.

If rifampicin resistance is detected, treat with at least six agents to which the my-cobacterium is likely to be sensitive. Test for resistance to 2nd-line drugs. Remember infection control measures. Seek expert advice for all drug-resistant cases.

Fig 9.10 Percentage of new TB cases with multi-drug resistant TB.

Health Organization 2016. http://www.who.int/tb/publications/global_report/en/

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Influenza is common throughout the world, affecting r5–10% of adults, and 20–20% of children each year. In most, it is a self-limiting illness. Complications can be life-threatening in the elderly, pregnant women, and those with chronic disease. There are r4 million cases of severe influenza and r500 000 deaths worldwide/yr.

Seasonal influenza

Presentation: Incubation: 1–4d. Fever, dry cough, sore throat, coryzal symptoms, headache, malaise, myalgia, conjunctivitis, eye pain ! photophobia. Complications include pneumonia, exacerbation of chronic lung disease, croup, otitis media, D&V, myositis, encephalitis, Reye syndrome (encephalopathy + fatty degenerative liver failure).

Diagnosis: Clinical: acute onset + cough + fever has positive predictive value >79%. Testing limited to outbreaks, and public health surveillance. Includes: viral PCR, rapid antigen testing, viral culture of clinical samples (throat swab, nasal swab, naso-pharyngeal washings, sputum).

Treatment:

Uncomplicated influenza symptomatic treatment eg paracetamol. Antivirals only if high risk:

Chronic disease: lung, heart, kidney, liver, CNS, DM

Immunosuppression: immunodeficiency, current or planned or within 6

months of immunosuppressive therapy, ↶CD4 (<200 in adults, <500 if child <5yr)

Pregnancy • >65yr

BMI>40 • <6 months old.

Complicated influenza includes lower respiratory tract infection, exacerbation of any underlying medical condition, all needing hospital admission. Give antiviral' inhibitors of influenza neuraminidase:

Oseltamivir: PO or NG. Adult dose: 75mg BD. 5d course. 1st line in UK.

Zanamivir: inhaled (10mg BD, 5d course, confirm technique), nebulized, or IV (res-piratory disease affecting nebulizer delivery, ITU). Used if: oseltamivir resistance (eg A(H1N1)), poor clinical response to oseltamivir, concerns re GI absorption of oseltamivir.

Retrospective observational data, and animal studies of oseltamivir and zanamivir show no evidence of harm in pregnancy. Supportive treatment for all. Extracorpor-eal membrane oxygenation (ECMO) has been used to support gas exchange in severe acute lung injury due to influenza.

Prevention:

Post-exposure prophylaxis: if high risk (see 'Treatment') AND not protected by vac-cination: oseltamivir PO OD (inhaled zanamivir OD if oseltamivir resistance) for 10d.

Annual vaccination in UK: all high risk (see 'Treatment'), children>2yrs, healthcare workers. See p407.

Pandemic influenza

Epidemiology

r27 million adults and children are estimated to be living with HIV worldwide (fig 9.11), with 1.2 million deaths/yr. Africa has most of the disease (r26 million), most of the mortality (790 000/yr), and r1% of the world's wealth.

UK: estimated r100 000 living with HIV (=1.9/1000) including 5% of men who have sex with men (MSM). r17% of those with HIV in UK are unaware of their infection.

WHO region'. ©World Health Organization 2016.

http://gamapserver.who.int/mapLibrary/Files/Maps/HIV_adult_prevalence_2015.png.

Pathophysiology

Prevention

Sexual transmission: Consistent and correct use of (male and female) condoms ‘ transmission by r90%. Serosorting is the restriction of (unprotected) sex depend-ing on HIV status. It is unsafe due to inaccuracies in HIV status (which is only as reliable as a person’s last test) and failure to disclose. It does not consider transfer of treatment resistance, other STIs, or hepatitis.

Presentation

With symptoms of early HIV infection:

Persistant generalized lymphadenopathy = swollen/enlarged lymph nodes >1cm in two or more non-contiguous sites (not inguinal) persisting for >2 months. Due to follicular hyperplasia caused by HIV infection. Exclude TB, infection, and malignancy.

In the asymptomatic, latent phase of chronic HIV infection:

In the UK there is universal testing in sexual health clinics, antenatal services, drug dependency programmes, and in patients with TB/hepatitis B/hepatitis C/lymphoma. Where HIV prevalence is >2/1000 universal testing by GPs and medical admissions units should be considered. Any request for a HIV test should be met.

With complications of immune system dysfunction: See pp400–1.

HIV testing

ELISA for HIV antibody and antigen (p24): 4th-generation assays test for HIV an-tibody and p24 antigen. This reduces the ‘window period’ (time of false-negative testing between infection and the production of measurable antigen/antibody) to average r10 days. Diagnosis in UK is confirmed by a confirmatory assay.

Rapid point-of-care testing: Immunoassay kit which gives a rapid result from a finger-prick or mouth swab. Only CE-marked kits should be used. Needs serological confirmation.

Viral load: Quantification of HIV RNA. Used to monitor response to ART. Not diag-nostic due to possibility of a false-positive result m care if used to test for sympto-matic primary HIV in the ‘window period’—confirmation of seroconversion is still required.

Nucleic acid testing/viral PCR: Qualitative test for the presence of viral RNA. Used to test for vertical transmission in neonates as placental transfer of maternal anti-bodies can affect ELISA antibody testing up to 18 months of age.

CD4 count: Cannot diagnose HIV. Used to monitor immune system function and disease progression in patients with HIV. <200 cells/microlitre is one of the defining criteria for AIDS.

See www.aidsmap.com for available HIV testing and country-specific resources.

Complications of HIV can be divided into:

complications of immune dysfunction (opportunistic infection/malignancy)

complicating comorbidity

complications of treatment, ie adverse drug effects (see pp402–2).

The differential diagnosis for symptoms presenting in a person with HIV is given in table 9.11. This is not exhaustive.

►Do not forget your usual differentials, the presen-tation may not relate to the patient’s HIV status.

Opportunistic disease

ART is part of the treatment regimen of all opportunistic infections (see pp402–2).

Pneumocystis jirovecii: (‘yee-row-vet-zee’) Presentation: progressive SOB on exertion, malaise, dry cough. Haemoptysis and pleuritic pain rare. Examination:

Candidiasis: Oral or oesophageal. Pain in the tongue, dysphagia, odynophagia. Di-agnosed clinically or endoscopically. Treated with systemic ‘-azole’, eg fluconazole.

Toxoplasma gondii: Toxoplasma abscesses are commonest cause of intracranial mass lesions when CD4 <200 cells/microlitre. Presentation: focal neurological signs

Cryptosporidium: Common cause of chronic diarrhoea in HIV pre-ART. Presen-tation: acute or sub-acute non-bloody, watery diarrhoea. Also cholangitis, pan-creatitis. Investigation: stool microscopy (multiple samples as oocyst excretion intermittent), PCR, enzyme immunoassay, direct fluorescent antibody. Treatment: supportive, ART.

Lymphoma: Increased risk of non-Hodgkin’s lymphoma in HIV. Includes: diffuse large B-cell lymphoma, Burkitt’s lymphoma, primary CNS lymphoma. Presentation: dependent upon area of involvement. Includes lymphadenopathy, cytopenia, CNS symptoms. Treatment: combined ART and chemotherapy. Rituximab for non-CNS disease. Whole-brain radiotherapy for CNS disease if excess toxicity with chemo-therapy.

Table 9.11 Differential diagnoses in HIV

| Antibiotic | Standard course for active disease | Notes | | — | — | — | | Rifampicin | 2 months intensive 4 months continuation | Enzyme inducer: care with warfarin, calcineurin inhibitors, oestrogens, phenytoin; body secre-tions coloured orange-red (includes contact lens staining); altered liver function. | | Isoniazid Pyrazinamide | 2 months intensive 4 months continuation 2 months intensive | Inhibits formation of active pyridoxine (Vit B6) which causes a peripheral neuropathy (⚠risk with DM, CKD, HIV, malnutrition) m give with prophylactic pyridoxine; hepatitis. Idiosyncratic hepatotoxicity, ↻dose if eGFR<20. | | Ethambutol | 2 months intensive | Colour blindness, ↻visual acuity, optic neuritis. Check visual acuity at start of treatment, moni-tor for symptoms. Monthly visual check if treat-ment > 2 months. Monitor levels if eGFR<20. |

2 Antiretroviral therapy (ART)7 is recommended for everyone with HIV, regardless of CD4 count.

Aims of ART To reduce the HIV viral load to a level undetectable by standard labora-tory techniques leading to immunological recovery, reduced clinical progression, and reduced mortality. These aims should be met with the least possible side-effects.

Mechanism of action (See fig 9.12.)

CCR5 antagonists inhibit the entry of the virus into the cell by blocking the CCR5 co-receptor.

Integrase strand transfer inhibitors (INSTIs) inhibit integrase and prevent HIV DNA

Pharmacokinetic enhancers/boosters increase the effectiveness of antiretroviral drugs allowing lower doses eg cobicistat, ritonavir.

CCR5 antagonist T T

HIV RNA

T PI

Proteins

Baseline tests: CD4, viral load, FBC, LFT, electrolytes, creatinine, pregnancy test, viral genotype for drug resistance.

Herpes simplex virus (HSV) (human herpesvirus 1 and 2)

Genital herpes: predominantly HSV2 (see p412).

Herpetic whitlow: painful vesicles on distal phalanx due to inoculation through a break in the skin.

Treatment: Aciclovir: ↓symptoms and viral shedding, will not prevent latent infec-tion. ►Give empirical IV aciclovir as soon as HSV encephalitis is suspected, mortality r70% in untreated disease (see p824).

Varicella zoster virus (VZV) (human herpesvirus 3)

Chicken pox: prodrome 1–2d: fever, malaise, headache, abdominal pain. Then rash (fig 9.14): pruritic, erythematous macules↝vesicles, crust in r48h. Infectious 1–2d pre-, to 5d post-rash development (lesions scabbed). Complications

Diagnosis: Clinical diagnosis unless immunosuppressed: viral PCR, culture, immuno-fluorescence.

Epstein-Barr virus (EBV) (human herpesvirus 4)

Diagnosis:

Serology: IgM to EBV viral capsid antigen in acute infection. IgG if past infection.

Cytomegalovirus (CMV) (human herpesvirus 5)

Other herpes viruses

Human herpesvirus 8 (HHV8): Oncogenic (see BOX ‘Oncogenic viruses’), Castleman’s disease.

Respiratory tract viruses

Human papilloma virus (HPV)

Anogenital warts (HPV 6, 11). Treatment: topical podophyllin, imiquimod; ablation.

Polyomavirus

r100% exposure. Disease only with immunosuppression: BK virus causes renal trans-plant nephropathy; JC virus causes progressive multifocal leucoencephalopathy.

Measles

Secondary infection: bacterial pneumonia, otitis

Subacute sclerosing panencephalitis: 5–10yr af-

Mumps

Rubella (German measles)

Passive immunity uses pre-formed antibody to protect against infection. It of-fers immediate but short-lived protection. Natural passive immunity occurs in the placental transfer of maternal antibodies to the fetus; acquired passive immunity includes treatment with immunoglobulin eg hepatitis B, rabies, tetanus, varicella-zoster.

Table 9.14 Additional vaccination of specific groups in UK (*=live vaccine)

| Hide and seek In 2009, there was justifiable global concern about a ‘swine flu’ pandemic. Based on a Cochrane review in 2008, which showed reduced complications with os-eltamivir, billions were spent stockpiling the drug worldwide. In fact, the positive conclusion was driven mainly by data from an industry-funded summary of 10 trials, of which only two had been published.™ Cochrane needed ac-cess to these missing data. The ensuing fight for information was to take 5 years. The offer of a secret contract, with secret terms, and secrecy about methods, was declined. These are not acceptable methods for meta-analysis. Inconsistencies be-gan to arise in conclusions about effectiveness. Were people seeing different data, or was this simply a close call with two sides separated by a very small fence? Either way, being able to see all the data started to become increasingly important. But even the largest phase three trial of the drug had never been published. And was self-reported pneumonia a useful outcome measure? In December 2009, Cochrane could only declare that paucity of data undermined previous findings. This battle for data became part of ‘Alltrials’: a campaign for transparency in clinical trials. r50% of all clinical trials remain unpublished. The hunt goes on to find them. You can run your own drug trial, choose what to publish, and watch how the data become skewed at: www.alltrials.net/news/the-economist-publication-bias. After half a decade, under ceaseless demand, and with the withholding of data become increasingly indefensible, the clinical study reports were released. These are normally used to provide authorities with a detailed trial report. They are not easy fodder for meta-analysis. Assessing 160 000 pages was uncharted territory for Cochrane. And the conclusion: oseltamivir shortens symptoms by <1d and hos-pitalization is not reduced. Other complications were unreliably reported. The WHO includes oseltamivir on its WHO Model List of Essential Medicines (19th edition, 2015) which means it is considered efficacious, safe, cost-effective, and a minimum requirement for basic healthcare. Does this stand up to independent scrutiny? The evidence base is certainly tarnished. But a pandemic is not a RCT. And the threshold of evidence to reverse policy decisions may be different from the threshold needed to introduce them. If a new pandemic looms, millions more will be thrown in, for now. | | — |

Travel

Worldwide ⛆ in fungal infection with new pathogenicity, ⛆virulence, and new infec-tive mechanisms. Incidence data limited by failures in recognition and diagnosis. Divided into superficial/cutaneous and systemic/invasive.

Superficial/cutaneous mycoses

Superficial candidiasis: Usually Candida albicans (fig 9.17), a commensal in mouth, vagina, and GI tract. Risk factors: immunosuppression, antibiotic treat-ment. Presentation: Oropharyngeal: white patches on erythematous background (plaque type); sore, in-

Malassezia: Commensals of greasy skin. Presentation: Pityriasis versicolor: scaly hypo/hyperpigmented rash with scaling (fig 9.20). Seborrhoeic dermatitis: scaling of face, scalp (dandruff), anterior chest. Malassezia folliculitis: itchy, follicular rash on back and shoulders (⛯⛯ acne).

Systemic/invasive mycoses

Healthcare-associated, or nosocomial, infections include diseases which occur:

Outside a healthcare setting but are brought in by patients, staff, or visitors and transmitted to others.

Clostridium difficile colitis)

Management of healthcare-associated infection

Treat: Refer to local guidelines, seek expert help. Initial antibiotic choice may differ for healthcare-associated infection.

Personal attire. In the UK there is a bare-below-the-elbows policy. Long hair should be tied back. In areas where infec-tion risk is particularly high (theatre, ICU),

Environment. Should be clean and safe, with effective decontamination.

Taking a sexual history

Examination

Urethritis/vaginal discharge See table 9.16.

Genital ulcer(s)

Genital herpes: HSV. Presentation: flu-like prodrome, then vesicles/papules around genitals, anus, throat. These burst, forming painful shallow ulcers. Also urethral discharge, dysuria, urinary retention, proctitis. Diagnosis: PCR. Treatment: analge-sia, topical lidocaine. Antivirals within 5d: aciclovir, valaciclovir, famciclovir.

Secondary: dissemination r4–10wks after chancre. Rash (maculopapular in 50–75%, on palms/soles in 11–70%), mucous patches, condyloma lata (raised, pale plaques, often flexural), fever, headache, myalgia, lymphadenopathy, hepatitis.

Lymphogranuloma venerum: Chlamydia trachomatis. Presentation: mostly MSM in UK. Painless papule/ulcer↬lymphadenopathy, fever, arthritis, pneumonitis. Direct transmission to rectal mucosa causes haemorrhagic proctitis: pain, rectal bleed-ing/discharge, tenesmus. Diagnosis: PCR. Treatment: doxycycline.

Exclude malaria in all travellers from the tropics (p416–9).

Look for tropical infection[a] but don't forget your usual differentials.

Respiratory: S. pneumoniae, H. influenzae, legionella, influenza, viral respiratory disease (SARS, MERS), TB, HIV-associated disease, melioidosis.2

Table 9.18 Differential diagnosis according to incubation time

| Needle-stick injury Risk of HIV transmission from a single needle-stick exposure from a person with HIV not on ART is r1 in 200 (lower than risks of hepatitis B and C transmission). Prevent: Use 'safer sharps' (incorporates a mechanism to minimize accidental injury). Do not recap unprotected medical sharps. When using sharps, ensure there is a disposal container nearby. Manage: Encourage the wound to bleed, ideally under running water (do not suck). Wash with soap and running water, do not scrub. Seek advice from occupational health/infection control (or A&E outside of work-ing hours) regarding source testing and post-exposure prophylaxis (p298). | | — |

Examination

Rash:

Ulcer: trypanosomiasis, Yesinia pestis, tick typhus, anthrax, tropical ulcer. Jaundice: Viral hepatitis, severe falciparum malaria, enteric fever, leptospirosis, re-lapsing fever, typhus, VHF, bartonellosis

Investigation

Directed by travel history and examination. In undifferentiated fever:

FBC: lymphopenia in viral infection including HIV; eosinophilia in parasitic/fungal eg soil-transmitted helminths, filiariasis, schistosomiasis, hydatid disease; platelets in malaria, dengue, HIV, typhoid, severe sepsis.

Consider: save serum, specific serology, or EDTA sample for PCR.

Support

Disease notification: www.gov.uk/health-protection-team.

Hospital for Tropical Diseases 0202 456 7890.

Epidemiology

2.2 billion people at risk in 95 countries = half the world's population (fig 9.22).

Most common tropical disease imported into UK, r2000 cases/yr.

Plasmodium vivax is the dominant parasite outside of sub-Saharan Africa.

Malaria parasites

University Press.

Initial human host

Liver infection

Next human host

transmission

Clinical features

Examination: Fever, otherwise unremarkable. If diagnosis is delayed or severe dis-ease then may present with jaundice, confusion, seizures.

Diagnosis

Microscopy of thick and thin blood smear. Sensitive and specific in experienced hands.

If P. falciparum (or P. knowlesi) estimated % parasitized red cells should be given:

Other: FBC (anaemia, thrombocytopenia), creatinine and urine output (AKI), clotting (DIC), glucose (hypoglycaemia), ABG/lactate (acidosis), urinalysis (haemoglobinuria). Malaria is notifiable to public health: www.gov.uk/health-protection-team

Falciparum malaria

Impaired consciousness/seizures (consider LP) • Hb <80g/L

Hypoglycaemia (<2.2mmol/L) • Haemoglobinuria

Artemisinin combination therapies (ACT) achieve rapid clearance of parasites by combined action at different stages of the parasite cycle (p417):

Dihydroartemisinin (DHA)-piperaquine: 4 tablets OD for 2d (if weight >60kg). Take >2h before and after food to prevent excessive peak levels. Possible ΔQTC m avoid in arrhythmia.

GI side-effects in r25%.

ΔFailure rates with antifolate drugs mean Fansidar® is no longer used.

Non-falciparum malaria

P. vivax, P. ovale, P. malariae, P. knowlesi:

If uncomplicated disease, treat with ACT as uncomplicated falciparum. Chloroquine can be used for non-falciparum disease. Dosing in adult: 620mg base at

be aware that ACT may work more quickly on both fever and parasite count

P. vivax: adult 20mg (0.5mg/kg) daily for 14d.

Mosquito-borne diseases[12] are transmitted by the bite of a mosquito infected with a virus, bacteria, or parasite. The mosquito acts as the disease vector. Mosquitoes are arthropods (see table 9.21). Mosquito-borne diseases can therefore also be de-scribed as vector-borne or arthropod-borne disease. When a virus is transmitted by an arthropod it is termed an arbovirus (ARthropod-BOrne VIRUS).

Dengue

Most important arbovirus in humans. Dengue viruses (Flaviviruses DENV 1–4) are transmitted by day-biting Aedes mosquito. 120 countries (fig 9.28). Symptoms in 100 million/yr. UK: r500 imported cases/yr.

permission from Oxford University Press.

Consider transfusion if haematocrit without clinical improvement. Avoid NSAIDs.

Chikungunya

Yellow fever

3 In UK testing done via Rare and Imported Pathogens Laboratory (RIPL): www.gov.uk/government/collec-tions/rare-and-imported-pathogens-laboratory-ripl

Lymphatic filariasis (elephantiasis)

Wuchereria bancrofti (fig 9.29) r90% of disease.

S. Upton,

bendazole plus either diethylcarbamazine (DEC) or ivermec- Reproduced with permis-

sion from World Health Organization. © World Health Organization.

Vector borne diseases are infections transmitted by the bite of infected arthropod species including mosquitoes, ticks, flies, and bugs (table 9.21).

Lyme disease (Lyme borreliosis)

Tick-borne multisystem disease caused by the spirochaete Bor-relia burgdorferi (or related Borrelia spp). rAll cases limited to northern hemisphere (mainly Europe and US). r2000–2000 cases/ yr in UK. Risk of infection from tick bite

is 2–12% in Europe. Pres-entation: ►≤75% remember the tick bite. Peak infection with 48–72h of attachment. Disease stages:

bluish-red plaque/nodule: check earlobes, nipples, genitals. Neuroborreliosis: lymphocytic meningitis, ataxia, amnesia, facial/cranial nerve palsies, neuropathy (severe pain, worse at night), encephalomyelitis. Carditis: acute onset 2nd/2rd-degree heart-block, myocarditis.

Late disseminated (months–yr): acrodermatitis chronic atrophicans = focal inflam-mation then atrophic skin; Lyme arthritis.

Rickettsial disease

Rickettsiae are obligate, intracellular coccobacillary forms lying between bacteria and viruses. Mammals and arthropods are natural hosts. ⚠Risk with rural activities eg camping, hiking, hunting. Divided into:
Other emerging illnesses: eg ehrlichia, anaplasma.

Leishmaniasis

Mucocutaneous (fig 9.22): leads to tissue destruction of nose, mouth, throat. 90% occurs in Bolivia, Brazil, Peru. Reproduced with permis-sion from World Health Organization. © World Health Organization. http:// www.who.int/leishmaniasis/mu-cocutaneous_leishmaniasis/en/
Rhodesiense HAT, incubation <21d, high fever, GI disturbance, lymphadenopathy, headache. Chancre at bite site in r84%, maculopapular rash. Progresses to myo-pericarditis, arrhythmias, and neurological symptoms.

Chagas' disease (American trypanosomiasis)

Anthrax (Bacillus anthracis)
Inhalation: fever, cough, myalgia, SOB, pleural effusion (haemorrhagic mediastini-tis), stridor, death.
Bartonella
B. bacilliformis (bartonellosis): from infected sandflies in Andes mountains. Oroya fever = fever, headache, myalgia, haemolysis. Later nodular➪vascular skin lesions. Diagnosis: Clinical, blood culture (fastiduous m needs prolonged culture), serology. Treatment: Cat-scratch disease often self-limiting. Azithromycin, aminoglycoside.

Brucellosis

Coxiella burnetii (Q fever)

Leptospirosis (Weil's disease)

Bubonic: most common form. Yersinia pestis enters at bite and travels via lym-phatics. Inflamed, painful lymph node is termed 'bubo' and can suppurate.

Diagnosis: Culture bubo fluid, blood, sputum. Rapid antigen testing available.

Table 9.22 VHF classification (HF=haemorrhagic fever)

| Complicating comorbidity Cardiovascular disease: Increased risk of CVD in HIV. Includes individuals where risk traditionally lower: younger age, normotensive, no DM, non-obese. Contrib-uting factors: dyslipidaemia caused by ART, acceleration of pro-atherosclerotic inflammatory processes by HIV. Management of CV risk factors although no out-come data to guide specific lipid and BP targets in HIV. Bone disease: Increased risk of low bone-mineral density and fragility fractures in HIV. Contributing factors: side effect of ART, increased prevalence of risk fac-tors, eg poor nutrition, smoking, alcohol, low vitamin D levels. Risk assess and consider bisphosphonate. TB: All patients with TB and HIV need ART (as soon as TB treatment tolerated and within 2 weeks if CD4 <100 cells/microlitre). Seek expert advice and refer to local guidelines. Consider Truvada® plus efavirenz as 1st line in UK (serum levels of integrase inhibitors are decreased by rifampicin). See ART, pp402–2, TB, pp294–5. Hepatitis B (HBV): Co-infection requires an ART regimen including antivirals with anti-HBV activity, eg tenofovir plus emtricitabine (not lamivudine or emtricit-abine as a single agent due to potential for emergence of HBV resistance). Hepatitis C (HCV): Assess all with HIV for HCV treatment. Pegylated interferon effi-cacy is less with lower CD4 count. Aim for CD4 >500 cells/microlitre with ART first. | | — |

Ebola

Undifferentiated (0–2d). Fever (>28℃ axillary), myalgia, weakness, anorexia, head-ache, sore throat. May not look unwell.

Post-infection: arthralgia, hepatitis, orchitis, transverse myelitis, meningitis, uveitis, vision/hearing impairment, social isolation, psychological effects.

fluid resuscitation, correct electrolytes/coagulation/glu-cose, treat secondary infection, nutrition. Ribavirin not effective. Trace contacts, support family. Experimental:

use of experimental therapy sufficient to guide clinical decisions? Where should re-sources be directed to improve outcome: drug development or basic healthcare provi-sion, eg would the capacity to check K+ improve mortality?

Marburg, Lassa, and CCHF

Table 9.22 Differentiation between VHF

| Presentation | Differential diagnosis | | — | — | | Fever | Intraoral abscess, sinusitis, pneumonia, TB, endocarditis, meningitis, encephalitis, pyomyositis, lymphoma, immune-reconstitution after commencement of ART, any non-HIV cause. | | Lymphadenopathy | Persistent generalized lymphadenopathy (p299), TB, syphilis, histoplas-mosis, cryptococcus, lymphoma, Kaposi's sarcoma, local infection. | | Rash | Drug reaction, herpes zoster, scabies, cutaneous cryptococcus or histoplasmosis, Kaposi's sarcoma, seborrhoeic dermatitis. | | Cough/SOB | Community-acquired pneumonia, Pneumocystis jirovecii, TB, bron-chial compression (TB, lymphoma, Kaposi's sarcoma), pulmonary Kaposi's sarcoma (uncommon), cardiac failure (HIV cardiomyopa-thy, infective pericardial effusion, HIV vasculopathy). | | Diarrhoea | Salmonella, Shigella, Clostridium difficile, amoebiasis, Giardia, Cryptosporidia, CMV, HIV enteropathy is a diagnosis of exclusion. | | Abdominal pain | TB, CMV colitis, pancreatitis (CMV, TB or secondary to ART). Do not forget a pregnancy test. | | Dysphagia | Candidiasis, HSV. | | ⛆Liver enzymes | Viral hepatitis (A, B, C,

CMV, HSV, EBV), drug-induced liver injury (anti-TB or ART), HIV cholangiopathy, lymphoma, congestion due to cardiac disease (pericardial effusion?). | | AKI | Pre-renal due to sepsis/dehydration, interstitial nephritis secondary to medication, HIV-associated nephropathy (proteinuria, CKD). | | Headache/ seizures/focal neurology | Meningitis (bacterial, TB, cryptococcal, syphilis), empyema, space-occupying lesion (toxoplasmosis, lymphoma, tuberculoma), adverse drug reaction, HIV encephalopathy, progressive multifocal leukoen-cephalopathy (PML), stroke (HIV vasculopathy). See p517. | | Eye disease | Herpes zoster, CMV retinitis. See pp428–9. | | Peripheral neuropathy | ART, CMV, HIV neuropathy, nutritional deficiency. |

Transmission: 1 Travel to endemic area (rural for Lassa fever; caves/primates/ante-lopes/bats for Ebola/Marburg; tick/ animal slaughter for CCHF). 2 Travel to known outbreak (http://www.promedmail.org). 3 Contact with infected specimen.

Gastroenteritis = diarrhoea (! vomiting) due to enteric infection with viruses, bac-teria, or parasites.

dysentery: infectious gastroenteritis with bloody diarrhoea

food poisoning: disease (infection or toxin) caused by consumption of food/water.

Table 9.24 Gastroenteritis by infectious aetiology

| Adherence Adherence to ART is associated with drug resistance, disease progression, and death. Adherence support should be integral to ART provision. Assess: Ask about adherence in a non-judgemental way. Do not blame. Explain the reasoning behind your questions. Is non-adherence due to practical problems or healthcare beliefs? Be ready to address both. What help would your patient like? Intervene: Normalize the situation—doubts and concerns about ART are common. Find time for discussion/information. Address concerns. Simplify the dosage regi-men, offer a multicompartment medication system. Link the taking of medication to a regular daily activity. Discuss side-effects: what are the risks/benefits to changing dose or ART regimen? | | — |

Diarrhoea without blood

2010–2015. ©World Health

Diarrhoea with blood (dysentery)

Giardiasis

Cryptosporidium

pogammaglobulinaemia, immunosuppressive therapy.

Amoebiasis

population, mortality r100 000/yr. Faecal–oral spread. .Boil water to destroy cysts.

Intestinal amoebiasis: dysentery (often insidious onset/relapsing), pain, colitis, ap-pendicitis, toxic megacolon.

Amoeboma = inflammatory abdominal mass, usually caecal/RIF ! obstruction.

Cyclospora

Coccidean protozoan Cyclospora cayetanensis. r50 imported cases/yr in UK. Presentation: Flu-like prodrome, watery diar-rhoea, weight loss, marked fatigue, low-grade fever in r25%. Self-limiting after 7–9wks in immunocompetent.

Diagnosis: Autofluorescent oocytes in stool (appear blue-green under UV

Roundworm: Ascaris lumbricoides r1 billion affected,

million affected.

One of most common infections world-wide, affects poor and deprived (fig 9.42). Parasites live in intestines, producing 1000s egg/day in faeces. Humans infected by eggs (ascariasis, trichinosis) or larvae (Ancylostoma) in contaminated food; or via direct penetration of the skin (hook-worm, Strongyloides). Presentation: Diarrhoea, abdominal pain, blood/protein

Table 9.26 Anthelmintic drugs

| Resource-limited settings In many resource-limited settings, universal access to ART remains an objective yet to be achieved. r50% of those in need of treatment for HIV do not receive it. Interim prioritization of those with symptomatic HIV or CD4 count <250 cells/ microlitre may be appropriate as these patients are at high risk of mortality and have most short-term benefit from ART. Equality in the treatment of HIV requires: Effective, acceptable, and reliable methods to reduce HIV transmission, including treatment as prevention. Rapid, accurate, and low-cost diagnosis and monitoring. Standardization and simplification of ART regimens. Evidence-based ART to prevent the use of sub-standard protocols which compro-mise treatment and lead to the emergence of drug-resistant strains. Reduced ART costs and/or effective allocation of resources. | | — |

Taeniasis (tapeworm)

Schistosomiasis (bilharzia)

www.clinmed.rcpjournal.org/content/15/1/67.full.pdf+html. Repro-

Presentation: r50% asymptomatic or non-specific symptoms. Clinical syndromes:

Host response to egg deposition:

Lung disease: pulmonary hypertension and cor pulmonale.

Echinococcosis (hydatid disease)

Alveolar echinococcosis: E. multilocularis. Found in northern hemisphere. Usual hosts are foxes and rodents.

Lung:: dyspnoea, chest pain, cough, haemoptysis.

Silent disease in breast, kidney, adrenals, bladder, heart, psoas.

Other liver flukes: opisthorchiasis and clonorchiasis

Botulism

Tetanus

Caused by anaerobic Clostridium tetani spores universally present in soil. Enters body via a breach in skin. Produces a neurotoxin (tetanospas-min) which disseminates via blood/ lymphatics and interferes with neu-rotransmitter release causing unop-posed muscle contraction and spasm

© Centers for Disease Control and Prevention.

Poliomyelitis

geal swabs, blood, CSF. PCR can differentiate wild-type from vaccine. Paired serol-ogy. Treatment: None. Vaccination: Salk (inactivated, IM) or Sabin (live, oral). In previously endemic areas 200 million volunteers have vaccinated 2 billion children, preventing 1.5 million deaths in the last 20y.

Rabies

'Furious rabies': hyperactivity and terror (hydrophobia, aerophobia)

Japanese encephalitis virus

West Nile virus

Mosquito-borne flavivirus with transmission in Europe, Middle East, Africa, Asia, Australia, and Americas. Presentation: Incubation 2–14d. Asymptomatic in r80%. West Nile fever in r20%: fever, headache, N&V, lymphadenopathy. Neuroinvasive in r1%: encephalitis, meningitis, flaccid paralysis, mortality r10%. Diagnosis: Serum/ CSF IgM, viral PCR. Treatment: Supportive. Vaccine awaited (2016).

Neurocysticercosis

guage disturbance, invol-untary movements. Also headache, visual loss, meningitis, hydrocepha-lus, cognitive impairment. Diagnosis: CT/MRI imaging plus serology. Treatment: Seizure control (evidence on drug choice, length of treatment or prophylaxis).

sponse provoked by treatment—consider dexamethasone.

Conjunctivitis

Common in tropical areas. Vision is normal. Differentiation between common infec-tious causes is outlined in table 9.29.

Trachoma

Immunosuppression and the eye

Herpes zoster ophthalmicus:

CMV retinitis:

©Prof Trobe.

Causes posterior uveitis. Presentation: blurred vision/ floaters. Diagnosis: clinical. Fundoscopy (fig 9.50): focus of choroiditis, chorioretinal scar from previous infection, overlying vitreal haze due to inflammatory response.

PCR. Treatment: atovaquone (toxicity), sulfadiazine, and pyrimethamine.

Adapted from Travel Medicine and Infectious Disease, 7(3), O'Brien, BM, 'A practical approach to com-mon skin problems in returning travellers', 125–46.

Scabies

turnal) pruritus, papular/scaly rash, burrows may be visible (fig 9.55). Itch can lead to secondary bacterial infection.

Diagnosis: Clinical. Skin scraping for mite/ eggs/faeces. Treatment: Topical permethrin 5% or malathion 0.5%.

Ivermectin given for filariasis (p421) may effectively treat concurrent scabies.

Cutaneous leishmaniasis

disease (South America) needs treating due to risk of mucocutaneous disease: pentava-lent antimony, eg meglumine antimoniate, sodium stibogluconate. (See p422.)

University Press.

Myiasis

Tungiasis

https://commons.wikimedia.org/wiki/

Leprosy (Hansen's disease)

Paucibacillary ('tuberculoid'): ♙immune response, granulomata with 🐕bacilli, smears may be –ve.

🐕blinking and lagophthalmos (VII nerve palsy), trauma from eyelashes (trichiasis). Diagnosis: Clinical, +ve skin smear, skin biopsy. Serology unreliable. Treatment: WHO multidrug therapy:

Paucibacillary: rifampicin 600mg/month, dapsone 100mg OD. Duration: 6 months. If single lesion: rifampicin 600mg+ofloxacin 400mg+minocycline 100mg (single dose).

Yaws

Diagnostic criteria

Pyrexia >3wks with no identified cause after evaluation in hospital for 3d or ?3 out-patient visits.

Immunodeficient PUO: Pyrexia in patient with <500 neutrophils/microlitre.

History

Most PUO are due to common diseases with atypical presentation. Consider all de-tails as potentially relevant.

Include: travel (p414), diet, animal contact, changes in medication, recreational drug use, obstetric/sexual history, family history (table 9.20).

Examination

Investigation

Microscopy and culture: Blood ≈2, urine, sputum (including AFB), stool, CSF.

Table 9.20 PUO differential according to history and examination findings

| An HIV vaccine? Vaccines are the most effective way to prevent infectious disease. They can also be therapeutic, clearing a virus after infection. HIV vaccines to date have failed to induce an immune response sufficient to confer

protection. Research is ongoing into neutralizing HIV antibodies, peptides, genes, viral vectors, physiological 'boosters', and mechanisms to counter the mutational evolution of HIV. See www.hvtn.org | | — |

Ask Do not expect to find apposite questions such as these in any other textbook: 1 'Have you delivered any septic babies in the last year?' Impress and cure your obstetrician friends who tell you that 'I'm so depressed about not being able to shake off this flu', and who have forgotten about transfer of brucellosis

'Could that be a Hyalomma tick bite from when you rode your ostrich in last week's race?' Crimean-Congo haemorrhagic fever.

'Has your pet hedgehog lost weight?' Salmonella.

'Did you sample the local frog paella?' Angiostrongyliasis causing eosino-philic meningitis.

'Can I see your pet lobster: he may be the cause of your bad hand?' Lob-sterman's hand, an erysipeloid infection from Erysipelothrix. Pet lobsters have a grand pedigree. Gérard de Nerval used to take his

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Bettmann / Bettmann collection / Getty Images.

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This is an important first question to ask and depends on recognizing characteristic patterns of cognitive, cranial nerve, motor, and sensory deficits. Locating a focal le-sion can be aided by features such as asymmetry (eg one pupil dilated, one upgoing plantar response) or a spinal level (effects may be symmetrical below the lesion).

Note that sometimes there is no single lesion, rather, a general insult causing a falsely localizing sign, eg abducens nerve palsy in •ICP. Other generalized causes of specific local effects are: trauma, encephalitis, anoxia, poisoning, or post-ictal states.

Patterns of loss Are crucial in locating the lesion (see BOX 'From findings to neu-roanatomy'):

Lower motor neuron (LMN): Lesions are caused by damage anywhere from the ante-rior horn cells distally, including the nerve roots, plexuses, and peripheral nerves. The pattern of weakness corresponds to the muscles supplied by the involved neurons.

Affected muscles show wasting ! fasciculation (spontaneous involuntary twitching).

There is hypotonia/flaccidity: the limb feels soft and floppy, providing little resistance to passive stretch. •Reflexes are reduced or absent; the plantars remain flexor. ►The chief differential for LMN weakness is a primary muscle disease (p510)—but here there is symmetrical loss, reflexes are normal or lost late, and there is no sensory component.

Mixed LMN and UMN signs: Can occur, eg in MND, 'B12, taboparesis (see p466).

Most of the fibres of the corticospinal (=pyramidal) tract decussate at the medullary pyramids, hence the name and the contralateral nature of symptoms. Extrapyramidal denotes damage to the basal ganglia and presents as parkinsonism (see p468, 494).

Whereas with rigidity, • tone is not velocity-dependent but constant throughout passive movement.

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The brain is a gland that secretes both thoughts and molecules: both products are modulated by neurotransmitter systems. Some target sites for drugs:

Precursor of the transmitter (eg levodopa).

Interference with the storage of transmitter in vesicles within the pre-synaptic neuron (eg tetrabenazine).

Binding to the post-synaptic receptor site (eg bromocriptine).

Binding to the receptor-modulating site (eg benzodiazepines).

Interference with the breakdown of neurotransmitter within the synaptic cleft (eg acetylcholinesterase inhibitors; monoamine oxidase inhibitors—MAOIS).

Reduce reuptake of transmitter from synaptic cleft into pre-synaptic cell (eg selective serotonin reuptake inhibitors—SSRIS, eg fluoxetine; or serotonin and noradrenaline reuptake inhibitors—SNRIS, eg mirtazapine).

Binding to pre-synaptic autoreceptors (eg pindolol, a β-blocker with partial 5HT au-toreceptor antagonist effects, can be used to augment antidepressant therapy).

Important neurotransmitters (and some associated drugs) are listed in table 10.1.

Table 10.1 Major neurotransmitters and associated drugs

| Muscle weakness grading (MRC classification) Grade 0 No muscle contraction • Grade 2 Active movement against gravity Grade 1 Flicker of contraction • Grade 4 Active movement against resistance • Grade 2 Some active movement • Grade 5 Normal power (allowing for age) Grade 4 covers a big range: 4–, 4, and 4+ denote movement against slight, moder-ate, and stronger resistance; avoid fudging descriptions—'strength 4/5 through-out' suggests a mild quadriparesis or myopathy. It is better to document 'poor effort' and the maximum grade for each muscle tested. ►Distribution of weakness tells us more than grade of weakness (grade does help document improvement). | | — |

*Agonism at D2 receptor agonists may cause pathological behavioural patterns, eg hypersexuality, patho-logical gambling or hobbying, and disorders of impulse control in people having no history of these.

†In chronic alcohol use, GABA receptors are downregulated; acamprosate, used in alcoholism, may help to maintain GABA signalling after alcohol withdrawal.

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Knowledge of the anatomy of the blood supply of the brain helps diagnosis and management of cerebrovascular disease (pp470–8). Always try to identify the area of brain that correlates with a patient's symptoms and identify the affected artery.

Internal carotid arteries Supply the majority of blood to the anterior two-thirds of the cerebral hemispheres and the basal ganglia (via the lenticulostriate arteries). At worst, internal carotid artery occlusion causes fatal total infarction of these areas. More often, the picture is like middle cerebral artery occlusion (see later in topic).

Anterior cerebral artery: (fig 10.2) Supplies the frontal and medial part of the cerebrum. Occlusion may cause a weak, numb contralateral leg ! similar, if milder, arm symptoms. The face is spared. Bilateral infarction is a rare cause of paraplegia and an even rarer cause of akinetic mutism.

Posterior cerebral artery: (figs 10.2, 10.4) Supplies the occipital lobe. Occlusion gives contralateral homonymous hemianopia (often with macula sparing).

Subclavian steal syndrome: Subclavian artery stenosis proximal to the origin of the vertebral artery may cause blood to be stolen by retrograde flow down this vertebral artery down into the arm, causing brainstem ischaemia typically after use of the arm. Suspect if the BP in each arm differs by >20mmHg.

Optic chiasm Internal

carotid artery

Anterior cerebral artery Anterior communicating artery Middle cerebral artery

Superior cerebellar artery

Pontine arteries

Basilar artery

Vertebral artery

Anterior spinal artery

Posterior communicating artery Posterior cerebral artery

Anterior inferior cerebellar artery

Posterior inferior cerebellar artery (PICA)

Fig 10.2 Berry aneurysm at junction of posterior communicating artery with internal carotid (p478). ©Dr D Hamoundi.

While there is some anatomical variation between individuals in ascribing particular nerve roots to muscles, tables 10.2–10.4 represent a reasonable compromise. Der-matomes and sensory nerve roots are shown in figs 10.5-10.9, pp454-5.

Remember to test proximal muscle power: ask the patient to sit from lying, to pull you towards him/herself, and to rise from squatting (if reasonably fit).

Observe walking—easy to forget, even if the complaint is of difficulty walking!

Don’t be caught out by weakness secondary to musculoskeletal pathology—the traditional neurological examination relies on the musculoskeletal system being intact. Ruptured tendons and fractures may mimic focal neurological lesions (espe-cially in patients who can’t give a clear history).

Table 10.2 Assessment of peripheral nerve function in the lower limb

| Storms on the sea of neurotransmission The complex and subtle mixture of chemicals that bathes our hundreds of trillions of synapses has been likened to a ‘sea’ of neurotransmitters. If so, it is surely a seascape of exquisite beauty, no matter how disturbed cognition may become by the storms that whip the waves on the surface. A well-chosen prescription may offer a lifeboat from such storms, but before prescribing any drug that modu-lates neurotransmission, consider that you are about to release a blunt and poorly understood force into a delicate environment. • The drug (or a metabolite) must be able to pass through the blood–brain barrier to have an effect. •The conse-quences of any sedative effects may be severe. • There will be short- and long-term SEs (eg tardive dyskinesia with neuroleptic drugs). • Most drugs affect many neurotransmitters, increasing therapeutic scope (and uncertainty) eg risperidone (blocks D2, 5HT2, a1 and a2 receptors). • Metabolites of drugs may have equal or more important pharmacological effects resulting in clinically important interactions with drugs affecting eg hepatic metabolism. •

One neurotransmitter may have many effects, eg dopaminergic neurons go awry in Parkinson's disease, schizo-phrenia, and addiction to drugs and gambling, by affecting motor control, motiva-tion, effort, reward, analgesia, stress, learning, attention, and cognition. | | — |

Table 10.2 Rapid screening tests for peripheral nerve roots

| Drugs increasing activity (≈ agonists) Drugs decreasing activity (≈ antagonists) | | — | | Dopamine Acts on receptors D1–5; affects mood and reward-seeking behaviour. | | Pramipexole, ropinirole, levodopa, Chlorpromazine (schizophrenia, OHCS p240) apomorphine (Parkinson's*) Metoclopramide (nausea) Cabergoline (hyperprolactinaemia; Inhibition of dopamine signalling may lead to acromegaly). drug induced parkinsonism. | | Serotonin (5-hydroxytryptamine; 5HT) Many receptor types 5HT1–7; multiple effects. | | Lithium (mood stabilizer) Ondansetron (nausea) Sumatriptan (migraine) Mirtazapine (depression) Buspirone (partial agonist; anxiety) Olanzapine, clozapine (schizophrenia) Fluoxetine, sertraline (reuptake inhibi- tors; depression). | | Amino acids Glutamate and aspartate act as excitatory transmitters on NMDA and non-NMDA receptors—relevant in epilepsy and CNS ischaemia. ffk-aminobutyric acid (GABA) is mostly inhibitory. | | Gabapentin, valproate (GABA agonists; Memantine (glutamate antagonist; epilepsy and neuropathic pain) dementia) Benzodiazepines (GABA agonists; seda-tion) Baclofen (GABA agonists; spasticity) Alcohol (GABA agonist)† | | Acetylcholine Multiple receptors classed into muscarinic and nicotinic types. Peripher-al agonists used in glaucoma (pilocarpine); myasthenia (anticholinesterases). Peripheral antagonists used in asthma (ipratropium); incontinence; to dry secretions pre-op; to dilate pupils; to • heart rate (atropine). Centrally acting drugs include: | | Donepezil, galantamine, rivastigmine Procyclidine, trihexyphenidyl (drug-induced (acetylcholinesterase inhibitors; parkinsonism) dementia) | | Histamine and purines (eg ATP) | | Cyclizine (antihistamine; nausea) Purinergic receptor blockers (emerging role in chronic pain). | | Neuropeptides Multiple and growing list; includes opioids and substance P | | Exogenous opioids (wide-ranging anal- Aprepitant ('chemotherapy-related nausea gesic and mood-related effects). by blocking substance P receptors). | | Noradrenaline, adrenaline (=norepinephrine, epinephrine) 4 receptor types: a1–2, β1–2. Noradrenaline is more specific for a-receptors but both transmitters affect all receptors. In the periphery, a-receptors drive arteriolar vasoconstriction and pupillary dilation; β1 stimulation leads to • pulse and myocardial contractility; β2 stimulation leads to bronchodi-latation, uterine relaxation, and arteriolar vasodilation. Centrally acting drugs include: | | Clonidine (refractory hypertension) Tricyclic antidepressants and venlafax-ine (5HT and noradrenaline reuptake inhibitors; depression) MAOIs |

Table 10.4 Assessment of peripheral nerve function in the upper limb

| Thomas Willis Thomas Willis (1621–1675) is one of those happy Oxford heroes who hold a bogus DM degree, awarded in 1646 for his Royalist sympathies while at Christ Church, the most loyally royal college in the University. He had a busy life inventing terms such as 'neurology' and 'reflex'. Not only has his name been given to his famous circle, but he was the first to describe myasthenia gravis, whooping cough, and the sweet taste of diabetic urine. He was the first person (few have followed him) to know the course of the spinal accessory nerve. He is unusual among Oxford neurologists in that he developed the practice of giving his lunch away to the poor. He also espoused iatrochemistry: a theory of medicine ac-cording to which all morbid conditions of the body can be explained by distur-bances in the fermentations and effervescences of its humours. | | — |

*Also, metacarpophalangeal joint flexion may be more on the affected side as flexor tendons are recruited—the basis of Froment's paper sign. Wartenberg's sign is persistent little finger abduction.

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Trigeminal

V

Ophthalmic division

Maxillary division V

Mandibular division

Great auricular, C2, C2 V2

Cervical plexus, superficial branches

Intercostal nerves

T2–T11

Axillary

Lateral cutaneous br. Anterior cutaneous br.

C2

C4

T2

T4 T2 C5

Brachial

Intercostobrachial Medial cutaneous Musculocutaneous

Posterior

T5

T6 T7

T8 T1

C6 & C7

plexus

Radial

Median Ulnar

cutaneous

Superficial branch

T9

T10 T11

T12

L1

C7

Lumbar plexus

Ilioinguinal L2

Genitofemoral

Lateral cutaneous

Anterior

Femoral

cutaneous rami

Saphenous

L2

L4 L5

Sacral Sciatic

Lateral cutaneous nerve of calf

Superficial and deep peroneal

S1

Sural

Medial plantar

Fig 10.5 The white areas denote terra incognita: considerable inter-individual variation exists, and no single best option can be given.

Fig 10.6 Pain in a dermatomal dis-tribution suggests a problem with a cranial nerve or dorsal root ganglion (radiculopathy)—where the cell bod-ies of sensory fibres live. What is the dermatome? What is the lesion? See

Ophthalmic division

C2 Maxillary division Trigeminal Mandibular division

Mastoid branch, C2, C2

C2 Great auricular branch, C2, C2

Occipital, C2

Superficial cervical plexus

C4

C5 T2

Occipital, C2 Occipital, C4 Occipital, C5–C8

Dorsal branches

T2

T5 T4

Supraclavicular, C2, C4

Dorsal rami of thoracic nerves

C7

T12

T6

T7 T8

T9 T10 T11

Cutaneous branch of axillary

Lateral cutaneous branches of intercostal nerves

Medial and lateral cutaneous br. of radial Medial cutaneous Intercostobrachial

Musculocutaneous

T1

C6 & C7 C8 S2 S5

L1 L2 L2

L2

Superficial branch of radial

Gluteal branch of 12th intercostal Lateral cutaneous br. of iliohypogastric Lateral branches of dorsal

lumbar and sacral rami

Medial branches of dorsal rami, L.1–S.6

S1 Perforating branch of

posterior cutaneous

S2 Lateral cutaneous

Obturator

L4 Medial cutaneous Femoral

Saphenous

Posterior cutaneous

Pudendal plexus

Lumbar plexus

S1

Superficial peroneal Sural

Tibial

Lateral plantar

Common peroneal

Sacral plexus

Cutaneous innervation of the foot

Saphenous nerve 1 Deep peroneal nerve 2 Superficial peroneal nerve 2 Medial plantar nerve 4

Lateral plantar nerve 5

Dorsal surface

1

2

Plantar surface

5 4

7 7

Calcaneal branch 6 2

(tibial nerve) 6

Sural nerve 7

Superficial branch of radial

Median

Palmar surface

Dorsal surface

Ulnar

Dorsal cutaneous branch of ulnar

Median

Every day, thousands of people visit the doctor complaining of headache. Tension headaches are the most common, but beware the disabling and treatable (migraine, cluster headache), and the sinister (space-occupying lesions, meningitis, subarach-noid haemorrhage). A good history is the key. Ask about:

Onset Rapid onset headaches are concerning; the key diagnosis to rule out here is •subarachnoid haemorrhage (SAH, p478), sudden-onset, ‘worst ever’ headache, often occipital, stiff neck, focal signs, consciousness. Other differentials include:

Meningitis (p822): fever, photophobia, stiff neck, purpuric rash, coma. May be associated with neck stiffness (≈ meningeal irritation). Do an LP, start antibiotics.

Tropical illness: eg malaria: travel history, flu-like illness (p416); typhus (p415). •In-tracranial hypotension: CSF leakage, eg iatrogenic after LP or epidural anaesthesia. Suspect if headaches worse on standing;. treat with epidural blood patch over leak, if conservative management with IV fluids and caffeine fails.

Character Tight band? Think tension headache (the usual cause of bilateral, non-pulsatile headache ! scalp muscle tenderness). Throbbing/pulsatile/lateralizing? Think migraine (p458).

Duration Chronic, progressive headaches can indicate • ICP. Typically worse on waking, lying, bending forward, or coughing. Also: vomiting, papilloedema, sei-zures, false localizing signs, or odd behaviour. Do imaging to exclude a space-occupying lesion, and consider idiopathic intracranial hypertension. ►LP is contraindicated until after imaging.

Precipitating causes Head trauma: Commonly causes localized pain but can be more generalized. It lasts r2wks; often resistant to analgesia. Do CT to exclude subdural or extradural haemorrhage if drowsiness ! lucid interval, or focal signs (p482). Also ask about: Analgesia, sex, food (eg chocolate, cheese, coffee).

Red flags See p780.

Social history Ask about stress or recent life events; may not explain the pathol-ogy, but will help you appreciate the context in which symptoms are experienced.

15% of us suffer from migraines, (φ:ð ғ 2: 1); the economic costs extend to £billions/yr. Symptoms Classically: •Visual or other aura (see below) lasting 15–20min followed within 1h by unilateral, throbbing headache. Or: •Isolated aura with no headache;

Partial triggers Seen in 50%: CHOCOLATE or: chocolate, hangovers, orgasms, cheese/caffeine, oral contraceptives, lie-ins, alcohol, travel, or exercise.

Associations Obesity, family history.

Diagnosis Clinical, based on the history. Diagnostic criteria if no aura: ?5 head-aches lasting 4–72h + nausea/vomiting (or photo/phonophobia) + any 2 of: • Unilateral

Pulsating • Impairs (or worsened by) routine activity.

Differentials Cluster or tension headache, cervical spondylosis, •BP, intracranial pa-thology, sinusitis/otitis media, dental caries. TIAS may mimic migraine aura.

the head, or rebreathing into paper bag (•Pa CO_2) may help abort attacks. Butterbur extracts or riboflavin supplementation may have a role. NICE recommend 10 sessions of acupuncture over 5–8 weeks if both topiramate and propranolol are unsuitable or ineffective. Transcutaneous nerve stimulation may help.

Causes of collapse ! loss of consciousness (LOC) are many; take a careful history (BOX).

Situation syncope Symptoms as for vasovagal syncope but with a clear precipi-tant: cough syncope occurs after a paroxysm of coughing; effort syncope is brought on by exercise; there is usually a cardiac cause, eg aortic stenosis, HCM; micturition syncope happens during or after urination: mostly men, at night.

Carotid sinus syncope Hypersensitive baroreceptors cause excessive reflex brady-cardia ! vasodilation on minimal stimulation (eg head-turning, shaving).

Epilepsy (p490) Features suggestive of this diagnosis include: attacks when asleep or lying down; aura; identifiable triggers (eg TV); altered breathing; cyanosis; typical tonic–clonic movements; incontinence of urine; tongue-biting; prolonged post-ictal drowsiness, confusion, amnesia, and transient focal paralysis (Todd's palsy).

Examination Cardiovascular, neurological. Measure BP lying and standing.

While the cause is being elucidated, advise against driving (see p158).

Patient is subject to continuous ECG and BP monitoring while strapped to a table and moved rapidly from resting horizontal position to vertical. Induction of symptoms with inappropriate BP drop >20mmHg or bradycardia suggests neurally mediated syncope. Consider pacing.

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Associated symptoms: Difficulty walking or standing (may fall suddenly to the ground), relief on lying or sitting still (vertigo is almost always worsened by move-ment); nausea, vomiting, pallor, sweating. Associated hearing loss or tinnitus implies labyrinth or VIIIth nerve involvement.

Causes

Benign positional vertigo: Occurs on head movement due to disruption of debris in the semicircular canal of the ears (canalolithiasis). Fatiguable nystagmus on performing the Hallpike manoeuvre is diagnostic; Epley manoeuvres clear the debris (OHCS p555).

Acute labyrinthitis (vestibular neuronitis): Abrupt onset of severe vertigo, nausea, vomiting ! prostration. No deafness or tinnitus. Causes: virus; vascular lesion. Severe vertigo subsides in days, complete recovery takes 2–4wks. 9: reassure. Sedate.

Ototoxicity: Aminoglycosides, loop diuretics, or cisplatin can cause deafness ! vertigo. Acoustic neuroma: (figs 10.12, 10.12) Doubly misnamed: it is a Schwannoma (not neuroma) arising from the vestibular (not auditory) nerve. They account for 80%

Traumatic damage: If trauma affects the petrous temporal bone or the cerebello-pontine angle then the auditory nerve may be damaged, causing vertigo, deafness, and/or tinnitus.

Herpes zoster: Herpetic eruption of the external auditory meatus; facial palsy ! deaf-ness, tinnitus, and vertigo (Ramsay Hunt syndrome, see p501).

Others: Vertiginous epilepsy; MS; stroke/TIA; migraine; motion sickness; alcohol in-toxication.

University Press.

Whisper test A simple but effective crude assessment of hearing: whisper numbers in one ear while blocking the other. Ask your patient to repeat the number. Make sure that failure is not from misunderstanding.

Conductive deafness Causes: Wax (remove, eg by syringing with warm water af-ter softening with olive oil drops), otosclerosis, otitis media, or glue ear (OHCS p546).

Chronic sensorineural deafness Often due to accumulated environmental noise toxicity, presbyacusis, or inherited disorders. Presbyacusis: Loss of acuity for high-frequency sounds starts before 20yrs old. We do not usually notice it until hearing of speech is affected. Hearing is most affected in the presence of background noise. Hearing aids are the usual treatment.

Sudden sensorineural deafness ►Get an ENT opinion today (steroids may cure)! Causes: Noise exposure; gentamicin/ other toxin; mumps; acoustic neuroma; MS; stroke; vasculitis; TB. Tests: ESR, FBC, LFT, pANCA, viral titres and TB (see BOX ‘Diagnostic tests for TB’, p294); evoked response audiometry; CXR; MRI; lymph node and naso-pharyngeal biopsy for culture.

This ringing or buzzing in the ears is common, and may cause depression or insomnia.

Investigate unilateral tinnitus fully to exclude an acoustic neuroma (p462).

Causes Inner ear damage and hearing loss (leading to auditory cortex hyper-excit-ability), wax, excess noise, head injury, otitis media, post-stapedectomy, Ménière’s, anaemia (if pulsatile then think of carotid artery stenosis or

dissection, AV fistulae, and glomus jugulare tumours). Drugs: Aspirin (reversible), loop diuretics, aminogly-cosides. Mean age at onset: 40–50yrs. ♂:♀≈1:1.

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cauda equina.)

- What is the lesion? • Was onset sudden or rapidly progressive? ►This is an emer-gency; it suggests cord compression so get urgent help (see next paragraph). • Are there any signs of infection (eg tender spine, •T°, •WCC, •ESR, •CRP: extradural abscess)? Cord compression (See also p528.) Symptoms: Bilateral leg weakness (arm weak-

ness—often less severe—suggests a cervical cord lesion, see p508) a sensory level ! preceding back pain (see p542). Bladder (and anal) sphincter involvement is late and manifests as hesitancy, frequency, and, later, as painless retention. Signs: Look for

Other patterns of leg weakness

Unilateral foot drop: DM, common peroneal nerve palsy, stroke, prolapsed disc, MS.

Weak legs with no sensory loss: MND, polio, parasagittal meningioma (an exception to the rule that weak legs mean cord or distal lesion).

Chronic spastic paraparesis: MS, cord primary malignancy/metastasis, MND, syrin-gomyelia, subacute combined degeneration of the cord (p224), hereditary spastic paraparesis, taboparesis (tertiary syphilis, see p412), histiocytosis X, parasites (eg schistosomiasis).

Chronic flaccid paraparesis: Peripheral neuropathy, myopathy.

Absent knee jerks and extensor plantars: (Ie combined LMN or UMN signs.) Com-bined cervical and lumbar disc disease, conus medullaris lesions, MND, myeloradicu-litis, Friedreich's ataxia, subacute combined degeneration of the cord, taboparesis.4

Tertiary syphilis (p412): in tabes dorsalis the afferent pathways from muscle spindles are lost, with reduced tone and tendon reflexes (without weakness). Later, additional involvement of the pyramidal tracts causes taboparesis—a spastic paraparesis with the peculiar combination of extensor plantars (from the taboparesis) and absent tendon reflexes (from the tabes dorsalis).

These are characterized by impairment of the planning, control, or execution of movement. They can have multiple manifestations:

Tremor Note frequency, amplitude, and exacerbating factors (stress; fatigue).

Rest tremor: Abolished on voluntary movement. Cause: parkinsonism (p494).

Intention tremor: Irregular, large-amplitude, worse at the end of purposeful acts, eg finger-pointing or using a remote control. Cause: cerebellar damage (eg MS, stroke).

Postural tremor: Absent at rest, present on maintained posture (arms outstretched) and may persist (but is not worse) on movement. Causes: benign essential tremor (autosomal dominant; improves with alcohol), thyrotoxicosis, anxiety, β-agonists.

Re-emergent tremor: Postural tremor developing after a delay of r10s. Causes:

Parkinson's disease (don't mistake for essential tremor).

Chorea Non-rhythmic, jerky, purposeless movements flitting from one place to another—eg facial grimacing, raising the shoulders, flexing/extending the fingers. Causes: Huntington's disease, Sydenham's chorea (rare complication of group A streptococcal infection). Worsened by levodopa.

Hemiballismus Large-amplitude, flinging hemichorea (affects proximal muscles) contralateral to a vascular lesion of the subthalamic nucleus (often elderly diabet-ics). Recovers spontaneously over months.

Athetosis Slow, sinuous, confluent, purposeless movements (especially digits, hands, face, tongue), often difficult to distinguish from chorea. Causes: Common-est is cerebral palsy (OHCS p214). Most other 'athetoid' patterns may now be better classed as dystonias. Pseudoathetosis: Caused by severe proprioceptive loss.

Tics Brief, repeated, stereotyped movements which patients may suppress for a while. Tics are common in children (and usually resolve). In Tourette's syndrome (p700), motor and vocal tics occur. Consider psychological support, clonazepam or clonidine if tics are severe (haloperidol may help but risks tardive dyskinesia).

Infarction or bleeding into the brain manifests with sudden-onset focal CNS signs. Someone in the UK has a stroke every 2.5 minutes; 1 in 4 of those will die within a year and half of survivors will have a permanent disability.

Causes • Small vessel occlusion/cerebral microangiopathy or thrombosis in situ.

Differentials Head injury, hypo/hyperglycaemia, subdural haemorrhage, intracranial tumours, hemiplegic migraine, post-ictal (Todd's palsy), CNS lymphoma, Wernicke's encephalopathy, hepatic encephalopathy, encephalitis, toxoplasmosis, cerebral ab-scesses, mycotic aneurysm, drug overdose (if comatose).

Modifiable risk factors ↑BP, smoking, DM, heart disease (valvular, ischaemic, AF), peripheral vascular disease, ↑PCV, carotid bruit, combined OCP, •lipids, •alcohol use,

clotting (eg •plasma fibrinogen, ↷antithrombin III, p274), •homocysteine, syphilis.

Acute management ► Protect the airway: This avoids hypoxia/aspiration.

Maintain homeostasis: Blood glucose: keep between 4–11 mmol/L. Blood pressure: only treat if there is a hypertensive emergency (eg encephalopathy or aortic dissec-tion) or thrombolysis is considered (ideally aim for ≤185/110) as treating even very high BPS may impair cerebral perfusion.

Screen swallow: 'Nil by mouth' until this is done (but keep hydrated).

Antiplatelet agents: Once haemorrhagic stroke is excluded, give aspirin 200mg (continue for 2 weeks, then switch to long-term antithrombotic treatment, p472).

Thrombectomy: Intra-arterial mechanical thrombectomy provides additional benefit for those with large artery occlusion in the proximal anterior circulation.

Admit to an acute stroke unit: multidisciplinary care improves outcomes (p474).

Fig 10.15 The T2-weighted (p746) image on the left shows oedema in the right occipital lobe. Dif-ferentials: infarct (right PCA), inflammation, or tumour. The diffusion-weighted image on the right shows limited diffusion in the region, indicating this is an infarct. ©Prof Peter Scally.

Cerebral Autosomal Dominant Arteriopathy with Subcortical Infarcts & Leucoencephalopathy: the main genetic cause of stroke (there is also an autosomal recessive form).

If +ve, register at SITS, www.sitsinternational.org

Primary prevention (Ie before any stroke.)

Control risk factors (p470): look for and treat hypertension, DM, •lipids (p690), cardiac disease (see BOX 'Cardiac causes of stroke') and help quit smoking (see p92). Exercise helps (•HDL, •glucose tolerance). Use lifelong anticoagulation in AF (see BOX 'Cardiac causes of stroke') and prosthetic heart valves. • For prevention post-TIA see p476.

Secondary prevention (Ie preventing further strokes.)

Tests (See p470 for imaging.) Investigate promptly to identify risk factors for fur-ther strokes, but consider whether results will affect management. Look for:

Hypertension. Look for retinopathy (p560), nephropathy, or cardiomegaly on CXR.

Cardiac source of emboli. (See BOX ‘Cardiac causes of stroke’.) 24h ECG to look for AF (p120). CXR may show an enlarged left atrium. Echocardiogram may reveal mural thrombus due to AF or a hypokinetic segment of cardiac muscle post-MI. It may also show valvular lesions in infective endocarditis or rheumatic heart disease. Transoesophageal echo is more sensitive than transthoracic.

Hypoglycaemia, hyperglycaemia, dyslipidaemia, and hyperhomocysteinaemia.

Vasculitis. •ESR, ANCA (p556). VDRL to look for active, untreated syphilis (p412).

Prothrombotic states, eg thrombophilia (p274), antiphospholipid syndrome (p554).

Hyperviscosity, eg polycythaemia (p266), sickle-cell disease (p240).

Thrombocytopenia and other bleeding disorders.

Genetic tests. CADASIL (p470); Fabry disease (p698).

Prognosis Overall mortality: 60 000/yr; UK 20% at 1 month, then ≡10%/yr. Full recov-ery: ≤40%. Drowsiness ≈ poor prognosis. Avoid pressure ulcers (fig 10.16).

Images (a) to (d) reproduced from Gosney et al., Oxford Desk Reference: Geriatric Medicine, 2012, with permission from Oxford University Press.

Setting achievable goals and acknowledging the patient’s own agenda is key.

Imperatives for re-enablement

Avoid further injury: minimize falls risk and take care when lifting the patient not to damage their shoulders.

Ensure good bladder and bowel care through frequent toileting. Avoid early cath-eterization which may prevent return to continence.

Position to minimize spasticity (occurs in r40%). Get prompt physiotherapy. Splints and botulinum toxin injections are helpful for focal spasticity.

Monitor progress: eg measure time taken to sit up and transfer to chair.

Monitor mood: in pseudo-emotionalism/emotional lability (sobbing unprovoked by sorrow, from failure of cortical inhibition of the limbic system), tricyclics or fluoxetine may help.

Engage the patient in their own recovery by making physiotherapy fun. Swimming (a hemiplegic arm may be supported on a special float), music, and video games are all enjoyable and • recovery through promoting cerebral reorganization. Constraint of the good arm may be helpful.

Involve the carer/spouse with all aspects of care-giving. Good rehab saves lives.

End-of-life decisions ►See p12.

This is an ischaemic (usually embolic) neurological event with symptoms lasting <24h (often much shorter). ► Without intervention, more than 1 in 12 patients will go on to have a stroke within a week, so prompt management is imperative.

Causes (See p470.) • Atherothromboembolism from the carotid is the chief cause: listen for bruits (though not a sensitive test). • Cardioembolism: mural thrombus post-MI or in AF, valve disease, prosthetic valve (p472). • Hyperviscosity: eg poly-cythaemia, sickle-cell anaemia, myeloma. • Vasculitis is a rare, non-embolic cause of TIA symptoms (eg cranial arteritis, PAN, SLE, syphilis, etc.).

Differentials Hypoglycaemia, migraine aura (p458), focal epilepsy (symptoms spread over seconds and often include twitching and jerking), hyperventilation, reti-nal bleeds. Rare mimics of TIA: Malignant hypertension, MS (paroxysmal dysarthria), intracranial tumours, peripheral neuropathy, phaeochromocytoma, somatization.

Tests FBC, ESR, U&Es, glucose, lipids, CXR, ECG, carotid Doppler ! angiography, CT or diffusion-weighted MRI, echocardiogram.

Treatment

Control cardiovascular risk factors: Optimize: BP (cautiously lower; aim for <140/85mmHg, p140); hyperlipidaemia (p690); DM (p206); help to stop smoking (p92).

Antiplatelet drugs: As with stroke, give aspirin 200mg OD for 2wks, then switch to clopidogrel 75mg OD. If this is contraindicated or not tolerated, give aspirin 75mg OD combined with slow-release dipyridamole.

Anticoagulation indications: Cardiac source of emboli (see p472).

Driving Prohibited for at least 1 month, see p158.

Prognosis Long-term risks of stroke or cardiovascular events following TIAs are de-pendent on underlying vascular risk factors: calculate using the ABCD2 score (see BOX and table 10.6).

Spontaneous bleeding into the subarachnoid space, often catastrophic (table 10.7).

Incidence 9/100 000/yr; typical age: 25–65.

Symptoms Sudden-onset excruciating headache, typically occipital—like a 'thun-derclap'. Vomiting, collapse, seizures, and coma often follow. Coma/drowsiness may last for days. Some patients report a preceding, 'sentinel' headache, perhaps due to a small warning leak from the offending aneurysm (r6%).

Signs Neck stiffness; Kernig's sign (takes 6h to develop); retinal, subhyaloid and vit-reous bleeds (=Terson's syndrome; •mortality ≈ 5). Focal neurology at presentation may suggest site of aneurysm (eg pupil changes indicating a IIIrd nerve palsy with a posterior communicating artery aneurysm) or intracerebral haematoma. Later deficits suggest complications (see later in topic).

Risk factors Previous aneurysmal SAH (new aneurysms form, old ones get bigger), smoking, alcohol misuse, •BP, bleeding disorders, SBE (mycotic aneurysm), family his-tory (2–5x •risk of SAH in close relatives). Polycystic kidneys, aortic coarctation, and Ehlers–Danlos syndrome (p149) are all associated with berry aneurysms.

Differentials Meningitis (p822), migraine (p458), intracerebral bleed, cortical vein thrombosis (p480), dissection of a carotid or vertebral artery, benign thunderclap headache (triggered by Valsalva manoeuvre, eg cough, coitus).

Tests • Urgent CT: Detects >95% of SAH within the 1st 24h (fig 10.18). • Consider LP: If CT –ve but the history is very suggestive of SAH (and no CI: p768). This needs to be done >12h after headache onset to allow breakdown of RBCs so that a positive sample is xanthrochromic (yellow, due to bilirubin: differentiaties between old blood from SAH vs a 'bloody tap').

Management ►Refer all proven SAH to neurosurgery immediately.

Re-examine CNS often; chart BP, pupils, and GCS (p788). Repeat CT if deteriorating.

Maintain cerebral perfusion by keeping well hydrated, but aim for SBP <160mmHg.

Nimodipine (60mg/4h PO for 2wks, or 1mg/h IVI) is a Ca2+ antagonist that reduces vasospasm and consequent morbidity from cerebral ischaemia.

Table 10.7 Mortality in subarachnoid haemorrhage

| Nerve root Muscle Test by asking the patient to: | | — | | Femoral nerve L1, 2, 2 Iliopsoas (also supplied via Flex hip against resistance with knee L1, 2, & 2 spinal nerves) flexed and lower leg supported: patient lies on back L2, 2, 4 Quadriceps femoris Extend at knee against resistance. Start with knee flexed Obturator nerve L2, 2, 4 Hip adductors Adduct leg against resistance Inferior gluteal nerve L5, S1, S2 Gluteus maximus Hip extension ('bury heel into the couch')—with knee in extension Superior gluteal nerve L4, 5, S1 Gluteus medius and minimus Abduction and internal

hip rotation with leg flexed at hip and knee Sciatic and common peroneal* nerves; sciatic and tibial** nerves *L4, 5 Tibialis anterior Dorsiflex ankle *L5, S1 Extensor digitorum longus Dorsiflex toes against resistance | | *L5, S1 Extensor hallucis longus Dorsiflex hallux against resistance *L5, S1 Peroneus longus and brevis Evert foot against resistance | | *L5, S1 Extensor digitorum brevis Dorsiflex proximal phalanges of toes | | (*)L5, S1, 2 Hamstrings (short head of Flex knee against resistance biceps femoris is from the common peroneal nerve) | | **L4, 5 Tibialis posterior Invert plantarflexed foot | | **S1, 2 Gastrocnemius Plantarflex ankle or stand on tiptoe | | **L5, S1, 2 Flexor digitorum longus Flex terminal joints of toes | | **S1, 2 Small muscles of foot Make the sole of the foot into a cup |

Most mortality occurs in 1st month. 90% of survivors of the 1st month, survive >1 year.

Fig 10.17 CT images can be manipulated to show only high-density structures such as bones and arteries containing contrast. Here is a middle cerebral artery aneurysm.

We thank Prof. Peter Scally for these CT images and

the commentaries on them.

Fig 10.18 Blood from a ruptured aneurysm oc-cupies the interhemispheric fissure (top arrow), a crescentic intracerebral area presumably near the aneurysm (2nd arrow), the basal cisterns, the lateral ventricles (temporal horns), and the 4th ventricle (bottom arrow).

We thank Prof. Peter Scally for these CT images and

the commentaries on them.

Thrombosis of the cerebral sinuses or veins causes cerebral infarction, though much less commonly than arterial disease. Seizures are common and focal; they can com-plicate diagnosis and post-ictal drowsiness may impair GCS assessment. Although r80% will make a good functional recovery, death is mainly due to transtentorial herniation from mass effect or oedema.9

Dural venous sinus thrombosis Most commonly sagittal sinus thrombosis (figs 10.19, 10.20; 47% of all IVT) or transverse sinus thrombosis (25%). Sagittal sinus thrombosis often coexists if other sinuses are thrombosed. Symptom onset is gradu-al (over days or weeks). Features are dependent on the sinus affected:

Sagittal sinus: Headache, vomiting, seizures, ‘vision, papilloedema.

Transverse sinus: Headache ! mastoid pain, focal CNS signs, seizures, papilloedema.

Sigmoid sinus: Cerebellar signs, lower cranial nerve palsies.

Inferior petrosal sinus: Vth and VIth cranial nerve palsies, with temporal and retro-orbital pain (Gradenigo’s syndrome,. suggesting otitis media is the cause).

Cavernous sinus: Often due to spread from facial pustules or folliculitis, causing headache, chemosis, oedematous eyelids, proptosis, painful ophthalmoplegia, fever.

Cortical vein thrombosis (CVT) Usually occurs with a sinus thrombus as it extends into the cortical veins, causing infarction in a venous territory (fig 10.21). These infarcts give rise to stroke-like focal symptoms that develop over days. There are often seizures, and an associated headache which may come on suddenly (thunderclap headache).

Differential diagnosis Subarachnoid haemorrhage, meningitis, encephalitis, in-tracranial abscess, arterial infarction.

Predictors of poor prognosis include: GCS score on admission <9, deep CVT location, CNS infection, malig-nancy, intracranial haemorrhage, mental status abnormality, age >27 years, and ð.

Fig 10.19 This magnetic resonance venogram (MRV) could look normal at first glance: the hardest thing to see in imaging is often that which is not there. Much of the superior sagittal sinus is not seen because it is filled with clot—a superior sagittal sinus thrombo-sis. The arrows point to where it should be seen. Posteriorly, the irregularity of the vessel indicates non-occlusive clot.

of Prof. P. Scally.

Fig 10.21 Venous territories (compare with arterial territories on p451). SSS—superior sagittal sinus; TS—transverse sinus; SV—Sylvian veins; ICV—internal cortical veins.

There is much greater variation in venous anatomy between individuals than there is in arterial anatomy, so this diagram is only a rough guide. The key point is to realize that infarction that crosses boundaries between arterial territories may be venous in origin.

Symptoms Fluctuating level of consciousness (seen in 25%) ! insidious physical or intellectual slowing, sleepiness, headache, personality change, and unsteadiness.

Signs •ICP (p820), seizures. Localizing neurological symptoms (eg unequal pupils, hemiparesis) occur late, often >1 month after the injury.

Differentials Stroke, dementia, CNS masses (eg tumours, abscesses).

Imaging (fig 10.22) CT/MRI shows clot ! midline shift (but beware bilateral isodense clots). Look for crescent-shaped collection of blood over 1 hemisphere. The sickle-shape differentiates subdural blood from extradural haemorrhage.

Management Reverse clotting abnormalities urgently. Surgical management de-pends on the size of the clot, its chronicity, and the clinical picture: generally those

>10mm or with midline shift >5mm need evacuating (via craniotomy or burr hole washout). Address the cause of the trauma (eg falls, abuse).

Beware deteriorating consciousness after any head injury that initially produced no loss of consciousness or after initial drowsiness post injury seems to have re-solved. This lucid interval pattern is typical of extradural bleeds.

Cause ►Suspect after any traumatic skull fracure. Often due to a fractured temporal or parietal bone causing laceration of the middle meningeal artery and vein, typically after trauma to a temple just lateral to the eye. Any tear in a dural venous sinus will also result in an extradural bleed. Blood accumulates between bone and dura.

Management Stabilize and transfer urgently (with skilled medical and nursing escorts) to a neurosurgical unit for clot evacuation ! ligation of the bleeding ves-sel. Care of the airway in an unconscious patient and measures to ‘ICP often re-quire intubation and ventilation (+ mannitol IVI, p821).

We thank Prof. Peter Scally for these CT images and commentary.

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Risk factors >65y, dementia/previous cognitive impairment, hip fracture, acute ill-ness, psychological agitation (eg pain).

Causes

Systemic infection: pneumonia, UTI, malaria, wounds, IV lines.

Alcohol withdrawal (2–5d post-admission; •LFTS, •MCV; history of alcohol abuse).

Vascular: stroke, myocardial infarction.

Tests Look for the cause (eg UTI, pneumonia, MI): do FBC, U&E, LFT, blood glucose, ABG, septic screen (urine dipstick, CXR, blood cultures); also consider ECG, malaria films, LP, EEG, CT.

Encourage visits from friends and family.

Practise sleep hygeine: restrict daytime napping, minimize night-time disturbance.

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Other causes Alcohol/drug abuse; repeated head trauma; pellagra (p268), Whip-ple's disease (p716); Huntington's (p702); CJD (p696); Parkinson's (p494); HIV; crypto-coccosis (p408); familial autosomal dominant Alzheimer's; CADASIL (p470).

Dilated ventricles without enlarged sulci. Signs: gait apraxia, incontinence, dementia; CSF shunts help.

Refer to a specialist memory service. • Acetylcholinesterase inhibitors (see BOX 'Pharmacological treatment'). • BP control (in heart failure there is a 2x •risk of AD; extra risk halves with BP control).

'Enduring' doesn't mean unfluctuating: cognition comes and goes, allowing poetic insights, as in Iris Murdoch's poignant self-diagnosis: 'I am sailing into the dark'.

>40 different types of epilepsy). NICE estimate 5–20% of people with 'epilepsy' have been wrongly diagnosed. ► All patients with a seizure must be referred for specialist assessment and investigation in <2 wks.

Counselling After any 'fit', advise about dangers, eg swimming, driving, heights until the diagnosis is known; then give individualized counselling on employment, sport, insurance, and conception (OHCS p28). The patient must contact DVLA and avoid driving until seizure-free for >1yr (p159).

Focal (partial) seizures: 1st line: carbamazepine or lamotrigine. 2nd line: leveti-racetam, oxcarbazepine, or sodium valproate.12

Myoclonic seizures: 1st line: sodium valproate.12 2nd line: levetiracetam, or topira-mate (but •SE). Avoid carbamazepine and oxcarbazepine—may worsen seizures.

Stopping AEDS: May be done under specialist supervision if the patient has been seizure-free for >2yrs and after assessing risks and benefits for the individual (eg the need to drive). The dose must be decreased slowly: over at least 2–2 months, or

Sudden unexpected death in epilepsy (SUDEP) More common in uncontrolled epilepsy, and may be related to nocturnal seizure-associated apnoea or asystole. Those with epilepsy have 2x •mortality. >700 epilepsy-related deaths are recorded/ yr in the UK; up to 17% are SUDEPs. The charity SUDEP Action may be of some help to families.

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Diagnosis This is clinical, made by a consultant neurologist using established crite-ria (eg McDonald, see table 10.9) and after alternative diagnoses have been excluded.

Table 10.8 Clinical features of MS

| Shoulder | Abduction | C5 | | Flexion | L1–L2 | | — | — | — | — | — | — | | Shoulder | Adduction | C5–C7 | Hip | Adduction | L2–2 | | Elbow | Flexion | C5–C6 | | Extension | L5–S1 | | Elbow | Extension | C7 | Knee | Flexion | L5–S1 | | Wrist | Flexion | C7–8 | Knee | Extension | L2–L4 | | Wrist | Extension | C7 | | Dorsiflexion | L4 | | | Flexion | C8 | Ankle | Eversion | L5–S1 | | Fingers | Extension | C7 | | Plantarflexion | S1–S2 | | | Abduction | T1 | Toe | Big toe extension | L5 |

Evolving focal neurology: See BOX for localizing signs. •ICP causes false localizing signs: VIth nerve palsy is commonest (p70) due to its long intracranial course.

Differentials Stroke, head injury, venous sinus thrombosis, vasculitis, MS, encephali-tis, post-ictal, metabolic, or idiopathic intracranial hypertension.

Prognosis Poor but improving (<50% survival at 5yrs) for CNS primaries; 40% 20yr survival for cerebellar haemangioblastoma; benign tumours are curable by excision.

Associations Endocrine abnormalities (Cushing's syndrome, hypoparathyroidism,

Prognosis Often self-limiting. Permanent significant visual loss in 10%.

Lyme disease, Guillain–Barré, sarcoid, and trauma often cause bilateral weakness.

Sphenopalatine ganglion 16 Chorda tympani

Spinal nucleus of V

Porus acusticus internus

Lacrimal gland

Nose/palate gland nerves

18 Stylomastoid foramen

20 Submandibular gan-glion (and gland, 21)

Lesions of individual peripheral or cranial nerves. Causes are usually local, such as trauma,or entrapment (eg tumour), except for carpal tunnel syndrome (see BOX 'Car-pal tunnel syndrome').

Radial nerve C5–T1 This nerve opens the fist. It may be damaged by compression against the humerus. Signs: Test for wrist and finger drop with elbow flexed and arm pronated; sensory loss is variable—the dorsal aspect of the root of the thumb (the anatomical snuff box) is most reliably affected. Muscles involved: ('BEAST') brachioradialis; extensors; abductor pollicis longus; supinator; triceps.

Lateral cutaneous nerve of the thigh L2–L3 Meralgia paraesthetica is antero-lateral burning thigh pain from entrapment under the inguinal ligament.

Tibial nerve L4–S3 Originates from sciatic nerve just above knee. Lesions lead to an inability to stand on tiptoe (plantarflexion), invert the foot, or flex the toes, with sensory loss over the sole.

Motor neuropathy: (Eg Guillain–Barré syndrome, lead poisoning, Charcot–Marie–Tooth syndrome.) Often progressive (may be rapid); weak or clumsy hands; diffi-culty in walking (falls, stumbling); difficulty in breathing (↓vital capacity). Signs: LMN lesion: wasting and weakness most marked in the distal muscles of hands and feet (foot or wrist drop). Reflexes are reduced or absent.

'Bulbar and corticobulbar ('pseudobulbar') palsy'.

signs, marked spastic leg weakness and pseudobulbar palsy. No cognitive decline.

Prognosis Poor, <2yrs post onset in half of patients.

Degeneration of the cervical spine with age is inevitable, and has a wide clinical spec-trum, ranging from asymptomatic to progressive spastic quadriparesis and sensory loss due to compression of the cord (myelopathy).

Signs Limited, painful neck movement ! crepitus (examine gently). Neck flexion may produce tingling down the spine (Lhermitte's sign, p497). NB: this does not distinguish between cord or roots (or both) involvement.

Which nerve root is affected? See table 10.11.

Table 10.11 Clinical patterns of nerve root impingement

| Nerve root | Muscle | Test by asking the patient to: | | — | — | — | | C2, 4 | Trapezius | Shrug shoulder (via accessory nerve) | | C5, 6, 7 | Serratus anterior | Push arm forward against resistance; look for scapula winging (p511) if weak | | C5, 6 | Pectoralis major (P major) | Adduct arm from above horizontal, | | | clavicular head | and push it forward | | C6, 7, 8 | P major sternocostal head | Adduct arm below horizontal | | C5, 6 | Supraspinatus | Abduct arm the first 15° | | C5, 6 | Infraspinatus | Externally rotate semi-flexed arm, elbow at side | | C6, 7, 8 | Latissimus dorsi | Adduct arm from horizontal position | | C5, 6 | Biceps | Flex supinated forearm | | C5, 6 | Deltoid | Abduct arm between 15° and 90° | | Radial nerve (p502) | Radial nerve (p502) | | | C6, 7, 8 | Triceps | Extend elbow against resistance | | C5, 6 | Brachioradialis | Flex elbow with forearm half way between pronation and supination | | C5, 6 | Extensor carpi radialis longus | Extend wrist to radial side | | C6, 7 | Supinator | Arm by side, resist hand pronation | | C7, 8 | Extensor digitorum | Keep fingers extended at MCP joint | | C7, 8 | Extensor carpi ulnaris | Extend wrist to ulnar side | | C7, 8 | Abductor pollicis longus | Abduct thumb at 90° to palm | | C7, 8 | Extensor pollicis brevis | Extend thumb at MCP joint | | C7, 8 Extensor pollicis longus Median nerve (p502) C6, 7 Pronator teres | C7, 8 Extensor pollicis longus Median

nerve (p502) C6, 7 Pronator teres | Resist thumb flexion at IP joint Keep arm pronated against resistance | | C6, 7 | Flexor carpi radialis | Flex wrist towards radial side | | C7, 8, T1 | Flexor digitorum superficialis | Resist extension at PIP joint (with prox-imal phalanx fixed by the examiner) | | C7, 8 | Flexor digitorum | Resist extension at index DIP joint of | | | profundus I & II | index finger | | C7, 8, T1 | Flexor pollicis longus | Resist thumb extension at interphalangeal joint (fix proximal phalanx) | | C8, T1 | Abductor pollicis brevis | Abduct thumb (nail at 90° to palm) | | C8, T1 | Opponens pollicis | Thumb touches base of 5th fingertip (nail parallel to palm) | | C8, T1 | 1st lumbrical/interosseus | Extend PIP joint against resistance | | (median and ulnar nerves) | (median and ulnar nerves) | with MCP joint held hyperextended | | Ulnar nerve (p502) | Ulnar nerve (p502) | | | C7, 8, T1 | Flexor carpi ulnaris | Flex wrist to ulnar side; observe tendon | | C7, C8 | Flexor digitorum profundus III & IV | Resist extension of distal phalanx of 5th finger while you fix its middle phalanx | | C8, T1 | Dorsal interossei | Finger abduction: cannot cross the middle over the index finger (tests index finger adduction too) | | C8, T1 | Palmar interossei | Finger adduction: pull apart a sheet of paper held between middle and ring finger DIP joints of both hands; the paper moves on the weaker side* | | C8, T1 | Adductor pollicis | Adduct thumb (nail at 90° to palm) | | C8, T1 | Abductor digiti minimi | Abduct little finger | | C8, T1 | Flexor digiti minimi | Flex little finger at MCP joint |

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Acquired myopathies of late onset Often part of systemic disease—eg hyper-thyroidism, malignancy, Cushing's, hypo- and hypercalcaemia.

Table 10.12 Genetics of some commoner congenital myopathies

| Trigeminal neuralgia Symptoms: Paroxysms of intense, stabbing pain, lasting seconds, in the trigemi-nal nerve distribution. It is unilateral, typically affecting mandibular or maxillary divisions. The face screws up with pain (hence tic douloureux). Triggers: Wash-ing affected area, shaving, eating, talking, dental prostheses. Typical patient: ♂ >50yrs old; in Asians ♀:♂ ≈ 2:1. Secondary causes: Compression of the trigemi- nal root by anomalous or aneurysmal intracranial vessels or a tumour, chronic meningeal inflammation, MS, zoster, skull base malformation (eg Chiari). MRI: Is necessary to exclude secondary causes (r14% of cases). 9: Carbamazepine (start at 100mg/12h PO; max 400mg/6h; lamotrigine; phenytoin 200–400mg/24h PO; or gabapentin (p504). If drugs fail, surgery may be necessary. This may be directed at the peripheral nerve, the trigeminal ganglion, or the nerve root. Microvascular decompression: Anomalous vessels are separated from the trigeminal root. Ste-reotactic gamma knife surgery can work, but length of pain relief and the time to treatment response are limiting factors. Facial pain ☣☣ : p65. | | — | Differentials Polymyositis/other myopathies (p510); SLE; Takayasu's arteritis (fati-gability of the extremities); botulism (see BOX).

>2min—a neat, non-invasive test (but not diagnostic). The Tensilon® (edrophoni-um) test may not give clear answers and has dangers, so is rarely used.

Treatment Pyridostigmine, 2,4-diaminopyridine or IVIg (get specialist help). ►Do regular CXR/high-resolution CT as symptoms may precede the cancer by >4yrs.

Disruption of pre-synaptic transmission affects release of ACh in autonomic nervous system as well as at neuromuscular junction, explaining the prominence of dysautonomia in LEMS unlike in MG.

Type 1 neurofibromatosis (NF1, von Recklinghausen's disease)

Signs: Café-au-lait spots: flat, coffee-coloured patches of skin seen in 1st year of life (clearest in UV light), increasing in size and number with age. Adults have ?6,

Type 2 neurofibromatosis (NF2)

Autosomal dominant inheritance, though 50% are de novo, with mosaicism in some (NF2 gene locus is 22q11). Rarer than NF1 with a prevalence of only 1 in 25 000.

A syrinx is a tubular cavity in or close to the central canal of the cervical cord. Mean age of onset: 20yrs. Incidence: 8/100 000/yr. Symptoms may be static for years, but then worsen fast—eg on coughing or sneezing, as •pressure causes extension, eg into the brainstem (syringobulbia, see later in topic).

hypertrophy, or unilateral odo- or chiromegaly (enlarged hand or foot), perhaps from release of trophic fac-tors via anterior horn cells; Char-cot's joints in the shoulder/wrist due to lost joint proprioception (see fig 5.11, p212).

Surgery Don't wait for gross dete-rioration to occur. Decompression at the foramen magnum may be tried in Chiari malformations to promote free flow of CSF, and so prevent syrinx dilatation. Surgery may reduce pain

Acute infection: May be associated with transient aseptic meningoencephalitis (typically self-limiting), myelopathy, and neuropathy.

Neuropathies: Common in HIV, and may be a result of the disease itself or antiret-roviral therapy. Up to 20% of patients have a peripheral neuropathy, which is pain-ful and predominantly sensory. Other clinical pictures include polyradiculopathy, mononeuritis multiplex, and proximal myopathy.

Fig 10.26 Cryptococcosis: (a) Chronic meningitis involving the basal leptomeningitis with multiple small intraparenchymal cysts seen in the cerebral cortex. (b) Under the microscope we see these cysts as dilatation of the perivascular space to form cavities filled with colonies of cryptococci, which appear as round basophilic structures.

Oncology and Palliative Care

The Other Side, by Kate Granger MBE, FRCP,

1981–2016.

Kate Granger, a medical registrar then consult-ant geriatrician, was diagnosed with a desmo-plastic small round cell tumour at the age of

Looking after people with cancer

point, you may fail to be effective in your guidance and support.

Tips for the discussion of a cancer diagnosis

Set the environment up carefully. Choose a quiet place where you will not be disturbed. Make sure family or friends are present according to your patient's wishes. Anticipate likely questions and be sure of your facts.

Find out what the patient already knows and believes (often a great deal). 'What are you worried about today?'

Give some warning: 'There is some bad news for us to address'.

Ascertain how much the person wants to know. 'Are you someone who likes to know all the details about your condition?' Although information is a prior-ity for the majority of cancer patients, this may change with the individual, and the course of the disease. 'Monitors' will seek information, 'blunters' will distract themselves.

Share information about diagnosis and treatments. Specifically list support-ing people (oncology multidisciplinary team) and institutions (hospices). Break information down into manageable chunks and check understanding for each.

Invite questions patients may feel they cannot ask. 'Is there anything else you want me to explain?' Do not hesitate to go over the same ground repeatedly. Allow denial, don't force the pace, give time. Listen to any concerns raised, encourage the airing of feelings. Empathize.

Address prognosis. Be honest. Doctors are often too optimistic. Encourage an appropriate level of hope (see BOX 'Spiritual pain', p525), refer to an expert.

Summarize, and offer availability. Record details of your conversation includ-ing the language used.

Follow through. ►Leave your patient with the knowledge that you are with them, and that your unwritten contract will not be broken.

No rules guarantee success. Use whatever your patient gives you—closely observe both verbal and non-verbal cues. Getting to know your patient, seeking out the right expert for each stage of treatment, and making an agreed management plan, are all required.

For any situation which involves the communication of bad news, consider SPIKES:3

Setting up the interview.

Assess the patient's Perception of the situation.

Obtain an Invitation (asking the patient's permission to explain).

Give Knowledge and information to the patient.

Address the Emotional response with Empathy.

Strategy and Summary: aim for consensus with patient and family.

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520 The genetics of cancer

Reprinted from Cell, 144(5), Hanahan et al., Hallmarks of Cancer: the Next Generation, 646–74, 2011, with permission from Elsevier.

Hereditary cancer syndromes

A hereditary cancer' is suggested by:

unusual early age or presentation (eg male breast cancer)

multiple primary cancers or bilateral/multifocal cancers

clustering of cancers in relatives

cancers in multiple generations

rare tumours (eg retinoblastoma) or histology (medullary thyroid cancer, p222)

ethnicity (eg Ashkenazi heritage and breast cancer).

Genetic testing is appropriate if the sensitivity and specificity of the test are good enough, and if the result of the test will impact diagnosis and management (see p27).

Breast/ovarian cancer

1st-degree relative with: breast cancer <40yrs, male breast cancer, bilateral breast cancer <50yrs

1st- and 2nd-degree relative with breast cancer or ovarian cancer

three 1st- or 2nd-degree relatives with breast cancer

risk assessment calculation of >2% risk in 10yrs or lifetime risk ≥17%

Colorectal cancer

r25% have a family history. r5% have identified mutations. Refer to specialist genet-ic service if: two 1st-degree relatives with colorectal cancer at average age <60yrs, or if criteria for an autosomal dominant colorectal cancer syndrome is met:

Familial adenomatous polyposis: Due to mutations in the APC tumour suppressor gene (5q) (fig 11.2). <1% of colorectal cancer. Causes multiple colorectal adenomas (>100 in classical disease) which undergo malignant transformation. Gene pen-etrance approaches 100% by 50yrs. Surveillance sigmoidoscopy from 12yrs, with prophylactic surgery usually <25yrs guided by polyp number, size, and dysplasia.

Peutz–Jeghers syndrome: 1 in 25 000–280 000. Hamartomatous polpys. 10–20% risk of colorectal cancer, r50–60% risk of GI cancer, r60% risk of breast cancer. Due to germline mutations in STK11, a tumour suppressor gene (19p14). Surveillance in all (see p708).

Prostate cancer

5–10% (r50% disease <55yrs) estimated to be due to inherited factors. Genes include BRCA1, BRCA2, mismatch repair, and HOXB12 which interacts with androgen receptor. Age and race contribute. See p520 for screening.

Other familial cancer syndromes Von Hippel–Lindau (p220, p712), Carney complex (p222), MEN (p222), neurofibromatosis (p514).

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A variety of clinical signs and symptoms should alert you to the possible presence of malignancy. The following list is based on clinical features with a 2% positive predic-tive value for cancer.[a] It is by no means exhaustive and does not negate the value of clinical judgement. Urgent = within 2 weeks.

Lung

Admit if: symptomatic superior vena caval obstruction (p528), stridor.

Urgent referral if: >40yrs with unexplained haemoptysis, CXR suggestive of cancer.

Urgent CXR if >40yrs and:

persistent/recurrent chest infection

finger clubbing

supraclavicular/cervical lymphadenopathy

thrombocytosis

two of: cough, fatigue, SOB, chest pain, weight loss, ↓appetite, smoker, asbestos.

Upper GI

Urgent endoscopy if: dysphagia, or >55yrs with weight loss and upper abdominal pain/reflux/dyspepsia.

Urgent referral if: >40yrs plus jaundice, or upper abdominal mass.

Urgent CT of the pancreas if >60yrs plus weight loss plus any of: diarrhoea, back pain, abdominal pain, nausea, constipation, new-onset diabetes.

Non-urgent endoscopy if:

>55yrs and one of: treatment-resistant dyspepsia, upper abdominal pain plus low Hb, ↓plts, or N&V plus upper GI symptoms/weight loss

haematemesis.

Lower GI

PR examination and FBC in all.

Urgent referral if: positive faecal occult blood, >40yrs with abdominal pain plus weight loss, >50yrs with unexplained rectal bleeding, >60yrs with iron-deficient anaemia or change in bowel habit.

Consider urgent referral if: rectal/abdominal mass, anal ulceration, <50yrs with rectal bleeding plus lower GI symptoms or weight loss or iron-deficiency anaemia.

Faecal occult blood testing if: >50yrs plus abdominal pain or weight loss, <60yrs with change in bowel habit or iron-deficiency anaemia, >60yrs and anaemia.

Gynaecological

Urgent referral if: ascites, pelvic mass (fibroid excluded), >55yrs with post-meno-pausal bleeding.

Breast

Urgent referral if: >20yrs with unexplained breast lump, >50yrs with symptoms or change to one nipple.

Consider urgent referral if: skin changes, >20yrs with axillary lump.

Urology

Urgent referral if:

irregular prostate on PR, abnormal age-specific PSA (see p520)

>40yrs with unexplained visible haematuria, >60yrs with unexplained non-visible haematuria plus dysuria or ΔWCC

non-painful enlargement or change in shape/texture of testicle.

Central nervous system

Urgent MRI in progressive, sub-acute loss of central neurological function.

Unexplained weight loss, ↶appetite, and DVT can be non-specific signs of cancer. Assess for any additional risk factors, symptoms, signs, and refer accordingly.

See also haematology (p252); thyroid (p600); skin (p596).

Chemotherapy

Single-agent: Rarely curative as genetically resistant cells are selected out.

Combination chemotherapy: A combination of drugs with different mechanisms of action and different side-effect profiles reduces the likelihood of resistance and toxicity. The drugs used should have:

cytotoxic activity for that tumour, preferentially able to induce remission

different mechanisms of action, ideally additive or synergistic effects

non-overlapping toxicity to maximize benefit of full therapeutic doses

different mechanisms of resistance.

Adjuvant: After other initial treatment to reduce the risk of relapse, eg following surgical removal of, eg breast, bowel cancer.

Neoadjuvant: Used to shrink tumours prior to surgical or radiological treatment. May allow later treatment to be more conservative.

Palliative: No curative aim, offers symptom relief, may prolong survival.

Classes of cytotoxic drugs

Alkylating agents: Anti-proliferative drugs that bind via alkyl groups to DNA lead-ing to apoptotic cell death, eg cyclophosphamide, chlorambucil, busulfan.

Angiogenesis inhibitors: Eg bevacizumab, aflibercept, sunitinib.

Antimetabolites: Interfere with cell metabolism including DNA and protein synthe-sis, eg methotrexate, 5-fluorouracil.

Antioestrogens: Aromatoase inhibitors (eg letrozole, anastrozole), oestrogen receptor antagonists (eg tamoxifen, raloxifene) used in breast cancer treatment.

Antitumour antibiotics: Interrupt DNA function, eg dactinomycin, doxorubicin, mitomycin, bleomycin.

Monoclonal antibodies: Antibodies to a specific tumour antigen can slow tumour growth by enhancing host immunity, or be conjugated with chemotherapy/radioac-tive isotopes to allow targeted treatment. Expect more of these in future.

Topoisomerase inhibitors: Interrupt regulation of DNA winding, eg etoposide.

Vinca alkaloids and taxanes: 'Spindle poisons' which target mechanisms of cell division, eg vincristine, vinblastine, docetaxel.

Side-effects

Due to cytotoxic effects on non-cancer cells. Greatest effect seen on dividing cells, ie gut, hair, bone marrow, gametes (see BOX 'Fertility and cancer', p525).

Vomiting: Prophylaxis given with most cytotoxic regimens (see p251).

Alopecia: May profoundly impact quality of life. Consider 'cold-cap', wig services.

Neutropenia: Most commonly seen 7–14d after chemotherapy. 2Neutropenic sep-sis is life-threatening and needs urgent assessment and empirical treatment (p252).

Radiotherapy[1o] is used in >50% of all cancer and forms part of treatment in 40% of those considered cured. It uses ionizing radiation to cause damage to DNA. This prevents cell division and leads to cell death. The aim of radiotherapy treatment is to inactivate cancer cells without causing a severe reaction in normal tissue.

Radical treatment Given with curative intent. Total dose ranges from 40–70 gray (Gy) in up to 40 fractions. Some regimens involve several smaller fractions a day with a gap of 6–8h. Combined chemoradiation is used in some sites, eg anus and oesophagus, to increase response rate.

Palliative radiotherapy Aims to relieve symptoms, may not impact on survival. Doses are smaller and given in fewer fractions to offer short-term tumour control with minimal side-effects. Palliation is used for brain metastases, spinal cord com-pression, visceral compression, and bleeding, eg haemoptysis, haematuria. Bone pain from metastases can be reduced or eliminated in 60% of cases.

Early reactions

Occur r2 weeks into treatment, peak r2–4 weeks after treatment.

Tiredness: r80%. Improves r4 weeks after treatment completed but chronic in r20%. Advise patients to stay as active as possible.

Skin reactions: Include erythema, dry desquamation, moist desquamation, and ulceration. Aqueous cream can be used on unbroken areas.

Mucositis: All patients receiving head and neck treatment should have a dental check-up before therapy. Avoid smoking. Antiseptic mouthwashes may help. Aspi-rin gargle and other soluble analgesics can be tried. Treat oral thrush with flucona-zole 50mg/24h PO, nystatin may exacerbate nausea.

Nausea and vomiting: Occur when stomach, liver, or brain treated. Try metoclo-pramide 10mg/8h PO (dopamine antagonist), domperidone 10mg/8h PO (blocks the central chemoreceptor trigger zone), or ondansetron 4–8mg/8h PO/IV (serotonin 5HT2 antagonist) (see p251).

Diarrhoea: Usually after abdominal or pelvic treatments. Maintain good hydration. Avoid high-fibre agents. Try loperamide 2mg PO after loose stools (max 16mg/24h).

Dysphagia: Following thoracic treatments. Speech and language input, nutrition.

Cystitis: After pelvic treatments. Drink plenty of fluid.

Late reactions

Months-years after treatment.

CNS/PNS: Somnolence: 4–6wks after brain radiotherapy. Consider ♙steroid dose. Spinal cord myelopathy: progressive weakness. MRI to exclude cord compression. Brachial plexopathy: numb, weak, or painful arm after axillary radiotherapy.

Lung: Pneumonitis can occur 6–12wks after thoracic treatment causing dry cough

! dyspnoea. Bronchodilators and tapered steroids may help.

GI: Xerostomia = reduced saliva. Dental care and nutrition important. Treat with water, saliva substitutes, salivary stimulants. Benign strictures of oesophagus or bowel. Treat with dilatation. Seek a specialist surgical opinion regarding fistulae. Radiation proctitis may be a problem after prostate irradiation.

GU: Urinary frequency: small fibrosed bladder after pelvic treatment. Vaginal stenosis, dyspareunia, erectile dysfunction can occur after pelvic radiotherapy. ‘fertility: due to pelvic radiotherapy (see p525).

Endocrine: Panhypopituitarism following radical treatment involving pituitary fossa. Check hormone profile in children: growth hormone replacement may be required. Hypothyroidism in r50% after neck treatment: check TFTs annually.

Secondary cancers: Risk (2–4 per 10 000 person-years) is usually insignificant

compared to recurrence/death from primary lesion. More important for younger patients after curative treatment. Women <25yrs receiving radiotherapy for Hodg-kin’s lymphoma should be offered breast screening from 8yrs after treatment.

♙ Cancer survival means ♙ numbers living with poor health or disability after treat-ment (r625 000 in UK). Remember the emotional and physical impact of cancer ex-tends beyond the prescribed course of radiotherapy/ chemotherapy.

Methods of delivering radiotherapy

Conventional external beam radiotherapy (EBRT): Is the most common form of treatment. Delivers beams of ionizing radiation to the patient from an external linear accelerator.

Stereotactic radiotherapy is a highly accurate form of EBRT used to target small lesions with great precision—most frequently in treating intracranial tumours. It is often referred to by the manufacturer's name, eg Gamma Knife®, Truebeam®.

Brachytherapy involves a radiation source being placed within or close to a tumour, allowing a high local radiation dose. Implants may be placed within a cavity (eg uterus, post-surgical space) or within tissue (eg prostate, breast).

Radioisotope therapy uses tumour-seeking radionuclides to target specific tissues. For example, 121I (radioiodine) to ablate remaining thyroid tissue after thyroidectomy for thyroid cancer.

Emergencies[11],[12] in oncology include:

Neutropenic sepsis

Temperature >28oC and neutrophil count <0.5≈109/L. Suspect in all patients who are unwell and within 6wks of receiving chemotherapy. Localizing signs may be absent. Examine indwelling catheter sites. 2Immediate treatment saves lives. Use local guidelines or treat empirically with piperacillin/tazobactam (see p252).

Spinal cord compression

2–5% of cancer patients have spinal metastases. r15% of those with advanced can-cers develop metastatic spinal cord compression. Most commonly associated with lung, prostate, breast, myeloma, melanoma. 2Urgent treatment is required to pre-serve neurological function and relieve pain.

Causes: Collapse or compression of a vertebral body due to metastases (common), direct extension of a tumour into vertebral column (rare).

Signs and symptoms: Back pain in r95%. Ask about nocturnal pain and pain with straining. Worry if there is cervical/thoracic pain. Also limb weakness, difficulty walking, sensory loss, bowel/bladder dysfunction. Maintain a high index of suspicion.

Superior vena cava (SVC) syndrome

Reduced venous return from head, neck, and upper limbs. Due to extrinsic compres-sion (most common), or venous thrombosis (consider if current or past central ve-nous access). 2SVC syndrome with airway compromise requires urgent treatment.

Causes: >90% of SVC syndrome results from malignancy. Most common cancers: lung (r75%), lymphoma, metastatic (eg breast), thymoma, germ cell.

Signs and symptoms: Diagnosis is made clinically. SOB, orthopnoea, stridor, pleth-ora/cyanosis, oedema of face and arm, cough, headache, engorged neck veins (non-pulsatile •JVP), engorged chest wall veins. Pemberton's test:1 elevation of the arms to the side of the head causes facial plethora/cyanosis.

Management: Prop up. Assess for hypoxia (pulse oximetry, blood gas) and give oxygen if needed. Dexamethasone 16mg/24h. CT is used to define the anatomy of the obstruction. Balloon venoplasty and SVC stenting provide the most rapid relief of symptoms (see BOX 'Interventional oncology', p527). Treat with radiotherapy or chemotherapy depending on the sensitivity of the underlying cancer.

Malignancy-associated hypercalcaemia

Most common metabolic abnormality in cancer patients: r10–20% of patients with cancer, r40% of myeloma. It is a poor prognostic sign: 75% mortality within 2 months. Calcium is highly protein-bound and needs correcting to the serum albumin concentration. PTH levels should be suppressed (see pp676–7).

Causes: PTH-related protein produced by the tumour (see p529), local osteolysis, eg myeloma, tumour production of calcitriol.

Signs and symptoms: Weight loss, anorexia, nausea, polydipsia, polyuria, constipa-tion, abdominal pain, dehydration, weakness, confusion, seizure, coma.

Management: Aggressive rehydration. Bisphosphonates (if eGFR ?20), eg zoledronic acid IV, usually normalize calcium within 2 days and can be given as a repeated infu-sion. Calcitonin produces a more rapid (2h) but short-term effect and tolerance can develop. Long-term treatment is by control of the underlying malignancy.

- Pemberton described this 'useful' sign of venous obstruction due to a goitre in 1946.

Brain metastases

Affect up to r40% of patients with cancer. Most commonly: lung, breast, colorectal, melanoma. Poor prognosis: median survival 1–2 months; better prognosis with single lesion, breast cancer (see also p820).

Signs and symptoms: Headache (r50%, often worse in the morning, when cough-ing or bending), focal neurological signs (r20%), ataxia (r21%), fits (r18%), nausea, vomiting, papilloedema.

Management: Urgent CT/MRI depending on underlying diagnosis, disease stage, and performance status. Dexamethasone 16mg/24h to reduce cerebral oedema. Stereo-tactic radiotherapy (see p527). Discuss with neurosurgery, especially if large lesion or associated hydrocephalus.

Tumour lysis syndrome

Chemotherapy for rapidly proliferating tumours (leukaemia, lymphoma, myeloma) leads to cell death and •urate, •K+, •phosphate, ↶calcium. Risk of arrhythmia and renal failure (see p214).

Management: Prevent with hydration and uricolytics, eg rasburicase, allopurinol.

Tumour markers[1'] are specific molecules (usually glycoproteins) that may be found in higher concentrations in the serum, tissue, or urine in patients with certain cancers.

Tumour markers in diagnosis

—— Tumour markers are insufficiently sensitive or specific to be diagnostic in isolation.

Many tumour markers are ⛄ in several cancers and benign conditions (table 11.4).

Measuring ≥1 tumour marker is unlikely to aid diagnosis unless suspecting a germ cell tumour.

Do not make opportunistic requests for panels of tumour markers in patients with non-specific symptoms: they are not helpful and lead to potentially unnecessary investigation. This includes testing PSA in women and CA 125 in men.

In carefully selected patients, in whom cancer is suspected, highly raised levels of a tumour marker may be helpful:

a-fetoprotein (aFP) and human chorionic gonadotrophin (hCG) in testicular/ germ cell tumours.

CA 125 in combination with USS and menopausal status.

aFPin those at high risk of hepatocellular carcinoma.

PSA >100ng/mL usually indicates metastatic prostate cancer.

Tumour markers in monitoring

The main value of tumour markers is in monitoring patients known to have cancer. This includes the course of the disease, the effectiveness of treatment, and the detec-tion of cancer recurrence. The following markers may be useful:
aFP and hCG in testicular/germ cell tumours.
CEA in colorectal cancer.
CA 125 in ovarian cancer.
A cautious interpretation of PSA within the limits of its specificity and sensitivity.

Table 11.4 Summary of tumour markers

Cancer staging Staging systems are used to describe the extent of a cancer. This is vital to deter-mine the most appropriate treatment, to assess prognosis, and to identify relevant clinical trials. A cancer is always referred to by the stage given at diagnosis. The TNM system is most widely used and is based on the extent of tumour (T), spread to lymph nodes (N), and the presence of metastases (M) (table 11.1). Table 11.1 TNM cancer staging Other prefixes may also be used: c refers to clinical stage; p is the stage after pathological examination; y refers to stage after neoadjuvant therapy; r is used if a tumour is re-staged after a disease-free interval; a indicates stage at autopsy. The TNM staging may be converted to an overall, less detailed classification of cancer stage: 0–IV. Stage 0 refers to carcinoma in situ; Stages I–III describe the size of cancer and/or nearby spread; Stage IV indicates metastatic disease. Some cancers may have alternative staging systems such as Duke's classifica-tion for colorectal cancer (p616). See also lung cancer (p176); breast cancer (p602); oesophageal cancer (p618); bladder cancer (p647).		—

Source data from 'Serum tumour markers: how to order and interpret them', Sturgeon C M, Lai L C, Duffy M J, 2012, BMJ Publishing Ltd.

You matter because you are you and you matter to the last moment of your life. We will do all we can to help you, not only to die peacefully, but to live until you die.

Dame Cicely Saunders (1918–2005), founder of the modern hospice.

Palliative care is the active, holistic care of patients with advanced progressive ill-ness. It combines management of pain and other symptoms, with the provision of psychological, social, and spiritual support.

Palliative care is not just for the end of life and it is not just for patients with cancer.

Palliative care should run in parallel with other medical treatments. Good symptom control is important in any disease for improving quality of life and may even pro-long survival.16 Take time to find out exactly what is troubling your patient using a problem-based approach. Consider:

physical
psychological
spiritual
social.

Remember, each person comes with a set of emotions, preconceptions, and a family already attached. ►Most hospitals now have a dedicated palliative care team for help and advice (including out of hours). Use their expertise.

Assessment of pain

Pain is one of the most feared sequelae of a terminal diagnosis and yet it is not inevitable. However, pain is a complex phenomenon. While the aim of management is for the patient to be pain free, this may not be achievable in all cases so do not promise this.

Management of pain

Aim to modify the underlying pathology where possible, eg radiotherapy, chemo-therapy, surgery. Use analgesia to relieve background pain and provide additional PRN doses for 'breakthrough' pain. Effective analgesia is possible in the majority of patients by combining five principles:

By the mouth—give orally whenever possible.

By the clock—give at fixed intervals to offer continuous relief.

By the ladder—following the WHO stepwise approach (see fig 12.5, p575).

For the individual—there are no standard doses for opioids, needs vary.

Attention to detail—communicate, set times carefully, warn of side-effects.

Opioids

The amount of opioid required to relieve pain varies and should be titrated on an individual basis. Oral morphine is 1st-line. If the oral route is unavailable, use mor-phine or diamorphine SC (see table 11.5, and tables 11.6, 11.7, p526). Explanation and regular review are important. Prescribe anti-emetics and laxatives for all patients.

Start low, go slow: For an opioid-naïve patient with moderate to severe pain, con-sider oral morphine 5mg every 4 hours plus 5mg PRN (maximum hourly). Consider a lower starting dose if elderly, ↓BMI, or renal impairment. If pain is not controlled,

dose by 20–50% every 24h.

Convert to modified release: When pain is controlled, calculate the total daily dose including PRN and divide into two 12h doses of a modified-release preparation (eg MST Continus® 12h). Transdermal preparations are available: seek expert help for dose, check adhesion, and rotate site.

Use a PRN dose for breakthrough pain: 1/10th–1/6th of the total daily dose as an immediate-release preparation, eg Oramorph® or Sevredol®.

Side-effects: Drowsiness, nausea/vomiting (usually ↓ after 5 days), constipation, dry mouth. If difficulty tolerating morphine, or pain plus toxicity, consider an opi-oid switch (eg oxycodone) and ↓ dose by 25–20%.

Renal failure: Patients with renal impairment (eGFR <20) are at risk of toxicity due to accumulation of renally excreted opioids and metabolites. Monitor closely. Fentanyl, alfentanil, and buprenorphine have predominantly hepatic metabolism

—seek expert advice.

Concerns: Patients may shrink from using opioids. Misconceptions are common: they are addictive, for the dying, if they use morphine now it will not work when they really need it. Respiratory depression is very rare when opioids are correctly titrated but opioids often get blamed when a patient deteriorates. Reassure pa-tients that opioids are effective and safe when used appropriately.

Morphine-resistant pain: Seek expert help. Consider methadone, ketamine, and adjuvants such as NSAIDS, steroids, muscle relaxants, anxiolytics, nerve blocks. If neuropathic pain is suspected, try amitriptyline, pregabalin, or topical lidocaine. Consider the effect of psychological and spiritual well-being on pain (see p525).

Rapid analgesia: Most PRN medication takes time to have an effect. If this is a problem, seek expert help regarding rapid-release preparations (eg sublingual, intranasal, or buccal fentanyl). Try to pre-empt times of high pain (eg dressing changes) and give analgesia in advance.

Table 11.5 Opioid dose equivalents: conversions are not exact, potency can vary. If in doubt, use a dose below your estimate. Practice is variable: always defer to local guidelines first.

Non-pain symptoms[1a] include:

Nausea and vomiting

Causes: Chemotherapy, constipation, hypercalcaemia, oral candidiasis, GI obstruc-tion, drugs, severe pain, infection, renal failure.

Cyclizine 50mg/8h: antihistamine, anticholinergic, central action so good for in-tracranial disorders.

Metoclopramide 10–20mg/8h: blocks central chemoreceptor trigger zone, peripheral prokinetic effects so good in gastroparesis, monitor for extra-pyramidal side-effects.

Domperidone 10–20mg/8h PO: peripheral antidopaminergic so no dystonic effects.

Haloperidol 1.5mg PO initially 1–2 times daily: dopamine antagonist, effective in drug- or metabolically induced nausea, use lower doses IV/SC as twice as potent.

Ondansetron 4–8mg/8h: serotonin antagonist, good for chemo/radiotherapy-relat-ed nausea, may cause constipation.

Levomepromazine 6.25mg, initially 1–2 times daily: broad spectrum, but can sedate, may be very effective if fear/anxiety are contributing to symptoms.

Antisecretory drugs, such as hyoscine butylbromide or octreotide may be required for patients with vomiting and bowel obstruction: seek expert advice.

Constipation

Causes: Very common side-effect of opioids. Better to prevent than treat so pre-scribe laxatives for all patients starting opioids. Also hypercalcaemia (see p528), dehydration, drugs, or intra-abdominal disease.

Treatment: Treat reversible causes. Good fluid intake. Ensure privacy and access to toilet. Medication options include the following:

Stimulant (eg senna 2–4 tablets or bisacodyl 5–10mg) at night ! a softener (eg sodium docusate 100mg BD).

Osmotic laxative (eg macrogol).

Rectal treatments: bisacodyl/glycerol suppositories, phosphate enema.

Breathlessness

Causes: Look for reversible causes including infection, effusion, anaemia, arrhyth-mia, thromboembolism. If stridor or signs of superior vena cava syndrome, treat urgently (see p528).

Oral problems

Causes: Poor oral hygiene, radiation, drugs (anticholinergics, chemotherapy, diuret-ics), infection (candidiasis, herpes simplex).

Insomnia

Causes: Terminally ill patients may experience physical and emotional exhaustion. Often multifactorial. Poor sleep can increase symptom burden.

Pruritus

Causes: Systemic disease (renal failure, hepatitis, polycythaemia), cancer-related (cholestasis, lymphoma, leukaemia, hepatoma, myeloma, paraneoplastic), primary-skin disease, drug reaction (opioids, SSRI, chemotherapy).

Treatment: Underlying causes where possible: cholestasis (biliary stenting, colestyr-amine, sertraline, rifampicin), opioid-induced (antihistamine, opioid switch), para-neoplastic (paroxetine). Topical emollients regularly and as a soap substitute. Avoid topical antihistamines due to risk of contact dermatitis.

Venepuncture

Repeated venepuncture with the risk of painful extravasation and phlebitis may be avoided by use of a central catheter (eg Hickman® tunnelled line or PICC). Problems: infection, blockage (flush with 0.9% saline or dilute heparin every week), axillary thrombosis, and line slippage.

Agitation See ‘Care in the last days of life’, p526.

Respiratory tract secretions See ‘Care in the last days of life’, p526.

Once it is recognized that a patient is entering the final days of their illness (see ‘Di-agnosing dying’, p12), the focus of care should be the relief of distressing symptoms.[22]

An individualized care plan should be made and discussed with your patient, their family, and relevant medical staff. Continue to treat reversible problems as appropriate (eg urinary retention). Stop observations and blood tests (unless you are going to act on them). Rationalize medi-cations but keep any that provide ongoing symptom benefit.

Prescribe as required subcutaneous end of life drugs. Prescribe PRN SC medica-tions before they are needed, in anticipation of symptoms (see table 11.6).

Start a syringe driver when symptom control drugs are needed regularly (see table 11.7). Practice is variable, some drugs may be used outside of licensed indications. Always defer to local guidelines first. If pain relief is insufficient, review regular dose and recalculate the PRN requirement (1/10th–1/6th of 24h dose).

Respond to changes in the clinical situation. Patients who are thought to be at the end of their life occasionally improve. Be alert to signs of improvement, and be prepared to switch back to active treatment when appropriate.

Communicate.[2]3 The importance of clear and regular written and verbal commu-nication with dying patients and their families cannot be overemphasized. Find out what is important to your patient. How much information do they know and want to know about their situation and prognosis? Be sensitive to social, religious, and cultural issues.

Rheumatology

We also thank Dr Susie Higgins for her contribution to this chapter.

Three screening questions for musculoskeletal disease

Are you free of any pain or stiffness in your joints, muscles, or back?

Can you dress yourself without too much difficulty?

Can you manage walking up and down stairs?

If yes to all three, serious inflammatory muscle/joint disease is unlikely.

Presenting symptoms: Related diseases:

Pattern of involved joints.

Symmetry (or not).

Morning stiffness >20min (eg RA).

Pain, swelling, loss of function, erythema, warmth.

Extra-articular features:

Rashes, photosensitivity (eg SLE).

Raynaud's (SLE; systemic sclerosis; polymyositis and dermatomyositis).

Dry eyes or mouth (Sjögren's).

Red eyes, iritis (eg AS).

Diarrhoea/urethritis (reactive arthri-tis).

Nodules or nodes (eg RA; TB; gout).

Mouth/genital ulcers (eg Behçet's,

SLE).

Weight loss (eg malignancy, any sys-temic inflammatory disease).

Crohn's/UC (in ankylosing spondylitis), preceding infections, psoriasis.

Current and past drugs:

NSAIDS, DMARDS (p547).

Biological agents (eg TNFa inhibitors).

Family history:

Arthritis, psoriasis, autoimmune disease.

Social history:

Age.

Occupation.

Sexual history.

Ethnicity (eg SLE is commoner in African-Caribbeans and Asians).

Ability to function (eg dressing, groom-ing, writing, walking).

Domestic situation, social support, home adaptations.

Smoking (may worsen RA).

IBD.

This aims to screen for rheumatological conditions primarily affecting mobility (as a consequence of underlying joint disease). It is based on the GALS locomotor screen (Gait, Arms, Legs, Spine).[1]

Essence 'Look, feel, and move' (active and passive). If a joint looks normal to you, feels normal to the patient, and has full range of movement, it usually is normal. Make sure the patient is comfortable, and obtain their consent before examination. The GALS screening examination should be done in light underwear.

Gait: Observe walking: is the gait smooth? Good arm swing? Stride length OK? Nor-mal heel strike and toe off? Can they turn quickly?

Range of joint movement Is noted in degrees, with anatomical position being the neutral position—eg elbow flexion 0°–150° normally, but with fixed flexion and lim-ited movement, range may be reduced to 20°–90°. A valgus deformity deviates later-ally (away from the mid-line, fig 12.2); a varus deformity points towards the mid-line.

University Press.

BMJ Publishing Group Ltd.

Schober's test: make a mark on the lumbar spine at the level of the posterior iliac spine. Measure out a line from 5cm below to 10cm above the mark. Ask to bend forward as far as they can. If the line does not lengthen by at least 5cm in flexion, there is reduced lumbar flexion, eg in ankylosing spondylitis.

Loss of joint space Osteophytes Subarticular sclerosis Subchondral cysts

Back pain is very common, and often self-limiting, but be alert to sinister causes, ie malignancy, infection, or inflammatory causes.

Examination

With the patient standing, gauge the extent and smoothness of lumbar forward/ lateral flexion and extension (see p540).

Test for sacroiliitis: palpate posteriorly down the length of the spine, including over spinous processes, paraspinal muscles, and the sacroiliac joints; examining for tenderness.

Neurological deficits (see BOX): test lower limb sensation, power, and deep tendon and plantar reflexes. Digital rectal examination for perianal tone and sensation.

Signs of generalized disease—eg malignancy. Examine other systems (eg abdo-men) as pain may be referred.

Causes Age determines the most likely causes:

15–20yrs: Prolapsed disc, trauma, fractures, ankylosing spondylitis (AS; p550), spondylolisthesis (a forward shift of one vertebra over another, which is congenital or due to trauma), pregnancy.

20–50yrs: Degenerative spinal disease, prolapsed disc, malignancy (primary or secondary from lung, breast, prostate, thyroid, or kidney ca).

>50yrs: Degenerative, osteoporotic vertebral collapse, Paget's (see p685), malig-nancy, myeloma (see p268), spinal stenosis.

Rarer: Cauda equina tumours, psoas abscess, spinal infection (eg discitis, usually staphy-lococcal but also Proteus, E. coli, S. typhi, and TB—there are often no systemic signs).

Table 12.2 Nerve root lesions

| A rapidly advancing speciality Rheumatology originates from the Greek word 'rheuma' meaning that which 'flows as a river or stream'. The British Society of Rheumatology defines rheuma-tology as a 'multidisciplinary branch of medicine that deals with the investigation, diagnosis and management of patients with arthritis and other musculoskeletal conditions…incorporating over 200 disorders affecting joints, bones, muscles and soft tissues, including inflammatory arthritis and other systemic autoimmune dis-orders, vasculitis, soft tissue conditions, spinal pain and metabolic bone disease'. Rheumatological diseases affect over 10 million UK adults and 12 000 children. Re-cent advances owe largely to new discoveries about the immunology of these disorders and the discovery of biologic DMARDS. | | — |

Osteoarthritis is the most common joint condition worldwide, with a clinically signif-icant impact on >10% of persons aged >60 years.™ It is usually primary (generalized), but may be secondary to joint disease or other conditions (eg haemochromatosis, obesity, occupational).

Tests Plain radiographs show: Loss of joint space, Osteophytes, Subarticular sclerosis and Subchondral cysts (fig 12.4 p541). CRP may be slightly elevated.[a]

2Consider septic arthritis in any acutely inflamed joint, as it can destroy a joint in under 24h and has a mortaility rate up to 11%. Inflammation may be less overt if im-munocompromised (eg from medication) or if there is underlying joint disease. The knee is affected in >50% cases.

Risk factors Pre-existing joint disease (especially rheumatoid arthritis); diabetes mellitus, immunosuppression, chronic renal failure, recent joint surgery, prosthetic joints (where infection is particularly difficult to treat), IV drug abuse, age >80yrs.[a]

Investigations Urgent joint aspiration for synovial fluid microscopy and culture is the key investigation (p541), as plain radiographs and CRP may be normal. The main differential diagnoses are the crystal arthropathies (p548). Blood cultures are es-sential (prior to antibiotics).

Ask yourself 'How did the organism get there?' Is there immunosuppression, or another focus of infection, eg from indwelling IV lines, infected skin, or pneumonia (present in up to 50% of those with pneumococcal arthritis)?[10]

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RA is a chronic systemic inflammatory disease, characterized by a symmetrical, de-forming, peripheral polyarthritis. It increases the risk of cardiovascular disease by 2–2 fold. Epidemiology Prevalence is r1% (• in smokers). φ:ð >2:1. Peak onset: 5th–6th decade. HLA DR4/DR1 linked (associated with •severity).

Diagnostic criteria See table 12.4.

Management ►Refer early to a rheumatologist (before irreversible destruction).

Disease activity is measured using the DAS28.2 Treatment should be escalated until satisfactory control is achieved: 'treat to target'.

Early use of DMARDS and biological agents improves long-term outcomes (see BOX 'Influencing biological events in RA').

Steroids rapidly reduce symptoms and inflammation. Avoid starting unless appro-priately experienced. Useful for acute exacerbations, eg IM depot methylpredniso-lone 80–120mg. Intra-articular steroids have a rapid but short-term effect (OHCS pp706–9). Oral steroids (eg prednisolone 7.5mg/d) may control difficult symptoms, but side effects preclude routine long-term use.

NSAIDS (see p545) are good for symptom relief, but have no effect on disease pro-gression. Paracetamol and weak opiates are rarely effective.

Offer specialist physio- and occupational therapy, eg for aids and splints.

Surgery may relieve pain, improve function, and prevent deformity.

There is •risk of cardiovascular and cerebrovascular disease, as atherosclerosis is accelerated in RA.[21] Manage risk factors (p92). Smoking also • symptoms of RA.

- In rheumatological palindromes, arthritis lasting hours or days runs to and fro, visiting and revisiting three or more sites, typically knees, wrists, and MCP joints. It may presage RA, SLE, Whipple's, or Behçet's disease. Remissions are (initially) complete, leaving no radiological mark.

3 28-joint Disease Activity Score—assesses tenderness and swelling at 28 joints (MCPS, PIPS, wrists, elbows, shoulders, knees), ESR/CRP, and patient's self-reported symptom severity.

Table 12.4 Criteria for diagnosing RA[22]

| Arthritides The pattern of joint involvement can provide clues to the underlying cause (table 12.1). Table 12.1 Patterns of presentation of arthritis *Connective tissue disease (eg SLE and relapsing polychondritis), sarcoidosis, malignancy (eg leukaemia), endocarditis, haemochromatosis, sickle-cell anaemia, familial Mediterranean fever, Behçet's. 2Exclude septic arthritis in any acutely inflamed joint, as it can destroy a joint in under 24h (p544). Inflammation may be less overt if immunocompromised (eg from the many immunosuppressive drugs used in rheumatological conditions) or if there is underlying joint disease. Joint aspiration (p541) is the key investigation, and if you are unable to do it, find someone who can. | | — |

*= MCPJ, PIPJ, 2nd–5th MTPJ, wrists, and thumb IPJ; †= with or without involvement of large joints.

Differential diagnoses Exclude septic arthritis in any acute monoarthropathy (p544). Then consider reactive arthritis, haemarthrosis, CPPD (see following topic) and palindromic RA (p546).

Prevention Lose weight, avoid prolonged fasts, alcohol excess, purine-rich meats, and low-dose aspirin. Prophylaxis: Start if >1 attack in 12 months, tophi, or renal stones. The aim is to attacks and prevent damage caused by crystal deposition. Use allopurinol and titrate from 100mg/24h, increasing every 4 weeks until plasma urate

Acute CPPD crystal arthritis Acute monoarthropathy usually of larger joints in elderly. Usually spontaneous but can be provoked by illness, surgery, or trauma.

Chronic CPPD Inflammatory RA-like (symmetrical) polyarthritis and synovitis.

Osteoarthritis with CPPD chronic polyarticular osteoarthritis with superimposed acute CPP attacks.

Fig 12.10 Needle-shaped monosodium urate crystals found in gout, displaying Negative birefringence under polarized light.

University Press.

Fig 12.11 Rhomboid-shaped calcium pyrophos-phate dihydrate crystals in Pseudogout, showing Positive birefringence in polarized light.

Sección de Reumatología, Hospital General Universitario de Alicante.

Fig 12.12 Don't underestimate the severity of pain caused by gout—as illustrated by satirical artist and gout sufferer James Gillray (1756–1815).

© Lordprice collection / Alamy Stock Photo.

The spondyloarthropathies (SpA) are a group of related chronic inflammatory condi-tions. They tend to, although not always, affect the axial skeleton with shared clinical features:

Seronegativity (= rheumatoid factor –ve).

HLA B27 association—see BOX.

'Axial arthritis': pathology in spine (spondylo-) and sacroiliac joints.

Asymmetrical large-joint oligoarthritis (ie <5 joints) or monoarthritis.

Enthesitis: inflammation of the site of insertion of tendon or ligament into bone, eg plantar fasciitis, Achilles tendonitis, costochondritis.

Dactylitis: inflammation of an entire digit ('sausage digit'), due to soft tissue oedema, and tenosynovial and joint inflammation.

Extra-articular manifestations: eg iritis (anterior uveitis), psoriaform rashes (psoriatic arthritis), oral ulcers, aortic valve incompetence, inflammatory bowel disease.

NB: BehÇet's syndrome (p694) can also present with uveitis, skin lesions, and arthritis and is not always associated with gross oral or genital ulcerations.

Enteric arthropathy Associations: Inflammatory bowel disease, GI bypass,

coeliac and Whipple's disease (p716). Arthropathy often improves with the treat-ment of bowel symptoms (beware NSAIDS). Use DMARDS for resistant cases.

4 Sacroiliitis on imaging plus ≥1 SpA feature or HLA B27 positive plus ≥2 SpA features.

Psoriatic arthritis (OHCS p594.) Occurs in 10–40% with psoriasis and can present before skin changes. Patterns are:

•symmetrical polyarthritis (like RA) •DIP joints

asymmetrical oligoarthritis •spinal (similar to AS) •psoriatic arthritis mutilans (rare, r2%, severe deformity).

Radiology: Erosive changes, with 'pencil-in-cup' deformity in severe cases. Associated with nail changes in 80%, synovitis (dac-tylitis), acneiform rashes and palmo-plantar pustulosis. Management: NSAIDS, sulfasalazine, methotrexate. Anti-TNF agents are also effective.

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Management: Currently no cure. Immunosuppressive regimens, including IV cyclophosphamide, are used for organ involvement or progressive skin disease. Trials of antifibrotic tyrosine kinase inhibitors are ongoing.'[1] Monitor BP and renal function. Regular ACE-i or A2RBS ' risk of renal crisis (p214). Raynaud's phenomenon: (see p708).

Mixed connective tissue disease Combines features of systemic sclerosis, SLE, and polymyositis and the presence of high titres of anti-U1-RNP antibodies.

Dermatomyositis Myositis plus skin signs: •Macular rash (shawl sign is +ve if over back and shoulders). •Lilac-purple (heliotrope) rash on eyelids often with oedema (fig 12.26, p562). •Nailfold erythema (dilated capillary loops). •Gottron's papules: roughened red papules over the knuckles, also seen on elbows and knees (pathognomonic if •CK + muscle weakness). Malignancy in 20% cases.

Extra-muscular signs In both conditions include fever, arthralgia, Raynaud's, in-terstitial lung fibrosis and myocardial involvement (myocarditis, arrhythmias).

Tests Muscle enzymes (ALT, AST, LDH, CK, & aldolase) • in plasma; EMG shows char-acteristic fibrillation potentials; muscle biopsy confirms diagnosis (and excludes mimicking conditions). MRI shows muscle oedema in acute myositis.

Autoantibody associations: (see BOX) anti-Mi2, anti-Jo1—associated with acute onset and intersti-tial lung fibrosis that should be treated aggressively.

Differential diagnoses Carcinomatous myopathy, inclusion-body myositis, mus-cular dystrophy, PMR, endocrine/metabolic myopathy (eg steroids), rhabdomyoly-sis, infection (eg HIV), drugs (penicillamine, colchicine, statins, or chloroquine).

Management Start prednisolone. Immunosuppressives (p276) and cytotoxics are used early in resistant cases. Hydroxychloroquine/topical tacrolimus for skin disease.

Plasma autoantibodies (Abs): disease associations

Always interpret in the context of clinical findings: Different antibodies have different disease asssociations.

Rheumatological: Rheumatoid factor (RhF) positive in:

Sjögren's syndrome ≤100% Mixed connective tissue disease 50%

Felty's syndrome ≤100% SLE ≤40%

RA 70% Systemic sclerosis 20%

Infection (SBE/IE; hepatitis) ≤50% Normal 2–10% Anticyclic citrullinated peptide Ab (anti-CCP):5 rheumatoid arthritis (r96% specificity) Antinuclear antibody (ANA) positive by immunofluorescence in:

SLE >95% Systemic sclerosis 96%

Autoimmune hepatitis 75% RA 20%

Sjögren's syndrome 68% Normal 0–2%

ANA titres are expressed according to dilutions at which antibodies can be detected, ie 1:160 means antibodies can still be detected after the serum has been diluted 160 times. Titres of 1:40 or 1:80 may not be significant. The pattern of staining may indicate the disease (although these are not specific):

Homogeneous SLE • Nucleolar Systemic sclerosis

Speckled Mixed CT disease • Centromere Limited systemic sclerosis Anti-double-stranded DNA (dsDNA): SLE (60% sensitivity, but highly specific). Antihistone Ab: drug-induced SLE (r100%).

Antiphospholipid Ab (eg anti-cardiolipin Ab): antiphospholipid syndrome, SLE.

Anticentromere Ab: limited systemic sclerosis.

Anti-extractable nuclear antigen (ENA) antibodies (usually with +ve ANA):

Anti-Ro (SSA) SLE, Sjögren's syndrome, systemic sclerosis.

Associated with congenital heart block.

Anti-La (SSB) Sjögren's syndrome, SLE (15%).

Anti-Sm SLE (20–20%).

Anti-RNP SLE, mixed connective tissue disease.

Anti Jo-1; Anti-Mi-2 Polymyositis, dermatomyositis.

Anti-Scl70 Diffuse systemic sclerosis.

Gastrointestinal: (For liver autoantibodies, see p284.)

Antimitochondrial Ab (AMA): primary biliary cholangitis (>95%), autoimmune hepatitis (20%), idiopathic cirrhosis (25–20%).

Anti-smooth muscle Ab (SMA): autoimmune hepatitis (70%), primary biliary chol-angitis (50%), idiopathic cirrhosis (25–20%).

Gastric parietal cell Ab: pernicious anaemia (>90%), atrophic gastritis (40%), 'normal' (10%).

Intrinsic factor Ab: pernicious anaemia (50%).

a-gliadin Ab, antitissue transglutaminase, anti-endomysial Ab: coeliac disease. Endocrine: Thyroid peroxidase Ab: Hashimoto's thyroiditis (r87%), Graves' (>50%). Islet cell Ab (ICA), glutamic acid decarboxylase (GAD) Ab: type 1 diabetes mellitus (75%).

Renal: Glomerular basement membrane Ab (anti-GBM): Goodpasture's disease (100%).

Antineutrophil cytoplasmic Ab (ANCA):

Cytoplasmic (cANCA), specific for serine proteinase-3 (PR3 +ve). Granulomatosis with polyangiitis (Wegener's) (90%); also microscopic polyangiitis (20%), pol-yarteritis nodosa (11%).

Perinuclear (pANCA), specific for myeloperoxidase (MPO +ve). Microscopic poly-angiitis (45%), Churg–Strauss, pulmonary-renal vasculitides (Goodpasture's).

Unlike immune-complex vasculitis, in ANCA-associated vasculitis no complement consumption or immune complex deposition occurs (ie pauci-immune vasculitis).'[2] ANCA may also be +ve in UC/Crohn's, sclerosing cholangitis, autoimmune hepatitis,

Felty's, RA, SLE, or drugs (eg antithyroid, allopurinol, ciprofloxacin). Neurological: Acetylcholine receptor Ab: myasthenia gravis (90%)(see p512). Anti-voltage-gated K +-channel Ab: limbic encephalitis.

Anti-voltage-gated Ca2+-channel Ab: Lambert–Eaton syndrome (see p512).

Anti-aquaporin 4: neuromyelitis optica (Devic's disease, p698).

Most centres now use anti-CCP antibodies for the initial workup of suspected RA.

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Management Refer: complex cases should involve specialist SLE/nephritis clinics.

General measures: High-factor sunblock. Hydroxychloroquine, unless contraindi-cated, reduces disease activity and improves survival. Screen for co-morbidities and medication toxicity. For skin flares, first trial topical steroids.

Maintenance: NSAIDs (unless renal disease) and hydroxychloroquine for joint and skin symptoms. Azathioprine, methotrexate, and mycophenolate as steroid-spar-ing agents. Belimumab (monoclonal antibody) used as an add-on therapy for auto-antibody positive disease where disease activity is high.'' (See BOX.)

Mild flares: (No serious organ damage.) Hydroxychloroquine or low-dose steroids.

Moderate flares: (Organ involvement.) May require DMARDs or mycophenolate.

——— Severe flares: If life- or organ-threatening, eg haemolytic anaemia, nephritis, severe pericarditis or CNS disease; urgent high-dose steroids, mycophenolate, rituxi-mab, cyclophosphamide. MDT working vital for neuropsychiatric lupus (psychometric testing, lumbar puncture may be indicated).

Lupus nephritis: (p214.) May require more intensive immunosuppression with ster-oids and cyclophosphamide or mycophenolate. BP control vital (e.g. ACE-i). Renal re-placement therapy (p206) may be needed if disease progresses; nephritis recurs in r50% post-transplant, but is a rare cause of graft failure.'™

Prognosis: r80% survival at 15 years.'3 There is an increased long-term risk of CVD

and osteoporosis.

Fiorentino, MD, PhD; by kind permission of Skin & Aging.

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The vasculitides are inflammatory disorders of blood vessels; commonly classified using the modified Chapel Hill criteria.'8 They can affect any organ, and presentation depends on the organs involved. It may be a primary condition or secondary to other diseases, eg SLE, RA, hepatitis B & C, HIV. Categorized by size of blood vessels affected.

Large Giant cell arteritis, Takayasu's arteritis (see p712).

Medium Polyarteritis nodosa, Kawasaki disease (OHCS p646).

Small • ANCA-associated: microscopic polyangiitis; granulomatosis with polyangiitis (Wegener's granulomatosis); and eosinophilic granulomatosis with polyangiitis (Churg Strauss syndrome; 40–60% are ANCA positive). • Immune complex vasculitis: Goodpas-ture's disease; cryoglobulinaemic vasculitis; IgA vasculitis (Henoch-Schonlein purpura).

Variable vessel vasculitis Behçet's (p694) and Cogan's syndrome.

Microscopic polyangiitis A necrotizing vasculitis affecting small- and medium-sized vessels. Symptoms: Rapidly progressive glomerulonephritis usually features; pulmonary haemorrhage occurs in up to 20%; other features are rare. Tests: pANCA (MPO) +ve (p552). Treatment: Steroids plus eg methotrexate. For maintenance: meth-otrexate, rituximab, or azathioprine.

Hypocomplementaemic urticarial vasculitis A lupus-like illness with urticaria and antibodies to complement (C1q).

Low-dose aspirin has been shown to decrease the rate of visual loss and cerebrovascular accidents in GCA

but there are also conflicting reports regarding its efficacy at preventing ischaemic events in GCA.

Fig 12.17 Livedo reticularis: pink–blue mottling caused by capillary dilata-tion and stasis in skin venules. Causes: physiological, cold, or vasculitis.

Fibromyalgia

Fibromyalgia comprises up to 10% of new referrals to the rheumatology clinic.™' Prevalence: 0.5–4%. φ:ð≈6:1 but varies depending on which diagnostic criteria are used .

Risk factors: BOX 'Risk factors'. Female sex, middle age, low household income, di-vorced, low educational status.

Associations: Other somatic syndromes such as chronic fatigue syndrome, irritable bowel syndrome (p266), and chronic headaches syndromes (see OHCS p502). Also found in r25% of patients with RA, AS, and SLE.

Investigations: All normal. Diagnosis is clinical. Over-investigation can consolidate illness behaviour; but exclude other causes of pain and/or fatigue (eg RA, PMR see p557, vasculitis see p556, hypothyroidism see p220, myeloma see p268).

Chronic fatigue syndrome (AKA myalgic encephalomyelitis)

The eye is host to many diseases: the more you look, the more you'll see, and the more you'll enjoy, not least because the eye is as beautiful as its signs are legion. Behçet's (p694.) Systemic inflammatory disorder, HLA B27 association. Causes a uvei-

tis amongst other systemic manifestations. Cause unknown.

Granulomatous disorders Syphilis, TB, sarcoidosis, leprosy, brucellosis, and tox-oplasmosis may inflame either the front chamber (anterior uveitis/iritis) or back chamber (posterior uveitis/choroiditis). Refer to an ophthalmologist.

Systemic inflammatory diseases May manifest as iritis in ankylosing spondylitis and reactive arthritis; uveitis in Behçet's; conjunctivitis in reactive arthritis; scle-ritis or episcleritis in RA, vasculitis, and SLE. Scleritis in RA and granulomatosis with polyangiitis (Wegener's) may damage the eye. ——— Refer urgently if eye pain. GCA causes optic nerve ischaemia presenting as sudden blindness.

Keratoconjunctivitis sicca A reduction in tear formation, tested by the Schirmer filter paper test (<5mm in 5min). It causes a gritty feeling in the eyes, and a dry mouth (xerostomia from 'saliva production). It is found on its own (Sjögren's syn-drome), or with other diseases, eg SLE, RA, sarcoidosis. 9: Artificial tears/saliva.

Haematological disorders Retinal haemorrhages occur in leukaemia; comma-shaped conjunctival haemorrhages and retinal new vessel formation may occur in sickle-cell disease. Optic atrophy is seen in pernicious anaemia (and also MS).

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Table 12.5 Differential diagnosis of a red eye

| | Appearance | Viscosity | WBC/mm3 | Neutrophils | | — | — | — | — | — | | Normal | Clear, colourless | — | ≤200 | None | | Osteoarthritis | Clear, straw | ≤1000 | ≤1000 | ≤50% | | Haemorrhagic* | Bloody, xanthochromic | Varies | ≤10 000 | ≤50% | | Acutely inflamed | Acutely inflamed | Acutely inflamed | Acutely inflamed | Acutely inflamed | | RA | Turbid, yellow | ' | 1000–50 000 | Varies | | Crystal | Turbid, yellow | ' | 5000–50 000 | r80% | | Septic | Turbid, yellow | ' | 10 000–100 000 | >90% |

Conjunctiva Iris Pupil Cornea Anterior

Intraocular Treatment Appearance

Acute glaucoma Both ciliary and conjunctival vessels injected.

Injected Dilated,

Steamy, Very shal-

Very high Refer.

Entire eye is red. See OHCS p420.

fixed, oval

hazy

low

IV acetazolamide + pilocarpine drops (miotic); peripheral iridotomy.

Anterior uveitis (iritis)

Redness most marked around cornea, which doesn't blanch on pressure. Usually unilateral. Causes: AS, RA, Reiter's sarcoidosis, herpes simplex, herpes zoster, and Behçet's disease. NB: a similar scleral appearance but without papillary or anterior chamber involvement may be scleritis (eg RA, SLE, vasculitis).

Injected Small,

irregular due to adhesions between the anterior lens and the pupil margin

Normal Turgid Normal Refer.

Steroid eye drops (eg 0.5% predniso-lone) + mydriatic (eg cyclopentolate 0.5%).

Conjunctivitis Often bilateral.

Conjunctival vessels injected, greatest toward fornices, but blanching on pressure.

Mobile over sclera. Purulent discharge.

Normal Normal Normal Normal Normal Most do not require

treatment.

Consider chloram-phenicol ointment or drops.

Subconjunctival haemorrhage

Bright red sclera with white rim around limbus. Causes: •BP; leptospirosis; bleeding disorders; trauma; snake venom; haemor-rhagic fevers.

Normal Normal Normal Normal Normal Looks alarming but

resolves spontane-ously. Check BP

if elderly; refer if traumatic; on warfarin?

Erythema nodosum (fig 12.21) Painful, blue-red, raised lesions on shins (! thighs/ arms). Causes: sarcoidosis, drugs (sulfonamides, contraceptive pill, dapsone), strep-tococcal infection. Less common: IBD, BCG vaccination, leptospirosis, Mycobacte-rium (TB, leprosy), Yersinia, or various viruses and fungi. Cause unknown in 20–50%.

Erythema migrans (fig 12.22) Presents as a small papule at the site of a tick bite which develops into a spreading large erythematous ring, with central fading. It lasts from 48h to 2 months and there may be multiple lesions in disseminated disease. Cause: the rash is pathognomonic of Lyme disease and occurs in r80% of cases (p422).

Erythema marginatum Pink coalescent rings on trunk which come and go. It is seen in rheumatic fever (or rarely other causes, eg drugs). See fig 2.42, p142.

Pyoderma gangrenosum (fig 12.24) Recurring nodulo-pustular ulcers, r10cm wide, with tender red/blue overhanging necrotic edge, purulent surface, and healing with cribriform scars on leg, abdomen, or face. Associations: IBD, autoimmune hepatitis, granulomatosis with polyangiitis (Wegener's), myeloma, neoplasia. φ>ð. Treatment: get help. Oral steroids ! ciclosporin should be 1st-line therapy.[a']

Vitiligo (fig 12.25) Vitellus is Latin for spotted calf: typically white patches ! hyper-pigmented borders. Sunlight makes them itch. Associations: autoimmune disorders; premature ovarian failure. Treat by camouflage cosmetics and sunscreens (! steroid creams ! dermabrasion). UK Vitiligo Society: 0800 018 2621.

Crohn's Perianal/vulval/oral ulcers; erythema nodosum; pyoderma gangrenosum.

Dermatomyositis Gottron's papules (rough red papules on the knuckles/extensor surfaces); shawl sign; heliotrope rash on eyelids (fig 12.26). It may be associated with lung, bowel, ovarian, or pancreatic malignancy (p552).

Diabetes mellitus Ulcers, necrobiosis lipoidica (shiny yellowish area on shin ! tel-angiectasia; fig 12.27), granuloma annulare (OHCS p586), acanthosis nigricans (pig-mented, rough thickening of axillary, neck, or groin skin with warty lesions; fig 12.28).

Hyperthyroidism Pretibial myxoedema: Red oedematous swellings above lateral malleoli, progressing to thickened oedema of legs and feet, thyroid acropachy—clubbing + subperiosteal new bone in phalanges. Other endocrinopathies: p202.

Liver disease Palmar erythema; spider naevi; gynaecomastia; decrease in pubic hair; jaundice; bruising; scratch marks. Malabsorption Dry pigmented skin, easy bruising, hair loss, leuconychia. Neoplasia Acanthosis nigricans: (See 'Diabetes mellitus' and fig 12.28.) Associ-ated with gastric cancer. Dermatomyositis: see earlier in topic. Thrombophlebitis

migrans: Successive crops of tender nodules affecting blood vessels throughout

the body, associated with pancreatic cancer (especially body and tail). Acquired ichthyosis: Dry scaly skin associated with lymphoma. Skin metastases: Especially melanoma, and colonic, lung, breast, laryngeal/oral, or ovarian malignancy.

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Surgery

The language of surgery

Right upper quadrant (RUQ) or hypochondrium.

Epigastrium.

Left upper quadrant (LUQ) or hypochondrium.

Right flank (merges posteriorly with right loin, p57).

Periumbilical or central area.

Left flank (merges posteriorly with left loin, p57).

Right iliac fossa (RIF).

Suprapubic area.

Left iliac fossa (LIF).

-ectomy Cutting something out.

-gram A radiological image.

-pexy Anchoring of a structure to keep it in position.

-plasty Surgical refashioning in order to regain function/cosmesis.

-scopy Procedure with instrumentation for looking into the body.

-stomy An artificial union between a conduit and the outside or another conduit.

-tomy Cutting something open to the outside world.

-tripsy Fragmentation of an object.

angio- Tube or vessel lith- Stone

appendic- Appendix mast- Breast chole- Relating to gall/bile meso- Mesentery colp- Vagina nephr- Kidney

cyst- Bladder orchid- Testicle

-doch- Ducts oophor- Ovary

enter- Small bowel phren- Diaphragm eschar- Dead tissue, eg from burn pyloro- Pyloric sphincter gastr- Stomach

pyel- Renal pelvis

hepat- Liver proct- Anal canal

hyster- Uterus salping- Fallopian tube

lapar- Abdomen splen- Spleen

abscess A cavity containing pus. Remember: if there is pus about, let it out. colic Intermittent pain from over-contraction/obstruction of a hollow viscus. cyst Fluid-filled cavity lined by epi/endothelium.

fistula An abnormal connection between two epithelial surfaces. Fistulae often close spontaneously, but will not in the presence of malignant tissue, dis-tal obstruction, foreign bodies, chronic inflammation, and the formation of a muco-cutaneous junction (eg stoma).

hernia The protrusion of a viscus/part of a viscus through a defect of the wall of its containing cavity into an abnormal position.

ileus Used in this book as a term for adynamic bowel.

sinus A blind-ending tract, typically lined by epithelial or granulation tissue, which opens to an epithelial surface.

stent An artificial tube placed in a biological tube to keep it open.

stoma (p582) An artificial union between conduits or a conduit and the outside. ulcer (p660) Interruption in the continuity of an epi/endothelial surface. volvulus (p611) Twisting of a structure around itself. Common GI sites include the

sigmoid colon and caecum, and more rarely the stomach.

epi- Upon pan- Whole peri- Around end- Inside para- Alongside sub- Beneath mega- Enlarged per- Going through trans- Across

Aims To provide diagnostic and prognostic information. Ensures the patient under-stands the nature, aims, and expected outcome of surgery. To allay anxiety and pain:

Ensure that the right patient gets the right surgery. Have the symptoms and signs changed? If so, inform the surgeon.

Assess/balance risks of anaesthesia, and maximize fitness. Comorbidities? Drugs? Smoker? Optimizing oxygenation before major surgery improves outcome.

Obtain informed consent (p568).

Check proposed anaesthesia/analgesia with anaesthetist.

Family history May be relevant, eg in malignant hyperpyrexia (p572); dystrophia myotonica (p510); porphyria; cholinesterase problems; sickle-cell disease.

Drugs Any drug/plaster/antiseptic allergies? ►Inform the anaesthetist about all

drugs even if 'over-the-counter'. ►Steroids: see p590; diabetes: see p588.

Antibiotics: Tetracycline and neomycin may • neuromuscular blockade.

Anticoagulants: ►Tell the surgeon. Avoid epidural, spinal, and regional blocks. Aspirin should probably be continued unless there is a major risk of bleeding. Dis-cuss stopping clopidogrel therapy with the cardiologists/neurologists. See p590.

Anticonvulsants: Give as usual pre-op. Post-op, give drugs IV (or by NGT) until able to take orally. Valproate: give usual dose IV. Phenytoin: give IV slowly (<50mg/min, on cardiac monitor). IM phenytoin absorption is unreliable.

p-blockers: Continue up to and including the day of surgery as this precludes a labile cardiovascular response.

Contraceptive pill: See BNF. Stop 4wks before major/leg surgery; ensure alterna-tive contraception is used. Restart 2wks after surgery, provided patient is mobile.

Digoxin: Continue up to and including morning of surgery. Check for toxicity (ECG; plasma level); do plasma K+ and Ca2+ (suxamethonium can •K+ and lead to ventricu-lar arrhythmias in the fully digitalized).

Diuretics: Beware hypokalaemia, dehydration. Do U&E (and bicarbonate).

Eye-drops: β-blockers get systemically absorbed.

HRT: As with contraceptive pill there may be an increased risk of DVT/PE.

Levodopa: Possible arrhythmias when patient under GA.

Lithium: Get expert help; may potentiate neuromuscular blockade and cause ar-rhythmias. See OHCS p249.

MAOIS: Get expert help as interactions may cause hypotensive/hypertensive crises.

Thyroid medication: see p600.

Tricyclics: These enhance adrenaline (epinephrine) and arrhythmias.

Preparation ►Starve patient; NBM ?2h pre-op for clear fluids and ?6h for solids.[1]

Is any bowel or skin preparation needed, or prophylactic antibiotics (p570)?

Start DVT prophylaxis as indicated, eg graduated compression stockings (CI in pe-ripheral arterial disease); LMWH (p250): eg moderate risk, 20mg r2h pre-op then 20mg/24h; high risk (eg orthopaedic surgery), 40mg 12h pre-op then 40mg/24h; or heparin 5000U SC 2h pre-op, then every 8–12h SC for 7d or until ambulant.

Ensure necessary premedications (p572), regular medications, analgesia, anti-emetics, antibiotics are all prescribed as appropriate. Confirm NBM.

Book any pre-, intra-, or post-operative x-rays or frozen sections.

Book post-operative physiotherapy.

If needed, site IV cannula, catheterize (p762), and/or insert a Ryle's tube (p759).

Meta-analyses have shown no benefit to patients from mechanical bowel cleansing before colonic surgery—this is no longer considered good practice. There is also no evidence for the use of enemas prior to rectal surgery.

- If within the last 6 months, the perioperative risk of re-infarction (up to 40%) makes most elective surgery too risky. ECHO, and stress testing (+ exercise ECG or MUGA scan, p741) should be done.

In which of the following situations would you seek 'informed written consent' from a patient? 1 Feeling for a pulse. 2 Taking blood. 3 Inserting a central line.

Removing a section of small bowel during a laparotomy for division of adhesions.

Orchidectomy after a failed operation for testicular torsion.

For consent to be valid

It can be given any time before the intervention/treatment is initiated. Earlier is better as this will give the patient time to think about the risks, benefits, and al-ternatives—he or she may even bring forward questions on issues that you had not considered relevant. Think of consent as an ongoing process throughout the patient's time with you, not just the moment of signing the form.

The proposed treatment or test must be clearly understood by the patient, taking into account the benefits, risks (including complication rates if known), additional procedures, alternative courses of action, and their consequences.

It must be given voluntarily. This can be difficult to evaluate—eg when live organ donation is being considered—see BOX 'Special circumstances for consent' for other difficult situations.

The doctor providing treatment or undertaking the test needs to ensure that the patient has given valid consent. The act of seeking consent is ultimately the respon-sibility of the doctor looking after the patient, though the task may be delegated to another health professional, as long as they are suitably trained and qualified.

When taking consent

Does the patient currently have capacity for the decision in question?

Are you the right person to be obtaining consent?[2]

Use words the patient understands and avoid jargon and abbreviations.

Ensure that they believe your facts and can retain ‘pros’ and ‘cons’ long enough to inform their decision. Fact sheets/ diagrams for individual operations help.

Make sure their choice is free from pressure from others, and explain that they can withdraw consent at any time after the form is signed. Some patients may view the consent form as a contract from which they cannot renege.

If the patient is illiterate, a witnessed mark does endorse valid consent. Similarly, if the patient is willing but physically unable to sign the consent form, then a wit-nessed entry into the medical notes stating so is valid.

Remember to discuss further procedures that may become necessary during the proposed treatment. This avoids waking up to a nasty surprise (eg a missing testi-cle as in scenario 5 earlier in this topic).

If in any doubt, turn to: your senior/consultant; your employing organization; your legal defence organiza-tion; your national medical association; your local research ethics committee.

Time administration correctly (eg IV prophylaxis should be given 20min prior to surgery to maximize skin concentration; metronidazole PR is given 2h before).

Use antibiotics which will kill anaerobes and coliforms.

Consider use of peri-operative supplemental oxygen. This is a practical method of reducing the incidence of surgical wound infections.

Practise strictly sterile surgical technique. (Ask for a hand with scrubbing up if you are not sure—theatre staff will be more than pleased to help!)

Antibiotic regimens ►Adhere to local guidelines. BNF examples include:

Appendicectomy; colorectal resections and open biliary surgery: A single dose of IV piperacillin/tazobactam 4.5g/8h or gentamicin 1.5mg/kg + metronidazole 500mg or co-amoxiclav 1.2g alone.

Oesophageal or gastric surgery: 1 dose of IV gentamicin or piperacillin/tazo-bactam or co-amoxiclav (doses as for appendicectomy).

Vascular surgery: 1 dose of IV piperacillin/tazobactam or flucloxacillin 1–2g + gentamicin. Add metronidazole if risk of anaerobes (eg amputations, gangrene, or diabetes).

MRSA: For high-risk patients add teicoplanin or vancomycin to the above-listed regimens.

Table 12.1 Classification of surgical procedures and wound infection risk

| American Society of Anesthesiologists (ASA) classification Class I Normally healthy patient. Class II Mild systemic disease. Class III Severe systemic disease that limits activity but is not incapacitating. Class IV Incapacitating systemic disease which poses a constant threat to life. Class V Moribund: not expected to survive 24h even with operation. You will see a space for an ASA number on most anaesthetic charts. It is a health. index at the time of surgery. The suffix E is used in emergencies, eg ASA 2E. | | — |

Data from MRCS Core Modules: Essential Revision Notes, S. Andrews, Pastest.

Before anaesthesia, explain to the patient what will happen and where they will wake up, otherwise the recovery room or ITU will be frightening. Explain that they may feel ill on waking. The premedication aims to allay anxiety and to make the anaesthesia itself easier to conduct (see BOX). Typical regimens might include:

Anxiolytics: Benzodiazepines, eg lorazepam 2mg PO; temazepam 10–20mg PO. In children, use oral premeds as first choice, eg midazolam 0.5mg/kg (tastes bitter so often put in paracetamol suspension). Give oral premedication 2h before surgery (1h if IM route used).

Analgesics: See p574. Pre-emptive analgesia is not often used and effects are hard to determine. The aim is to dampen pain signals before they arrive. In children or anxious adults, local anaesthetic cream may be used on a few sites before inserting an IVI (►the anaesthetist may prefer to site the cannula themselves!).

Anti-emetics: Post-operative nausea and vomiting is experienced by r25% of all patients. 5HT2 antagonists (eg ondansetron 4mg IV/IM) are the most effective agents; others, eg metoclopramide 10mg/8h IV/IM/PO, are also used—see p251.

Antacids: Ranitidine 50mg IV or omeprazole 40mg PO/IV if aspiration risk.

Antisialogues: Glycopyrronium (200–400mcg in adults, 4–8mcg/kg in children; given IV/IM 20–60min before induction) is sometimes used to decrease secretions that may cause respiratory obstruction in smaller airways.

Antibiotics: See p570.

Side-effects of anaesthetic agents

Hyoscine, atropine: Anticholinergic, m tachycardia, urinary retention, glaucoma, sedation (especially in the elderly).

Opioids: Respiratory depression, ‘cough reflex, nausea and vomiting, constipation.

Thiopental: (Induction agent.) Laryngospasm.

Propofol: (Induction agent.) Respiratory/cardiac depression, pain on injection.

Volatile agents, eg isoflurane: Nausea and vomiting, cardiac depression, respira-tory depression, vasodilation, hepatotoxicity (see BNF).

The complications of anaesthesia are due to loss of:

Pain sensation: Urinary retention, pressure necrosis, local nerve injuries (eg radial nerve palsy from arm hanging over the table edge).

Consciousness: Cannot communicate ‘wrong leg/kidney’. NB: in some patients (eg 0.15%) retained consciousness is the problem. Awareness under GA sounds like a contradiction of terms, but remember that anaesthesia is a process rather than an event. Such awareness can lead to ill-defined, delayed neuroses and post-traumatic stress disorder (OHCS p252).

Muscle power: Corneal abrasion (m tape the eyes closed), no respiration, no cough (leads to pneumonia and atelectasis—partial lung collapse causing shunting ! im-paired gas exchange: it starts minutes after induction, and may be related to the use of 100% O2, supine position, surgery, age, and to loss of power).

Local/regional anaesthesia If unfit/unwilling to undergo general anaesthesia, lo-cal nerve blocks (eg brachial plexus) or spinal blocks (contraindication: anticoagula-tion, local infection) using long-acting local anaesthetics such as bupivacaine may be indicated. See table 12.4 for doses and toxicity effects.

Drugs complicating anaesthesia ►Inform anaesthetist. See p566 for lists of spe-cific drugs, and actions to take.

Give 1mg/kg every 5min IV—up to 10mg/kg in total (OHCS p628).

Approach to management (fig 12.5 and p522.) Review and chart each pain carefully and individually.

Identify and treat the underlying pathology wherever possible.

Give regular doses rather than on an ‘as-required’ basis.

Choose the best route: PO, PR, IM, epidural, SC, inhalation, or IV.

Explanation and reassurance contribute greatly to analgesia.

Allow the patient to be in charge. This promotes wellbeing, and does not lead to overuse. Patient-controlled continuous IV morphine delivery systems are useful.

Liaise with the Acute Pain Service, if possible.

Epidural analgesia Opioids and anaesthetics are given into the epidural space by infusion or as boluses. Ask the advice of the Pain Service. SES are thought to be less, as the drug is more localized: watch for respiratory depression and local anaesthetic-induced autonomic blockade (‘BP).

Not to be confused with hyoscine hydrobromide; used for drying secretions and in motion sickness.

WHO’s Pain relief ladder

Confusion may manifest as agitation, disorientation, and attempts to leave hospital, especially at night. Gently reassure the patient in well-lit surroundings. See p484 for a full work-up. Common causes are:

hypoxia (pneumonia, atelectasis, LVF, PE) • infection (see earlier)

drugs (opiates, sedatives, and many others) • alcohol withdrawal (p280)

urinary retention • liver/renal failure.

MI or stroke

Occasionally, sedation is necessary to examine the patient; consider lorazepam 1mg PO/IM (antidote: flumazenil) or haloperidol 0.5–2mg IM. Reassure relatives that post-op confusion is common (seen in up to 40%) and reversible.

Dyspnoea or hypoxia Any previous lung disease? Sit patient up and give O2, monitor peripheral O2 sats by pulse oximetry (p162). Examine for evidence of: •pneumonia, pulmonary collapse, or aspiration •LVF (MI; fluid overload) •pulmonary embolism (p190) •pneumothorax (p190; due to CVP line, intercostal block, or mechanical ventila-tion). Tests: FBC; ABG; CXR; ECG. Manage according to findings.

BP may be from pain, urinary retention, idiopathic hypertension (eg missed medi-cation), or inotropic drugs. Oral cardiac medications (including antihypertensives) should be continued throughout the perioperative period even if NBM. Treat the cause, consider increasing the regular medication, or if not absorbing orally try 50mg labetalol IV over 1min (see p140).

Review fluid chart and examine for signs of volume depletion.

Urinary retention is also common, so examine for a palpable bladder.

Establish normovolaemia (a CVP line may help); you may need 1L/h IVI for 2–2h. A ‘fluid challenge’ of 250–500mL over 20min may also help.

Catheterize bladder (for accurate monitoring)—see p762; check U&E.

If intrinsic renal failure is suspected, stop nephrotoxic drugs (eg NSAIDS, ACE-i) and refer to a nephrologist early.

Nausea/vomiting Any mechanical obstruction, ileus, or emetic drugs (opiates, dig-oxin, anaesthetics)? Consider AXR, NGT, and an anti-emetic (►not metoclopramide because of its prokinetic property). See p251 for choice of anti-emetics.

Na+ Pre-op level? Excess IV fluids may exacerbate the situation. Correct slowly (p672). SIADH (p672) can be precipitated by pain, nausea, opioids, and chest infection.

DVTS occur in 25–50% of surgical patients, and many non-surgical patients. All hospi-tal inpatients should be assessed for DVT/PE risk and offered prophylaxis if appropri-ate. 65% of below-knee DVTs are asymptomatic; these rarely embolize to the lung.

Risk factors •Age, pregnancy, synthetic oestrogen, trauma, surgery (especially pel-vic/orthopaedic), past DVT, cancer, obesity, immobility, thrombophilia (p274).

Signs •Calf warmth/tenderness/swelling/erythema. •Mild fever. •Pitting oedema.

Cellulitis; ruptured Baker’s cyst. Both may coexist with a DVT.

Tests ►Calculate Wells score (see table 12.5) before ordering D-dimer. D-dimer is sen-sitive but not specific for DVT (also • in infection, pregnancy, malignancy, and post-op).

Wells score: ≤1 point = DVT unlikely: Perform D-dimer. If negative, DVT excluded. If positive, proceed to USS (if USS negative, DVT excluded; if positive, treat as DVT).

≥2 points = DVT likely: Do D-dimer and USS. If both negative, DVT excluded. If USS posi-tive, treat as DVT. If D-dimer positive and USS negative, repeat USS in 1 week.

Do thrombophilia tests (p274) before commencing anticoagulant therapy if there are no predisposing factors, in recurrent DVT, or if DVT in unusual site. Look for un-derlying malignancy: Urine dip; FBC, LFT, Ca2+; CXR ! CT abdomen/pelvis (and mam-mography in φ) if >40yrs.

Nine questions to ask

- Is it both legs? 2 Is she pregnant? 3 Is she mobile?

4 Any trauma? 5 Any pitting (p25)? 6 Past diseases/on drugs? 7 Any pain? 8 Any skin changes? 9 Any oedema elsewhere? Tests ►Look for proteinuria (+hypoalbuminaemia ≈nephrotic syndrome). CCF?

Treatment of leg oedema Treat the cause. Diuretics for all is not an answer. Elevat-ing legs for dependent oedema (ankles higher than hips—do not just use footstools); raise the foot of the bed. Graduated support stockings may help (CI: ischaemia).

Mastectomy Arm lymphoedema in up to 20% of those undergoing axillary node sampling or dissection. The risk of lymphoedema increases with the level of axillary dissection: risk is lower with level 1 dissection (remains inferior to pec. minor) com-pared to level 2 dissection (goes superior to pec. minor, rarely done); skin necrosis.

Arterial surgery Bleeding; thrombosis; embolism; graft infection; MI; AV fistula formation. Complications of aortic surgery: Gut ischaemia; renal failure; respira-tory distress; trauma to ureters or anterior spinal artery (leading to paraplegia); ischaemic events from distal emboli from dislodged thrombus; aorto-enteric fistula.

Colonic surgery Early: Sepsis; ileus; fistulae; anastomotic leak (11% for radical rectal surgery); haemorrhage; trauma to ureters or spleen. Late: Adhesional obstruction (BOX).

Tracheostomy Mediastinitis; surgical emphysema. Later: stenosis.

Splenectomy (p272.) Acute gastric dilatation (a serious consequence of not using a NGT, or to check that the one in place is working); thrombocytosis; sepsis. ►Lifetime sepsis risk is partly preventable with pre-op vaccines—ie Haemophilus type B, meningococcal, and pneumococcal (p407 & p167) and prophylactic penicillin.

Genitourinary surgery Septicaemia (from instrumentation in the presence of in-fected urine)—consider a stat dose of gentamicin; urinoma—rupture of a ureter or renal pelvis leading to a mass of extravasated urine.

Gastrectomy See p622. Prostatectomy p642. Haemorrhoidectomy p622.

582

A stoma (Greek=mouth) is an artificial union between a conduit and the outside world—eg a colostomy, in which faeces are made to pass through an opening in the abdominal wall when a loop of colon is brought out onto the skin. NB: a stoma can also be made between two internal conduits (eg a choledochojejunostomy).

Colostomies (Usually left illiac fossa and flush with the skin—fig 12.8.) May be tem-porary or permanent. Are they suitable for a laparoscopic operation?

Loop colostomy: A loop of colon is exteriorized and partially divided, forming two stomas that are joined together (the proximal end passes stool, the distal end passes mucus, see fig 12.6). A rod under the loop prevents retraction and may be removed after 7d. A loop colostomy is often temporary and performed to protect a distal anastomosis, eg after anterior resection.

End colostomy: The bowel is divided and the proximal end brought out as a stoma; the distal end may be: 1 resected, eg abdominoperineal (AP) resection (inspect the perineum for absent anus when examining a stoma) 2 closed and left in the abdo-men (Hartmann's procedure) 2 exteriorized, forming a 'mucous fistula'.

Paul–Mikulicz colostomy: A double-barrelled colostomy in which the colon is di-vided completely (eg to excise a section of bowel). Each end is exteriorized as two separate stomas.

Output: Colostomies ideally pass 1–2 formed motions/day into an adherent plastic pouch. Some may be managed with irrigation, thus avoiding a pouch.

Incidence: 21 000 stomas/yr in UK (>50% are permanent). Most manage their stomas well. The cost for appliances is r£1500/yr. If there is a skin reaction to the adhesive or pouch, a change of device may be all that is needed. Contact the stoma nurse.

Alternative (non-stoma forming) surgery Low/ultralow anterior resection: All or part of the rectum is excised and the proximal colon anastomosed to the top of the anal canal (the lower the level of anastomosis, the higher the risk of complication). Transanal endoscopic microsurgery: Allows excision of small tumours within the rectum with preservation of sphincter function.

Urostomies are fashioned after total cystectomy, bringing urine from the ureters to the abdominal wall via an ileal conduit that is usually incontinent. Formation of a catheterizable valvular mechanism may retain continence.

Advances in urological surgery have seen an increase in continence-saving procedures such as orthotopic neobladder reconstruction, with good long-term continence rates.

When choosing a stoma site, avoid:

Bony prominences (eg anterior superior iliac spine, costal margins).

The umbilicus.

Old wounds/scars—there may be adhesions beneath.

Skin folds and creases.

The waistline.

The site should be assessed pre-operatively by the stoma nurse, with the patient both lying and standing.

584

Over 25% of hospital inpatients may be malnourished. Hospitals can become so focused on curing disease that they ignore the foundations of good health—mal-nourished patients recover more slowly and experience more complications. See table 12.6.

Why are so many hospital patients malnourished?

Increased nutritional requirements (eg sepsis, burns, surgery).

Increased nutritional losses (eg malabsorption, output from stoma).

Decreased intake (eg dysphagia, nausea, sedation, coma).

Effect of treatment (eg nausea, diarrhoea).

Enforced starvation (eg prolonged periods nil by mouth).

Missing meals (eg due to investigations—minimize meal time disruption).

Difficulty with feeding (eg lost dentures; no one available to assist).

Unappetizing food.

Identifying at-risk patients Assess nutrition state (using eg Malnutrition Univer-sal Screening Tool3) and weight on admission; reassess weekly thereafter. Involve dieticians early in those at risk.

History: Recent ‘weight (>20%, accounting for fluid balance); recent reduced in-take; diet change (eg recent change in consistency of food); nausea, vomiting, pain, diarrhoea which might have led to reduced intake.

Investigations: Generally unhelpful. Low albumin suggestive, but is affected by many things other than nutrition.

•Albumin can be helpful in monitoring recovery.

Guidelines for success

Use fine-bore (9Fr) nasogastric feeding tube when possible.

Check position of nasogastric tube (pH testing) before starting feeding (p759); the positioning of a nasojejunal tube can be checked on abdominal X-ray.

Build up feeds gradually to avoid diarrhoea and distension.

Weigh at least weekly.

Check blood glucose and plasma electrolytes (monitor for refeeding syndrome if previously malnourished—p587).

Treat underlying conditions vigorously, eg sepsis may impede +ve nitrogen balance.

Table 12.6 Daily energy and nutritional requirements

| The right to refuse treatment Theirs not to make reply, Theirs not to reason why, Theirs but to do and die. Alfred, Lord Tennyson from The Charge of the Light Brigade, 1854. The rights of a patient are something of an antithesis to this military macabre of Tennyson, and it is our responsibility to respect the legal and ethical rights of those we treat. We do this not only for the sake of the individual, but also for the sake of an enduring trust between the patient and doctor, remembering that it is the patient’s right to refuse treatment (if a fully competent adult) even when this may result in the death of the patient, or even the death of an unborn child, whatever the stage of pregnancy. The only exceptions apply to treatment of mental health disorders (see eg in England and Wales the Mental Health Act 2007). | | — |

Administration Nutrition must be given via a dedicated central venous line (or peripherally inserted central catheter—PICC line) or via a dedicated lumen of a multi-lumen catheter (see figs 12.9 and 12.10).

Requirements There are many different regimens for parenteral feeding. Most provide 2000kCal and 10–14g nitrogen in 2–2L; this usually meets a patient’s daily requirements (see table 12.6, p585). r50% of calories are provided by fat and r50% by carbohydrate. Regimens comprise vitamins, minerals, trace elements, and elec-trolytes; these will normally be included by the pharmacist.

Complications

Thrombosis: Central vein thrombosis may occur, resulting in pulmonary embolus or superior vena caval obstruction (p528).

Metabolic imbalance: Electrolyte abnormalities—see BOX ‘Refeeding syndrome’; deranged plasma glucose; hyperlipidaemia; deficiency syndromes (table 6.9, p268); acid-base disturbance (eg hypercapnia from excessive CO2 production).

Mechanical: Pneumothorax; embolism of IV line tip.

Guidelines for success

Liaise closely with line insertion team, nutrition team, and pharmacist.

Meticulous sterility. Do not use central venous lines for uses other than nutrition. Remove the line if you suspect infection. Culture its tip.

Review fluid balance at least twice daily, and requirements for energy and elec-trolytes daily.

Check weight, fluid balance, and urine glucose daily during establishment of par-enteral nutrition. Check plasma glucose, creatinine and electrolytes (including cal-cium and phosphate), and FBC daily until stable. Check LFT and lipid clearance three times a week until stable. Check zinc and magnesium weekly.

Do not rush. Achieve the maintenance regimen in small steps.

Treat underlying conditions vigorously—eg sepsis may impede +ve nitrogen balance.

Enteral feeding promotes integrity of the gut mucosal barrier, thus preventing bacterial and endotoxin translocation across the gut wall, which can lead to multiple organ dysfunction and perpetuation of a sys-temic inflammatory response—even when the gut is not the primary source of pathology.

Insulin-treated diabetes mellitus

Try to place the patient first on the list in order to minimize the fasting period.

Give all usual insulin the night before surgery.

Long-acting (basal) insulin is usually continued at normal times (eg glargine; de-temir), even when patients are on a variable rate intravenous insulin infusion (VRIII)—previously known as a ‘sliding scale’ (see BOX).

If on AM list, ensure no subcutaneous rapid-acting (bolus) or mixed insulin is given on the morning of surgery. If PM list, give the normal morning bolus insulin, or half the mixed insulin dose.

If eating and drinking post-operatively, resume the usual insulin with evening meal. If AM list (or early PM) and eating a late lunch, give half the morning insulin dose with this meal. If not eating until evening, a VRIII may be needed if the capillary glucose readings are high.

Omit all rapid-acting and mixed insulin while the patient is on a VRIII.

It not eating or drinking post-op, start a VRIII 2 hrs prior to surgery. Aim for serum glucose levels of 6–10mmol/L and check finger-prick glucose every 2 hrs. When ready to eat, give normal dose of rapid acting or mixed insulin with the 1st meal and stop the VRIII 20–60min later.

IV fluid is required while the patient is on a VRIII: see BOX.

A glucose–potassium–insulin (GKI) infusion can be used as an alternative to a VRIII, although it is no longer used as standard in the UK.

Tablet-treated diabetes mellitus

If diabetes is poorly controlled (eg fasting glucose >10mmol/L), treat as for patients on insulin (see earlier in topic).

Give usual medications the night before surgery, except long-acting sulfonylureas (eg glibenclamide) which can cause prolonged hypoglycaemia when fasting and may need to be substituted 2–2 days pre-operatively. Discuss with the diabetes team.

If eating and drinking post-operatively: on AM list, omit morning medication and take any missed drugs with lunch, after surgery. If PM list, take normal medications with breakfast, omit midday doses, and take any missed drugs with a late lunch. The dose of these may need reducing, depending on dietary intake.

If not eating or drinking post-op, start a VRIII 2 hours prior to surgery. Once eating and drinking, oral hypoglycaemics can be restarted.

Some patients may need a phase of subcutaneous insulin following major sur-gery—refer to the diabetes team if serum glucose levels are persistently raised.

Metformin and iodine IV contrast: Metformin can be continued after IV contrast in patients with normal serum creatinine and/or eGFR >60mL/min. To minimize the risk of nephrotoxicity, if serum creatinine is raised or eGFR <60mL/min, stop met-formin for 48h after contrast and check renal function has returned to baseline before restarting.

Diet-controlled diabetes There are usually no problems; patients should be treated as if not diabetic (and do not need to be first on the list). Check capillary blood glucose peri-operatively. Avoid 5% glucose IVI as this increases blood glucose levels.

Avoid operating in patients with obstructive jaundice–consider prior ERCP to relieve. There is •risk of bleeding, peri-operative infection, and renal failure.

Coagulopathy Vitamin K ‘ in obstruction (requires bile in order to be absorbed. If no history of chronic liver disease, give parenteral vitamin K (consider even if clotting is normal). FFP may be required in liver disease or active bleeding.

Sepsis •Risk due to ••bacterial translocation •bacterial colonization of biliary tree

‘neutrophil function. If cholangitis present, give antibiotics. Antibiotic prophylaxis for ERCP not recommended unless biliary decompression fails, or there is a history of biliary disorders; liver transplantation; presence of a pancreatic pseudocyst; or neu-tropenia, in which case give oral ciprofloxacin or IV gentamicin (check local policy).

Minor procedures under local anaesthetic: No supplementation required.

Moderate procedures: (Eg joint replacement.) Give 50mg hydrocortisone before induction and 25mg every 8h for 24h. Resume normal dose thereafter.

Major surgery: Give 100mg hydrocortisone before induction and 50mg every 8h for

24h. After 24h, halve this dose each day until the level of maintenance.

Patients with primary adrenal insufficiency will need extra cover—discuss with an endocrinologist. The major risk with adrenal insufficiency is hypotension, so if this is encountered without an obvious cause, consider a stat dose of hydrocortisone. —— See p826 for treatment of Addisonian crisis and BNF section 6.2 for steroid dose equivalents.

Laparoscopy was developed within gynaecology and is now in widespread use for diagnostic purposes and surgical procedures such as appendicectomy, fundoplica-tion, splenectomy, adrenalectomy, hernia repair, colectomy, prostatectomy, and nephrectomy. Minimally invasive surgery is also used for thyroidectomy and par-athyroidectomy.

Challenges of minimal access surgery The 2-dimensional visual representation and different surgical approach alters the normal appearance of familiar anatomy. Palpation is impossible and it may be harder to locate lesions prior to resection. As a result, pre-operative imaging may need to be more extensive. A new skill has to be learned and taught.

Bear in mind that the following groups of patients may not be suitable for day-case surgery: •Severe dementia. •Severe learning difficulties. •Living alone (and no helpers).

Children if supervision difficult—changes in expectation, delays, and pain relief can be problematic. •BMI >22 (p244). •ASA category ? III (p567) thus potentially unstable comorbidities—discuss with the anaesthetist. •Infection at the site of the operation.

Examine the regional lymph nodes as well as the lump. If the lump is a node, examine its area of drainage. Always examine the circulation and nerve supply distal to any lump.

History How long has it been there? Does it hurt? Any other symptoms, eg itch? Any other lumps? Is it getting bigger? Ever been abroad? Otherwise well?

Physical exam Remember the 6 S's: site, size, shape, smoothness (consistency), sur-face (contour/edge/colour), and surroundings. Other questions: Does it transilluminate (see next paragraph)? Is it fixed/tethered to skin or underlying structures (see BOX)? Is it fluctuant/compressible? Temperature? Tender? Pulsatile (US duplex may help)?

Transilluminable lumps After eliminating as much external light as possible, place a bright, thin 'pen' torch on the lump, from behind (or at least to the side), so the light is shining through the lump towards your eye. If the lump glows red it is said to transilluminate—a fluid-filled lump such as a hydrocele is a good example.

Lymph nodes Causes of enlargement: Infection: Glandular fever; brucellosis; TB; HIV; toxoplasmosis; actinomycosis; syphilis. Infiltration: Malignancy (carcinoma, lymphoma); sarcoidosis.

Rheumatoid nodules (fig 12.12) Collagenous granulomas which appear in established rheumatoid arthritis on the extensor aspects of joints—especially the elbows (fig 12.12).

Fibromas These may occur anywhere in the body, but most commonly under the skin. These whitish, benign tumours contain collagen, fibroblasts, and fibrocytes.

Dermoid cysts Contain dermal structures and are found at the junction of embry-onic cutaneous boundaries, eg in the midline or lateral to the eye. See fig 12.14.

Malignant tumours of connective tissue Fibrosarcomas, liposarcomas, leiomyo-sarcomas (smooth muscle), and rhabdomyosarcomas (striated muscle). These are staged using modified TNM system including tumour grade. Needle-core (Trucut®) biopsies of large tumours precede excision. Any lesion suspected of being a sarcoma should not be simply enucleated. ►Refer to a specialist.

Neurofibromas See p514.

Keloids Caused by irregular hypertrophy of vascularized collagen forming raised edges at sites of previous scars that extend outside the scar (fig 12.15). Common in dark skin. Treatment can be difficult. Intralesional steroid injections are a mainstay.

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Malignant tumours

Malignant melanoma: (See fig 12.16.) φ : ð ≈ 1.2:1. UK incidence: ?10:100 000/yr (up

If in doubt, refer.

Table 12.8 Glasgow 7-point checklist (refer if ?3 points, or with 1 point if suspicious)

| Category | Description | Infection risk | | — | — | — | | Clean | Incising uninfected skin without opening a viscus | <2% | | Clean-contaminated | Intra-operative breach of a viscus (but not colon) | 8–10% | | Contaminated | Breach of a viscus + spillage or opening of colon | 12–20% | | Dirty | The site is already contaminated with pus or faeces, or from exogenous contagion, eg trauma | 25% |

Pre-malignant tumours

Bowen's disease: Slow-growing red/brown scaly plaque, eg on lower legs. Histology: full-thickness dysplasia (carcinoma in situ). It infrequently progresses to squamous cell cancer. Penile Bowen's disease is called Queyrat's erythroplasia. Treatment: cryo-therapy, topical fluorouracil (see as for 'Solar keratoses') or photodynamic therapy. See also Kaposi's sarcoma (p702); Paget's disease of the breast (p708).

Leprosy: Suspect in any anaesthetic hypopigmented lesion (p441). •Syphilis: Any genital ulcer is syphilis until proved otherwise. Secondary syphilis: papular rash—in-cluding, unusually, on the palms (p412).

Don't biopsy lumps until tumours within the head and neck have been excluded by an ENT surgeon. Culture all biopsied lymph nodes for TB.

Tests Do virology and TB tests (p294). US shows lump consistency: cystic, solid, com-plex, vascular. CT defines masses in relation to their anatomical neighbours. CXR may show malignancy or, in sarcoid, reveal bilateral hilar lymphadenopathy. Consider fine-needle aspiration (FNA).

Submandibular triangle (Bordered by the mental process, mandible, and the line between the two angles of the mandible.) •If <20yrs, self-limiting lymphadenopathy is likely. If >20yrs, exclude malignant lymphadenopathy (eg firm and non-tender).

Is TB likely? •If it is not a node, think of submandibular salivary stone, sialadenitis, or tumour (see BOX for Salivary gland pathology).

Cystic hygromas (usually infants) arise from jugular lymph sac. These macrocystic lymphatic malformations transilluminate brightly. Treat by surgery or hypertonic sa-line sclerosant injection. Recurrence can be troublesome.

•Pancoast's tumour (see p708). •Subclavian artery aneurysm will be pulsatile.

External carotid
artery
Mylohyoid Submandibular
nodes
Ant. and Post.
digastric
Submandibular
gland
Hyoid bone Sternohyoid Sternothyroid
Common carotid
artery
Sternomastoid
Parotid gland Parotid nodes Sternomastoid Hypoglossal nerve Tonsillar node Occipital nodes Scalenus medius
Internal jugular vein
Omohyoid Scalenus anterior Vagus nerve
Subclavian artery Subclavian vein

If the thyroid (fig 12.21) is enlarged (=goitre), ask yourself: 1 Is the thyroid diffusely en-larged or nodular? 2 Is the patient euthyroid, thyrotoxic (p218), or hypothyroid (p220)?

Diffuse goitre: Causes: Endemic (iodine deficiency); congenital; secondary to goi-trogens (substances that ↓ iodine uptake); acute thyroiditis (de Quervain's); physi-ological (pregnancy/puberty); autoimmune (Graves' disease; Hashimoto's thyroiditis).

What should you do if high-resolution ultrasound shows impalpable nodules?

Such thyroid nodules can usually be observed provided they are:

<1cm across (which accounts for most; ultrasound can detect lumps <2mm; such 'incidentalomas' occur in 46% of routine autopsies) and are asymptomatic.

There is no past history of thyroid cancer or neck irradiation.

No family history of medullary cancer (if present, do USS-guided FNA).

Thyroid cancer

Papillary: (60%.) Often in younger patients. Spread: lymph nodes and lung (jugulo-digastric node metastasis is the so-called lateral aberrant thyroid). R: total thyroid-ectomy to remove non-obvious tumour ! node excision ! radioiodine (121I) to ablate residual cells. Give levothyroxine to suppress TSH. Prognosis: better if young and φ.

Follicular: (≤25%.) Occurs in middle-age and spreads early via blood (bone, lungs). Well-differentiated. R: total thyroidectomy + T4 suppression + radioiodine ablation.

Medullary: (5%.) Sporadic (80%) or part of MEN syndrome (p222). May produce

calcitonin which can be used as a tumour marker. They do not concentrate iodine.

Perform a phaeochromocytoma screen pre-op. R: thyroidectomy + node clear-ance. External beam radiotherapy may prevent regional recurrence.

Lymphoma: (5%.) φ:ð≈2:1. May present with stridor or dysphagia. Do full stag-ing pre-treatment (chemoradiotherapy). Assess histology for mucosa-associated lymphoid tissue (MALT) origin (associated with a good prognosis).

Anaplastic: Rare. φ:ð≈2:1. Elderly, poor response to any treatment. In the absence of unresectable disease, excision + radiotherapy may be tried.

Fig 12.21 The anatomy of the region of the thyroid gland. The important structures that must be considered when operating on the thyroid gland include:

Recurrent laryngeal nerve

Superior laryngeal nerve

Parathyroid glands

Trachea

Common carotid artery

Internal jugular vein (not depicted—see fig 12.22).

Fig 12.22 Radionuclide study of the thyroid showing changes consistent with Graves' dis-ease (see also hot and cold nodules (p216) and nuclear medicine, p738). There is increased up-take of the radionuclide trace diffusely through-out both lobes of the gland.

Epidemiology Affects 1 in 8 φ; nearly 60 000 new cases per year in UK (incidence increasing). Rare in men (r1% of all breast cancers).

Investigations (See p82 for history and examination.) ►All lumps should undergo 'triple' assessment: Clinical examination + histology/cytology + mammography/ul-trasound; see fig 12.24.

As with any mass (see p594), determine size, site, shape, and surface. Find out if it is pulsatile and if it is mobile. Examine supraclavicular and inguinal nodes. Is the lump ballotable (like bobbing an apple up and down in water)?

Left upper quadrant mass Is it spleen, stomach, kidney, colon, pancreas, or a rare cause (eg neurofibroma)? Pancreatic cysts may be true (congenital; cystadenomas; retention cysts of chronic pancreatitis; cystic fibrosis) or pseudocysts (fluid in lesser sac from acute pancreatitis).

Smooth hepatomegaly Hepatitis, CCF, sarcoidosis, early alcoholic cirrhosis (a small liver is typical later); tricuspid incompetence (↬ pulsatile liver).

Investigating lumps Check FBC (with film); CRP; U&E; LFT; Ca2+; tumour markers only as appropriate. Imaging by CT or US (transvaginal approach may be useful); MRI also has a role, eg in assessment of liver masses (p286). Others: TB tests (p294). Biopsy to give a tissue diagnosis may be obtained using a fine needle guided by CT, US, or endoscopy.

606

Clinical syndromes that usually require laparotomy

Rupture of an organ (Spleen, aorta, ectopic pregnancy.) Shock is a leading sign—see table 12.10 for assessment of blood loss. Abdominal swelling may be seen. Any history of trauma: blunt trauma ↬ spleen; penetrating trauma ↬ liver? Delayed rupture of the spleen may occur weeks after trauma. Peritonism may be mild.

Syndromes that may not require a laparotomy

Local peritonitis: Eg diverticulitis, cholecystitis, salpingitis, and appendicitis (the lat-ter will need surgery). If abscess formation is suspected (swelling, swinging fever, and •WCC) do US or CT. Drainage can be percutaneous (US or CT-guided), or by lapa-rotomy. Peritoneal inflammation can cause localized ileus with a ‘sentinel loop’ of intraluminal gas visible on plain AXR (p729).

Tests U&E; FBC; amylase; LFT; CRP; lactate (is there mesenteric ischaemia?); urinalysis.

Plan Bed rest, keep NBM; assess volume status (BOX) and treat shock (p790); cross-match/group and save; analgesia (p574); arrange imaging; consider need for IVI, blood cultures, and antibiotics (eg piperacillin/tazobactam 4.5g/8h IV); ECG.

Bornholm disease Tuberculosis (p292) Typhoid fever (p415) Pneumococcal peritonitis Porphyria (p692) Cholera (p420) ƨleaking AAA (p654) are the Unterseeboote of the acute abdomen—unsuspected, undetectable unless carefully looked for, and underestimatedly deadly. They may have non-specific symptoms and signs that are surprisingly mild, so always think of them when assessing the acute abdomen and hopefully you will ‘spot’ them!

Fig 12.26 Erect CXR showing air beneath the right hemidiaphragm, indicating pres-ence of a pneumoperitoneum.

Causes:

Iatrogenic, eg laparoscopic surgery (de-tectable on CXR up to 10d post-op).

608

Variations in the clinical picture

Inflammation in a retrocecal/retroperitoneal appendix (2.5%) may cause flank or

The shocked, confused octogenarian who is not in pain.

Hints

If a child is anxious, use their hand to press their tummy.

Expect diagnosis to be wrong half the time. If diagnosis is uncertain, re-examine often. A normal appendix is removed in up to 20% of patients.

Complications

Perforation is commoner if a faecolith is present and in young children, as the diagnosis is more often delayed.

There is a second peak between 60–70yrs; older adults may present later with atypical symptoms.

Cardinal features of intestinal obstruction •Vomiting,9 nausea and anorexia.•Colic occurs early (‘ in long-standing obstruction). •Constipation may be absolute (ie no faeces or flatus passed) in distal obstruction; less pronounced if obstruction is high.

The key decisions

Is it obstruction of the small or large bowel? In small bowel obstruction, vomiting occurs early, distension is less, and pain is higher in the abdomen; in large bowel obstruction, pain is more constant. The AXR plays a key role (fig 12.28 & p728).

Causes See table 12.12.

Management

General principles: Cause, site, speed of onset, and completeness of obstruction determine definitive therapy: strangulation and large bowel obstruction require surgery; ileus and incomplete small bowel obstruction can be managed conserva-tively, at least initially.

Surgery: ►Strangulation needs emergency surgery. Closed loop obstruction may be managed with surgery or endoscopic decompression attempted. Endoscopic stenting may be used for obstructing large bowel malignancies either in palliation or as a bridge to surgery in acute obstruction (p616). Small bowel obstruction secondary to adhesions should rarely lead to surgery—see BOX, p581.

(c) Sigmoid volvulus: there is a characteristic AXR with an ‘inverted U’ loop of bowel that looks a bit like a coffee bean.

Fig 12.29 Unenhanced axial CT of the ab-domen showing multiple loops of dilated, fluid-filled small bowel in a patient with small bowel obstruction.

Irreducible: contents cannot be pushed back into place (see p614 for technique).

Incarceration: contents of the hernial sac are stuck inside by adhesions.

Paraumbilical hernias occur just above or below the umbilicus. Risk factors are obesity and ascites. Omentum or bowel herniates through the defect. Surgery in-volves repair of the rectus sheath (Mayo repair).

Spigelian hernias occur through the linea semilunaris at the lateral edge of the rectus sheath, below and lateral to the umbilicus.

Maydl’s hernias involve a herniating ‘double loop’ of bowel. The strangulated por-tion may reside as a single loop inside the abdominal cavity.

Sciatic hernias pass through the lesser sciatic foramen (a way through various pelvic ligaments). GI obstruction + a gluteal mass suggests this rare possibility.

abdominal contents through a defect in the anterior abdominal wall to the right of the umbilicus. Prompt surgical repair required. Exomphalos: Abdominal contents are found outside the abdomen, covered in a three-layer membrane consisting of peritoneum, Wharton’s jelly, and amnion. Surgical repair less urgent because the bowel is protected by these membranes.

614

Anterior superior

Deep inguinal node

Direct inguinal hernia Course of spermatic cord

Artery Vein

This is the 2rd most common cancer and 2nd most common cause of UK cancer deaths (16 000 deaths/yr). Usually adenocarcinoma. 86% of presentations are in those >60yrs old. Lifetime UK incidence: ð = 1:15; φ = 1:19.

Presentation depends on site: Left-sided: Bleeding/mucus PR; altered bowel habit or obstruction (25%); tenesmus; mass PR (60%). Right: 'Weight; 'Hb; abdominal pain; obstruction less likely. Either: Abdominal mass; perforation; haemorrhage; fistula. See p522 for a guide to urgent referral criteria. See fig 12.24 for distribution.

Prognosis Survival is dependent on age and stage; for stage 1 disease, 5yr survival is r75% but this drops to just 5% with diagnosis at stage 4, hence the imperative for effective screening (BOX).

Hepatic flexure 2%

Descending colon 2%

Anus 2%

Other and unspecified 9%

Site 20% occur in the upper part, 50% in the middle, and 20% in the lower part. They may be squamous cell (proximal) or adenocarcinomas (distal; incidence rising).

Tests Gastroscopy + multiple ulcer edge biopsies. ►Aim to biopsy all gastric ul-cers as even malignant ulcers may appear to heal on drug treatment. Endoscopic ultrasound (EUS) can evaluate depth of invasion; CT/MRI helps staging. Staging lapa-roscopy is recommended for locally advanced tumours. Cytology of peritoneal wash-ings can help identify peritoneal metastases.

There are three main types of bowel ischaemia: ►AF with abdominal pain should always prompt thoughts of mesenteric ischaemia.

Prognosis: Poor for arterial thrombosis and non-occlusive disease (<40% survive), though not so bad for venous and embolic ischaemia.

Physical complications of gastrectomy

Abdominal fullness: Feeling of early satiety (! discomfort and distension) im-proving with time. Advise to take small, frequent meals.

Gastric tumour: A rare complication of any surgery which ' acid production.

Metabolic complications

Anaemia: Usually from lack of iron, hypochlorhydria, and stomach resection. B12

Oesophagus Stomach Bile duct

(stomach joined to jejunum)

Complications Dysphagia (if the wrap is too tight), 'gas-bloat syndrome' (inability to belch/vomit), and new-onset diarrhoea.

Stomach

the lower oesophageal sphincter

Oesophagus Oesophagus

Causes •Iatrogenic, eg endoscopy/biopsy/dilatation (accounts for 85–90% of per-forations). •Trauma, eg penetrating injury/ingestion of foreign body. •Carcinoma

4 Iatrogenic perforations are less prone to mediastinitis and sepsis and may be man-aged conservatively with NG tube, PPI, and antibiotics. Others require resuscitation, PPI, antibiotics, antifungals, and surgery (debridement of mediastinum and place-ment of T-tube for drainage and formation of a controlled oesophago-cutaneous fistula).
BMI ?40 (or ?25 with significant comorbidities that could improve with ‘weight).
Intensive management in tier 2 service (provides guidance on diet, physical ac-tivity, and psychosocial concerns, as well as lifelong medical monitoring).
Comparison with medical therapy Surgery is more effective in achieving weight loss than non-surgical management and weight loss is more likely to be maintained in the longer term. Adverse events are more common following surgery, and vary from one procedure to another.
Sleeve gastrectomy: (fig 12.41) Involves division of the stomach vertically, reduc-ing it in size by about 75%. The pyloric valve at the bottom of the stomach is left
Oesophagus
Subcutaneous access port
Pylorus
Diverticular disease Altered bowel habit ! left-sided colic relieved by defecation; nausea and flatulence. High-fibre diets do not help symptoms; try antispasmodics, eg mebeverine 125mg/8h PO. Surgical resection occasionally resorted to.
Stage 1 Pericolic or mesenteric abscess Surgery rarely needed Stage 2 Walled off or pelvic abscess May resolve without surgery Stage 2 Generalized purulent peritonitis Surgery required
Complications ►►Perforation: There is ileus, peritonitis ! shock. Mortality: 40%. Manage as for an acute abdomen. At laparotomy a Hartmann’s procedure may be performed (p582). Primary anastomosis is possible in selected patients. Emergency laparoscopic management is an emerging alternative.
Abscesses, eg with swinging fever, leucocytosis, and localizing signs, eg boggy rectal mass (pelvic abscess—drain rectally). If no localizing signs, remember the aphorism: pus somewhere, pus nowhere = pus under the diaphragm. A sub-phrenic abscess is a horrible way to die, so do an urgent ultrasound. Antibiotics ! ultrasound/CT-guided drainage may be needed.
Anorectal abscesses Usually caused by gut organisms (rarely staphs or TB). ð:φ≈1:8. Perianal (r45%), ischiorectal (≤20%), intersphincteric (>20%), supralevator (r5%) (fig 12.42). 9: Incise & drain under GA. Associations: DM, Crohn’s, malignancy, fistulae.
Anal ulcers Consider Crohn’s, anal cancer, lymphogranuloma venerum, TB, syphilis.
Causes Constipation with prolonged straining is a key factor. In many the bowel habit may be normal. Congestion from a pelvic tumour, pregnancy, CCF, or portal hypertension are important in only a minority of cases. ►Elicit red flags in history.
An abdominal examination to rule out other diseases.

- Non-operative: (2nd & 2rd degree, or 1st degree if medical therapy failed.) •Rubber

cholesterol). Mixed stones: Faceted (calcium salts, pigment, and cholesterol). Gall-stone prevalence: 8% of those over 40yrs. 90% remain asymptomatic. Risk factors for stones becoming symptomatic: smoking; parity.

Other presentations

Cholangitis: (Bile duct infection.) Causing RUQ pain, jaundice, and rigors (Charcot’s triad, BOX ‘Complications of gallstones’). Treat with, eg piperacillin/tazobactam

Pancreatitis: See p626.

Mirizzi's syndrome: A stone in the GB presses on the bile duct causing jaundice.

Common abbreviations used in this section: CBD, common bile duct; GB, gallbladder.

Causes The one mnemonic we can all agree on: 'GET SMASHED'. Gallstones (r25%), Ethanol (r25%), Trauma (r1.5%), Steroids, Mumps, Autoimmune (PAN), Scorpion venom, Hyperlipidaemia, hypothermia, hypercalcaemia, ERCP (r5%) and emboli, Drugs. Also pregnancy and neoplasia or no cause found (r10–20%).

Nil by mouth, consider NJ feeding (decrease pancreatic stimulation). Set up IVI and give lots of crystalloid, to counter third-space sequestration, until vital signs are satisfactory and urine flow stays at >20mL/h. Insert a urinary catheter and consider CVP monitoring.

If worsening: ITU, O2 if 'PaO2. In suspected abscess formation or pancreatic ne-crosis (on CT), consider parenteral nutrition ! laparotomy & debridement ('necro-sectomy'). Antibiotics may help in severe disease.

Any acute abdomen (p606), myocardial infarct.

Indications for urgent intervention (delay kills glomeruli): Presence of infec-tion and obstruction—a percutaneous nephrostomy or ureteric stent may be needed to relieve obstruction (p640); urosepsis; intractable pain or vomiting; impending AKI; obstruction in a solitary kidney; bilateral obstructing stones.

Clinical features

Chronic upper tract obstruction: Flank pain, renal failure, superimposed infection. Polyuria may occur due to impaired urinary concentration.

Do venepuncture for PSA before PR, as PR can • total PSA by r1ng/mL (free PSA • by 10%). It's difficult to know if acute retention raises PSA, but relieving obstruction does cause it to drop.

Drugs are useful in mild disease, and while awaiting surgery. • a-blockers are 1st line (eg tamsulosin 400mcg/d PO; also alfuzosin, doxazosin, terazosin). 'Smooth muscle tone (prostate and bladder). SE: drowsiness; depression; dizziness; 'BP; dry mouth; ejaculatory failure; extra-pyramidal signs; nasal congestion; •weight.

Transurethral resection of prostate (TURP) ≤14% become impotent (see BOX). Crossmatch 2U. Beware bleeding, clot retention, and post TURP syndrome: absorp-tion of washout causing CNS & CVS disturbance. r12% need redoing within 8yrs.

Transurethral laser-induced prostatectomy (TULIP) may be as good as TURP.

Associations Drugs (eg β-blockers, bromocriptine, methysergide, methyldopa), autoimmune disease (eg thyroiditis, SLE, ANCA+ve vasculitis), smoking, asbestos.

Treatment Retrograde stent placement to relieve obstruction (removed after 12 months) ! ureterolysis (dissection of the ureters from the retroperitoneal tissue). Immunosuppression (in idiopathic RPF) with low-dose steroids has good long-term results.

with permission from Oxford University Press.

Wilms' tumour (nephroblastoma) is a childhood tumour of primitive renal tubules and mesenchymal cells.

Prevalence: 1:100 000—the chief abdominal malignancy in children. It presents with an abdominal mass and haematuria. 9: OHCS p122.

Presentation Painless haematuria; recurrent UTIs; voiding irritability.

Tests

Cystoscopy with biopsy is diagnostic.

Bimanual EUA helps assess spread.

Treating TCC of the bladder

T4: Usually palliative chemo/radiotherapy. Chronic catheterization and urinary di-versions may help to relieve pain.

Survival This depends on age at surgery. For example, the 2yr survival after cystec-tomy for T2 and T2 tumours is 60% if 65–75yrs old, falling to 40% if 75–82yrs old (op-erative mortality is 4%). With unilateral pelvic node involvement, only 6% of patients survive 5yrs. The 2yr survival with bilateral or para-aortic node involvement is nil.

7th Edition. New York: Springer; 2010.

Is the bladder palpable after voiding (retention with overflow)?

Incontinence in women Often under-reported with delays before seeking help.

Check for: UTI; DM; diuretic use; faecal impaction; palpable bladder; GFR.

Do urodynamic assessment (cystometry & urine flow rate measurement) before any surgical intervention to exclude detrusor overactivity or sphincter dyssynergia.

*Mind over bladder: •Void when you DON'T have urge; DON'T go to the bathroom when you do have urge.

Acute, tender enlargement of testis = torsion (p652) until proved otherwise.

Cannot get above: inguinoscrotal hernia (p614) or hydrocele extending proximally.

Testicular and cystic: hydrocele.

Haematocele Blood in tunica vaginalis, follows trauma, may need drainage/excision.

Staging: 1 No evidence of metastasis. 2 Infradiaphragmatic node involvement (spread via the para-aortic nodes not inguinal nodes). 3 Supradiaphragmatic node involvement. 4 Lung involvement (haematogenous).

Encourage regular self-examination (prevents late presentation).

The aim is to recognize this condition before the cardinal signs and symptoms are fully manifest, as prompt surgery saves testes. If surgery is performed in <6h the salvage rate is 90–100%; if >24h it is 0–10%.

Signs: Inflammation of one testis—it is very tender, hot, and swollen. The testis may lie high and transversely. Torsion may occur at any age but is most common at 11–20yrs. With intermittent torsion the pain may have passed on presentation, but if it was severe, and the lie is horizontal, prophylactic fixing may be wise.

Treatment: ►Ask consent for possible orchidectomy + bilateral fixation (orchi-dopexy)—see p568. At surgery expose and untwist the testis. If its colour looks good, return it to the scrotum and fix both testes to the scrotum.

Retractile testis: The genitalia are normally developed but there is an excessive cremasteric reflex. The testis is often found at the external inguinal ring. 9: reassur-ance (examining while in a warm bath, for example, may help to distinguish from maldescended/ectopic testes).

Complications of maldescended and ectopic testis Infertility; ≈40 increased risk of testicular cancer (risk remains after surgery but in cryptorchidism may be ↓ if or-chidopexy performed before aged 10), increased risk of testicular trauma, increased risk of testicular torsion. Also associated with hernias (due to patent processus vagi-nalis in >90%, p612) and other urinary tract anomalies.

Hormonal: Hormonal therapy, most commonly human chorionic gonadotrophin (hCG), is sometimes attempted if an undescended testis is in the inguinal canal.

Common sites Aorta (infrarenal most common), iliac, femoral, and popliteal arteries.

Occurs due to atherosclerosis causing stenosis of arteries (fig 12.52) via a multi-factorial process involving modifiable and non-modifiable risk factors. 65% have coexisting clinically relevant cerebral or coronary artery disease.

—— Cardiovascular risk factors should be identified and treated aggressively. The chief feature of PAD is intermittent claudication (= to limp). Prevalence = 10%.

Signs Absent femoral, popliteal, or foot pulses; cold, white leg(s); atrophic skin; punched out ulcers (often painful); postural/dependent colour change; Buerger's an-gle (angle that leg goes pale when raised off the couch) of <20° and capillary filling time >15s are found in severe ischaemia.

4 1 Risk factor modification: Quit smoking (vital). Treat hypertension and high choles-terol. Prescribe an antiplatelet agent (unless contraindicated), to prevent progression and to reduce cardiovascular risk. Clopidogrel is recommended as 1st-line.

Percutaneous transluminal angioplasty (PTA) is used for disease limited to a sin-gle arterial segment (a balloon is inflated in the narrowed segment). 5-year paten-cy is 79% (iliac) and 55% (femoral). Stents can be used to maintain artery patency.

Fig 12.54 CT angiogram showing (a) normal lower limb vasculature and (b) heavily diseased arteries with previ-ous left femoral anterior tibial bypass.

Examination See p78.

Education—avoid prolonged standing and elevate leg(s) whenever possible; sup-port stockings (compliance is a problem); lose weight; regular walks (calf muscle action aids venous return).

Endovenous laser ablation (EVLA) is similar but uses a laser. Outcomes are similar to surgical repair after 2yrs (in terms of quality of life and recurrence).

Saphena varix Dilatation in the saphenous vein at its confluence with the femoral vein (the SFJ). It transmits a cough impulse and may be mistaken for an inguinal or femoral hernia, but on closer inspection it may have a bluish tinge.

History Ask about number, pain, trauma. Explore comorbidities—eg VVS, peripheral arterial disease, diabetes, vasculitis. Length of history? Is the patient taking steroids? Are self-induced ulcers, dermatitis artefacta, a possibility? Has a biopsy been taken?

Management Managing ulcers is often difficult and expensive. Treat the cause(s) and focus on prevention. Optimize nutrition. Are there adverse risk factors (drug addiction, or risk factors for arteriopathy, eg smoking)? Get expert nursing care. Consider referral to specialist community nurse or leg ulcer/tissue viability clinic:

Routine use of antibiotics does not improve healing. Only use if there is infection (not colonization).

Oxford University Press.

Practical Procedures

Fig 18.1 NHS 'clean your hands' campaign poster. Contains public sector information licensed under Open Government Licence v3.0. https://www.whatdotheyknow.com/request/21861/response/56086/attach/2/04072%20Hand%20Hygiene%205%201.1.pdf

Nasogastric tubes

These tubes are passed into the stomach via the nose. Large (eg 16F) are good for drainage but can be uncomfortable for patients. Small (eg 10F) are more comfort-able for feeding but can be difficult to aspirate and are poor for drainage. Used:

To decompress the stomach/gastrointestinal tract especially when there is ob-struction, eg gastric outflow obstruction, ileus, intestinal obstruction.

For gastric lavage.

To administer feed/drugs, especially in critically ill patients or those with dys-phagia, eg motor neuron disease, post CVA.

Passing the tube Nurses are experts and will ask you (who may never have passed one) to do so only when they fail—so the first question to ask is: 'Have you asked the charge-nurse from the ward next door?'

Wear non-sterile gloves and an apron to protect both you and the patient.

Explain the procedure. Take a new, cool (hence less flexible) tube. Have a cup of water to hand. Lubricate well with aqueous gel.

Use the tube, by holding it against the patient's head, to estimate the length required to get from the nostril to the back of the throat.

Place lubricated tube in nostril with its natural curve promoting passage down, rather than up. The right nostril is often easier than the left but, if feasible, ask the patient for their preference. Advance directly backwards (not upwards).

When the tip is estimated to be entering the throat, rotate the tube by r180° to discourage passage into the mouth. Ask the patient to swallow a sip of water, and advance as they do, timing each push with a swallow. If this fails: Try the other nostril.

The tube has distance markings along it: the stomach is at r25–40cm in adults, so advance > this distance, preferably 10–20cm beyond. Tape securely to the nose.

Confirming position This is vital prior to commencing any treatment through the tube. Misplaced nasogastric tubes have led to a number of preventable deaths, and feeding via a misplaced tube is considered an NHS Never Event (a serious, largely preventable patient safety incident that should never occur if the available preventative measures have been implemented).

If the pH is >5.5 and the NGT is needed for drug or feed administration then the position must be checked radiologically. Request a CXR/abdo X-ray (tell the radi-ologist why you need it). Look for the radio-opaque line/tip (this can be hard to see, look below the diaphragm, but if in doubt, ask for help from the radiologist).

The 'whoosh' test is NOT an accepted method of testing for tube position.

Either spigot the tube, or allow to drain into a dependent catheter bag secured to clothing (zinc oxide tape around tube to form a flap, safety pin through flap).

Do not pass a tube nasally if there is any suspicion of a facial fracture.

Get senior help if the patient has recently had upper GI surgery—it is not good practice to push the tube through a fresh anastomosis.

Complications •Pain, or, rarely: •Loss of electrolytes •Oesophagitis •Tracheal or duodenal intubation •Necrosis: retro- or nasopharyngeal •Stomach perforation.

Weaning When planning removal of an NGT in situ for decompression or relief of obstruction, it is wise to wean it so that the patient manages well without it. Drainage should be <750mL/24 hours for successful weaning.

First it should be on free drainage with, eg, 4hrly aspirations.

Then spigot with 4hrly aspirations.

Then spigot only. If this is tolerated along with oral intake then it is probably safe to remove the tube; if not, then take a step backwards.

760

Preparation is key, remember the following before you start

- Equipment: Set up a tray with cleaning swabs, gauze, cannulae (swallow your pride and take at least three of different sizes, see table 18.1), dressings, 0.9% sa-line, 10mL syringe, needle-free adaptor (eg octopus with bionector), blood tubes if required, portable sharps bin 🐟 needlestick injuries do happen.
- Patient: Have them lying down, explain procedure, obtain verbal consent, place tourniquet around arm, rest the arm below the heart to aid venous filling.

3 Site: Look for the best vein—it should be palpable; some of the best veins are not easily visible, some of the most visible collapse on insertion. Tapping gently helps.

Never cannulate: AV fistulae arms, limbs with lymphoedema. ►Avoid: Sites crossing a joint (if possible), the cephalic vein in a renal patient.

4 Consider: EMLA® cream, cold spray, or 1% lidocaine for children or those with needle phobia. EMLA® takes 45min to work, but can save you hassle later.

Insertion care bundle

- Aseptic technique. 2 Hand hygiene. 3 Apron + non-sterile gloves. 4 Skin prepara-tion—2% chorhexidine in 70% isopropyl alcohol (allow to dry for 30 seconds). Do not repalpate vein after cleaning unless wearing sterile gloves. 5 Dressing—sterile and transparent so that insertion site can be observed.

After insertion

- Take blood with syringe or adaptor. 2 Remove tourniquet. 3 Attach needle-free de-vice (if appropriate) and flush with 10mL 0.9% saline. 4 Apply dressing. 5 Let nursing staff know that cannula is in place and ready for use. 6 Document insertion according to local policy. 7 Write up appropriate fluids or parenteral medication.

When seeing your patient on the daily ward round (and to avoid being called to review or replace cannulae at 6pm) do a RAID[1] assessment: consider if the drip is:

Required—can the patient manage with oral medication/fluids?

Appropriate—should you consider a PICC, central line, long-term line, etc?

Infected—any signs of inflammation or infection? Remove if yes. Peripheral cannulae should be replaced every 72–96 hours.

Dressed properly—many drips are replaced early because they have ‘fallen out’, or are kinked from poor dressings.

Tissued or infected cannulae need replacing, either with another peripheral cannula, or with a longer-term access device, such as a PICC line.

If you fail after three attempts ►Shocked patients need fluid quickly: if you are having trouble putting in a drip, call your senior. The following advice assumes that the drip is not immediately life-saving. 2If it is, see BOX.

Ask for help—from colleagues or seniors—do not be ashamed, everyone has to learn and even senior doctors have bad days; a fresh pair of eyes can be all it takes. As a house officer, one of us was asked to place a drip when a very shame-faced consultant had ‘had a go’ to prove he still could, and found out that he couldn’t!

Help yourself—try putting the hand in warm water, using a small amount of GTN

paste over the vein, or using ultrasound if available to help you identify the vein.

If there is no one else to help, take a break and come back in half an hour. Veins come and go, and coming back with fresh eyes can make all the difference.

Table 18.1 Intravenous cannulae sizes and UK colour conventions

| Gauge | Colour | Diameter (mm) | Length (mm) | Flow rate (mL/min) | | — | — | — | — | — | | 14G | ORANGE/BROWN | 2.0 | 45 | 250 | | 16G | GREY | 1.7 | 42 | 170 | | 18G | GREEN | 1.2 | 40 | 90 | | 20G | PINK | 1.0 | 22 | 55 | | 22G | BLUE | 0.28 | 25 | 25 | | 24G | YELLOW | 0.07 | 19 | 24 |

Flow rate is given as maximum flow rate under gravity; faster rates may be achievable with rapid infusion devices.

According to Poiseuille's law1 the flow rate (Q) of a fluid through a tubular structure is inversely proportional to viscosity (η) and length (l) and proportional to the pressure difference across it (Pi – Po) and the radius to the power of 4(r 4). Hence:

(Pi – Po) r 4

η l

- Poiseuille's law is a neat piece of physiology and worth remembering—it is applicable in some form to almost every system in the body. Note that it is a 4th-power law: a small change in the radius makes a huge difference to flow.

Urinary tract infections are the second commonest health care-associated infection, and urinary catheters are frequently to blame. Think, does the patient really need a catheter? If so, use the smallest you can and take out as soon as possible.

Per urethram Aseptic technique required.

Indications • Relieve urinary retention. • Monitor urine output in critically ill pa-tients. • Collect uncontaminated urine for diagnosis. ► It is contraindicated in ure-thral injury (eg pelvic fracture) and acute prostatitis.

Explain the procedure, and obtain verbal consent. Prepare a catheterization trolley: gloves, catheter, lidocaine jelly, cleaning solution, drape, kidney dish, gauze swabs, drainage bag, 10mL water and syringe, specimen container.

Put sterile lidocaine 1–2% gel on the catheter tip and ≤10mL into the urethra (≤5mL if φ). In men, lift and gently stretch the penis upwards to eliminate any urethral folds that may lead to false passage formation.

If you are having trouble getting past the prostate, try: more lubrication, a gentle twisting motion; a larger catheter; or call the urologists, who may use a guidewire.

Remember to reposition the foreskin in uncircumcised men after the catheter is inserted to prevent oedema of the glans and paraphimos.

Documentation: In the notes be sure to document the indication for catheterization, size of catheter, whether insertion was difficult or straightforward, any complica-tions, residual volume and colour of urine. It is good practice to document that the foreskin has been replaced. Sign with your name, date, and designation.

Suprapubic catheterization: Sterile technique required ►Absolutely contraindi-cated unless there is a large bladder palpable or visible on ultrasound, because of the risk of bowel perforation. Be wary, particularly if there is a history of abdominal or pelvic surgery. Suprapubic catheter insertion is high risk and you should be trained before attempting it, speak to the urologists first.

Fig 18.2 A size 14F latex Foley catheter with the balloon inf-lated via the topmost port of the outer end (green).

© Dr Tom Turmezei (not to scale).

Fig 18.2 The external end of a size 20F 3-way catheter. The lowest port is for the bladder irrigation fluid and the upper-most port (yellow) is for bal-loon inflation.

© Dr Tom Turmezei (not to scale).

Fig 18.4 A size 10F catheter for self-catheterization. They are usually smaller than indwelling catheters, eg 10F compared to 14F. Note that this catheter also has no balloon.

© Dr Tom Turmezei (not to scale).

For patients with refractory or recurrent ascites that is symptomatic, it is possible to drain the ascites using a long pig-tail catheter. Paracentesis in such patients even in the presence of spontaneous bacterial peritonitis may be safe. Learn at the bedside from an expert.

Procedure Requires sterile technique.

Ensure you have good IV access—eg 18G cannula in the antecubital fossa.

Explain the procedure including the risks of infection, bleeding, hyponatraemia, renal impairment, and damage to surrounding structures (such as liver, spleen, and bowel), and obtain consent from the patient. Serious complications occur in less than 1 in 1000 patients. Ask the patient to empty their bladder.

Examine the abdomen carefully, evaluating the ascites and checking for orga-nomegaly. Mark where you are going to enter. If in doubt, ask the radiology de-partment to ultrasound the abdomen and mark a spot for drainage. Approach from the left side unless previous local surgery/stoma prevents this—call a senior for support and advice if this is the case.

Prepare a tray with 2% chlorhexidine solution, sterile drapes, 1% lidocaine, sy-ringes, needles, sample bottles, and your drain. Clean the abdomen thoroughly and place sterile drapes, ensure you maintain sterile technique throughout. Infil-trate the local anaesthetic.

Perform an ascitic tap (see p765) first so that you know you are in the correct place: remove 20mL fluid for MC&S.

Away from the patient, carefully thread the catheter over the (large and long) needle using the guide so that the pig-tail has been straightened out. Remove the guide.

With the left hand hold the needle r2.5cm (1 inch) from the tip—this will stop it from advancing too far (and from performing an aortic biopsy). With the right hand, hold the other end.

Gently insert the needle perpendicular to the skin at the site of the ascitic tap up to your hold with your left hand—ascites should now drain easily. If necessary, advance the needle and catheter a short distance until good flow is achieved.

Advance the catheter over the needle with your left hand, keeping the needle in exactly the same place with your right hand. ►Do not re-advance the needle be-cause it will go through the curled pig-tail and do not withdraw it because you won't be able to thread in the catheter.

When fully inserted, remove the needle, connect the catheter to a drainage bag (keep it below the level of the abdomen), and tape it down securely to the skin.

The patient should stay in bed as the ascites drains.

Document clearly in the notes the indication for the procedure, that consent was obtained, clotting and U&ES checked pre-procedure, how much lidocaine was re-quired, how much fluid was removed for investigations, and whether there were any complications to the procedure.

Replenish intravascular volume with human albumin (see 'Contraindications' ear-lier in topic).

Ask the nursing staff to remove the catheter after 6h or after a pre-determined volume has been drained (up to 20L can come off in 6h) and document this clear-ly in the medical notes. Drains are removed after 4–6h to prevent infection.

Check U&ES after the procedure and re-examine the patient.

Indications

Pneumothorax (p814): ventilated; tension; persistent/recurrent despite aspiration (eg <24h after 1st aspiration); large 2nd spontaneous pneumothorax if >50yrs old.

Malignant pleural effusion, empyema, or complicated parapneumonic effusion.

Pleural effusion compromising ventilation, eg in ICU patients.

Traumatic haemopneumothorax.

Post-operatively: eg thoracotomy; oesophagectomy; cardiothoracic surgery.

Pleural effusions are best drained under US guidance using a Seldinger tech-nique. This technique is also used for pneumathoraces (except in traumatic or post-operative situations) without US guidance; for this reason it is detailed here.

Sterile procedure

Identify the point for drainage. In effusions, this should be done with US, ideally under direct guidance or with a marked spot. For pneumothoraces, check the drainage point from CXR/CT/examination.

Preparation: trolley with dressing pack; 2% chlorhexidine; needles; 10mL syring-es; 1% lidocaine; scalpel; suture; Seldinger chest drain kit; underwater drainage bottle; connection tubes; sterile H2O; dressings. Incontinence pad under patient.

Maintain sterile technique—clean and place sterile drapes. Scrub for insertion.

Prepare your underwater drain by filling the bottle to the marked line with ster-ile water. Ensure this is kept sterile until you need it.

Infiltrate down to pleura with 10mL of 1% lidocaine and a 21G needle. Check that air/fluid can be aspirated from the proposed insertion site; if not, do not proceed.

Attach the Seldinger needle to the syringe containing 1–2mL of sterile saline. The needle is bevelled and will direct the guidewire; in general advance bevel up for pneumothoraces, bevel down for effusions.

Make a nick in the skin where the wire enters, and slide the dilators over the wire sequentially from smallest to largest to enlarge the hole, keep gauze on hand. Slide the Seldinger drain over the wire into the pleural cavity.

Remove the wire and attach a 2-way tap to the drain, then connect to the underwater drainage bottle.

Suture the drain in place using a drain stitch—make a stitch in the skin close to the drain site, tie this fairly loosely with a double knot. Then tie the suture to the drain. It is usually best to be shown this before attempting it for yourself. Dress the drain, and ensure it is well taped down.

Check that the drain is swinging (effusion) or bubbling (pneumothorax) and ensure the water bottle remains below the level of the patient at all times. If the drain needs to be lifted above the patient, clamp it briefly. ►You should never clamp chest drains inserted for pneumothoraces. Clamping for pleural effusions can con-trol the rate of drainage and prevent expansion pulmonary oedema.

Request a CXR to check the position of the drain.

Removal In pneumothorax: Consider when drain is no longer bubbling and CXR shows re-inflation. Give analgesia beforehand, eg morphine. Smartly withdraw dur-ing expiration or Valsalva. There is no need to clamp the drain beforehand as reinser-tion is unlikely. In effusions: Generally the drain can be removed when drainage is <200mL/24h, but for cirrhotic hydrothoraces the chest drain is treated similarly to the ascitic drain (see p764) with HAS supplementation and removal at 4–6 hours.

Method Explain to the patient what sampling CSF entails, why it is needed, that co-operation is vital, and that they can communicate with you at all stages.

Place the patient on his or her left side, with the back on the edge of the bed, fully flexed (knees to chin). A pillow under the head and another between the knees may keep them more stable.

Landmarks: plane of iliac crests through the level of L 2/4 (see fig 18.8). In adults, the spinal cord ends at the L 1/2 disc (fig 18.9). Mark L 2/4 intervertebral space (or one space below, L 4/5), eg by a gentle indentation of a needle cap on the overlying skin (better than a ballpoint pen mark, which might be erased by the sterilizing fluid).

Use aseptic technique (hat, mask, gloves, gown) and 2% chlorhexidine in 70% alco-hol to clean the skin, allow to dry and then place sterile drapes.

Open the spinal pack. Assemble the manometer and 2-way tap. Have three plain sterile tubes and one fluoride tube (for glucose) ready.

Using a 25G (orange) needle, raise a bleb of local anaesthetic, then use a 21G (green) needle to infiltrate deeper.

Wait 1min, then insert spinal needle (22G, stilette in place) perpendicular to the body, through your mark, aiming slightly up towards the umbilicus. Feel resistance of spinal ligaments, and then the dura, then a 'give' as the needle enters the suba-rachnoid space. NB: keep the bevel of the needle facing up, parallel with dural fibres.

Withdraw stilette. Check CSF fills needle and attach manometer (2-way tap turned off towards you) to measure 'opening' pressure.

Catch fluid in three sequentially numbered bottles (10 drops per tube).

Reinsert stilette then remove needle and apply dressing. Document the procedure clearly in the notes including CSF appearance and opening pressure.

Send CSF promptly for microscopy, culture, protein, lactate, and glucose (do plas-ma glucose too)—call the lab to let them know. If applicable, also send for: cytol-ogy, fungal studies, TB culture, virology (! herpes and other PCR), syphilis serology, oligoclonal bands (+serum sample for comparison) if multiple sclerosis suspected. Is there xanthochromia (p478)?

If you fail; ask for help—try with the patient sitting or with radiological guidance.

CSF composition Normal values: Lymphocytes <5/mm2; no polymorphs; protein <0.4g/L; glucose >2.2mmol/L (or ?50% plasma level); pressure <200mm CSF. In men-ingitis: See p822. In multiple sclerosis: See p496.

Bloody tap: This is an artefact due to piercing a blood vessel, which is indicated (unreliably) by fewer red cells in successive bottles, and no yellowing of CSF (xan-thochromia). To estimate how many white cells (W) were in the CSF before the blood was added, use the following:

W = CSF WCC − [(blood WCC ≈ CSF RBC) ÷ blood RBC].

Adapted with permission from Vakil et al., Diagnosis and Management of Medical Emergencies, 1977 Oxford University Press.

Fig 18.9 Axial T2-weighted MRI of the lumbar spine. The conus ends at the L1/L2 level with continuation of the cauda equina. Lumbar punc-ture below the L2 level will not damage the cauda equina as the nerve roots will part around an LP needle.

Do not wait for a crisis before familiarizing yourself with the defibrillator, as there are several types. All hospitals should include this information in your induction but check how the machine on your ward works.

Indications To restore sinus rhythm if VF/VT; AF, flutter, or supraventricular tachy-cardias if other treatments (p126) have failed, or there is haemodynamic compro-mise (p120 & p806). This may be done as an emergency,

eg VF/VT, or electively, eg AF.

Aim To completely depolarize the heart using a direct current.

Procedure 2For VF/pulseless VT follow the ALS algo-rithm on p894 and call the arrest team!

Unless critically unwell, conscious patients require a

general anaesthetic or monitored heavy sedation.

If elective cardioversion of AF ensure adequate antico-agulation beforehand.

Almost all defibrillators are now paddle-free and use 'hands-free' pads instead (less chance of skin arc than jelly).

Place the pads on chest, one over apex (p29) and one below right clavicle. The positions are often given by a diagram on the reverse of the pad.

Cardioversion: Synchronize the shock with the rhythm by pressing the 'SYNC' button on the machine. This en-sures the shock does not initiate a ventricular arrhyth-mia. However, this only works for cardioversion; if the sync mode is engaged in VF, the defibrillator will not discharge.

Monophasic defibrillators: (fig 18.10) Set the energy

Fig 18.11 Rectilinear bipha-sic waveform with truncated exponential decay. Most new external defibrillators use this waveform.

level at 260J for VF/VT (2arrest situation); 200J for AF; 50J for atrial flutter.

Biphasic defibrillators: (fig 18.11) Impedance is less with a biphasic shock and 120–200J is used for shocks for VF/VT. They use less energy and are just as effective as monophasic defibs in cardioversion. 120–200J will cardiovert most arrhythmias.

Automatic external defibrillators: (AEDS) Can be used by anyone who can turn them on and apply the pads. Follow the instructions given by the AED.

Shocking

Consider anticoagulation in AF (see p120).

Clearly state that you are charging the defibrillator.

Make sure no one else is touching the patient, the bed, or anything is in turn touching these.

Clearly state that you are about to shock the patient.

Give the shock. If there is a change in rhythm before you shock and the shock is no longer required, turn the dial to 'discharge'. Do not allow anyone to ap-proach until the reading has dropped to 0J.

After a shock: 2in resuscitation, resume CPR immediately and do not reassess rhythm until the end of the cycle (see p894, fig A2); in cardioversion, watch ECG; consider need to repeat the shock. Up to three are usual for AF/flutter.

Get an up-to-date 12-lead ECG.

In children, use 4J/kg in VF/VT; see OHCS p229.

Taking arterial blood gas (ABG) samples

Having an artery sampled is more unpleasant for the patient than venepuncture: explain that it is going to feel different and is for a different purpose (p162 for indications and analysis). The usual site is the radial artery at the wrist.

Check with the patient that they do not have an arterio-venous fistula for haemodial-ysis. Never, ever sample from a fistula.

Procedure:

Get kit ready; include: port-able sharps bin; pre-heparin-ized syringe; needle (blue size (22G) is good, although many

Fig 18.12 The ideal position for the wrist, slightly hyperex-tended, resting on an unopened litre bag of fluid or a bandage is ideal. In an unconscious patient or for arterial line insertion, taping the thumb to the bed can hold the wrist in the perfect position if you do not have an assistant.

syringes now come pre-made with needle); gloves; 2% chlorhexidine/70% alco-hol swab; gauze; tape.

Feel thoroughly for the best site. Look at both sides.

Wipe with cleaning swab. Let the area dry. Get yourself comfortable.

If the patient is drowsy or unconscious, ask an assistant to hold the hand and arm with the wrist slightly extended (fig 18.12).

Before sampling, expel any excess heparin in the syringe. Infiltration over the artery with a small amount of 1% lidocaine (p572) through a 25G (orange) needle makes the procedure painless.

Hold the syringe like a pen, with the needle bevel up. Let the patient know you are about to take the sample. Feel for the pulse with your other hand and enter at 45°, aiming beneath the finger you are feeling with.

In most syringes, the plunger will move up on its own in a pulsatile manner if you are in the artery; rarely, entry into a vein next to the artery will give a similar result. Colour of the blood is little guide to its source.

Allow the syringe to fill with 1–2mL, then remove the needle and apply firm pressure for 5 minutes (10 if anticoagulated).

Expel any air from the syringe as this will alter the oxygen-ation of the blood. Cap and label the sample, check the patient's temperature and FiO2 (0.21 if on air). Take the sample to the nearest analysis ma-

chine or send it by express de-livery to the lab (which may be by your own feet, get someone

sampling.

else to apply pressure) as it should be analysed within 15 minutes of sampling.

Syringes and analysis machines differ, so get familiar with the local nuances. The other site that is amenable to ABG sampling is the femoral artery (fig 18.12). Surprisingly this may be less uncomfortable as it is a relatively less sensitive area

and because, when supine, the patient cannot see the needle and thus may feel less

apprehensive. The brachial artery can also be used, but be aware that the median nerve sits closely on its medial side and it is an end-artery. Normal values: p752.

Cricothyroidotomy Thsi is an emergency procedure to overcome airway obstruc-tion above the level of the larynx. It should only be done in absolute 'can't intubate, can't ventilate' situations, ie where ventilation is impossible with a bag and mask (! airway adjuncts) and where there is an immediate threat to life. If not, call anaes-thetics or ENT for immediate help.

Indications Upper airway obstruction when endotracheal intubation not possible, eg irretrievable foreign body; facial oedema (burns, angio-oedema); maxillofacial trauma; infection (epiglottitis).

you can access the trachea directly ap-proximately halfway between the cricoid cartilage and the suprasternal notch. Ideally use a purpose-designed kit (eg QuickTrach∏, MiniTrach∏), all hos-pitals will stock one version. If no kit is available then a cannula (needle crico-thyroidotomy) can buy time, and in out-of-hospital situations a blade and empty biro case have saved lives. 2Needle and

kit cricothyroidotomies are temporary measures pending formal tracheostomy.

Cricothyroidotomy kit: Most contain a guarded blade, and a large (4–6mm) shaped cannula (cuffed or uncuffed depending on brand) over an introducer, plus a connector and binding tape. The patient will have to be ventilated via a bag, as the resistance is too high to breathe spontaneously. This will sustain for 20–45min.

Surgical cricothyroidotomy: Smallest tube for prolonged ventilation is 6mm. Introduce high-volume, low-pressure cuff tracheostomy tube through a horizontal incision in membrane. Take care not to cut the thyroid or cricoid cartilages.

Complications Local haemorrhage ! aspiration; posterior perforation of trachea

! oesophagus; subglottic stenosis; laryngeal stenosis if membrane over-incised in childhood; tube blockage; subcutaneous tunnelling; vocal cord paralysis or hoarse-ness (the recurrent laryngeal nerve runs superiorly in the tracheo-oesophageal groove).

Table 18.2 Contraindications to central venous cannulation

| A last throw of the dice Just once it may come down to you. For some, this is one of the challenges and thrills in medicine. There may be no one else available to help when there is an absolute and urgent indication for IV drugs/fluids/blood—and all of the previously discussed measures have been tried, and have failed. Think of lonesome night shifts, over-run emergency departments, a disaster scene, war, or medicine in the field. The following measures are not recommended for non-life-threatening scenarios: 2Don't worry. Have a good look again. Feet (avoid in diabetics)? Inside of the forearm? Upper arm? 2Have you really exhausted all of your options for help from a colleague? Maybe the anaesthetist or ICU registrar is approachable—they do have remarkable skills. 2Is the patient familiar with his/her own veins (eg previous IV drug abuser)? 2If there is only a small amount of IV medication required and a small, short vein, you may be able to gain access with a carefully placed butterfly needle that is taped down. Some drugs cannot be passed this way (eg amiodarone, K+). 2The external jugular vein may become prominent when the patient is head down (Trendelenburg) by 5–10° (►not in situations of fluid overload, LVF, •ICP). Only at-tempt cannulation of this vein if you are not going to jeopardize future central line insertion, and if you can clearly determine the surrounding anatomy. 2In an arrest situation, the 2015 Advanced Life Support Guidelines recommend the intraosseous route in both adults and children if venous access is not pos-sible; access devices should be available within resuscitation settings (eg emer-gency department). Only do the following if you have had the appropriate training/experience: 2In children, consider cannulating a scalp vein. 2Central venous catheterization (p775). This may be just as hard in a profoundly hypovolaemic arrest patient, and a good knowledge of local anatomy and of the procedure (! ultrasound guidance) will be invaluable. If you don't have an intraosseous access device, a cut down to the long saphen-ous vein may be attempted, in extremis, even if you have no prior experience (at this site you won't kill by being ham-fisted). 2Make a transverse incision 1–2cm anterior and superior to the medial malleolus. 2Free vein with forceps. 2Cannulate it under direct vision. ►Here, 'first do no harm' is trumped by 'nothing ventured, nothing gained'. Hopefully, it shouldn't ever have to come to these measures, but one day… | | — |

Complications (r20%.) Insertion is not without hazard, so decide whether the pa-tient requires a line first, and then ask for help if you are inexperienced.

Position the patient slightly head down to avoid air embolism and fill the veins to improve your chances of success. ►This can compromise cardiac function and pre-cipitate acute LVF so check if your patient has a cardiac history. Minimize the time the patient is head down; if they are unable to lie flat, consider a femoral approach. Turn their head slightly to the left.

This should be a sterile procedure so use full aseptic technique (hat, mask, gloves, gown) and clean with 2% chlorhexidine in 70% isopropyl alcohol before draping. Ensure your equipment is prepared, flush the catheter lumens with saline.

If US is unavailable, the landmark procedure can be used to identify insertion point—approximately at the junction of the two

heads of sternocleidomastoid at about the level of the thyroid cartilage (fig 18.17). Feel gently for the carotid pulse, then infiltrate with 1% lidocaine just lateral to this. The vein is usually superficial (fig 18.18).

Insert the introducer needle with a 5mL syringe

attached, advance gently at a 45° angle, aiming for the ipsilateral nipple and aspirating continu-ously. If you are using US, watch the needle tip enter the vein, if the landmark approach keep your fingers on the carotid pulse.

Fig 18.17 Position of internal jugular and subclavian veins (red) compared to the clavicle (yellow).

Remove the dilator, keeping hold of the wire, thread the flushed catheter over the wire, then remove the wire. The line should sit at about 12cm on the right side (17cm on the left). Check you can aspirate blood from each lumen, then flush them.

Suture the catheter in place (many have little 'wings' for suturing) and dress. Re-quest a CXR to confirm position and exclude pneumothorax. The tip of the catheter should sit vertically in the SVC.

Subclavian vein Should be taught by an expert and should ideally be carried out under US guidance. Some physicians prefer this approach, but even in experienced hands there is a •risk of complications compared to US-guided internal jugular lines.

Often it is wiser to liaise with a specialist pacing centre to arrange prompt, definitive pacing than to try temporary transvenous pacing, which often has complications (see later in topic) and therefore may delay a definitive procedure.

Possible indications in the acute phase of myocardial infarction

Complete AV block:

With inferior MI (right coronary artery occlusion) pacing may only be needed if symptomatic; spontaneous recovery may occur.

With anterior MI (representing massive septal infarction).

Second-degree block:

Wenckebach (p99; implies decremental AV node conduction; may respond to atropine in an inferior MI; pace if anterior MI).

Mobitz type 2 block is usually associated with distal fascicular disease and car-ries high risk of complete heart block, so pace in both types of MI.

First-degree block: Observe carefully: 40% develop higher degrees of block.

Bundle branch block: Pace prophylactically if evidence of trifascicular disease (p100) or non-adjacent bifascicular disease.

Sino-atrial disease + serious symptoms: Pace unless responds to atropine.

Other indications where temporary pacing may be needed

Pre-op: if surgery is required in patients with type 2 or complete heart block (whether or not MI has occurred); do 24h ECG; liaise with the anaesthetist.

Drug poisoning, eg with β-blockers, digoxin, or verapamil.

Symptomatic bradycardia, uncontrolled by atropine or isoprenaline.

Suppression of drug-resistant VT and SVT (overdrive pacing; do on ICU).

Asystolic cardiac arrest with P-wave activity (ventricular standstill).

During or after cardiac surgery—eg around the AV node or bundle of His.

Technique for temporary transvenous pacing Learn from an expert.

Preparation: Monitor ECG; have a defibrillator to hand, ensure the patient has pe-ripheral access; check that a radiographer with screening equipment is present. If you are screening, wear a protective lead apron.

Setting the pacemaker: Set the output to 2V or over 2 times the threshold value (whichever is higher) in 'demand' mode. Set the rate as required. Suture the wire to the skin, and fix with a sterile dressing.

Check the position of the wire (and exclude pneumothorax) with a CXR.

Recurrent checks of the pacing threshold are required over the next few days. The formation of endocardial oedema can raise the threshold by a factor of 2–2.

Complications Pneumothorax; sepsis; cardiac perforation; pacing failure: from loss of capture, loss of electrical continuity in pacing circuit, or electrode displacement.

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Emergencies

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Colour lithograph by Sir L. Ward [Spy], 1896.

We thank Dr Andrew Johnston and Bernard Ho, our Junior Reader, for their contribution.

The vast majority of headaches are benign, but when taking a history do not forget to ask about the following[1] (early diagnosis can save lives):

Worrying features or ‘red flags’

First and worst headache—subarachnoid haemorrhage (p478).

Thunderclap headache—subarachnoid haemorrhage (p478: p480 for other causes).

Unilateral headache and eye pain—cluster headache, acute glaucoma (p456).

Unilateral headache and ipsilateral symptoms—migraine, tumour, vascular (p458).

Cough-initiated headache—•ICP/venous thrombosis (p480).

Worse in the morning or bending forward—•ICP/venous thrombosis (p480).

Persisting headache ! scalp tenderness in over-50s—giant cell arteritis (p556).

Headache with fever or neck stiffness—meningitis (p822).

Change in the pattern of ‘usual headaches’ (p456).

Decreased level of consciousness (p456).

Two other vital questions:

Where have you been? (Malaria, p416).

Might you be pregnant? (Pre-eclampsia; especially if proteinuria and •BP, p458.) Always examine a patient presenting with a severe headache; if nothing about his-tory or examination is concerning, both you and the patient will be reassured, but

subtle abnormalities are important not to miss.

No signs on examination

Tension headache (p456).

Migraine (p458).

Cluster headache (p457).

Post-traumatic (p456).

Drugs (nitrates, calcium-channel antagonists) (p114).

Carbon monoxide poisoning or anoxia. (p842)

Subarachnoid haemorrhage (p478).

Signs of meningism?

Meningitis (may not have fever or rash—p822).

Subarachnoid haemorrhage (p478—examination may be normal).

Decreased conscious level or localizing signs?

Stroke (p470).

Encephalitis/meningitis (p822).

Cerebral abscess (p824).

Subarachnoid haemorrhage (pp478–9, figs 10.17, 10.18).

Venous sinus occlusion (p480—focal neurological deficits).

Tumour (p498).

Subdural haematoma (p482).

TB meningitis (p292).

Papilloedema?

Tumour (p498).

Venous sinus occlusion (p480—focal neurological deficits).

Malignant (accelerated phase) hypertension (p128).

Idiopathic intracranial hypertension (p498).

Any CNS infection, if prolonged (eg >2wks)—eg TB meningitis (p292).

Others

Giant cell arteritis (p556—•ESR and tender scalp over temporal arteries).

Acute glaucoma (p456—painful red eye—get pressures checked urgently).

Vertebral artery dissection (p470—neck pain and cerebellar/medullary signs).

Cervical spondylosis (p508).

Sinusitis.

Paget's disease (p685—•ALP).

Altitude sickness (OHCS p770).

Wheezing?

Asthma (p810).

COPD (p812).

Heart failure (p800).

Anaphylaxis (p794).

Stridor? (Upper airway obstruction.)

Foreign body or tumour.

Acute epiglottitis (younger patients).

Anaphylaxis (p794).

Trauma, eg laryngeal fracture.

Crepitations?

Heart failure (p800).

Pneumonia (p816).

Bronchiectasis (p172).

Fibrosis (p198).

Chest clear?

Pulmonary embolism (p818).

Hyperventilation.

Metabolic acidosis, eg diabetic ketoacidosis (p822).

Anaemia (p224).

Drugs, eg salicylates.

Shock (may cause 'air hunger', p790).

Pneumocystis jirovecii pneumonia (p400).

CNS causes.

Others

Pneumothorax (p814—pain, increased resonance, tracheal deviation if tension pneumothorax).

Pleural effusion (p192—'stony dullness').

Key investigations

Baseline observations—O2 sats, pulse, temperature, peak flow.

ABG if saturations <94% or concern about acidosis/drugs/sepsis.

ECG (signs of PE, LVH, MI?).

CXR.

Baseline bloods: glucose, FBC, U&E, consider drug screen.

First exclude any potentially life-threatening causes, by virtue of history, brief exami-nation, and limited investigations. Then consider other potential causes. For the full assessment of cardiac pain, see pp94, 118.

Life-threatening

Acute myocardial infarction (pp796–9).

Angina/acute coronary syndrome (pp796–9).

Aortic dissection (p655).

Tension pneumothorax (p814).

Pulmonary embolism (p818).

Oesophageal rupture (p820).

Others

Pneumonia (p816).

Chest wall pain:

Muscular.

Rib fractures.

Bony metastases.

Costochondritis.

Gastro-oesophageal reflux (p254).

Pleurisy (p166).

Empyema (p170).

Pericarditis (p154).

Oesophageal spasm (p250).

Herpes zoster (p404).

Cervical spondylosis (p508).

Intra-abdominal:

Cholecystitis (p624).

Peptic ulceration (p252).

Pancreatitis (p270).

Sickle-cell crisis (p240).

Key investigations

CXR.

ECG.

FBC, U&E, and troponin (p118). Consider D-dimer only if low probability of venous thromboembolism. See ‘Modified Wells’ score for PE, p191.

Just because the patient’s chest wall is tender to palpation, this doesn’t mean the cause of the chest pain is musculoskeletal. Even if palpation reproduces the same type of pain, ensure that you exclude all potential life-threatening causes. Although chest wall tenderness has discriminatory value against cardiac pain, it may be a feature of a pulmonary embolism.

Definition Unrousable unresponsiveness. Quantify using Glasgow Coma Scale (GCS).

Causes of impaired conscious level/coma

Metabolic:

Drugs, poisoning, eg carbon monoxide, alcohol, tricyclics.

Hypoglycaemia, hyperglycaemia (ketoacidotic, or HONK, pp822–4).

Hypoxia, CO2 narcosis (COPD).

Septicaemia (p792).

Hypothermia.

Myxoedema (p824), Addisonian crisis (p826).

Hepatic/uraemic encephalopathy (pp275, 298).

Neurological:

Trauma.

Infection: meningitis (p822); encephalitis (eg herpes simplex—p404), tropical: ma-laria (p416; do thick films), typhoid, typhus, rabies, trypanosomiasis.

Tumour: 1° or 2° (p528).

Vascular: stroke (p470), subdural (p482), subarachnoid (p478), hypertensive en-cephalopathy (p140).

Epilepsy: non-convulsive status (p484) or post-ictal state.

Immediate management See fig 19.2 (and coma CNS exam, p789).

Assess Airway, Breathing, and Circulation. Consider intubation if GCS <8 (p788). Sup-port the circulation if required (ie IV fluids). Give O2 and treat any seizures. Protect the cervical spine unless trauma is known not to be the cause.

Check blood glucose; give eg 200mL 10% glucose IV stat if hypoglycaemia possible.

IV thiamine if any suggestion of Wernicke's encephalopathy; see later in topic.

IV naloxone (0.4–2mg IV) for opiate intoxication (may also be given IM or via ET tube); IV flumazenil (p842) for benzodiazepine intoxication only if airway comp-romised as risk of seizures especially if concomitant tricyclic intoxication.

Examination ►Vital signs are vital—obtain full set, including temperature.

Signs of trauma—haematoma, laceration, bruising, CSF/blood in nose or ears, frac-ture 'step' deformity of skull, subcutaneous emphysema, 'panda eyes'.

Stigmata of other illnesses: liver disease, alcoholism, diabetes, myxoedema.

Skin for needle marks, cyanosis, pallor, rash (meningitis; typhus), poor turgor.

Smell the breath (alcohol, hepatic fetor, ketosis, uraemia).

Opisthotonus (fig 9.45, p426) ≈meningitis or tetanus. Decerebrate/decorticate (p788)?

Meningism (pp456, 822) ►but do not move neck unless cervical spine is cleared.

Pupils (p789) size, reactivity, gaze.

Heart/lung exam for BP, murmurs, rubs, wheeze, consolidation, collapse.

Abdomen/rectal for organomegaly, ascites, bruising, peritonism, melaena.

Are there any foci of infection (abscesses, bites, middle ear infection)?

Any features of meningitis: neck stiffness, rash, focal neurology?

Note the absence of signs, eg no pin-point pupils in a known heroin addict.

If the diagnosis is unclear:

Treat the treatable: O2; naloxone as above; glucose (eg 200mL of 10% IV); Pabrinex® IV for Wernicke's encephalopathy, p714; septic specifics: cefotaxime 2g/12h IV (men-ingitis, p822), artemether/quinine (malaria, p418), aciclovir (encephalitis, p824).

Do routine biochemistry, haematology, thick films, blood cultures, blood ethanol, drug screen, etc.

Arrange urgent CT head, if normal, and no CI, proceed to LP.

The diagnosis should now be clear, eg hypo/hyperglycaemia; alcohol excess; poison-ing; uraemia; pneumonia; subarachnoid; hypertensive/hepatic encephalopathy.

Fig 19.2 Managing coma. NB: check pupils every few minutes during the early stages, particularly if trauma is the likely cause. Doing so is the quickest way to find a lo-calizing sign (so helpful in diagnosis, but remember that false localizing signs do occur)—and observing changes in pupil behaviour (eg becoming fixed and dilated) is the quickest way of finding out just how bad things are.

This gives a reliable, objective way of recording the conscious state of a person. It can be used by medical and nursing staff for initial and continuing assessment. It has val-ue in predicting ultimate outcome. Three types of response are assessed, note in each case the best response (or best of any limb) which should be recorded (table 19.1).

Table 19.1 The Glasgow Coma Scale

| ABCDE preliminary assessment (primary survey) Airway Protect cervical spine, if injury possible. Assessment: any signs of obstruction? Ascertain patency. Management: establish a patent airway. Breathing Assessment: determine respiratory rate, check bilateral chest move-ment, percuss, and auscultate. Management: if no respiratory effort, treat as arrest (see p894, fig A2), intubate and ventilate. If breathing compromised, give high-concentration O2, manage according to findings, eg relieve tension pneumothorax. Circulation Assessment: check pulse and BP; check if peripherally shut down; check capillary refill; look for evidence of haemorrhage. Management: if shocked, treat as on p790. If no cardiac output, treat as arrest (see p894, fig A2). Disability Assess 'level of consciousness' with AVPU score (alert? responds to voice? to pain? unresponsive?); check pupils: size, equality, reactions. Glasgow Coma Scale, if time allows. Exposure Undress patient, but cover to avoid hypothermia. Quick history from relatives assists diagnosis: Events surrounding onset of illness, evidence of overdose/suicide attempt, any suggestion of trauma? Past medical history, especially diabetes, asthma, COPD, alcohol, opiate or street drug abuse, epilepsy or recent head injury, recent travel. Medication, current drugs. Allergies. Once ventilation and circulation are adequate, proceed to carry out history, examination, investigations, and management in the usual way. | | — |

Adapted from 'Assessment of coma and impaired consciousness: a practical scale', Graham Teasdale and Bryan Jennett, The Lancet, Vol 304, No. 7872, 81–84 (1974), Elsevier.

An overall score is made by summing the score in the three areas assessed.

No response to pain + no verbalization + no eye opening = 2.

Severe injury, GCS ≤8—consider airway protection.

Moderate injury, GCS 9–12.

Minor injury, GCS 12–15.

Causing pain is not a pleasant thing, there are acceptable and unacceptable meth-ods. Try fingernail bed pressure with a pen/pencil, sternal pressure (not a rub), or suprascapular squeeze. Abnormal responses to pain can help to localize the damage:

Flexion = decorticate posture (arms bent inwards on chest, thumbs tucked in a clenched fist, legs extended) implies damage above the level of the red nucleus in the midbrain.1

Extension = decerebrate posture (adduction and internal rotation of shoulder, pro-nation of forearm) indicates midbrain damage below the level of the red nucleus.

NB: an abbreviated coma scale, AVPU, is sometimes used in the initial assessment ('primary survey') of the critically ill:

A = alert

V = responds to vocal stimuli

P = responds to pain

U = unresponsive

NB: GCS scoring is different in young children; see OHCS p201.

- Red nucleus output reinforces upper limb antigravity flexion. When its output is damaged, the unregu-lated reticulospinal and vestibulospinal tracts reinforce extension tone of upper and lower limbs.

The Glasgow Coma Scale is reproduced from The Lancet, Vol. 204, Teasdale G & Jennet B, Assessment of Coma and Impaired Consciousness: A Practical Scale, ©1974, with permission from Elsevier.

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Circulatory failure resulting in inadequate organ perfusion. Often defined by 'BP—systolic <90mmHg—or mean arterial pressure (MAP) <65mmHg—with evidence of tissue hypoperfusion, eg mottled skin, urine output (UO) of <0.5mL/kg/h, serum lactate >2mmol/L. Signs: 'GCS/agitation, pallor, cool peripheries, tachycardia, slow capillary refill, tachypnoea, oliguria.

MAP = cardiac output (CO) ≈ systemic vascular resistance (SVR).

CO = stroke volume ≈ heart rate.

m shock can result from inadequate CO or a loss of SVR, or both.

Inadequate cardiac output

Hypovolaemia:

Bleeding: trauma, ruptured aortic aneurysm, GI bleed.

Fluid loss: vomiting, burns, 'third-space' losses, eg pancreatitis, heat exhaustion.

Pump failure:

Cardiogenic shock, eg ACS, arrhythmias, aortic dissection, acute valve failure.

Secondary causes, eg PE, tension pneumothorax, cardiac tamponade.

Peripheral circulatory failure (loss of SVR)

Anaphylaxis: p794.

Neurogenic: Eg spinal cord injury, epidural or spinal anaesthesia.

Endocrine failure: Addison's disease, p826 or hypothyroidism; see p824.

Other: Drugs, eg anaesthetics, antihypertensives, cyanide poisoning.

Assessment ►►ABCDE (p779). With shock we are dealing primarily with 'C ' so get large-bore IV access ≈2 and check ECG for rate, rhythm (very fast or very slow will compromise cardiac output), and signs of ischaemia.

General review: Cold and clammy suggests cardiogenic shock or fluid loss. Look for signs of anaemia or dehydration, eg skin turgor, postural hypotension? Warm and well perfused, with bounding pulse points to septic shock. Any features sug-gestive of anaphylaxis—history, urticaria, angioedema, wheeze?

CVS: Usually tachycardic (unless on β-blocker, or in spinal shock—OHCS p757) and hypotensive. But in the young and fit, or pregnant women, the systolic BP may remain normal, although the pulse pressure will narrow, with up to 20% blood volume depletion. Difference between arms (>20mmHg)—aortic dissection (p655)?

JVP or central venous pressure: If raised, cardiogenic shock likely.

Check abdomen: Any signs of trauma, or aneurysm? Any evidence of GI bleed?

Management ►If BP unrecordable, call the cardiac arrest team.

Septic shock: See p792.

Anaphylaxis: See p794.

Cardiogenic shock: See p802.

Hypovolaemic shock: Identify and treat underlying cause. Raise the legs.

Give fluid bolus 10–15mL/kg crystalloid via large peripheral line, if shock improves, repeat, titrate to HR (aim <100), BP (aim SBP >90) and UO (aim >0.5mL/kg/h).

If no improvement after 2 boluses, consider referral to ICU.

Haemorrhagic shock: Stop bleeding if possible. See table 19.2 for grading.

If still shocked despite 2L crystalloid or present with class III/IV shock then crossmatch blood (request O Rh–ve in an emergency, see p248).

Give FFP with red cells (1 : 1 ratio); aim for platelets >100 and fibrinogen >1 (guided by results, but eg 1 pool of platelets and 2 pools of cryoprecipitate per 6–8 units of red cells). Consider tranexamic acid 2g IV. Discuss with haematology early.

Heat exposure (heat exhaustion):

Tepid sponging + fanning; avoid ice and immersion. Resuscitate with IVI, eg 0.9% saline ! hydrocortisone 100mg IV. Lorazepam 1–2mg IV or chlorpromazine 25mg IM/IV may be used to stop shivering. Stop cooling when core temperature <29°C.

Table 19.2 Categorizing shock

| Ă | Best motor response | Best verbal response | Best verbal response | Eye opening | Eye opening | | — | — | — | — | — | — | | 6 | Obeying commands | 5 | Oriented (time, place, person) | 4 | Spontaneous | | 5 | Localizing to pain | 4 | Confused conversation | 3 | In response to speech | | 4 | Withdrawing to pain | 3 | Inappropriate speech | 2 | In response to pain | | 3 | Flexor response to pain | 2 | Incomprehensible sounds | 1 | None | | 2 | Extensor response to pain | 1 | None | | | | 1 | No response to pain | | | | |

792

Sepsis is a major killer. There are >150 000 cases of sepsis in the UK each year result-ing in >44 000 deaths, and much morbidity.

Sepsis Life-threatening organ dysfunction caused by a dysregulated host response to infection.

Septic shock Sepsis in combination with:

EITHER lactate >2mmol/L despite adequate fluid resuscitation

OR the patient is requiring vasopressors to maintain MAP ?65mmHg.

Acute management in sepsis ►Early recognition and treatment is key. See fig 19.2.[2] Antibiotics: These should be broad spectrum and start within 1h. Consider covering for non-bacterial microbes, eg give aciclovir if HSV encephalitis is suspected.

Fluids: Give within 1h if high risk with SBP <90, AKI, or lactate >2 (consider if <2).

Give 500mL boluses of crystalloids with 120–154mmol/L sodium (eg 0.9% saline) over 15mins. Caution in heart failure.

If no improvement after two boluses, speak with a senior.

Oxygen: Give oxygen as required for target saturations. These will be 94–98% (or

88–92% if the patient is at risk of CO2 retention, eg in severe COPD).

Critical care review: Speak with critical care early if intensive care support (eg inotropes, ventilation, haemofiltration, intensive monitoring) may be required.

Surgical involvement: Eg emergency wound debridement.

Manage acute complications: Shock (p790), AKI (p298), DIC (p252), ARDS (p186), ar-rhythmias (may spontaneously resolve when sepsis improves).

Recognize the need to assess for sepsis

Consider possibility of sepsis in any patient with signs or symptoms that are suggestive of infection (eg productive cough or offensive oozing wound)

Consider non-specific signs (patient/carer concern; altered mental state; feeling generally unwell), particularly when there is reason to have low threshold for sepsis assessment (see BOX)

Gather information

History: Full medical history: take time to clarify the time course of symp-toms and any travel history

Examination:

General: capillary refill time, mottled/ashen skin, conscious level

Localize source of infection: don't forget to assess wounds/ulcers which may be hidden by dressings

Bedside tests: Blood gas for lactate, observations incl. HR, BP, RR, sats, tem-perature; ECG; urine dip and urine output monitoring

High risk

Request immediate senior review

Investigations

Blood tests: Serial ABGS, or VBGS for lactate; blood cultures; U&E, CRP, FBC, LFT, clotting screen

Micro samples: Sputum and urine for MC&S; swab any wounds; con-sider LP; send fluid from drains and lines; joint aspirates; ascitic tap Imaging: CXR, consider CT/USS/MRI/echo of suspected source

Review

Immediately alert a consultant if, after 1h of antibiotics and fluids:

SBP <90 •RR >20 •Reduced GCS

Raised lactate not reduced by >20%

►►Consider critical care referral.

Fig 19.2 Management of sepsis in adults.

Type I IgE-mediated hypersensitivity reaction. Release of histamine and other agents causes: capillary leak; wheeze; cyanosis; oedema (larynx, lids, tongue, lips); urticaria. More common in atopic individuals. An anaphylactoid reaction results from direct release of mediators from inflammatory cells, without involving antibodies, usually in response to a drug, eg acetylcysteine.

Examples of precipitants

Drugs, eg penicillin, and contrast media in radiology.

Latex.

Stings, eggs, fish, peanuts, strawberries, semen (rare).

Signs and symptoms

Itching, sweating, diarrhoea and vomiting, erythema, urticaria, oedema.

Wheeze, laryngeal obstruction, cyanosis.

Tachycardia, hypotension.

Mimics of anaphylaxis

Carcinoid (p271).

Phaeochromocytoma (p228, p827).

Systemic mastocytosis.

Hereditary angioedema.

Management ►See fig 19.4.

Fig 19.4 Management of anaphylaxis.

2Adrenaline (=epinephrine) is given IM and NOT IV unless the patient is severely ill, or has no pulse. The IV dose is different: 100mcg/min—titrating with the response. This is 0.5mL of 1 : 10 000 solution IV per minute. Stop as soon as a response has been obtained.

If on a β-blocker, consider salbutamol IV in place of adrenaline.

Acute coronary syndrome (ACS) includes unstable angina, STEMI, and NSTEMI (p798). STEMI is a common medical emergency; prompt appropriate treatment saves lives.3 Initial treatment ►See fig 19.5. Take brief history, do a quick physical examination

and a 12-lead ECG. Observe on cardiac monitor or telemetry in case of dysrhythmia. Other tests on admission: U&E, troponin, glucose, cholesterol, FBC, CXR.

Aspirin: 200mg PO (if not already given); consider ticagrelor (180mg PO) or prasug-rel (60mg PO if no history of stroke/TIA and <75yrs) as newer alternatives to clopi-dogrel (200mg PO) as they have been shown to be superior in outcome studies.

Morphine: 5–10mg IV (repeat after 5min if necessary). Give anti-emetic with the 1st

dose of morphine: metoclopramide 10mg IV (1st line), or cyclizine 50mg IV (2nd line).

GTN: routine use now not recommended in the acute setting unless patient is hyper-tensive or in acute LVF. Useful as anti-anginal in chronic/stable patients.

Oxygen is recommended if patients have SaO2 <95%, are breathless or in acute LVF.

Restore coronary perfusion in those presenting <12h after symptom onset (see BOX).

Anticoagulation: An injectable anticoagulant must be used in primary PCI. Bivali-rudin is preferred, if not available use enoxaparin ! a GP IIb/IIIa blocker.

ß-blockers provide additional benefit when started early, eg bisoprolol 2.5mg PO OD.

Ensure no evidence of cardiogenic shock, heart failure, asthma/COPD, or heart block.

Right ventricular infarction Confirm by demonstrating ST elevation in rV2/4 and/ or echo. NB: rV4 means that V4 is placed in the right 5th intercostal space in the midclavicular line. Treat hypotension and oliguria with fluids (avoid nitrates and diu-retics). Monitor BP carefully, and assess early signs of pulmonary oedema. Intensive monitoring and inotropes may be useful in some patients.

ă Management of an acute STEMI

Attach ECG monitor and record a 12-lead ECG

IV access.

Bloods for FBC, U&E, glucose, lipids, troponin (p119, fig 2.22)

Brief assessment:

History of cardiovascular disease; risk factors for IHD

Examination: pulse, BP (both arms), JVP, murmurs, signs of CCF, upper limb pulses, scars from previous cardiac surgery, CXR if will not delay R

Contraindications to PCI or fibrinolysis?

Aspirin: 200mg (unless already given by GP/paramedics)

Ticagrelor: 180mg (or alternative antiplatelet—see 'Aspirin' in text)

STEMI on ECG and PCI available within 120min?

Fibrinolysis

For further management see p120

Fig 19.5 Management of an acute STEMI.

First stabilize with medical therapy; early risk stratification will identify those in need of further treatment and prompt angiography (involve cardiologists).'

Assessment

Brief history: (See p26.) Previous angina, relief with rest/nitrates, history of cardio-vascular disease, risk factors for IHD.

Examination: (See p28.) Pulse, BP, JVP, cardiac murmurs, signs of heart failure, pe-ripheral pulses, scars from previous cardiac surgery.

Investigations ECG: ST depression; flat or inverted T-waves; or normal; FBC, U&E, tro-ponin, glucose, random cholesterol; CXR.

Management ►See fig 19.6 for acute management, but p796 if ST elevation. The aim of therapy is to control pain then initiate anti-ischaemic and antiplatelet therapy. Oral antiplatelet therapy: Aspirin 200mg PO, followed by 75mg OD. For those with

confirmed ACS give a second antiplatelet agent, eg clopidogrel (200mg PO then 75mg

Anticoagulation: Ideally fondaparinux (factor Xa inhibitor) 2.5mg OD; if not avail-able, use low-molecular-weight heparin (LMWH, eg enoxaparin 1mg/kg/12h) or un-fractionated heparin (aim APTT 50–70s) until discharge.

ß-blockers: In higher-risk patients with no contraindications (consider diltiazem as alternative). ►Do not use β-blockers with verapamil—can precipitate asystole.

Nitrates (PO or IV): For recurrent chest pain.

ACE-i: Should be given to all patients unless there are CI (monitor renal function).

Lipid management: Start early, eg atorvastatin 80mg OD (see p120).

Prognosis Overall risk of death r1–2%, but r15% for refractory angina despite medi-cal therapy. Risk stratification can help predict those most at risk and allow interven-tion to be targeted at those individuals: calculate using GRACE score.2 The following are associated with an increased risk:

History of unstable angina.

ST depression or widespread T-wave inversion.

Raised troponin (except patients with ST elevation MI).

Age >70 years.

General comorbidity, previous MI, poor LV function or DM.

High-risk patients should be considered for inpatient coronary angiography. Symp-tomatic lesions may be addressed by coronary stenting or CABG (see p122).

Further measures

Wean off glyceryl trinitrate (GTN) infusion when stabilized on oral drugs.

Continue fondaparinux (or LMWH or heparin) until discharge.

Observe on cardiac monitor or telemetry in case of dysrhythmia. Check serial ECGS, and troponin >12h after pain.

Address modifiable risk factors: smoking, hypertension, hyperlipidaemia, diabetes.

Gentle mobilization.

Ensure patient on dual antiplatelet therapy, β-blocker, ACE inhibitor, and statin.

If symptoms recur, refer to cardiologist for urgent angiography & PCI or CABG.

- GRACE = Global Registry of Acute Coronary Events. Risk is scored based on age, heart rate, BP, re-nal function, Killip class of heart failure, and other events, eg raised troponin. Very complicated to calculate so recommendation by European Society of Cardiology is to use an online calculator, eg http://www.outcomes-umassmed.org/grace/.

Monitor closely; record ECG while in pain

If SaO2 <90% or breathless, low-flow O2

Nitrates: GTN spray or sublingual tablets as required

Aspirin: 200mg PO.

Consider need for second antiplatelet agent (see 'Oral antiplatelet therapy' in text)

Fondaparinux: 2.5mg OD SC or LMWH 1mg/kg/12h SC

Causes

Fluid overload.

Symptoms Dyspnoea, orthopnoea (eg paroxysmal), pink frothy sputum. NB: note drugs recently given and other illnesses (recent MI/COPD or pneumonia).

Investigations

CXR (p125, pp722–4): cardiomegaly, signs of pulmonary oedema: look for shadowing (usually bilateral), small effusions at costophrenic angles, fluid in the lung fissures, and Kerley B lines (septal linear opacities).

Consider echo.

Monitoring progress: BP; pulse; cyanosis; respiratory rate; JVP; urine output; ABG. Observe on cardiac monitor or telemetry in case of dysrhythmia.

Repeat CXR.

ACE-i if LVEF <40%. If ACE-i contraindicated, consider hydralazine and nitrate (may also be more effective in African-Caribbeans).

Optimize management of AF if present (p120); consider anticoagulation.

If the patient is worsening:

Increase nitrate infusion if able to do so without dropping systolic BP <100mmHg

Fig 19.7 Management of heart failure.

Causes

Pulmonary embolus (p818).

Myocarditis; myocardial depression (drugs, hypoxia, acidosis, sepsis) (p152).

Management ►See fig 19.8.

Investigation and treatment may need to be done concurrently.

Cardiac tamponade Pericardial fluid collects ↬ intrapericardial pressure rises ↬

Signs: ‘BP, •JVP, and muffled heart sounds (Beck’s triad); •JVP on inspiration (Kuss-maul’s sign); pulsus paradoxus (pulse fades on inspiration). Echocardiography may be diagnostic. CXR: globular heart; left heart border convex or straight; right cardio-phrenic angle <90°. ECG: electrical alternans (p154).

Differential diagnosis (See p128.)

Pre-excited tachycardias, eg AF, atrial flutter, or AV re-entry tachycardia, with un-derlying WPW (p122).

Connect patient to a cardiac monitor and have a defibrillator to hand.

Check for adverse signs. Low cardiac output (clammy, ‘consciousness, BP <90); oliguria; angina; pulmonary oedema.

Correct any hypokalaemia and hypomagnesaemia: up to 60mmol KCl at 20mmol/h, and 4mL 50% magnesium sulfate over 20min both via central line.

If haemodynamically stable VT: Correct hypokalaemia and hypomagnesaemia: as above.

After correction of VT: Establish the cause (via the history and tests described above).

Ventricular fibrillation: (ECG p129, fig 2.29) Use non-synchronized DC shock (there is no R wave to trigger defibrillation, p770): see p894, fig A2.

Torsade de pointes: A form of VT, with a constantly varying axis, often in the setting of long QT syndromes (ECG p129, fig 2.21). Causes (p711): congenital or from drugs (eg some antidysrhythmics, tricyclics, antimalarials, antipsychotics). Torsade in the set-ting of congenital long-QT syndromes can be treated with high doses of β-blockers.

Ă Management of broad complex tachycardia

Give O2 if SaO2<90%, get IV access & 12-lead ECG

Chest pain/ischaemia on ECG

Yes

ECG shows rate of >100bpm and QRS complex duration of <120ms (<2 small squares on

Sinus tachycardia: Normal P wave followed by normal QRS—not an arrhythmia! Do not attempt to cardiovert; if necessary (ie not a physiological response to fever/ hypovolaemia) rate control with β-blockers.

Atrial flutter: atrial rate r260–240bpm. Sawtooth baseline, due to a re-entrant circuit usually in the right atrium. Ventricular rate often 150bpm (2:1 block).

Junctional tachycardia: AV node is part of the pathway. P wave either buried in QRS

AV re-entry tachycardia, includes an accessory pathway, eg WPW (p122).

Otherwise, identify the underlying rhythm and treat accordingly. The most impor-tant thing is to decide whether the rhythm is regular or not (irregular is likely AF).

2min. NB: NOT if on a β-blocker. If no response, a further 5mg IV over 2min (if age

Atrial tachycardia: Rare; may be due to digoxin toxicity: withdraw digoxin, consider digoxin-specific antibody fragments. Maintain K+ at 4–5mmol/L.

Seek specialist advice if resistant junction tachycardia, or accessory pathway. Wolff-Parkinson-White (WPW) syndrome (ECG p122, fig 2.27.) Caused by congenital accessory conduction pathway between atria and ventricles. Resting ECG shows short

*Check your defibrillator: energies given are for a typical biphasic defibrillator (preferred); if a monophasic shock used, higher energies will be required.

Symptoms Often asymptomatic. Fatigue, nausea, dizziness. The presence of synco-pe, chest pain, or breathlessness is concerning and suggests the presence of adverse signs; sudden cardiac death can occur.

Causes

Physiological: Heart rates as low as 40bpm at rest and 20bpm in sleep can be ac-cepted in asymptomatic trained athletes.

Post-MI—particularly after an inferior MI (the right coronary artery supplies the sinoatrial node and atrioventricular node in most people).

Aortic valve disease, eg infective endocarditis (p150; do daily ECGs looking for heart block).

Vasovagal—very common (p460).

Other—hypothermia, •ICP (Cushing's triad: bradycardia, hypertension, and ir-regular breathing: 2 urgent senior input needed).

Management ►Follow a logical approach,[a] see fig 19.11. [a]

Connect patient to cardiac monitor/telemetry.

If atropine is insufficient and adverse signs persist, transcutaneous pacing should be considered (p777). If this cannot be initiated immediately (eg waiting for an anaesthetist), consider other medications such as isoprenaline infusion.

Fig 19.11 Management of bradycardia.

Presentation Acute breathlessness and wheeze.

Investigations PEF—but may be too ill; ABG if saturations <92% or life-threatening fea-tures; CXR (if suspicion of pneumothorax, infection or life-threatening attack); FBC; U&E.

Assessing the severity of an acute asthma attack

Unable to complete sentences in one breath.

PEF 22–50% of predicted or best.

Silent chest, cyanosis, feeble respiratory effort.

Arterial blood gases:

Management ►Rapid treatment and reassessment is key,7 see fig 19.12.

Monitor oxygen saturation, heart rate, and respiratory rate.

been stable on discharge medication for 24h

steroid (inhaled and oral) and bronchodilator therapy

respiratory clinic appointment within 4wks.

Drugs used in acute asthma

Hydrocortisone and prednisolone (steroid; reduces inflammation).

PEF, ability to speak, RR, pulse rate, O2 sats Warn ICU if severe or life-threatening attack

If severe/life-threatening add in ipratropium 0.5mg/6h to nebulizers Hydrocortisone 100mg IV or prednisolone 40–50mg PO

10mg/h continuously. Add ipratropium if not already given

Fig 19.12 Management of acute asthma.

History (See p48.) Ask about usual/recent treatments (especially home oxygen), smoking status, and exercise capacity (may influence a decision to ventilate the patient).

Investigations

ABG (p771).

ECG.

Management ►Ensure oxygenation then treat the reversible,8 see fig 19.12.

Treatment of stable COPD and more advanced disease: See pp184–5.

Salbutamol 5mg/4h and ipratropium 500mcg/6h

Start at 24–28%, aim sats 88–92%

IV hydrocortisone 200mg and oral prednisolone

Use if evidence of infection, eg amoxicillin 500mg/8h PO, alternatively clarithromycin or doxycycline (p287)

If no response:

Consider intubation and ventilation if pH <7.26 and Pa CO2 is rising despite non-invasive ventilation only where appropriate (see 'Consider the ceiling of care' in text)

Fig 19.12 Management of acute COPD.

Causes

Chronic lung disease: Asthma; COPD; cystic fibrosis; lung fibrosis; sarcoidosis.

Carcinoma.

Clinical features

Symptoms: Can be asymptomatic (especially in fit young people with small pneu-mothoraces) or sudden onset of dyspnoea and/or pleuritic chest pain. Patients with asthma or COPD may present with a sudden deterioration. Mechanically ventilated patients can suddenly develop hypoxia or an increase in ventilation pressures.

Management ►See fig 19.14. Depends on whether it is a primary pneumothorax or secondary (=underlying lung disease or smoker >50yrs old), size, and symptoms.[a]

Aspiration of a pneumothorax, see p767.

2This is a medical emergency. (See fig 16.42, p749.)

Treatment

2Then insert a chest drain. See p766.

Fig 19.14 Acute management of pneumothorax.

Common organisms

Streptococcus pneumoniae is the commonest cause (60–75%).

Staphylococcus aureus found more commonly in ICU patients.

Viruses including influenza account for up to 15%.

Management ►Ensure oxygenation then identify and treat reversible pathology,[10] see fig 19.15.

Oxygen saturation and ABG if SaO2 <92% or severe pneumonia.

Sputum cultures (if CURB-65 ?2 or if CURB-65 =2 and not had antibiotics yet).

Pleural fluid may be aspirated for culture (if CURB-65 ?2).

Age ?65. Score: 0–1: home treatment if possible; ?2: hospital therapy; ?2: indicates

Treatment: See antibiotic guidance (table 4.2 on p167). Most patients who require IV

Fig 19.15 Management of pneumonia.

Risk factors

Immobility; active inflammation (eg infection, IBD).

Signs and symptoms

With thromboprophylaxis, PE following surgery is far less common, but PE may oc-cur after any period of immobility, or with no predisposing factors. Breathlessness may be the only sign. Multiple small emboli may present less dramatically with pleuritic pain, haemoptysis, and gradually increasing breathlessness. ►Look for a source of emboli—especially DVT (is a leg swollen?).

ECG: commonly normal or sinus tachycardia; right ventricular strain pattern V1–V2 (p98), right axis deviation, RBBB, AF, may be deep S waves in I, Q waves in III, in-verted T waves in III ('SI QIII TIII').

Serum D-dimer: low specificity (• if thrombosis, inflammation, post-op, infection, malignancy) m check only in patients with low pre-test probability (p191).

If good story and signs, make the diagnosis. Start treatment (fig 19.16) before definitive investigations: most PE deaths occur within 1h.

Long-term anticoagulation: either DOAC (p250—switch directly from LMWH) or warfarin (continue LMWH until INR >2).

Prevention See p190.

Causes

Peptic ulcer disease (PUD) 35–50%.

Mallory–Weiss tear 15%.

Signs and symptoms (See pp56–62.) Haematemesis, or melaena, dizziness (espe-cially postural), fainting, abdominal pain, dysphagia? Hypotension (in young may be postural only), tachycardia (not if on β-blocker), 'JVP, 'urine output, cool and clammy, signs of chronic liver disease (p276), eg telangiectasia, purpura, jaundice. NB: ask about previous GI problems, drug use, alcohol.

Peripherally cool/clammy; capillary refill time >2s; urine output <0.5mL/kg/h.

Systolic BP <100mmHg; postural drop >20mmHg.

Insert two large-bore (14–16G) IV cannulae and take blood for FBC, U&E, LFT, clotting, and group & save.

Consider a urinary catheter and monitor hourly urine output.

If suspicion of varices (eg known history of liver disease or alcohol excess) then give terlipressin IV (1–2mg/6h for ≤2d) and initiate broad-spectrum IV antibiotics (eg piperacillin/tazobactam IV 4.5g/8h).

If endoscopic control fails, surgery or emergency mesenteric angiography/embo-lization may be needed. For uncontrolled oesophageal variceal bleeding, a Seng-staken–Blakemore tube may compress the varices, but should only be placed by someone with experience.

Signs of a rebleed:

Haematemesis or 'fresh' melaena (NB: it is normal to pass decreasing amounts of melaena for 24h post-haemostasis, as blood makes its way through the GI tract).

Primary care Prompt actions save lives. ₂If suspect meningitis arrange urgent transfer to secondary care. If a non-blanching rash is present, give benzylpenicil-lin 1.2g IM/IV before admitting.

Features

Meningism: neck stiffness, photophobia, Kernig's sign (pain + resistance on passive knee extension with hip fully flexed).

opisthotonus (p426, fig 9.46).

If • ICP, summon help immediately and inform ICU.

If features of menigism give dexamethasone 10mg/6h IV

Those who have kissed the patient's mouth. Give ciprofloxacin (500mg PO, 1 dose; child 5–12yrs: 250mg; child <5yrs: 20mg/kg to max 125mg).

Signs and symptoms

Fever.

Seizures.

Causes

Viral: HSV-1 & 2, arboviruses, CMV, EBV, VZV (varicella-zoster virus), HIV (seroconver-sion), measles, mumps, rabies, Japanese B encephalitis, West Nile virus, tick-borne encephalitis.

Investigations

Bloods: Blood cultures; serum for viral PCR (also throat swab and MSU); toxoplasma

LP: Typically moderately •CSF protein and lymphocytes, and ‘glucose (p822). Send

Management

Symptomatic treatment: eg phenytoin for seizures (p826).

Investigations CT/MRI p746 (eg ‘ring-enhancing’ lesion); •WCC, •ESR; biopsy. Treatment Urgent neurosurgical referral; treat •ICP (p820). If frontal sinuses or teeth are the source, the likely organism will be Strep. milleri (microaerophilic),

Investigations

ABG, U&E, Ca2+, FBC, ECG.

Management ►See fig 19.20. Basic life support—and these agents:[1’]

Seek ICU help: Paralysis and anaesthesia with eg propofol is required. Close moni-toring, especially respiratory function, is vital. Consider whether this could be pseudoseizures (p490), particularly if there are odd features (pelvic thrusts; resist-ing attempts to open lids and your attempts to do passive movements; arms and legs flailing around).

Open and secure the airway (adjuncts as necessary) Remove false teeth if poorly fitting

Give 2nd dose of lorazepam if no response after 10–20min

IV infusion: If seizures continue, start phenytoin,

General anaesthesia: Continuing seizures after 60–90mins of above therapies require expert help with paralysis (eg propofol infusion) and ventilation with continuous EEG monitoring in ICU. NB: ►never spend longer than 20min on someone with status epilepticus without having help at the bedside from an anaesthetist

Initial management ►See fig 19.21. Write full notes. Record times.

Involve neurosurgeons early, especially if ‘GCS, or if •ICP suspected.

Nurse semi-prone if no spinal injury; meticulous care to airway and bladder.

Focal neurological deficit.

Vomiting more than once.

>1m or >5 stairs •retrograde amnesia of >20min.

The patient has been intubated.

Clinical suspicion of cervical spine injury AND ANY OF: •age 65 years or older •high-impact injury •focal neurological deficit •paraesthesia in the upper or lower limbs.

Admit if: •new, clinically significant abnormalities on CT •GCS<15 after CT, regardless of result or continuing worrying signs (eg vomiting) •when CT indications met but CT unavailable •other concerns (eg drugs or alcohol, other injuries, CSF leak, shock, suspected non-accidental injury, meningism).

Complications Early: Extradural/subdural haemorrhage, seizures. Late: Subdural (p482), seizures, diabetes insipidus, parkinsonism, dementia.

Oxygen if saturations <92% or hypoxic on ABG

Treat seizures with lorazepam ! phenytoin (p826)

Investigations:

When? Where? How? Had a fit? Lucid interval? Alcohol?

If either is present, suspect basilar skull fracture: do CT

If detected, or if the patient has other indicators for neck imaging, im-mobilize the neck and get cervical spine X-ray or CT neck (see text)

Fig 19.21 Immediate management plan for head injury.

Causes

Primary or metastatic tumours.

Infection: meningitis, encephalitis, brain abscess.

Status epilepticus.

Signs and symptoms

Altered GCS: drowsiness, listlessness, irritability, coma.

Pupil changes (constriction at first, later dilation—do not mask these signs by using agents such as tropicamide to dilate the pupil to aid fundoscopy).

Investigations

CXR—any source of infection that might indicate abscess?

Management ►See fig 19.22. The goal is to ‘ICP and avert secondary injury. Urgent neurosurgery is required for the definitive treatment of •ICP from focal causes (eg haematomas). This is achieved via a craniotomy or burr hole. Also, an ICP monitor (or bolt) may be placed to monitor pressure. Holding measures are listed in fig 19.22.

Herniation syndromes

Cerebellar tonsil herniation is caused by •pressure in the posterior fossa forcing the cerebellar tonsils through the foramen magnum. Ataxia, VIth nerve palsies, and upgoing plantar reflexes occur first, then loss of consciousness,

irregular breathing, and apnoea. This syndrome may proceed very rapidly given the small size of, and poor compliance in, the posterior fossa.

Elevate the head of the bed to 20–40°

Give 20% solution 0.25–0.5g/kg IV over 10–20min (eg 5mL/kg). Effect is seen after r20min and lasts for 2–6h. Follow serum osmolality—aim for about 200mosmol/kg but don't exceed 210

Eg dexamethasone 10mg IV and follow with 4mg/6h IV/PO

Monitor the patient closely; consider monitoring ICP

Diagnosis

Hyperglycaemia (blood glucose >11.0mmol/L) or known DM.

Blood ketones >6mmol/L.

K <2.5mmol/L on admission.

Systolic BP <90mmHg.

Management ►Replace volume then correct metabolic defects,[1a] see fig 19.22.

Pitfalls in diabetic ketoacidosis

High WCC may be seen in the absence of infection.

Hyponatraemia is common, due to osmolar compensation for the hyperglycaemia.

Ketonuria does not equate with ketoacidosis. Anyone may have up to ++ketonuria after an overnight fast. Not all ketones are due to diabetes—consider alcohol if glucose normal. Always check venous blood ketones.

Serum amylase is often raised (up to ≈10) and non-specific abdominal pain is com-mon, even in the absence of pancreatitis.

ABC approach, 2 large-bore cannulae:

Insulin: Add 50 units human soluble insulin to 50mL 0.9% saline. Infuse con-tinuously at 0.1 unit/kg/h. Continue patient's regular long-acting insulin at usual doses and times; consider initiating long-acting insulin in newly diagnosed T1DM.

Check VBG (pH, HCO −, K+) at 2h, 4h, 8h, 12h, and 24h (or more frequent)

Avoid hypoglycaemia! When glucose <14mmol/L start 10% glucose at 125mL/h to run alongside saline and prevent hypoglycaemia

Find and treat infection/cause for DKA

Myxoedema coma The ultimate hypothyroid state before death.

Treatment: Preferably on ICU.

Correct any hypoglycaemia.

If infection suspected, give antibiotic, eg co-amoxiclav 1.2g/8h IV.

Further 9: T2 5–20mcg/4–12h IV until sustained improvement (r2–2d) then levothyrox-ine 50mcg/24h PO.

Hydrocortisone + IV fluids as needed (hyponatraemia is dilutional). Hyperthyroid crisis (thyrotoxic storm) Signs and symptoms: φ:ð≈4:1. Severe

Precipitants: Recent thyroid surgery or radioiodine; infection; MI; trauma.

IV access, fluids if dehydrated. NG tube if vomiting

High-dose digoxin may be needed to slow the heart, but ensure adequately β-blocked, give with cardiac monitoring

Continuing treatment: After 5d reduce carbimazole to 15mg/8h PO. After 10d stop propranolol and iodine. Adjust carbimazole (p218)

Precipitating factors Infection, trauma, surgery, missed medication.

Hydrocortisone 100mg IV stat.

Blood, urine, sputum for culture, then antibiotics if concern about infection.

Continuing treatment

Give IV fluids as guided by clinical state and to correct U&E imbalance.

Fludrocortisone may well be needed if the cause is adrenal disease: ask an expert.

Presentation Headache; ophthalmoplegia; ‘consciousness; hypotension; hypo-thermia; hypoglycaemia; signs of hypopituitarism (p222).

Only after hydrocortisone begun: liothyronine (L-tri-iodothyronine sodium), eg

Patients with phaeochromocytoma may have had undiagnosed symptoms for some time, but stress, abdominal palpation, parturition, general anaesthetic, or contrast media used in imaging can cause acute hypertensive crises.

Principle is combined a- and β-adrenoreceptor blockade, but a must be started first, as unopposed β-blockade can worsen hypertension.

Diagnosis Mainly from the history. The patient may not tell the truth about what has been taken. If there are any tablets with the patient, use MIMS Colour Index, EMIMS images, BNF descriptions, or the computerized system ‘TICTAC’ (ask pharmacy) to identify tablets and plan specific treatment.

Fast or irregular pulse: Salbutamol, antimuscarinics, tricyclics, quinine, or phe-nothiazine poisoning.

Hyperthermia: Amphetamines, MAOIS, cocaine, or ecstasy (p842).

Constricted pupils: Opiates (p842) or insecticides (organophosphates, p842).

Hypoglycaemia: (p824) Insulin, oral hypoglycaemics, alcohol, or salicylates.

- Osmolality: Alcohols (ethyl or methyl); ethylene glycol. See p668.

Empty stomach if appropriate (p840).

Continuing care Measure temperature, pulse, BP, and blood glucose regularly. Keep on cardiac monitor. If unconscious, nurse semi-prone, turn regularly. Catherize if the bladder is distended, or acute kidney injury (p298) is suspected, or forced diuresis undertaken. Consider ICU, eg if ‘respiration.

Present intentions. Do they still feel suicidal? Do they wish it had worked?

What are the patient’s resources (friends, family, work, personality)?

Mental Capacity Act or the Mental Health Act: (In England and Wales) may pro-vide for the detention of the patient against his or her will: see OHCS p406.

GI decontamination Recommended for many drugs. The treatment of choice is now activated charcoal rather than gastric lavage. If in doubt, consult Toxbase or Poisons Information Service.

Gastric emptying and lavage NB: we do not recommend gastric lavage is at-tempted unless specifically suggested by Toxbase or Poisons Information Service. If comatose, or no gag reflex, ask for an anaesthetist to protect airway with cuffed endotracheal tube. If conscious, get verbal consent.

Position the patient in left lateral position.

Confirm position in stomach (see p759).

Repeat until no tablets in siphoned fluid.

Haemodialysis This may be needed for poisoning from ethylene glycol, lithium, methanol, phenobarbital, salicylates, and sodium valproate.

Benzodiazepines Flumazenil (for respiratory arrest) 200mcg over 15s; then 100mcg at 60s intervals if needed. Usual dose range: 200–600mcg IV over 2–6min (up to 1mg; 2mg if on ICU). May provoke fits. Use only after expert advice. Mild: Dizziness, anxiety, tachycardia, nausea, drowsiness/confusion.

Digoxin Symptoms: 'Cognition, yellow-green visual halos, arrhythmias, nausea, and anorexia. If serious arrhythmias are present, correct hypokalaemia, and inactivate with digoxin-specific antibody fragments (DigiFab®). If load or level is unknown, give 20 vials (800mg)—adult or child >20kg.

Oral anticoagulants See p251. If major bleed, give vitamin K 5mg slow IV and prothrombin complex concentrate 50U/kg IV (or if unavailable, fresh frozen plasma 15mL/kg IVI). For abnormal INR with no (or minimal) bleeding, see BNF. If it is vital that anticoagulation continues, enlist expert help. Discuss with haematology. NB: coagulation defects may be delayed for 2–2d following ingestion.

HR, sweating, urinary incontinence), mutism, confusion, coma, •WCC, •CK; it may be treated with cooling. Dantrolene 1–2.5mg/kg IV (p572) (max 10mg/kg/day) can help, bromocriptine and amantadine are alternatives.

Paraquat poisoning (Found in weed-killers.) This causes D&V, painful oral ulcers, alveolitis, and renal failure. Diagnose by urine test. Give activated charcoal at once (100g followed by a laxative, then 50g/2–4h). ►Get expert help. Avoid O2 early on (promotes lung damage).

Monitor urine output and U&E (AKI pp298–9), LFT, CK, FBC, and coagulation (DIC p252). Metabolic acidosis may benefit from treatment with bicarbonate.

Hypertension can be treated with nifedipine 5–10mg PO or phentolamine 2–5mg IV. Treat hypotension conventionally (p790).

844

Management General measures: See p828. GI decontamination is recommended in those presenting <4h after overdose: give activated charcoal 1g/kg (max 50g).

If >8–24h and suspicion of large overdose (>7.5g) err on the side of caution and start acetylcysteine, stopping it if level below treatment line and INR/ALT normal.

Ongoing management

Consider referral to specialist liver unit guided by eg King's College criteria (BOX, p275).

Vomiting, dehydration, hyperventilation, tinnitus, vertigo, sweating.

Bloods. Paracetamol and salicylate level, glucose, U&E, LFT, INR, ABG, HCO2–, FBC. Salicylate level may need to be repeated after 2h, due to continuing absorption if a potentially toxic dose has been taken. Monitor blood glucose 1–2hrly, beware hypoglycaemia, if severe poisoning, monitor salicylate levels, serum pH, and U&E.

Dialysis may well be needed if salicylate level >700mg/L, and if AKI or heart failure, pulmonary or cerebral oedema, confusion or seizures, severe acidosis despite best medical therapy, or persistently •plasma salicylate. Contact nephrology early.

110

80

50

20

0 2 4 6 8 10 12 14 16 18 20 22 24

0.7

0.4

0.1

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>5% in children or the elderly, burns of special sites, chemical and electrical burns, and burns with inhalational injury.

>25% COHb or >20% if pregnant). SpO2 (oximetry) is unreliable in CO poisoning.

Acknowledgement

Definition Hypothermia implies a core (rectal) temperature <25℃.

Low room temperature: poverty, poor housing.

Autonomic neuropathy (p505); eg diabetes mellitus, Parkinson's.

Increased exposure to cold: falls, especially at night when cold.

The patient

If the patient is shivering then the hypothermia is mild, if they are not shivering despite temp <25℃ then the hypothermia is severe.

low-reading one PR. Is the rectal temperature <25℃? Infra-red ear thermometers

Treatment Use ABCDE approach (p779)—but don't expose to cold.

Warm IVI.

Consider urinary catheter (to monitor renal function).

<22℃ then mortality >50%.

Planning All hospitals have a detailed Major Incident Plan, but additionally the tasks of key personnel can be distributed on individual Action Cards.

Triage: See OHCS p800. There are several commercial systems available to label pa-tients so emergency personnel can see at a glance the scale of the incident. The key is to divide patients by the urgency of care/transfer to hospital:

Non-urgent (label GREEN = walking wounded/those who are stable and can wait)

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Early warning scores are scoring systems based on physiological parameters. The magnitude of the given score reflects how far the parameter varies from normal. The collated score from different parameters is used in:
the initiation of a timely and competent clinical response.

- Royal College of Physicians. National Early Warning Scores (NEWS): standardising the assessment of acute illness severity in the NHS. London: RCP, 2012.

Refer to local early warning scores where available.

2Ensure the safety of the patient and yourself.

2Airway: Head tilt (if no spine injury) and chin lift/jaw thrust.

Advanced life support See algorithm fig A2.

Shockable rhythm: VF/pulseless VT

2CPR is resumed immediately (20:2). Reassess pulse/rhythm only after 2 minutes of CPR.

2Continue CPR 20:2. Obtain IV access and secure airway. Once airway secure switch to continuous compressions and ventilation. Give adrenaline 1mg IV.

Drugs

Discontinuing resuscitation Needs clinical judgement: what is the likelihood of achieving a successful return of spontaneous circulation? If there is a shockable rhythm or a reversible cause then attempts are usually continued. It is reasonable to discontinue if asystole >20mins without a reversible cause. Ask for the opinion of others in the resuscitation team.

as part of end-of-life care (p12, p526)

Do not make judgements about the quality of life of others based on your own perception.

Drug Dose and frequency Notes

Anti-emetics

Drug Dose and frequency Notes

Antihistamines

Tranquillizers for non-psychotic behavioural disturbance